Academic English Mastery

A Comprehensive Textbook & Interactive Workbook

基于 LaTeX 的学术英语教学书与实战练习册

Academic English Curriculum Development Group

学术英语课程研制组

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Preface & Methodology / 前言与编写说明

本教材与配套实战练习册（*Academic English Mastery*）严格对标国际主流高水平期刊（Nature, Science, IEEE/ACM Transactions, Cell, Lancet, AER）及顶级学术会议（NeurIPS, ICML, ICLR, CVPR, ACL）的真实发表级学术英语规范，旨在打破中国学者与高年级研究生在学术写作中长久存在的“语法正确但生硬冗长、词汇孤立但缺乏语境支配价、句式散乱且头重脚轻、段落堆砌而信息断裂、过度依赖 AI 生成空洞套话”五大核心痛点。

The Four Pedagogical Pillars / 四大学术语言工程支柱

1. **词汇语境支配价与学术熟词生义（Lexical Valency & Polysemy）**：不再孤立背诵单词，而是透彻解析 1,577 组高频学术多义词的核心学术义项、配价框架（Valency Frames）与强搭配网络（Collocations）。
2. **句法骨架与抗头重脚轻工程（Syntactic Constituents & End-Weight Principle）**：以五大核心句型为基础，贯彻“短主语引导、谓语紧随、复杂修饰后置”的尾重原则（End-Weight），通过 36 条核心语法法则（G001–G036）和 24 组高分功能句型模板（EP001–EP024）重塑句法骨架。
3. **篇章信息流与图尔敏微论证体系（Given-to-New Flow & Toulmin Argumentation）**：严密实施“已知信息在前、新信息在后”的主位推进流（Given-to-New Theme Progression）与 Toulmin 论证模型（Claim, Data, Warrant, Backing, Rebuttal, Qualifier），系统杜绝机械连接词与 AI 空洞套话（L001–L030）。
4. **深度病段诊所与母语级重构图谱（Diagnostic Revision Clinic & Full-Text Anthology）**：提供 60 例涵盖真实学科场景的典型病段，通过七维病灶透视与四步重构手术实现 publication-ready 升华；精选 24 篇跨五大学科门类的完整学术范文（FT01–FT24），以段落蓝图链（BP01–BP48）深度解构全篇气韵。

How to Use This Book / 本书使用指南与学习路线图

本书采用模块化、双轨制编排，既可作为高校研究生与科研人员的学术写作系统教材，亦可作为高强度日常打卡训练的实战练习册。

Visual Hierarchy & Callout Legend / 排版视觉规范说明

ConceptBox (深海蓝)	核心理论定义、句法骨架法则、篇章论证机制与模型架构总览。
ExampleBox (雅致青)	真实高水平论文例句、经典发表级句式、病段与定稿对照呈现。
TipBox (暖琥珀)	常见学术陷阱警示、修辞去套话技巧、句法成分深度剖析。
ExerciseBox (常青绿)	课后实战演练、改写训练、每日打卡任务。
\DefectTarget	病段中的核心语病处高亮标记（玫瑰红粗体）。
\writinglines[11mm]	为手写习惯专门预留的 11mm 宽行距水平书写横线。

Four Specialized Study Tracks / 四大学习路径建议

- **路径 A：硕博论文发表快速攻坚（Fast-Track Publication）**：重点精读 Part II（句法架构）、Part III（篇章逻辑）、Part IV（48 段落蓝图）与 Part V（60 病段诊所），快速掌握学术正文各部分的标准段落展开套路与自查改写手术。
- **路径 B：60 天全景手写实战训练（Systematic Daily Practice）**：按 Part VII 练习册规划，每日完成填空、句子改写（SentenceRewriteItem）及段落重构（ParagraphRevision），在 11mm 书写线上扎实锤炼手写肌肉记忆，并每日对照 Part VIII 书末答案核验解析。
- **路径 C：国际会议演讲与答辩申请（Presentation & Rebuttal）**：精读 Part IV Track 6（蓝图 BP41–BP48）及 Part VI 第 19–20 章（Oral 演讲稿 FT07–FT08、ICLR 审稿回复 FT09、答辩应答全案 FT10、个人陈述 SoP FT11、课题申请书 FT12）。
- **路径 D：语言考试高分冲刺（IELTS 8.0+ / TOEFL 110+ / 六级 600+）**：研读 Part VI 第 21–22 章中的雅思小作文（FT13–FT15）、托福综合写作（FT16）、六级写作（FT19–FT20）及雅思/托福大作文（FT21–FT24）。

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Part I

Lexical Awakening & Academic Polysemy / 词汇激活与熟词生义精讲

Chapter 1

Academic Polysemy & Meaning Shifts in Context / 学术熟词生义与语境迁移

在学术英语写作中，最隐蔽、最普遍的错误并非罕见词拼写，而是“日常高频词”在进入科学语境后发生的语义迁移（Semantic Drift）。许多中国学者习惯用日常通识意义来推导学术文本，往往造成致命的语义偏离或修辞失位。本章系统解构学术熟词生义的认知模型，建立高频多义词核心矩阵，并深入四大核心语义簇进行全景辨析。

1.1 The Cognitive Paradigm of Academic Semantic Drift / 语义迁移机制与认知模型

学术语言要求极高的“指称明确性（Referential Precision）”与“配价严格性（Valency Rigor）”。当日常词汇进入学术语境时，通常经历以下三类认知重塑：

1. **隐喻映射与专业定域（Domain Restriction via Metaphor）**：将物理世界的动作投射到概念空间。例如，日常动作 *address*（寄信/写地址）投射到学术论文中，特指“系统性应对核心瓶颈与研究挑战”（address a critical bottleneck）。
2. **评价中立化与退情绪化（Neutralization of Affect）**：日常带有强烈情感倾向的词汇，在学术论文中转化为中性技术描述。例如 *critical*（批判的/吹毛求疵的）转变为“决定系统成败的关键阈值”（critical threshold）或“批判性实证审视”（critical examination）。
3. **安全降级与完整性丧失（Degradation of State）**：例如 *compromise*（日常为让步/妥协），在计算机与工程实验中转指“系统安全性或数据完整性遭受破坏”（compromise model robustness / data integrity）。

Core Principle: Everyday vs. Academic Valency & Semantic Frame

学术词汇的释义必须与语法配价（Valency Frame）强绑定：

`Everyday Sense` → [Informal / Transitive-Intransitive Flexible] ↔ `Academic Sense` → [Strict Complement / Fixed Preposition]

例如动词 *plant*（日常为种植物），在工业与具身智能控制中作为名词特指“受控工业装置或被控对象物理工厂（controlled plant）”，支配被动与控制闭环。

1.2 The High-Frequency Academic Polysemy Master Registry / 核心熟词生义对照总览

本节从 1,577 组学术多义条目中精选出最核心的 50 个经典高频熟词生义对照，涵盖词性、日常熟悉义、学术目标义以及易错使用陷阱，供读者快速查阅与自检。

Table 1.1: High-Frequency Academic Polysemy Master Registry (核心学术熟词生义对照总览)

Lemma	POS	Everyday Sense	Academic Target Sense	Trap & Boundary
a great deal of	quant. phr.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：大量的（quant.phr.）	一词多义与跨语境迁移对比：[V00014 quant.phr.] 大量的（例：The transcription required a great d
a handful of	quant. phr.	基础字面义：少量的	多义项/多词性完整映射：少量的；寥寥几个（quant.phr.）	词义与句法边界：模糊数量；方法和结果需精确计数时应给实际数字。
abandon	v.	v. 遗弃（人或地方）；半途而废	v. 因传感器校准失败、假设不成立或梯度发散而正式弃用（某种方法、假说或实验方案, abandon an approach）	学术写作中 abandon an approach 需交代具体实验/校准原因，不等于轻微修改（modify/alter），也不带情感贬义。
above	adv./adj.	基础字面义：上文	多义项/多词性完整映射：上文；上述的（adv./adj.）	词义与句法边界：只指确实在前出现的内容。
absence	n.	基础字面义：缺失	多义项/多词性完整映射：缺失；不存在（n.）	词义与句法边界：absence of；缺少记录未必表示现实事件不存在。
abstract	adj.	adj. 抽象的（与 concrete 具体相对）	(1) adj. 抽象概念的（an abstract concept，需给出操作性定义）；(2) n. 论文摘要（a structured abs	一词多词性多义：作形容词指抽象概念，作可数名词指论文摘要（摘要中的指标结论必须与正文实验表严格一致）。| 词表跨单元多义对比：[V00024 adj
access	n.	n. 通道；入口	(1) n. 获取/使用（算力、数据集、教育或医疗服务）的可及性或权限（access to，不可数）；(2) v. 访问、读取（数据库/内存，及物	词性与介词铁律：作名词必须接介词 to（access to services）；作动词直接接宾语（accessed the database），严禁写

Lemma	POS	Everyday Sense	Academic Target Sense	Trap & Boundary
accessible	adj.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：可获取的；易使用的（adj.）	一词多义与跨语境迁移对比：[V00031 adj.] 可获取的；易使用的（例：The online materials should be acces
accommodate	v.	v. 为……提供食宿；容纳（座位/乘客）	v. (1)（模型架构、理论框架或损失函数）容纳、适配新证据或异质性分布（accommodate new evidence / heterogene	熟词生义高频词：在机器学习与方法论语境中指“（模型/框架）兼容或纳入新特征/异质性”，切勿译为“提供住宿”；注意双 c 双 m 拼写。| 词表跨单元
according to	prep. phr.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：根据某人或某来源的说法（prep. phr.）	一词多义与跨语境迁移对比：[V00039 prep. phr.] 根据某人或某来源的说法（例：According to the authors, th
account for	phr. v.	phr. v. 说明……的去向	phr. v. (1) 解释、说明（误差、方差或观测差异）的原因（account for the discrepancy / variation）;	双核心学术义必须据宾语区分：接 discrepancy/variation 时表“解释原因”；接 proportion/percentage/ vari
accuracy	n.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：准确度（n.）	一词多义与跨语境迁移对比：[V00049 n.] 准确度（例：The team evaluated measurement accuracy.）；[V
accuracy / accurate	n./adj.	基础字面义：准确度	多义项/多词性完整映射：准确度／准确的（n. / adj.）	词义与句法边界：准确度不等于精密度 precision；含义取决于评估定义。
achievement	n.	基础字面义：成就	多义项/多词性完整映射：成就；达到（n.）	词义与句法边界：动词 achieve；不是所有 participation 都代表 achievement。
acknowledge	v.	v. 致谢；点头示意	v. (1) 在 Limitations 章节或 Rebuttal 中坦诚承认（样本局限、分布偏移或估计不确定性, acknowledge limit	acknowledge a limitation 体现科学严谨性，不等于推翻全文核心贡献。| 词表跨单元多义对比：[V00054 v.] 承认（例：

Lemma	POS	Everyday Sense	Academic Target Sense	Trap & Boundary
acknowledge / acknowledge-ment	v./n.	v. 致谢；点头示意	v. (1) 在 Limitations 章节或 Rebuttal 中坦诚承认（样本局限、分布偏移或估计不确定性, acknowledge limit	acknowledge a limitation 体现科学严谨性，不等于推翻全文核心贡献。
acquire	v.	基础字面义：获取	多义项/多词性完整映射：获取；习得（v.）	词义与句法边界：较正式；须避免学习效果保证。
acquisition	n.	基础字面义：获得	多义项/多词性完整映射：获得；习得（n.）	词义与句法边界：acquire 的名词形式。
adapt	v.	v. 改编小说；适应气候	v. (1) 将（预训练模型、控制策略或方法）调整、迁移以适配新领域或新任务（adapt a method for irregular observ	形近词必辨：adapt（调整修改以适配新分布/新工况）vs. adopt（直接采纳既有方法或超参数约定）。| 词表跨单元多义对比：[V00060 v
address	v.	n. 通信地址；演讲	v. 探讨、处理、针对性解决或回应（研究问题、模型局限或审稿人质疑, address a limitation / question / issue）	熟词生义第一高频词：作动词直接接宾语, 严禁写 *address to a problem；且 address a question 指“围绕该问题展开
adjust	v.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：调整；适应（v.）	一词多义与跨语境迁移对比：[V00070 v.] 调整；适应（例：Students needed time to adjust to the new
administration / administration building	n./phr.	n. 行政机关；校园行政楼（administration building）	n. (1)（问卷、评估量表或实验测试的）施测、发放与执行（the administration of a questionnaire）；(2)	方法章节熟词生义：the administration of the questionnaire 指“问卷的实施过程”, 绝非“管理问卷的行政机构”。
adopt	v.	v. 收养（儿童或宠物）	v. 采纳、采用（统一的评估协议、记号约定、基线架构或原则立场, adopt a convention / baseline / stance）	区分 adopt a convention（采用某约定）与 adapt a model（修改调整模型）。
adopt a stance	phr.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：采取立场（phrase）	一词多义与跨语境迁移对比：[V00076 phrase] 采取立场（例：The committee adopted a principled stan

Lemma	POS	Everyday Sense	Academic Target Sense	Trap & Boundary
adoption	n.	基础字面义：采用	多义项/多词性完整映射：采用；收养（n.）	词义与句法边界：动词 adopt；本例为采用，不是收养义。
advance	v.	v./n. 前进；预先（in advance）	v. 正式提出（待检验的科学假说、机制解释或理论论点，advance a hypothesis / theory）	The paper advances a hypothesis 意为“本文提出一项假说”，不是“让假说前进”。
advantage	n.	基础字面义：优点	多义项/多词性完整映射：优点；优势（n.）	词义与句法边界：an advantage of; advantage over 用于相对优势。
advent	n.	基础字面义：出现	多义项/多词性完整映射：出现；到来（n.）	词义与句法边界：常用 the advent of; advent 无对应动词。
adverse	adj.	adj. 逆风的	adj. 不利的、有害的；恶略工况或不良事件（adverse conditions / adverse events / adverse effect	形近词辨析：adverse（客观不利的，如具身智能在 adverse lighting/weather conditions 下测试） vs. aver
advocate	v.	n. 提倡者；辩护律师	v. 主张、提倡（某种评估规范或政策方案，直接接名词或动名词 advocate a policy / advocate providing clear	句法配价：接动作时用动名词 advocate doing sth，不用 *advocate to do sth。| 词表跨单元多义对比：[V0008
affect	v.	v. 感动；侵袭	v. 对（估计值、收敛稳定性或预测精度）产生影响（affect the estimate / outcome）	affect 为普通及物动词“影响”；effect 作名词为“效应（treatment effect）”，作正式及物动词时意为“使发生、实现（effe
affine	adj.	基础字面义：仿射的	多义项/多词性完整映射：仿射的；由线性映射加平移构成的（adj.）	词义与句法边界：严格线性映射保持原点，仿射映射可有平移项；affine 不能泛指所有非线性变换。
aggregate	n.	基础字面义：合计数	多义项/多词性完整映射：合计数；总量（n.）	词义与句法边界：也可作动词 aggregate observations。
aim	n.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：目标；目的（n.）	一词多义与跨语境迁移对比：[V00091 n.] 目标；目的（例：The aim of this paper is to compare annota

Lemma	POS	Everyday Sense	Academic Target Sense	Trap & Boundary
akin	adj.	基础字面义：类似的	多义项/多词性完整映射：类似的；相近的（adj.）	词义与句法边界：通常作表语，固定 akin to；说明相似点与区别，不能把类比当成严格等价。
albeit	conj.	基础字面义：虽然	多义项/多词性完整映射：虽然；尽管（conj.）	词义与句法边界：正式让步；常接简化短语。
allocate	v.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：分配（v.）	一词多义与跨语境迁移对比：[V00097 v.] 分配（例：The department allocated staff time to data c
allow	v.	v. 允许某人做某事	v. (1) 使（算法、研究者或系统）能够实现某功能（allow researchers to compare）；(2) allow for：将（	高频搭配区分：allow sb to do（使能够）vs. allow for variation/ curvature（顾及、容许某种变异或非线性）。
along with	prep. phr.	基础字面义：连同	多义项/多词性完整映射：连同；和……一起（prep. phrase）	词义与句法边界：不把后接成分当作改变主语单复数的 and。
alter	v.	v. 改衣服尺寸	v. 改变、更改（实验流程、估计趋势或参数设置，alter the procedure / estimated trend）	A drastic reduction in the observation period could alter the estimated tre
alternative	n./adj.	基础字面义：替代办法	多义项/多词性完整映射：替代办法；可供选择的（n./adj.）	词义与句法边界：alternative to 名词/动名词。
amenable	adj.	adj. 顺从的，好商量的	adj. （模型、方程或简化系统）适合接受某种数学/解析处理的（amenable to analytical treatment / convex o	理论推导高频词：固定接介词 to（be amenable to analytical treatment），意为“可用解析方法求解/处理”。
amount	n.	基础字面义：量	多义项/多词性完整映射：量；数量（用于不可数名词）（n.）	词义与句法边界：不用 an amount of participants；information 为不可数名词。
amount to	v. phr.	基础字面义：相当于	多义项/多词性完整映射：相当于；达到（v. phrase）	词义与句法边界：amount to 后接名词/动名词。
analogous to	adj. phr.	基础字面义：与……相似	多义项/多词性完整映射：与……相似；可类比于（adj. phr.）	词义与句法边界：介词 to；限定在所给分析中，不自动等同。

Lemma	POS	Everyday Sense	Academic Target Sense	Trap & Boundary
anarchist	n., adj.	基础字面义：无政府主义者	多义项/多词性完整映射：无政府主义者；无政府主义的（n., adj.）	词义与句法边界：教材词缀例词单列；例句为原创假设情境。词缀对应关系与拼写见同单元说明。
animated	adj.	基础字面义：活跃的	多义项/多词性完整映射：活跃的；热烈的（adj.）	词义与句法边界：不必含敌意；不能自动译成愤怒争吵。
annual	adj.	已学基础义或同词元其他语境义（共关联 2 个词表词义单元）	当前单元特定学术义：每年的；一年一次的（adj.）	一词多义与跨语境迁移对比：[V00128 adj.] 每年的；一年一次的（例：The association holds an annual conf
anomalous	adj.	基础字面义：异常的	多义项/多词性完整映射：异常的；偏离通常模式的（adj.）	词义与句法边界：对应名词 anomaly；异常不必然是假值或误差，应依据预先说明的规则处理。
anomaly	n.	基础字面义：异常现象	多义项/多词性完整映射：异常现象；偏离预期模式的情况（n.）	词义与句法边界：可数名词，复数 anomalies；异常可能来自真实事件、测量误差或模型局限，不能直接判为错误。

1.3 Deep-Dive Semantic Clusters / 核心多义动词与名词深度解析

1.3.1 Cluster A: Research Intent & Methodological Action / 研究意图与方法操作

Core Lemmas: `address`, `accommodate`, `treat`, `yield`, `plant`, `observe`

Lemma Focus: address (v.)

Everyday Sense: 写地址/发表演说 $\Longrightarrow$ **Academic Sense**: **系统性应对、攻克核心瓶颈与学术挑战**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

The proposed framework addresses the long-standing bottleneck of gradient vanishing in deep architectures.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 日常表地址，学术中作及物动词（直接接宾语，严禁使用 *address to），常接 challenge, issue, bottleneck。

Lemma Focus: accommodate (v.)

Everyday Sense: 提供住宿/容纳 $\Longrightarrow$ **Academic Sense**: **（模型架构、理论框架）适配、容纳异质性分布或变化**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

The attention mechanism is extended to accommodate irregular sampling intervals across sensor streams.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 学术中表示模型包容与适配复杂现实条件，避免使用口语词 fit into 或 deal with。

Lemma Focus: yield (v.)

Everyday Sense: 屈服/出产 $\Longrightarrow$ **Academic Sense**: **（实验或理论推导）产出定量结果、带来性能提升**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

Ablation experiments demonstrate that removing the recurrent unit yields a 14% increase in inference throughput.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 作及物动词时接实验结论或数据增益；不可混淆于日常“投降、让步”（yield to）。

1.3.2 Cluster B: Data Integrity, Vulnerability & Security / 数据完整性与安全脆弱性

Core Lemmas: `compromise, corrupt, contaminate, sound`

Lemma Focus: compromise (v./n.)

Everyday Sense: 折中/妥协 ⟹ **Academic Sense**: **破坏、削弱系统的安全性、鲁棒性或数据完整性**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

Out-of-distribution noise severely compromises the decision boundaries of standard classifiers.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 在学术论文中，compromise robustness/integrity 意为“使……受损”，绝非中性的政治妥协。

Lemma Focus: corrupt (v./adj.)

Everyday Sense: 腐败的/贪污的 ⟹ **Academic Sense**: **（数据包/特征表示）损坏、失真或被噪声污染**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

Adversarial perturbations corrupt high-frequency spectral features without altering human perception.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 专指数据丢失或比特翻转，在机器学习与分布式系统中具有极其精确的技术含义。

Lemma Focus: sound (adj./n.)

Everyday Sense: 声音/听起来 ⟹ **Academic Sense**: **（理论/证明/逻辑）严密成立的、可靠无懈可击的**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

The authors provide a mathematically sound proof demonstrating the asymptotic stability of the closed-loop system.

Usage Boundary & Diagnostic Warning

Reviewer Trap: a sound proof / sound methodology 指经过严密数学或实证检验的方案，不可用 good 或 valid 浅层替换。

1.3.3 Cluster C: Argumentation, Stance & Nuance / 论证立场与审慎修辞

Core Lemmas: `appreciate, critical, robust, reconcile, harbor`

Lemma Focus: appreciate (v.)

Everyday Sense: 感激/欣赏 $\Longrightarrow$ **Academic Sense**: **（学术认知）充分理解、领会深层机理与内在复杂性**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

To fully appreciate the trade-off between latency and accuracy, one must examine the memory access overhead.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 在文献综述或讨论中，appreciate the complexity 指“透彻认识到某问题的复杂本质”。

Lemma Focus: critical (adj.)

Everyday Sense: 批评的/爱挑剔的 $\Longrightarrow$ **Academic Sense**: **决定性的、关键极限阈值的；具有批判性实证视角的**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

Selecting the appropriate regularization hyperparameter is critical to preventing overfitting in sparse regimes.

Usage Boundary & Diagnostic Warning

Reviewer Trap: critical value / threshold 为统计与物理临界值；critical perspective 为审慎客观的批判视角。

Lemma Focus: reconcile (v.)

Everyday Sense: 和解/使调和 $\Longrightarrow$ **Academic Sense**: **调和两种看似冲突的实验观察或理论假设**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

Our empirical findings reconcile the discrepancy between previous asymptotic bounds and finite-sample behavior.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 学术论文中的破局利器：当两大学派各执一词时，本文提出新机制“调和分歧”。

1.3.4 Cluster D: Epistemic & Disciplinary Nominals / 核心学科名词与范式概念

Core Lemmas: `discipline, account, vehicle, charge, condition`

Lemma Focus: discipline (n./v.)

Everyday Sense: 纪律/惩罚 $\Longrightarrow$ **Academic Sense**: **学术学科、知识专业分支（academic discipline）**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

Interdisciplinary collaboration between computational biology and robotics has unlocked new diagnostic paradigms.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 作名词复数 disciplines 指各个学术专业领域，不可与 training/punishment 混淆。

Lemma Focus: account (n./v.)

Everyday Sense: 账户/账单 $\Longrightarrow$ **Academic Sense**: **（学术解释）系统性理论阐释与机理说明（give an account of）**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

The paper provides a unified theoretical account of how positional embeddings affect attention entropy.

Usage Boundary & Diagnostic Warning

Reviewer Trap: account for 既可表示“解释……的原因”，亦可表示“占……比例（account for 45%）”。

Lemma Focus: condition (v./n.)

Everyday Sense: 条件/状态 $\Longrightarrow$ **Academic Sense**: **以……为先验条件约束（conditioned on / conditional probability）**

Grammar & Valency Frame: 及物/不及物严格受约，后接限定性学术宾语或介词。

Authentic Literature Exemplar

The trajectory generator is conditioned on multimodal sensory inputs to ensure collision-free motion.

Usage Boundary & Diagnostic Warning

Reviewer Trap: 在生成模型与贝叶斯推断中，be conditioned on 是高频核心及物搭配。

Chapter 2

Academic Valency Shifts & Morphological Derivatives / 动词配价转换与形态派生

词汇不仅是孤立的语义单元，更是句法结构的指挥棒。动词的支配价（Verb Valency）决定了句子的核心骨架是 SV、SVO 还是 SVOC。同时，学术英语拥有高度能产的形态构词系统。本章完整剖析 24 大核心学术词缀体系，构建名、动、形、副四元派生图谱，并严格界定动词配价的及物/不及物边界。

2.1 The 24 Golden Academic Affixes & Structural Productivity / 24 大核心学术词缀体系

学术构词法是扩大高端学术词汇量与进行精准概念封装的基石。在标准学术训练集中，涵盖了最具学术生产力的 24 个黄金词缀（11 个后缀与 13 个前缀）。下表列出全部 24 个词缀的语义功能、学术派生典型范例及易错陷阱。

Table 2.1: The 24 Golden Academic Affixes Master Table (24 大核心学术词缀详解)

Affix	Type	Morphological Function	Academic Exemplars	Usage Trap & Notes
-able	Suffix (后缀)	能够被…	*predictable, identifiable, portable*	词义与句法边界：不是所有动词都能自由加 -able；predictable 不等于已被预测。
-ant	Suffix (后缀)	有某种作用的物质或人等	*coolant, antidepressant*	学术语境与搭配约束：anti-depressant / antidepressant 为药物类别词，不由词缀推断疗效。
-cy	Suffix (后缀)	状态	*accuracy, urgency*	词义与句法边界：accuracy 是名词，accurate 是形容词。
-ee	Suffix (后缀)	受某动作影响的人	*employee, trainee*	学术语境与搭配约束：并非所有 -ee 词都可机械解释为被动接受者。
-hood	Suffix (后缀)	状态	*childhood, adulthood, likelihood*	词义与句法边界：likelihood 的形容词相关形式是 likely；注意拼写。

Affix	Type	Morphological Function	Academic Exemplars	Usage Trap & Notes
-ify	Suffix (后缀)	使具有某种性质	*clarify, purify, unify*	学术语境与搭配约束：clarification / purification / unification 为相应名词。
-ise / -ize	Suffix (后缀)	使处于某状态	*modernise, modernize, colonise*	学术语境与搭配约束：英美拼写及出版社偏好不同；全文保持一致。
-ism / -ist	Suffix (后缀)	思想、学说、体系，或行为、性质	*heroism, modernism, anarchist*	词义与句法边界：modernist 可为名词或形容词；Marxism 大写。
-less	Suffix (后缀)	没有…的	*meaningless, colourless, fearless*	学术语境与搭配约束：词义由语境限定；colourless 的美式拼写为 colorless。
-ocracy / -ocrat	Suffix (后缀)	政体	*autocracy, autocrat, democracy*	词义与句法边界：autocrat 指独裁统治者；democrat 可指民主制度支持者。democratic、democratise/d
-proof	Suffix (后缀)	防…	*waterproof, soundproof, dustproof*	词义与句法边界：某些构词表示抗性；实际性能仍取决于规范与试验。
anti-	Prefix (前缀)	反对	*anti-bacterial, anti-pollution*	词义与句法边界：前缀；有些词写作一词，如 antibacterial；拼写以词典/格式要求为准。
bi-	Prefix (前缀)	二	*bilingual, biannual, bimonthly*	词义与句法边界：biannual 通常一年两次；biennial 每两年一次；bimonthly 有歧义，应写 every two
co- / col- / com- / con-	Prefix (前缀)	共同	*co-author, cooperate, collaborate*	词义与句法边界：构词形式已固定，不能任意替换前缀。
contra- / counter-	Prefix (前缀)	反对	*contradict, counter-claim*	词义与句法边界：counterclaim 也常连写；contradict 意为反驳或与…矛盾。

Affix	Type	Morphological Function	Academic Exemplars	Usage Trap & Notes
eco-	Prefix (前缀)	与生态或环境有关	*eco-tourism, eco-friendly*	学术语境与搭配约束：eco-friendly 是可被检验的评价，不能仅凭标签认定环保。
hyper-	Prefix (前缀)	过度	*hyperactive, hyperinflation*	词义与句法边界：hyperinflation 是经济学术语，不能任意指一般涨价。
il- / im- / in- / ir-	Prefix (前缀)	不	*illogical, impossible, indistinct*	词义与句法边界：否定前缀形式取决于固定词形，不能自由选择。
inter-	Prefix (前缀)	在…之间	*interrelated, interact, interconnection*	词义与句法边界：不要与 intra-（内部）混淆。
mal-	Prefix (前缀)	不良	*malfunction, malpractice*	词义与句法边界：malpractice 通常指专业人士的失职或不当行为。
multi-	Prefix (前缀)	多	*multilingual, multi-storey*	学术语境与搭配约束：multivariate 指多变量，不等于 multilingual。
over-	Prefix (前缀)	过度	*overload, overworked, overheat*	学术语境与搭配约束：可构成动词或分词；须检查具体词形。
post-	Prefix (前缀)	在…之后	*post-war, post-colonial*	学术语境与搭配约束：某些学科术语亦可连写，如 postcolonial。
pre-	Prefix (前缀)	在…之前	*pre-industrial, pre-war*	学术语境与搭配约束：时间界限须明确。

2.2 Four-Way Morphological Derivation Matrices / 名、动、形、副四元派生图谱

在学术改写（Paraphrasing）与遵循尾重原则（End-Weight）时，作者需要自由在不同词性间切换，将沉重的谓语从句转化为紧凑的介词短语或副词修饰。下表展示 12 组核心学术词族的四元形态派生矩阵：

Verb (动词)	Noun (名词)	Adjective (形容词)	Adverb (副词)
assume	*assumption*	*assumptive / assumed*	*assumptively*
define	*definition*	*definitive / defining*	*definitively*
evaluate	*evaluation*	*evaluative*	*evaluatively*
hypothesize	*hypothesis*	*hypothetical*	*hypothetically*
infer	*inference*	*inferential*	*inferentially*
methodize	*method / methodology*	*methodological*	*methodologically*
observe	*observation*	*observational*	*observationally*
quantify	*quantity / quantification*	*quantitative*	*quantitatively*
robustify	*robustness*	*robust*	*robustly*
signify	*significance*	*significant*	*significantly*
stabilize	*stability / stabilization*	*stable*	*stably*
validate	*validity / validation*	*valid*	*validly*

2.3 Verb Valency Frameworks & Complementation Constraints / 动词配价框架与介词搭配规约

动词是句子的引擎。学术写作中最严重的语法缺陷之一就是混淆动词的及物性（Transitivity）与介词搭配规约。

2.3.1 Strictly Transitive Verbs Misused with Prepositions (*discuss about, *address to)

受母语负迁移影响，许多初学者倾向于在纯及物动词后添加多余的介词，导致严重的句法违规：

- **discuss**: 严格及物，直接接宾语。***discuss about the mechanism** $\Longrightarrow$ **discuss the mechanism**.
- **address**: 严格及物，直接接挑战。***address to the problem** $\Longrightarrow$ **address the problem**.
- **emphasize**: 严格及物，直接接重点。***emphasize on the importance** $\Longrightarrow$ **emphasize the importance**.
- **investigate**: 严格及物。***investigate into the phenomenon** $\Longrightarrow$ **investigate the phenomenon**.

2.3.2 Strictly Intransitive & Ergative Verbs (*is occurred, *is arisen)

纯不及物动词与作格动词（Ergative Verbs）不具备被动语态形态，误用被动语态是审稿人一眼看穿的低级错误：

- **occur / arise / exist**: 绝对无被动语态。***The issue was occurred in Stage 2** $\Longrightarrow$ **The issue occurred in Stage 2**.
- **converge / deteriorate**: 作格动词，主干自然呈现。***The loss is converged** $\Longrightarrow$ **The loss converges to a local minimum**.

2.3.3 Prepositional Repertoire Restrictions (Fixed Prepositional Frames)

学术核心动词具有强烈的介词固定搭配，错置介词会彻底破坏句式语感：

- **attribute ... to ...**: 将……归因于……（The performance improvement is attributed to the decoupled attention head.）
- **correlate with**: 与……具有相关性（Sensor temperature correlates strongly with inference latency.）
- **preclude ... from ...**: 阻止/排除……免于……（The strict memory budget precludes the model from deploying full attention layers.）

Chapter 3

High-Frequency Academic Collocation Networks & Discipline Registries / 高频学术搭配网络与学科语域图谱

词汇从不单独作战，而是以搭配簇（Collocation Chunks）的形式成块存储与输出。地道的学术语感来源于对“互信息量高（High Mutual Information）”的黄金搭配的熟练调动。本章系统解构学术四大核心句法搭配类型，并按五大学科门类构建专属学术语域图谱。

3.1 The Grammar of Collocation & Mutual Information / 学术搭配强度的语言学原理

计算语言学中的搭配强度通常采用互信息量（Mutual Information, MI）或 *t*-score 衡量。高频地道的学术搭配不仅让阅读体验流畅，更能让审稿人迅速建立专业信任感。

3.2 Universal High-Impact Collocation Lexicon / 通识学术高频强力搭配矩阵

学术通识写作中，主要依赖四种核心句法搭配形态：

3.2.1 Verb + Noun Collocations (动宾强搭配)

Collocation Phrase	Chinese Meaning	Exemplar Sentence
advance a hypothesis	提出学术假说	*We advance a hypothesis that non-stationary drift originates in low-frequency harmonics.*
mitigate a risk / bias	缓解风险/偏差	*The proposed normalization mitigates the risk of gradient explosion.*
bridge a gap	弥合研究空白	*This work bridges the gap between theoretical sample complexity and empirical throughput.*
corroborate an observation	佐证观测结果	*Ablation metrics corroborate our initial theoretical derivation.*

3.2.2 Adjective + Noun Collocations (形名高阶搭配)

Collocation Phrase	Chinese Meaning	Exemplar Sentence
compelling evidence	令人信服的实证证据	*The benchmark trials provide compelling evidence of generalization capability.*
empirical rigor	实证严谨性	*Methodological validity hinges on empirical rigor across cross-validation splits.*
untenable assumption	站不住脚的假设	*Assuming independent noise across sensors is an untenable assumption in real grids.*
formidable challenge	巨大艰巨的技术挑战	*Latency constraints present a formidable challenge for edge deployment.*

3.2.3 Noun + Preposition Networks (名词介词网络)

Collocation Phrase	Chinese Meaning	Exemplar Sentence
implication for	对……的深远启示	*These findings carry substantial implications for clinical decision pipelines.*
discrepancy between	……之间的实测差异	*The discrepancy between simulated and real trajectories stems from friction.*
prerequisite for	……的先决必要条件	*High sample diversity is an indispensable prerequisite for policy stability.*

3.2.4 Verb + Preposition Networks (动介固定搭配)

Collocation Phrase	Chinese Meaning	Exemplar Sentence
adhere to	严格遵守（协议/准则）	*The training pipeline strictly adheres to Open-X Embodiment benchmarks.*
deviate from	偏离（理论基线）	*Empirical trajectory plots deviate from asymptotic curves under heavy load.*
reconcile with	与……达成一致调和	*Our modular decomposition reconciles with classic control theory proofs.*

3.3 Disciplinary Lexical Registries / 五大学科门类专属学术语域图谱

不同学术领域拥有截然不同的专业词库与语域要求。下述图谱列出五大核心学科门类的高频高阶搭配：

Domain I: Computer Science, AI & Quantitative Modeling (计算机与人工智能): *latent representation, loss convergence, attention entropy, spectral decomposition, benchmark fidelity, out-of-distribution generalization, parameter ablation, policy optimization.*

Domain II: **Biomedical & Life Sciences (生物医学与生命科学)**: *clinical cohort, phenotypic variance, therapeutic efficacy, biomarker expression, statistically significant, double-blind trial, pathological pathway, ethical clearance*.

Domain III: **Economics, Finance & Management (经济金融与管理科学)**: *econometric specification, endogenous instrument, equilibrium dynamics, marginal propensity, liquidity premium, empirical proxy, exogenous shock, causal identification*.

Domain IV: **Engineering, Materials & Physical Sciences (工程材料与物理科学)**: *finite element analysis, crystalline lattice, mechanical tensile strength, thermodynamic equilibrium, acoustic impedance, degradation threshold, constitutive relation*.

Domain V: **Humanities, Education & Social Sciences (人文社科与教育学)**: *epistemological framework, qualitative hermeneutics, socio-demographic disparity, pedagogical intervention, thematic coding, discourse hegemony, empirical ethnographic study*.

Part II

Syntactic Architecture & Anti-Top-Heavy Grammar / 句法骨架与防头重脚轻语法

Chapter 4

Five Foundational Sentence Patterns in Scholarly Writing / 学术写作五大核心基本句型

一切复杂的学术长难句，本质上均由五大基本句型（Five Canonical Clause Patterns）经过扩展、修饰后置与从属复合演化而来。掌握句子核心主干（Subject, Predicate Verb, Object, Complement）的定位与成分配比，是杜绝中式流水句与残缺句的第一道防线。

4.1 Constituent Foundations & Clause Architecture / 句子骨架与核心成分解析

英语是典型的“主干驱动型（Head-Driven）”语言。一个合格的学术句子必须具备清晰的主谓核心：

- **主语 (Subject, S)**: 承载核心学术实体，力求紧凑凝练（建议 ≤ 5 词）。
- **谓语动词 (Predicate Verb, V)**: 决定句型骨架与后置补足成分配价（Valency）。
- **宾语 (Object, O)**: 动作的直接或间接承受者。
- **补足语 (Complement, C)**: 对主语或宾语状态、性质进行定性定量的评测补充。
- **状语 (Adverbial, A)**: 交代时间、空间、因果、条件、程度等上下文边界。

4.2 The Five Canonical Clause Patterns in Research Literature / 五大基本句型学术深度应用

4.2.1 Pattern I: SV (Subject + Intransitive Verb) — 自然过程与客观演变

Structural Breakdown & Academic Utility

SV 句型由不及物动词引导，常用于客观描述系统自发变化、自然衰减或数学收敛过程。核心谓语动词后可后置介词短语状语。

Publication-Grade Exemplar

The training loss converges rapidly during the initial five epochs.

4.2.2 Pattern II: SVO (Subject + Transitive Verb + Object) — 主动干预与实验操作

Structural Breakdown & Academic Utility

SVO 是学术写作中最坚实、最常见的主动干预句型。主语代表模型、算法或作者团队，谓语动词直接施加动作于客体。

Publication-Grade Exemplar

The transformer architecture captures long-range temporal dependencies across sensor channels.

4.2.3 Pattern III: SVC (Subject + Linking Verb + Complement) — 属性归属与状态界定

Structural Breakdown & Academic Utility

SVC 用于刻画客体的内在性质。系动词（remain, prove, appear, be of）连接主语与表语，表语赋予主语学术价值评估。

Publication-Grade Exemplar

The estimated parameter remains stable across varied noise levels.

4.2.4 Pattern IV: SVOO (Subject + Dative Verb + Indirect Object + Direct Object) — 资源分配与功能赋予

Structural Breakdown & Academic Utility

双宾语句型体现资源供给或机制赋予，通常可转换为 SVO + Preposition 结构（provides X for Y / gives X to Y）。

Publication-Grade Exemplar

The platform provides researchers with an integrated evaluation environment.

4.2.5 Pattern V: SVOC (Subject + Complex Transitive + Object + Complement) — 批判性裁定与效能定性

Structural Breakdown & Academic Utility

SVOC 是学术评价中极具表现力的高阶句型。宾语补足语（suboptimal）直接定性动作导致的结果状态，论证力量极其强劲。

Publication-Grade Exemplar

Ablation trials render the initial baseline suboptimal under heavy computation load.

Chapter 5

Principle of End-Weight and Information Distribution / 尾重原则与信息焦点分布

初学者在学术写作中最致命的毛病就是“头重脚轻（Top-Heavy Subject）”：将一个长达十几词的动名词短语或主语从句堆在句首，谓语动词极度延后，迫使读者长时间挂起短时认知记忆，产生极其沉重的阅读负荷。本章系统讲解认知语言学中的尾重原则（End-Weight Principle）以及三项外科重构手术。

5.1 Cognitive Processing Load & The Anti-Top-Heavy Imperative / 认知负荷与防头重脚轻原理

英语母语读者的视觉聚焦模型遵循“短主语启动 → 强谓语连接 → 复杂信息尾部汇聚”的认知节奏。若句首主语超过 8 个词，读者大脑在定位到核心动作前已产生认知疲劳。遵循尾重原则即是将长短语、定语从句及复杂条件后移至谓语动词之后。

The Golden Rule of End-Weight (尾重黄金法则)

在学术正式书面语中，轻量主干居前，重量信息居后：

```
[Compact Subject (≤5 words)] + [Active Verb] + [Heavy Object / Clause / Prepositional Modifiers]
```

5.2 Three Reconstructive Surgeries for Top-Heavy Subjects / 头重脚轻病句的三大外科重构手术

5.2.1 Surgery 1: Extraposition with Anticipatory ’It’ (形式主语 it 替代法)

Defective Top-Heavy vs. Balanced Reconstruction

[Top-Heavy Defect]:

That modern deep reinforcement learning policies suffer from severe distribution shifts when deployed in real robots is widely recognized.

[End-Weight Reconstruction]:

It is widely recognized that modern deep reinforcement learning policies suffer from severe distribution shifts when deployed in real robots.

Surgical Rationale: 将沉重的主语从句 That ... 移至句尾，句首使用轻巧的形式主语 it 占位，主谓核心 it is widely recognized 在前 4 词内迅速建立。

5.2.2 Surgery 2: Active Reporting Subject Extraction (强施事主干提取法)

Defective Top-Heavy vs. Balanced Reconstruction

[Top-Heavy Defect]:

The observation that the gradient magnitude increases non-linearly with batch size was made by the team during training.

[End-Weight Reconstruction]:

The team observed that gradient magnitude increases non-linearly with batch size during training.

Surgical Rationale: 将隐藏在介词短语或被动语态中的真实施事者（The team）提取为全句主动语态主语，动词由 was made 升华为直接宾语从句引导词 observed。

5.2.3 Surgery 3: Inversion & Heavy-NP Shift (完全倒装与长成分后移法)

Defective Top-Heavy vs. Balanced Reconstruction

[Top-Heavy Defect]:

Several fundamental limitations regarding out-of-distribution generalization under non-stationary sensor dynamics lie at the heart of this dispute.

[End-Weight Reconstruction]:

At the heart of this dispute lie several fundamental limitations regarding out-of-distribution generalization under non-stationary sensor dynamics.

Surgical Rationale: 将句末简短的地点介词短语 At the heart of this dispute 置于句首充当逻辑主位，谓语动词 lie 紧随其后，超长并列主语后置平稳落地。

5.3 The Given-to-New Contract at the Clause Level / 句内“已知—新信息”尾端汇聚法则

尾重原则与篇章的“已知—新信息契约（Given-to-New Contract）”高度统一：已知信息作为简短主位引领句首，新信息包含大量技术细节与修饰，自然居于句尾承重。

Chapter 6

Academic Grammar Mastery: The 36 Golden Rules (G001–G036) / 学术语法全景图谱：36 条黄金法则

学术语法绝非机械死板的条条框框，而是确保科学交流准确、无歧义且具备严密学术严谨度的工程规约。本章完整收录标准学术体系的 36 条黄金语法法则（G001–G036），分为六大进阶教学集群，逐条提供原则阐释、正反例对比与警示。

6.1 Cluster 1: Clause Architecture & Basic Mechanics (G001–G006) / 句法架构与核心骨干

6.1.1 G001: 五大基本句型（SV, SVO, SVC, SVOO, SVOC）与动词配价边界

Core Grammar Principle & Theoretical Framework

Category: `C1_basic_clause_patterns`

Principle: 英语句子的核心骨架由谓语动词的及物性与配价决定：不及物动词构成 SV（Evidence accumulates / A problem arose），系动词构成 SVC（The label is meaningless），单及物动词构成 SVO（The team abandoned the approach），双及物动词构成 SVOO，复合宾语动词构成 SVOC（The pronoun makes the final sentence ambiguous）。严禁把不及物动词当及物动词用，或把及物动词误加介词。

Operational Rules:

- 不及物动词（arise, emerge, occur, differ, fluctuate, accumulate, oscillate）后不直接接名词宾语，需用状语或介词短语扩展。
- 单及物动词（address, discuss, access, affect, investigate, assess, warrant）直接接宾语，禁止按中文习惯误加介词（禁止写 *discuss about, *address to, *access to 作动词）。
- SVOC 结构中宾语补足语用形容词或分词说明宾语状态(keep the room temperature constant, make the task easier)。

Authentic Examples & Contrasts

SV: A problem arose when the data formats changed.

SVO: This paper addresses unequal access to educational resources.

Diagnostic Warning & Common Errors

- 误将不及物动词用于被动或直接加宾语：*A problem was arisen / *differ two groups。
- 在及物动词后添加冗余介词：*discuss about the findings / *address to the limitation。
- SVOC 中误用副词作宾补：*keep the room temperature constantly（应为 constant）。

6.1.2 G002: 可数名词、不可数名词及多义词在不同义项下的可数性切换

Core Grammar Principle & Theoretical Framework

Category: `C4_A2_nouns_countability_across_senses`

Principle: 不仅要区分绝对不可数名词（evidence, information, research, literature, equipment, apparatus, documentation, feedback, knowledge）与可数名词（study, finding, discrepancy, anomaly, covariate），更要掌握“同一单词在不同词义下的可数性切换”（Rule A.2）。

Operational Rules:

- 不可数学术名词不加 a/an 或复数 -s（严禁写 *an evidence, *researches, *equipments, *feedbacks）；表单项用 a piece/item of evidence, a research study, a piece of equipment。
- 一词多义下的可数性切换：discipline 表“纪律”不可数，表“学科”可数（two academic disciplines）；work 表“研究工作”不可数，workings 表“计算步骤”；experience 表“经验”不可数，表“经历”可数（experiences in childhood）；paper 表“纸”不可数，表“论文”可数。
- 量词精确匹配：修饰不可数用 a great deal of, a large amount of, much, less and less；修饰可数复数用 a large number of, a handful of, many, fewer and fewer。

Authentic Examples & Contrasts

The transcription required a great deal of effort.

A large number of records still require checking.

Diagnostic Warning & Common Errors

- 量词与可数性错配：*a great deal of participants / *an amount of respondents / *fewer time。
- 不可数名词误加复数：*many evidences / *new equipments / *useful feedbacks。

6.1.3 G003: 主谓一致、数量短语一致与邻近名词干扰排除

Core Grammar Principle & Theoretical Framework

Category: `C2_subject_verb_agreement`

Principle: 复杂主语的主谓一致取决于核心名词（Head Noun）而非后置介词短语中的邻近名词。尤其需区分 a number of 与 the number of，以及 along with / accompanied by / as well as 连接的附加成分。

Operational Rules:

- the number of + 复数名词作主语，谓语用单数；a (large/small) number of + 复数名词作主语，谓语用复数。
- 主语后接 along with, together with, accompanied by, as well as 时，谓语动词与前面的核心主语保持一致。
- the bulk of / a proportion of / a percentage of + 名词作主语时，谓语单复数取决于 of 后的名词是不可数/单数还是复数可数名词。
- means, series 单复数同形，谓语由限定词决定（a series of observations was vs two series were）。

Authentic Examples & Contrasts

The number of completed interviews was recorded separately.

A small number of respondents left this item blank.

Diagnostic Warning & Common Errors

- 被 of 短语中的复数名词误导：*The analysis of interview responses show（应为 shows）。
- 把 as well as 误当成 and 改变主语单复数。

6.1.4 G004: 冠词系统（a/an, the, 零冠词）与学术特指/泛指边界

Core Grammar Principle & Theoretical Framework

Category: `C4_articles_and_specificity`

Principle: 冠词决定名词短语的指称范围：单数可数泛指用 a/an；指代本研究特定样本/模型（the sample, the model）或带 of 后置限定的属性（the accuracy of the measurements, the nature of the task）用 the；复数可数或不可数表普遍概念用零冠词。

Operational Rules:

- 单数可数名词在句中不可裸奔（bare singular），必须带冠词或限定词（a limitation, the protocol, each trajectory）。
- 注意冠词改变词义：nature（零冠词）指“大自然”，而 the nature of the problem 指“问题的基本性质”；in charge of（主动负责）vs in the charge of（由他人看管负责）。

- 图表编号直接作专名指称时不加 the（see Table 1, Figure 2 shows），但普通名词需加冠词（in the table below）。

Authentic Examples & Contrasts

Equity is an abstract concept that requires an operational definition.

The nature of the task determines which measures are appropriate.

Diagnostic Warning & Common Errors

- 单数可数名词漏掉冠词：*We conducted survey / *Trained model achieved lower error。
- 图表编号前误加冠词：*See the Table 2（应为 See Table 2）。

6.1.5 G005: 词形变化、词缀构词与词性语法槽位匹配（名、动、形、副）

Core Grammar Principle & Theoretical Framework

Category: `A4_word_forms_and_morphology`

Principle: 掌握学术词族与常见词缀（-able, -cy, -ify, -ise/-ize, -ism/-ist, -less, anti-, co-, il-/im-/in-/ir-, multi-, over-, pre-, post-）的转换，确保名词充当主宾表、动词充当谓语、形容词修饰名词或作表语、副词修饰动词/形容词/全句。

Operational Rules:

- 严禁词性混填：accurate (adj.) / accuracy (n.) / accurately (adv.)；analyse (v.) / analysis (n., 复数 analyses) / analytical (adj.)；vary (v.) / variable (n./adj.) / variation (n.)。
- 警惕词缀语义陷阱：predictable 表示“可预测的”而非“已被预测的”；successive 表示“连续相继的”而非“成功的（successful）”；priceless/meaningless 语义受语境限定；bi-monthly 有歧义，学术写作应改写为 every two months 或 twice a month。
- 英美拼写（analyse/analyze, behaviour/behavior, parametrisation/parameterization）在同一篇文稿中必须统一。

Authentic Examples & Contrasts

Accurate records are essential for checking the analysis.

Attendance increased in three successive years.

Diagnostic Warning & Common Errors

- 形容词与名词混用：*assess the accurate of measurements（应为 accuracy）。
- 混淆同根异义派生词：把 successive（连续的）误作 successful（成功的），把 stationary（平稳的）误拼为 stationery（文具）。

6.1.6 G006: 核心介词规约搭配与名/动/形 + 介词固定结构

Core Grammar Principle & Theoretical Framework

Category: `B_collocations_and_prepositions`

Principle: 学术词汇的介词搭配高度固定，且同一词族的名词、动词和形容词常要求不同介词（如 access a database [v.] vs access to services [n.] vs accessible to users [adj.]；emphasise the need [v.] vs place emphasis on [n.]）。

Operational Rules:

- 动词 + 介词固定搭配：account for, attribute A to B, base A on B, confine A to B, consist of, contribute to, correlate with, decompose A into B, depend on, differ from, disentangle A from B, draw on, focus on, incorporate A into B, rely on, stem from。
- 名词/形容词 + 介词精确方向：change in (某指标变化) vs change of (更换)；increase/decrease in (指标) vs of (数值)；attributable to, akin to, amenable to, analogous to, compatible with, consistent with, invariant under, proportional to, superior to, synonymous with。
- 当 to 为介词时（an approach to, contribute to, be devoted to, prior to, with regard to, amenable to, receptive to, subject to），后接动词必须用动名词（-ing）！

Authentic Examples & Contrasts

The team developed an approach to handling missing observations.

The simplified model is amenable to analytical treatment.

Diagnostic Warning & Common Errors

- 把介词 to 误当成不定式符号：*an approach to handle / *contribute to improve / *be devoted to solve。
- 介词错配：*superior than（应为 superior to）、*synonymous to（应为 synonymous with）、*attributable for（应为 attributable to）。

6.2 Cluster 2: Tense, Aspect, Voice & Reference (G007–G012) / 时态语态与指代基石

6.2.1 G007: 时态、体与学术论文各章节的方法论时间分工

Core Grammar Principle & Theoretical Framework

Category: `C3_tense_aspect_time`

Principle: 学术论文的时态传递方法论状态：一般过去时描述本研究特定实验步骤与过去采样结果；一般现在时描述数学定义、图表展示内容、模型架构属性及本文篇章导引；现在完成时总结某领域迄今的研究演进或前文已完成论证。

Operational Rules:

- 描述本研究特定数据集采集、预处理步骤与单次实验观测用一般过去时（The team abandoned the approach; Two analysts analysed the data independently）。
- 描述数学性质、算法机制、图表内容及普遍事实用一般现在时（The affine function f(x) = ax + b includes an intercept b; Table 1 shows...）。
- 描述近十年研究趋势或前文已展现的内容用现在完成时（The field has developed in recent years; As we have seen, the two estimates use different denominators）。

Authentic Examples & Contrasts

The team accessed the database before the policy changed.

A weakly stationary process has a constant finite mean and autocovariance that depends only on the lag.

Diagnostic Warning & Common Errors

- 把尚未开展的 Future Work 写成已完成的一般过去时。
- 在明确过去时间状语（during the pilot）下误用现在完成时。

6.2.2 G008: 语序、否定辖域、部分否定与句首否定倒装

Core Grammar Principle & Theoretical Framework

Category: `C2_word_order_and_negation`

Principle: 精确控制否定与限定副词（only, solely, merely, by no means, under no circumstances, rarely, seldom）的辖域。当否定或唯一下界短语置于句首时，主句必须采用部分倒装。

Operational Rules:

- 区分全部否定（none of, in no way, under no circumstances, by no means）与部分否定（not all, not necessarily, not always）。
- 当 by no means, in no way, under no circumstances, in no circumstances, on no account, rarely, seldom, only when... 置于句首时，主句必须部分倒装（助动词/be/情态动词提至主语前）。
- 间接疑问从句（assess to what extent the results depend on...）一律保持陈述语序，不倒装。

Authentic Examples & Contrasts

*By no means is the result certain. (*对比句中：*A large sample is by no means sufficient to eliminate systematic bias.)*

Under no circumstances should test labels be used for model tuning.

Diagnostic Warning & Common Errors

- 句首否定强调漏掉倒装：*Under no circumstances test labels should be used。
- 间接疑问句误用倒装：*examine how does the system behave（应为 how the system behaves）。

6.2.3 G009: 代词指代一致性、比较替代词（that/those of）与消除指代悬空

Core Grammar Principle & Theoretical Framework

Category: `C5_pronouns_and_reference`

Principle: 代词必须具有唯一、明确的先行词。学术写作中严禁出现可同时指向多个名词的歧义代词（ambiguous pronoun）或缺乏总结名词的裸奔 this。

Operational Rules:

- 当句中有两个同数名词时，不用裸 it/they/this，应改用 the former / the latter 或 This + 总结性范畴名词（this discrepancy, this constraint, these observations）。
- 比较结构中必须用 that of 替代前文单数/不可数名词，用 those of 替代前文复数可数名词，确保比较对象同范畴（The cost is low relative to that of the earlier procedure）。

Authentic Examples & Contrasts

The pronoun makes the final sentence ambiguous.

The cost is low relative to that of the earlier procedure.

Diagnostic Warning & Common Errors

- 比较结构漏掉 that/those of，导致拿“误差”直接跟“基线模型”相比：*The error of our model is lower than the baseline（应为 lower than that of the baseline）。

6.2.4 G010: 不定式与动名词的补足语分工及非谓语动词搭配

Core Grammar Principle & Theoretical Framework

Category: `C7_nonfinite_infinitives_gerunds`

Principle: 动词后接不定式（to do）还是动名词（doing）取决于动词配价规约；同时注意 warrant, facilitate, advocate, involve, consider, anticipate 等核心学术动词的补足语限制。

Operational Rules:

- 接动名词（doing）或名词、不接不定式的典型动词：consider, involve, advocate, anticipate, recommend, suggest, warrant(warrant further investigation / warrant revising), facilitate(facilitate comparison，严禁写 *facilitate sb to do)。

- 接不定式（to do）的典型动词：aim, attempt, tend, seek, intend, resolve, enable sb to do, allow sb to do, force sb to do, oblige sb to do。

Authentic Examples & Contrasts

The report advocates providing clearer information to applicants.

The procedure involves checking every record twice.

Diagnostic Warning & Common Errors

- 误用不定式：*warrant to conclude / *facilitate researchers to compare / *advocate to provide / *involve to check。

6.2.5 G011: 主动语态与被动语态的基本结构、时态/情态变形与及物性限制

Core Grammar Principle & Theoretical Framework

Category: `D_active_and_passive_voice`

Principle: 被动语态由 be + 过去分词构成，可结合各种时态与情态动词。转换时必须保持原句时态、体、情态、否定和逻辑关系，且严禁将不及物动词或状态短语强行变为被动。

Operational Rules:

- 只有及物动词或带受事宾语的短语动词（analyse, allocate, carry out, account for, base on, expose to, confine to）才能转换为被动语态，且短语介词不可遗漏。
- 不及物动词及状态短语绝不可用于被动语态：consist of, occur, arise, emerge, happen, belong to, differ, appear, exist, take place, stem from, result from, coincide with（严禁写 *is consisted of, *was occurred, *was belonged to）。

Authentic Examples & Contrasts

*Active only: The questionnaire consists of three sections. (绝不可写 *is consisted of)*

Passive with modal: An anomalous observation should be investigated before it is removed.

Diagnostic Warning & Common Errors

- 将不及物动词被动化：*The index is consisted of three parts / *A problem was arisen / *The pattern was emerged。

6.2.6 G012: 数值、比例、分位数、矩阵量纲与增减幅度的精确语法

Core Grammar Principle & Theoretical Framework

Category: `A1_C4_numerical_and_quantitative_grammar`

Principle: 定量描述必须严格区分“差值(by)”、“终值(to)”、“比例(proportion/percentage)”、“比值（ratio)”、“百分点（percentage points)”以及“均值（mean)、中位数（median)、众数(mode)、中程数（midrange)”。

Operational Rules:

- 动词 + by + 差值（decreased by 5 units）vs 动词 + to + 终值（decreased to 15)；名词 + of + 变化量（growth of 10%）vs 名词 + in + 变化对象（growth in demand)。
- 从 20% 升至 25% 是 an increase of 5 percentage points（绝对百分点变化)，而相对增幅为 25%（relative percentage change)。
- double 表示增至两倍，halve 表示减半（不可写 *halve by half)；midrange 是 (min+max)/2, median 是排序后居中的分位数（0.5 quantile)，二者绝不可混同。

Authentic Examples & Contrasts

The count increased by ten, from twenty to thirty.

A rise from 20% to 25% is five percentage points.

Diagnostic Warning & Common Errors

- 介词 by 与 to 混用导致差值与终值颠倒。
- 把 percentage point（百分点）与 per cent（百分比）混同。

6.3 Cluster 3: Lexical Precision, Hedging & Modifiers (G013–G018) / 词汇精度、谨慎陈述与修饰从句

6.3.1 G013: 近义词与多义词深度辨析：语义范围、褒贬色彩、技术定义与语域

Core Grammar Principle & Theoretical Framework

Category: `A1_A5_A6_A7_A8_lexical_and_polysemy_precision`

Principle: 区分近义词与一词多义在语义范围、判断强度、褒贬色彩及学科专有定义上的差别。禁止把仅有部分重合的词当作完全同义词替换，禁止为堆砌大词而改变事实或判断强度。

Operational Rules:

- 测量与统计核心辨析：accuracy（准确度，接近真值）vs precision（精密度，重复测量一致)；reliable（稳定一致）vs valid（有效测量目标构念或推理成立)；efficacy（受控试验效力）vs effectiveness（真实场景效果)；equal（均等）vs equitable（按需求公平分配)；

seasonality（固定日历周期）vs cyclicality（非固定周期的波动）；isotonic（保序的/等渗的）vs isotropic（各向同性的）。

- 褒贬与语域辨析：simple（简明清晰的）vs simplistic（过度简化的，贬义）；exploit structure（中性：利用结构）vs exploit workers（贬义：剥削）；manipulate a variable（中性：实验操纵自变量）vs falsify data（贬义：伪造数据）。

Authentic Examples & Contrasts

An equitable allocation of resources may differ from an equal allocation.

Business cyclicality may involve expansions and contractions of varying duration, whereas seasonality follows a fixed calendar period.

Diagnostic Warning & Common Errors

- 把 suggest/indicate 机械替换为 prove/demonstrate 导致过度推断。
- 混淆形近学术词：adapt/adopt, adverse/averse, discrete/discreet, temporal/temporary, stationary/stationery, isotonic/isotropic。

6.3.2 G014: 情态动词、确定性梯度与科学谨慎表达（Epistemic Hedging）

Core Grammar Principle & Theoretical Framework

Category: `C6_E_modals_and_hedging`

Principle: 科学写作要求结论的语气强度与实验/数据证据的完备度严格对齐。通过情态动词（may, might, can, could）、认知副词（arguably, apparently, plausibly, relatively, typically）及谨慎动词（suggest, indicate, point to, appear to）控制断言边界。

Operational Rules:

- 当存在未测混杂（unmeasured confounding）、小样本或仿真环境限制时，必须使用 may be attributable to / suggests that / supports a tentative conclusion。
- 禁止在经验研究中滥用 irrefutable, undeniable, inevitable, universal, conclusively prove；只有经核实的数学反例推翻全称命题时才可用 disprove / irrefutable。

Authentic Examples & Contrasts

The observed discrepancy may be attributable to differences in measurement procedures.

A distributional forecast is arguably more informative when decisions depend on tail risk.

Diagnostic Warning & Common Errors

- 在改写或总结时删去原句的 may / arguably / in this sample 等限定词，造成过度绝对化。

6.3.3 G015: 分词短语（现在分词与过去分词）作后置定语与状语及悬垂修饰防范

Core Grammar Principle & Theoretical Framework

Category: `C7_C11_participles_and_dangling_modifiers`

Principle: 分词短语能将冗长的从句浓缩为紧凑的后置定语或状语；但句首分词短语的逻辑主语必须与主句主语完全一致，绝不可产生悬垂修饰（Dangling Modifier）。

Operational Rules:

- 后置分词定语：records exceeding the limit (= that exceed...)；predictions compared with ground-truth labels (= that are compared...)。
- 句首分词状语主语一致铁律：Using a 48-step look-back window, the model reduces MSE（the model 可以 use window；但不能写 *Using a 48-step window, the MSE decreases，因为 MSE 不会自己 use window!）。

Authentic Examples & Contrasts

Records exceeding the stated size limit were excluded.

Compared with earlier studies, this study used a larger sample.

Diagnostic Warning & Common Errors

- 悬垂分词：*Training on synthetic data, the generalization error remained high（应改为 When the model was trained on synthetic data, ... 或 Training on synthetic data left the model with high generalization error）。

6.3.4 G016: 限制性/非限制性定语从句与“介词 + which/whom”精确修饰

Core Grammar Principle & Theoretical Framework

Category: `C8_relative_clauses`

Principle: 定语从句是构建精确后置定语、避免中式前置长定语的核心工具。必须严格区分限制性（无逗号，界定子集）与非限制性（有逗号，补充评注，不可用 that），并掌握介词前置结构。

Operational Rules:

- 限制性定语从句缩窄先行词范围：The analogy breaks down where the two systems differ in the feature on which the comparison depends。
- 介词 + which 结构由从句动词或先行词搭配决定：the age at which, the point at which, the rate at which, the extent to which, a protocol by means of which, the assumptions on which the model rests。

Authentic Examples & Contrasts

We devised a protocol by means of which records could be checked.

The study examines the degree to which access varies by region.

Diagnostic Warning & Common Errors

- 非限制性定语从句逗号后误用 that。
- 介词 + which 遗漏固定介词：*the extent which the findings generalise（漏掉 to）。

6.3.5 G017: 名词性从句与同位语从句（the claim/assumption/fact that...）

Core Grammar Principle & Theoretical Framework

Category: `C8_noun_clauses`

Principle: 同位语从句（the claim/assumption/hypothesis/notion/contention/fact that + 完整句）用于精确展开抽象学术名词的内涵；介词后若需接完整陈述句，必须借助 the fact that 或 in that。

Operational Rules:

- 同位语从句连接词 that 不作成分但不可省略，也不可换为 which（The contention that all data are interchangeable needs justification）。
- 介词（account for, despite, due to, stem from）不能直接接 that 从句，必须用 account for the fact that / despite the fact that / stem from the fact that。

Authentic Examples & Contrasts

The model cannot account for the fact that rates differ across sites.

The scheme was adopted despite the fact that it was costly.

Diagnostic Warning & Common Errors

- 介词直接接 that 从句：*despite that it was costly / *due to that several records were missing。

6.3.6 G018: 状语从句、让步/条件结构与介词短语的紧凑转换

Core Grammar Principle & Theoretical Framework

Category: `C8_C9_adverbial_clauses`

Principle: 掌握从属连词（although, whereas, while, provided that, because, in that）、介词短语（despite, in spite of, due to, prior to, in light of, in the absence of）与连接副词（nevertheless, nonetheless, consequently, conversely）之间的句法转换。

Operational Rules:

- 让步压缩：Although the approach has limitations → Albeit with limitations / Despite its limitations（严禁写 *despite of）。
- 条件限定：provided that / as long as + 完整分句 <→ subject to / under + 名词短语。
- 对照结构：whereas / while 连接两个对称分句（The first model uses monthly data, whereas the second uses daily data）。

Authentic Examples & Contrasts

The approach is informative, albeit with limitations.

The result applies provided that the assumptions hold.

Diagnostic Warning & Common Errors

- 混淆介词与连词：*despite the sample is large。
- while 表对比时与时间含义发生歧义（在时序/并发场景中表对比优先用 whereas）。

6.4 Cluster 4: Comparisons, Parallelism & Mechanics (G019–G024) / 比较论证、平行结构与标点规约

6.4.1 G019: 比较结构、拉丁比较级（superior/prior to）与比较维度限定（in terms of）

Core Grammar Principle & Theoretical Framework

Category: `C9_comparison_structures`

Principle: 科学比较必须做到：(1) 比较主体语法同构，(2) 明确比较指标与条件（in terms of / with respect to / under comparable conditions），(3) 正确使用拉丁比较级与类比结构。

Operational Rules:

- superior to, inferior to, prior to, analogous to, akin to 一律用介词 to，严禁用 than。
- 声明某方法更优（superior / optimal / outperform）时，必须限定评估指标与数据集范围，禁止无条件宣称“处处最优”。

Authentic Examples & Contrasts

The method was superior to the baseline on this metric, but other outcomes were not assessed.

The procedure is akin to stratified sampling, although the allocation rule differs.

Diagnostic Warning & Common Errors

- 误用 *superior than / *prior than。
- 省略比较基准（baseline）或评价维度。

6.4.2 G020: 语态的语篇决策：基于研究对象、旧信息前置与施事责任的主动/被动选择

Core Grammar Principle & Theoretical Framework

Category: `D_voice_selection_strategy`

Principle: 破除“被动语态一定比主动语态更学术”的迷思。语态选择服务于：(1) 维持句间话题链连贯（把上句提到的受事对象作为本句被动主语，满足 Given-before-New），(2) 明确谁对设计决策或核验承担责任。

Operational Rules:

- 当描述标准化数据清洗、传感器部署或客观流程，且受事对象是段落主题时，选用被动语态（The records are arranged in chronological order before the time-series analysis）。
- 当阐述本文核心创新设计、范围限定决策或强调多名标注者独立操作时，选用主动语态（We confine the analysis to...; Two analysts analysed the data independently; The algorithm exploits the sparsity of the matrix）。

Authentic Examples & Contrasts

Passive for topic continuity: The predictions are compared with ground-truth labels that were checked by two annotators.

Active for clear methodological agency: We reformulate the constrained problem as an equivalent optimization problem.

Diagnostic Warning & Common Errors

- 通篇机械使用被动语态导致语句冗长、施事责任模糊（如分不清是前人文献做的还是本文做的）。

6.4.3 G021: 句子改写核心规范：语义等价、限定守恒与消除四大改写偏差

Core Grammar Principle & Theoretical Framework

Category: `F_sentence_rewriting_rules`

Principle: 句子改写必须在改变句法或词汇结构的同时，100% 保持原句的事实范围、否定、时间、程度和认知确定性，并能准确诊断语法错误、表达不自然、语域不当与语义扭曲。

Operational Rules:

- 限定守恒：原句的 provided that, in this sample, under Gaussian-kernel assumptions, may, arguably 等在改写后必须保留等价限定。
- 多路径自然改写：支持动词 <→ 派生名词转换、从句 <→ 介词短语转换、主动 <→ 被动焦点转换，提供多个自然可接受版本。

Authentic Examples & Contrasts

Source: Missing records account for the discrepancy between the totals. Rewrite 1: The discrepancy between the totals is attributable to missing records. Rewrite 2: Missing records explain why the two totals differ.

Diagnostic Warning & Common Errors

- 在改写中偷换概念或将相关关系升级为因果结论。

6.4.4 G022: 并列、从属、平行结构（Parallelism）与对比对称性

Core Grammar Principle & Theoretical Framework

Category: `C10_parallelism_and_subordination`

Principle: 由 and, or, rather than, as opposed to, as well as, both... and... 连接的成分必须在词性、句法层级与逻辑角色上严格平行。

Operational Rules:

- 对比排除连接词（rather than, as opposed to）两侧结构必须平行：reporting observed values as opposed to predictions；measured an association rather than a causal effect。
- 分别对应结构（A and B ... X and Y, respectively）两侧元素个数与顺序必须严格一一对应。

Authentic Examples & Contrasts

The table reports observed values as opposed to predictions.

The symbols x and y denote temperature and demand, respectively.

Diagnostic Warning & Common Errors

- 两侧词性不对称：*combines quantitative measurements with interviewing participants（名词短语与动名词不对称）。

6.4.5 G023: 学术标点规范、分号/冒号逻辑功能与逗号粘连（Comma Splice）根治

Core Grammar Principle & Theoretical Framework

Category: `C11_punctuation_and_clause_combining`

Principle: 根治中国学生写作中最普遍的标点语法错误——逗号粘连（Comma Splice，即仅用逗号连接两个独立句子，或把 nevertheless, therefore, however 当连词用逗号连接两句）。

Operational Rules:

- 连接副词（however, therefore, hence, nevertheless, nonetheless, consequently, moreover, furthermore, conversely, likewise）连接两个独立分句时，前句末尾必须用分号（;）或句号（.）！
- 前置复合修饰语加连字符（a small-scale study, an in-depth analysis, ground-truth labels, blue-collar occupations），后置作表语或状语时不加连字符（examine the issue in depth）。

Authentic Examples & Contrasts

The observations are dependent; hence, the usual independence assumption does not apply.

The sample is small; nevertheless, the measurements identify a useful question for further study.

Diagnostic Warning & Common Errors

- Comma Splice：*The sample is small, however, the protocol is reproducible（第一个逗号必须改为分号或句号）。

6.4.6 G024: 适度名词化（Nominalization）、复杂名词短语与动词活力平衡

Core Grammar Principle & Theoretical Framework

Category: `C12_nominalization_and_complex_np`

Principle: 名词化可将前句的过程浓缩为后句主语（如 The relaxation of this assumption...），实现紧凑衔接；但绝不能认为“名词化越多越学术”，过度名词化会导致句子僵化、头重脚轻。

Operational Rules:

- 用“The + 派生名词 + of + 核心对象”打包已知过程作主语，谓语仍使用清晰有力的动词（The elimination of duplicate entries reduced the file size）。
- 当主语和宾语都已是长抽象名词时，中间谓语直接用强动词（affects, constrains, clarifies），不要写 *conducts an influence on。

Authentic Examples & Contrasts

The emergence of new devices changed the monitoring procedure.

The quantification of uncertainty is central to this analysis.

Diagnostic Warning & Common Errors

- 僵尸名词堆砌（Zombie Nouns）：*The implementation of the reduction of parameters was carried out（应直接写 We reduced the number of parameters）。

6.5 Cluster 5: Ambiguity, Causality & Discourse Architecture (G025–G030) / 排歧机制、因果推断与篇章骨架

6.5.1 G025: 句法歧义消除（介词短语附着、修辞辖域与比较基准排歧）

Core Grammar Principle & Theoretical Framework

Category: `C11_syntactic_ambiguity_resolution`

Principle: 消除句子中的结构性多解（Structural Ambiguity），确保读者无需猜测修饰语究竟挂靠在哪个词上。

Operational Rules:

- 调整介词短语位置或改写为定语从句，消除“V + NP + PP”中的 PP 附着歧义。
- 明确区分绝对差值（in absolute terms）与相对变化（in relative terms），并交代基准分母（denominator）。

Authentic Examples & Contrasts

The absolute difference was reported alongside the relative change.

The revised definition is unambiguous.

Diagnostic Warning & Common Errors

- 修饰语远离核心词导致语义滑稽或产生双重解释。

6.5.2 G026: 因果关系、统计相关、混杂（Confounding）与反事实（Counterfactual）表达边界

Core Grammar Principle & Theoretical Framework

Category: `E_G_causality_vs_correlation`

Principle: 在机器学习、流行病学、经济学与社会科学写作中，必须严格区分预测准确率/统计相关（prediction accuracy / correlation / association）与因果效应（causal effect / causality）。

Operational Rules:

- 观察性关联用 correlate with, be associated with, coincide with；不能仅凭高预测精度或滞后项相关性（lagged correlation）声称确立了 causality。
- 讨论因果识别时，准确使用 confounding（混杂因素）、counterfactual outcome（反事实结局）、exogenous（外生的，需写明与扰动项关系）及 disentangle A from B（分离交织效应）。

Authentic Examples & Contrasts

Predictive accuracy alone cannot establish causality.

Unmeasured confounding may bias an estimated treatment effect.

Diagnostic Warning & Common Errors

- 在观察性数据或纯预测模型论文中滥用 cause, determine, prove a causal link。

6.5.3 G027: 文献综述、观点归属与报道动词（Reporting Verbs）的认知立场控制

Core Grammar Principle & Theoretical Framework

Category: `E_reporting_verbs_and_attribution`

Principle: 在 Related Work 与议论文引用中，通过精准选择报道动词（argue, claim, assert, contend, maintain, suggest, note, observe, point out, demonstrate, show, acknowledge, question, challenge）体现对前人工作的客观定位。

Operational Rules:

- 区分事实呈现（show, demonstrate, reveal）、作者主观论断（claim, assert, contend, maintain）与审慎提议（suggest, propose, hypothesize）。
- According to the authors 仅用于引述他人或外部记录，绝不可写 *According to me。

Authentic Examples & Contrasts

The authors argue that simpler baselines deserve attention.

The author asserts that the categories are exhaustive, whereas recent benchmarks challenge this assumption.

Diagnostic Warning & Common Errors

- 用一成不变的 *Author A says that... 贯穿整个 Related Work 章节。

6.5.4 G028: 学术段落逻辑架构：主题句—机制解释—实证/实例支撑—范围限定

Core Grammar Principle & Theoretical Framework

Category: `G_paragraph_structure_and_argumentation`

Principle: 无论是顶会论文 Discussion 还是六级/雅思托福主体段，高质量段落都必须具备“清晰主题句（Topic Sentence）→ 作用机制或理论解释（Mechanism/Warrant）→ 具体数据或实例（Concrete Evidence/Instance）→ 边界限定（Qualification）”四层逻辑，拒绝仅靠连接词填空。

Operational Rules:

- 防范逻辑跳跃（Logical Leap）：从“某现象存在或某指标提升”到“应全面推广某政策/模型”之间，必须补充成本、可行性（feasibility）或分布偏移（distribution shift）的权衡（trade-off）。
- 正确区分对比连接词：on the contrary 仅用于否定刚提及的命题（“恰恰相反”），而对比两个不同群体的表现必须用 by contrast / in contrast / on the other hand。

Authentic Examples & Contrasts

The report claims that every village was sampled. On the contrary, remote villages were excluded.

Urban use increased; by contrast, rural use remained stable.

Diagnostic Warning & Common Errors

- 把 on the contrary 误当作 on the other hand / by contrast 使用。
- 段落只有空洞观点句和夸张形容词，缺乏具体机制解释与可核验实例。

6.5.5 G029: 语篇信息流推进（Given-New Contract）与跨句主题链构建

Core Grammar Principle & Theoretical Framework

Category: `C12_G_information_packaging`

Principle: 地道英语段落之所以流畅自然、毫无生硬拼接感，核心在于遵守“旧信息在前（Given）、新信息在后（New）”的信息契约，用总结性指代或语态转换实现句间无缝咬合。

Operational Rules:

- 链式推进（Linear Progression）：第 N 句句末引入的新概念（如 an anomaly in the sensor readings），在第 N+1 句句首打包为旧信息主语（This anomaly / Such discrepancies...）。
- 避免连续多句使用毫无关联的全新长主语，造成读者认知断层。

Authentic Examples & Contrasts

An anomaly in the sensor readings prompted a check of the calibration procedure. This check revealed a drift in the reference voltage.

Diagnostic Warning & Common Errors

- 把旧信息放在句末，而把长达 15 词的全新复杂概念硬塞在句首主语位置。

6.5.6 G030: 让步反驳（Concession-Rebuttal）、局限性陈述与外推边界（Extrapolation Limits）

Core Grammar Principle & Theoretical Framework

Category: `E_G_concession_rebuttal_and_limitations`

Principle: 顶会论文 Limitations 章节、Rebuttal 及高分议论文的核心技能：坦诚承认方法或样本的约束条件（acknowledge constraints），同时清楚界定在什么范围内结论依然稳健（robust），并指出何时外推（extrapolation）会失效。

Operational Rules:

- 区分内插（interpolation，在已观测范围内估计）与外推（extrapolation，向观测年龄、工况或分布范围外推断，需更强假设）。
- 用 While / Although / Despite 承认局限，用 nevertheless / aún / provided that 说明受控范围内的有效性与后续改进方向（warrant further investigation）。

Authentic Examples & Contrasts

Extrapolation beyond the observed age range requires additional justification.

The discrepancy warrants further investigation, but it does not warrant a causal conclusion.

Diagnostic Warning & Common Errors

- 要么完全隐瞒局限，要么自相矛盾地声称局限毫无影响。

6.6 Cluster 6: Anti-Defect Diagnostics & Advanced Frontier Genres (G031–G036) / 病灶诊断与前沿文体适配

6.6.1 G031: 句子成分精细剖析（主、谓、宾、表、定、状、补、同位语）与核心主干定位

Core Grammar Principle & Theoretical Framework

Category: `C1_sentence_constituents_analysis`

Principle: 自然地道的英语写作建立在清晰的句子成分（Constituents）分工之上：主语（Subject，建议用简短核心名词短语）+ 谓语动词（Predicate Verb，承载核心动作或状态关系）+ 宾语/表语/补语（Object/Complement），而将限定范围、实验条件、时间窗口等次要信息合理分配给后置定语（Post-modifier）或状语（Adverbial）。

Operational Rules:

- 主语（S）与谓语（V）的距离应尽可能紧凑：避免在主语核心词和谓语动词之间插入过长的非必要修饰成分，防止读者丢失句子主干。

- 区分定语（修饰名词，回答 which/what kind of）与状语（修饰动词或全句，回答 how/when/where/under what condition）：英語长定语一律后置（介词短语、分词短语、定语从句），切忌按中文习惯把长串修饰语全部堆在名词前面。
- 同位语（Appositive，如 a random walk, an archetypal example of a nonstationary process）用于紧随名词后补充等价定义或量纲说明，比单独另起一句更紧凑。

Authentic Examples & Contrasts

[主语 The affine function f(x) = ax + b] [谓语 includes] [宾语 an intercept b].

[状语 When the relevant variances are finite and nonzero], [主语 lag-one Pearson autocorrelation] [谓语 measures] [宾语 the linear association] [后置定语 between observations one time step apart].

Diagnostic Warning & Common Errors

- 中式前置长定语堆砌：*The collected from three rural clinics during the two-year pilot study data were noisy（应改为 The data collected from three rural clinics during the two-year pilot study were noisy）。
- 把状语介词短语误当成主语：*In Table 1 shows the ablation results（应为 Table 1 shows... 或 In Table 1, we report...）。

6.6.2 G032: 消除“头重脚轻（Top-Heavy Subject）”与遵循尾重原则（End-Weight Principle）

Core Grammar Principle & Theoretical Framework

Category: `C10_C12_end_weight_and_anti_top_heavy`

Principle: 中国学习者与劣质机翻最常见的句法缺陷是“头重脚轻（Top-Heavy Sentence）”——主语长达十几个词，谓语却只是一个软弱的 is / occurs / exists。地道英语严格遵循“尾重原则（End-Weight）”：主语简短（旧信息），把复杂的从句、长列表或重信息放在谓语之后（句末焦点位置）。

Operational Rules:

- 诊断头重脚轻：当主语超过 8–10 个词且包含从句或多层 of 短语，而谓语仅为 be/have/exist 时，必须重构句子。
- 重构方法 1（换用核心实体作主语 + 强动词）：把长主语中的行动者、模型或实验设为短主语，将原抽象名词还原为谓语动词（如把 *The investigation of how irregular sampling intervals affect autocorrelation estimation is important 改为 *We investigate how irregular sampling intervals affect autocorrelation estimation* 或 *Irregular sampling intervals can distort autocorrelation estimates*）。
- 重构方法 2（右移重成分或使用先行 it / 存在结构）：当需要评价某个复杂命题时，用 It is critical that... 或将状语条件移至句首/句末。

Authentic Examples & Contrasts

头重脚轻病句：**Whether the learned policy can generalize from simulation to a physical robot with actuator delays and sensor noise is uncertain.** 地道改写：**It remains uncertain whether the learned policy can generalize from simulation to a physical robot with actuator delays and sensor noise.** 或 **Actuator delays and sensor noise make sim-to-real policy transfer uncertain.**

头重脚轻病句：**A discrepancy between the recorded timestamps and the sensor logs in the second rollout was found.** 地道改写：**The audit revealed a discrepancy between the recorded timestamps and the sensor logs in the second rollout.**

Diagnostic Warning & Common Errors

- 主语塞满三四个 of 嵌套短语，导致读者读到句尾才看到一个单薄的 is。
- 用 Whether / That 名词性从句作长主语而谓语只有一个形容词。

6.6.3 G033: 去除“AI 味（Anti-AI-ese）”与过度修辞：用具体机制、指标与实例替代空洞夸张

Core Grammar Principle & Theoretical Framework

Category: `E_anti_ai_style_and_concreteness`

Principle: 典型 AI 生成文本充斥着空洞的套话与浮夸副词/形容词（如 delve into, testament to, tapestry, underscores the paramount importance, revolutionize, unparalleled, groundbreaking），却缺乏具体技术细节。地道顶会论文（NeurIPS/ICLR/CoRL）与高分雅思/托福/六级写作追求“低修辞密度、高信息密度”：用具体动词、明确指标和真实实例说话。

Operational Rules:

- 删除无实质信息的“清嗓子”套话（Throat-clearing）：少用 *It is worth noting that / It is undeniable that / In the realm of modern artificial intelligence*，直接陈述事实主干。
- 禁止“只有夸张形容词而不给实例或数据支撑”：不要写 *The model achieves an enormous, unparalleled improvement*, 而要写 *The proposed parametrization reduces MSE by 8.4% relative to the seasonal naive baseline on hourly electricity demand*。
- 优先选用朴素、精准的单音节或核心学术动词（examine, test, compare, constrain, decompose, bound, mitigate）替代故作高深的修辞词。

Authentic Examples & Contrasts

AI 腔/空洞夸张：**This groundbreaking paradigm delves into the intricate tapestry of temporal dynamics and achieves unparalleled accuracy.** 地道学术改写：**This model decomposes the series into trend and seasonal components, achieving lower forecasting error than the linear baseline on three benchmarks.**

六级/雅思空洞句：**Technology has an enormous impact on education in many ways.** 地道具体展开：**Asynchronous learning platforms allow part-time students to review lecture recordings at their own pace, reducing scheduling conflicts.**

Diagnostic Warning & Common Errors

- 堆砌 paramount, unprecedented, revolutionize, multifaceted 等宣传式词汇，违反客观学术语域。
- 只下宏大判断（如 *This method is very efficient*）而不给出具体的计算复杂度、内存占用或延迟数据。

6.6.4 G034: 四大常见表达病灶诊断：口语化、过度表达（Over-expression）、不精确表达与指代悬空

Core Grammar Principle & Theoretical Framework

Category: `A_E_F_four_major_writing_defects`

Principle: 针对中国学习者在论文、雅思托福及六级写作中的四大高频失分点建立系统修正机制：(1) 口语化表达混入书面语，(2) 过度表达（语义重复或情态过满），(3) 不精确表达（缺少比较基准、分母或时间单位），(4) 悬空指代。

Operational Rules:

- 修正口语化（Colloquialism）：书面论文中将 a lot of 改为 substantial / a large number of；将 get 改为 obtain / acquire；将 big 改为 substantial / marked；将 do a study 改为 conduct / carry out a study；将 pretty up / dumb down 改为 refine the visual layout / oversimplify。
- 修正过度表达与语义冗余（Over-expression & Pleonasm）：禁止写 *halve by 50%*（halve 已含 50%）、*mutually collaborate with each other*、*completely eliminate all*、*future forthcoming*。
- 修正不精确表达（Imprecision）：报告 lag（滞后）、rate（速率）或 relative change（相对变化）时，必须指明采样时间单位（sampling interval）、基线（baseline）与分母（denominator）。

Authentic Examples & Contrasts

口语化病句：**We did a small study and got a lot of data, which pretty up our paper.** 修正：**We conducted a pilot study and collected a substantial amount of data to test the protocol.**

不精确病句：**The error dropped by 5%.** 精确修正：**The classification error rate decreased by 5 percentage points (from 20% to 15%) relative to the baseline.**

Diagnostic Warning & Common Errors

- 把口头演讲过渡语（如 *Getting back to the question / That is all very well, but*）直接写进正式期刊论文正文。

- 混淆相对百分比变化（per cent）与绝对百分点变化（percentage points）。

6.6.5 G035: 机器学习、时间序列、生成模型与具身智能（Embodied AI）的技术表达句法

Core Grammar Principle & Theoretical Framework

Category: `E_cs_ai_timeseries_embodied_syntax`

Principle: 面向深度学习、时间序列预测、生成式建模与具身智能（Embodied AI）写作，训练如何用精准、克制、零 AI 腔的英语句式描述数学符号、分布特性、物理/几何约束、感知—动作闭环（perception-action loop）及实验消融。

Operational Rules:

- 问题与符号定义：用 Let X denote... / Here, N denotes... / We parameterize the predictive distribution as... 保持全篇符号（notation）一致，严格区分标量（scalar）、向量（vector）与矩阵（matrix）。
- 时间序列与生成模型：严格区分平稳性（stationarity）与遍历性（ergodicity）、季节性（seasonality，固定日历周期）与周期波动（cyclicality，非固定时长）、概率校准（calibration，预测分布与观测频率一致）与锐度（sharpness，预测分布集中度）。
- 具身智能（Embodied AI）：准确使用 spatial orientation（空间方位）、trajectory（运动/策略轨迹）、velocity（速度矢量，含方向）、perturbation（外部扰动）、closed-loop feedback（闭环反馈）及 physical/kinematic constraints（物理与运动学约束），不把仿真结果（simulation）直接冒充真实物理部署（real-world deployment）。

Authentic Examples & Contrasts

Time Series & Generative: Forecast calibration concerns statistical consistency between predictive distributions and observations, whereas sharpness refers to the concentration of the predictive distribution.

Embodied AI: The controller enforces velocity and torque constraints while tracking the reference trajectory under external perturbations.

Diagnostic Warning & Common Errors

- 把仿真环境（simulation）中的策略表现直接写成真实世界因果有效，忽略 sim-to-real discrepancy。
- 混淆 covariance（协方差，受量纲缩放影响）与 correlation（标准化相关系数），或混淆 isotonic（保序的）与 isotropic（各向同性的）。

6.6.6 G036: 跨场景文体适配：顶会论文、学术演讲（Tech Talk）、个人陈述（SoP）与六级/雅思托福高分议论文

Core Grammar Principle & Theoretical Framework

Category: `E_G_multi_genre_adaptation`

Principle: 同一核心词汇库在不同应用场景下需要灵活切换语域与句式结构：学术论文重实证边界与方法可复现；口头演讲（Presentation/Seminar）重听众引导与简洁路标；六级/雅思托福（CET-6/IELTS/TOEFL）重观点清晰、让步权衡与具体民生/科技/教育实例。

Operational Rules:

- 顶会论文与审稿回复（Paper & Rebuttal）：用 We thank the reviewer for pointing out this limitation; to address this concern, we conducted an additional sensitivity analysis... 礼貌、有据地回应质疑（take issue with / respond to / clarify）。
- 学术演讲与海报展示（Tech Talk & Oral Presentation）：合理使用路标句（I would like to begin by explaining... / As this slide shows... / Let us now turn to... / I will leave five minutes for questions），句子宜短，避免听众无法跟上的多重嵌套从句。
- 六级与雅思托福写作（CET-6 & IELTS/TOEFL Writing）：摒弃万能废话模板，采用“明确立场句（adopt a principled stance）+ 具体机制解释（account for / facilitate / mitigate）+ 落地实例（concrete example，如远程教育、可再生能源补贴、公共交通可及性）+ 让步权衡（albeit / on balance / outweigh）”。

Authentic Examples & Contrasts

Tech Talk (Spoken Academic): As this slide shows, the naive baseline performs competitively when the look-back horizon is short; let us now turn to why complex models overfit noisy series.

IELTS/CET-6 Argumentative Essay: Subsidising public transport can facilitate access to education for rural students; however, municipal authorities must weigh this benefit against long-term infrastructure maintenance costs.

Diagnostic Warning & Common Errors

- 在雅思/六级作文中使用生硬虚假的统计编造（如 *According to a 2025 Harvard survey, 93.7% of people...*），而不是使用合理的机制论证与具体生活/政策实例。
- 在口头报告结束语与书面论文结论之间混用语域（如在论文结尾写 *That is all I have to say, thank you for listening*）。

Chapter 7

Expressive Academic Sentence Templates (EP001–EP024) / 学术功能句型高阶模板库

在学术论文的不同小节中, 作者需要高效复用符合国际同行惯例的功能型表达句型（Expression Patterns）。本章系统呈现 24 组高分学术句型模板（EP001–EP024），涵盖从研究意图、量化对比、因果推断到前沿 AI 技术的全周期写作需求。

7.1 Cluster A: Research Foundations, Scope & Definitions (EP001–EP006) / 研究起步、范围定义与概念界定

7.1.1 EP001: 明确陈述研究目标、问题定义与讨论边界

Template Formula & Rhetorical Function

Function: `stating_aims_and_scope` **Register**: `formal_academic`

Template Formula:

`This study/ paper examines [specific problem] and aims to [verb] [measurable objective] within [stated scope].`

Exposition: 开门见山说明研究问题与范围，主语简短，直接用强动词 examines / addresses / investigates。

Authentic Exemplars in Research Literature

This paper addresses unequal access to educational resources.

The aim of this paper is to compare annotation rules.

Academic Pitfall & Reviewer Perspective

- 使用冗长空洞的套话开头，或写 *address to a problem。

7.1.2 EP002: 精确区分可数样本量与不可数开销/证据

Template Formula & Rhetorical Function

Function: describing_methods_and_quantities **Register**: general_academic

Template Formula:

While a large number of [plural countable nouns] were collected, the procedure required only a small amount of [uncountable noun].

Exposition: 在数据集与实验开销描述中准确搭配可数与不可数量词。

Authentic Exemplars in Research Literature

A large number of records still require checking.

The protocol requires a small amount of additional information.

Academic Pitfall & Reviewer Perspective

- *a great deal of participants / *an amount of records。

7.1.3 EP003: 客观归属文献观点或档案记录来源

Template Formula & Rhetorical Function

Function: attributing_sources_and_claims **Register**: general_academic

Template Formula:

According to [authors/ records], [reported finding]; however, [methodological qualification].

Exposition: 准确区分外部文献说法与本文实证判断。

Authentic Exemplars in Research Literature

According to the authors, the evaluation uses synthetic sentences.

According to the records, the event began before sunset.

Academic Pitfall & Reviewer Perspective

- *According to me（第一人称严禁使用）。

7.1.4 EP004: 界定概念的操作性定义与数学符号表示

Template Formula & Rhetorical Function

Function: `defining_concepts_and_notation` **Register**: `formal_academic`

Template Formula:

```
For the purposes of this analysis, we define [term] as [operational criterion], where [symbol] denotes [quantity].
```

Exposition: 用于将抽象概念转化为可观测、可复现的操作性指标或符号。

Authentic Exemplars in Research Literature

We define the outcome as the number of visits per week.

Here, N denotes the number of observations.

Academic Pitfall & Reviewer Perspective

- 定义含糊或符号指代前后不一致。

7.1.5 EP005: 描述系统模块、数据集或分类体系的构成

Template Formula & Rhetorical Function

Function: `classification_and_composition` **Register**: `formal_academic`

Template Formula:

```
[Whole System/ Dataset] consists of / comprises [N] components, whereas [Subset] accounts for / constitutes [Proportion].
```

Exposition: 用主动语态的 consist of / comprise 表整体由部分组成，用 constitute / account for 表部分占整体的比例。

Authentic Exemplars in Research Literature

The questionnaire consists of three sections.

Missing responses account for a small proportion of the records.

Academic Pitfall & Reviewer Perspective

- *is consisted of / *comprise of。

7.1.6 EP006: 描述实验执行、基线核查与可复现流程

Template Formula & Rhetorical Function

Function: describing_methods_and_procedures **Register**: formal_academic

Template Formula:

We conducted [experiment/ survey] under controlled conditions and verified [readings/outputs] against [ground-truth/reference].

Exposition: 直接、清晰地交代实验操作与质量核验步骤，避免头重脚轻。

Authentic Exemplars in Research Literature

The team verified sensor readings against manual measurements.

Two analysts analysed the data independently.

Academic Pitfall & Reviewer Perspective

- *make an experiment / *do a research。

7.2 Cluster B: Quantitative Analysis, Graphics & Comparison (EP007–EP012) / 量化分析、图表汇报与多维对比

7.2.1 EP007: 同时报告绝对差值与相对变化（区分% 与百分点）

Template Formula & Rhetorical Function

Function: describing_trends_and_changes **Register**: formal_academic

Template Formula:

[Metric] decreased/ increased by [X] percentage points (from [A] % to [B] %), corresponding to a [Y]% relative change.

Exposition: 消除百分比歧义，同时给出基线值、绝对差值与相对变化率。

Authentic Exemplars in Research Literature

The absolute difference was reported alongside the relative change.

A rise from 20% to 25% is five percentage points.

Academic Pitfall & Reviewer Perspective

- 混淆 percentage points 与 per cent，或隐瞒基数分母。

7.2.2 EP008: 使用 be of + 抽象名词结构自然提升评价语域

Template Formula & Rhetorical Function

Function: academic_register_and_nominal_style **Register**: formal_academic

Template Formula:
[Factor/ Dataset] is of considerable/ particular [importance/ significance/ value/ relevance] to [target problem].

Exposition: 自然替换口语化的 be very important/useful。

Authentic Exemplars in Research Literature

The consent procedure is of particular significance to this study.

The evidence is of great relevance to the policy debate.

Academic Pitfall & Reviewer Perspective

- *be of important（of 后必须接名词 importance）。

7.2.3 EP009: 谨慎归因与多因素解释（May Account For / Be Attributable To）

Template Formula & Rhetorical Function

Function: hedging_and_attribution **Register**: formal_academic

Template Formula:
The observed discrepancy may be attributable to / stem from [Factor A] rather than [Factor B].

Exposition: 在分析实验误差或异常现象时，客观提出可能成因而不武断定论。

Authentic Exemplars in Research Literature

The observed discrepancy may be attributable to differences in measurement procedures.

Measurement error may account for part of the discrepancy.

Academic Pitfall & Reviewer Perspective

- *attributable for / 把可能成因直接写成唯一确定原因。

7.2.4 EP010: 图表、坐标轴、斜率与分布形态的精准描述

Template Formula & Rhetorical Function

Function: describing_figures_and_tables **Register**: formal_academic

Template Formula:

In Figure [N], plotting [Y] against [X] reveals [pattern/ slope], while the shaded band denotes [uncertainty interval].

Exposition: 用于论文结果部分与雅思小作文（Task 1），准确说明横纵轴、单位与误差带。

Authentic Exemplars in Research Literature

In the figure, a shaded band denotes the observation interval.

We plotted temperature against time.

Academic Pitfall & Reviewer Perspective

- plot A against B 中颠倒纵轴 A 与横轴 B 的关系。

7.2.5 EP011: 多方法跨维度对比（In Terms Of / With Respect To / Whereas）

Template Formula & Rhetorical Function

Function: comparing_and_contrasting **Register**: formal_academic

Template Formula:

The methods are compared in terms of [Metric A] and [Metric B]: [Method 1] excels in [Condition 1], whereas [Method 2] is more robust to [Condition 2].

Exposition: 有据、具体地对比不同模型或方案的优劣，拒绝一刀切吹嘘。

Authentic Exemplars in Research Literature

The methods are compared in terms of accuracy and computational cost.

The first model uses monthly data, whereas the second uses daily data.

Academic Pitfall & Reviewer Perspective

- 比较主语悬垂或未指明比较维度。

7.2.6 EP012: 显式声明模型假设、研究范围与成立前提

Template Formula & Rhetorical Function

Function: stating_assumptions_and_constraints **Register**: formal_academic

Template Formula:

We confine the analysis to [domain/ regime]; the bound holds provided that [regularity/stationarity assumption] is satisfied.

Exposition: 用于理论推导或实证边界说明，体现科学严谨性。

Authentic Exemplars in Research Literature

We confine the analysis to observations collected before the policy change.

A sufficient condition guarantees the stated result when that condition holds.

Academic Pitfall & Reviewer Perspective

- 隐瞒模型成立的强假设。

7.3 Cluster C: Scientific Nuance, Causality & Transitions (EP013–EP018) / 科学审慎、因果辨析与篇章过渡

7.3.1 EP013: 严辨统计相关与因果机制（Correlation vs. Causality & Confounding）

Template Formula & Rhetorical Function

Function: distinguishing_correlation_and_causation **Register**: formal_academic

Template Formula:

Although [Variable X] correlates with [Outcome Y], predictive accuracy alone cannot establish causality due to potential unmeasured confounding.

Exposition: 用于机器学习可解释性、医疗/社会科学数据分析，严格划清相关与因果的界限。

Authentic Exemplars in Research Literature

A correlation between two variables does not by itself establish a causal relationship.

Unmeasured confounding may bias an estimated treatment effect.

Academic Pitfall & Reviewer Perspective

- 将特征重要性或滞后相关性直接称为因果作用。

7.3.2 EP014: 使用 respectively 与平行结构实现高密度信息配对

Template Formula & Rhetorical Function

Function: `precision_and_correspondence` **Register**: `formal_academic`

Template Formula:

`[Symbol/ Group A] and [Symbol/ Group B] denote/ represent [Entity X] and [Entity Y], respectively.`

Exposition: 在不增加从句负担的前提下清晰交代多组符号或实验参数的一一对应关系。

Authentic Exemplars in Research Literature

The symbols x and y denote temperature and demand, respectively.

The slope and intercept of f(x) = ax + b are a and b, respectively.

Academic Pitfall & Reviewer Perspective

- 前后序列元素数量不匹配或顺序颠倒。

7.3.3 EP015: 章节过渡与 Given-New 指代衔接（This + 总结名词）

Template Formula & Rhetorical Function

Function: `discourse_signposting` **Register**: `formal_academic`

Template Formula:

`This [discrepancy/limitation/pattern] motivates our next analysis, where we turn to [next topic] in greater detail.`

Exposition: 用 This + 范畴名词总结上句结果并自然开启下一句或下一节。

Authentic Exemplars in Research Literature

An anomaly in the sensor readings prompted a check of the calibration procedure.

We now turn to the evaluation criteria.

Academic Pitfall & Reviewer Perspective

- 使用指代不清的裸 This 作主语。

7.3.4 EP016: 多目标权衡（Trade-offs）与成本收益约束评估

Template Formula & Rhetorical Function

Function: evaluating_tradeoffs_and_arguments **Register**: formal_academic

Template Formula:

The design involves a trade- off between [computational cost / coverage] and [accuracy / latency]; on balance, the benefits outweigh the costs when [condition].

Exposition: 用于系统设计、模型复杂度分析以及六级/雅思托福利弊权衡类作文。

Authentic Exemplars in Research Literature

The proposal involves a trade-off between coverage and cost.

The expected benefits must outweigh the costs before the intervention is recommended.

Academic Pitfall & Reviewer Perspective

- *outweigh than（outweigh 直接接宾语，不加 than）。

7.3.5 EP017: 坦诚说明研究局限与外推风险（Limitations & Extrapolation）

Template Formula & Rhetorical Function

Function: acknowledging_limitations_and_extrapolation **Register**: formal_academic

Template Formula:

We acknowledge that [dataset/ simulation limitation]; therefore, one should not extrapolate these findings to [unseen regime] without further validation.

Exposition: 用于论文 Limitations 小节，体现成熟严谨的科学态度。

Authentic Exemplars in Research Literature

The authors acknowledge that the sample excludes rural clinics.

We should not extrapolate from a single district to the entire country without justification.

Academic Pitfall & Reviewer Perspective

- 用空洞借口掩盖核心样本偏差。

7.3.6 EP018: 全篇总结与后续研究动议（Conclusion & Future Directions）

Template Formula & Rhetorical Function

Function: concluding_and_motivating_research **Register**: formal_academic

Template Formula:

```
In summary, the empirical evidence supports [claim] under [tested conditions],
while the remaining anomalies warrant further investigation in [future setting].
```

Exposition: 收束全文核心贡献，不夸大、不遗漏边界条件。

Authentic Exemplars in Research Literature

The discrepancy warrants further investigation, but it does not warrant a causal conclusion.

The authors conclude that further external validation is needed.

Academic Pitfall & Reviewer Perspective

- *warrant to do（应为 warrant + 名词/doing）。

7.4 Cluster D: Frontier Domains, Modern AI & High-Stakes Genres (EP019–EP024) / 前沿技术、人工智能与高阶文体

7.4.1 EP019: 机器学习与时间序列：问题定义、滞后项与符号约定

Template Formula & Rhetorical Function

Function: ml_timeseries_problem_formulation **Register**: technical_specialized

Template Formula:

```
Let [symbol] denote the observation at time t; we decompose the series into
[components] and evaluate predictions over horizon H.
```

Exposition: 用于机器学习与时间序列论文的 Problem Formulation 部分，清晰界定输入历史窗口、滞后项（lagged observations）与预测步长。

Authentic Exemplars in Research Literature

Here, N denotes the number of observations, and a lag of one step links the current state to the preceding observation.

Time-series decomposition separates a series into components such as trend, seasonality and remainder.

Academic Pitfall & Reviewer Perspective

- 未说明采样间隔（sampling interval）就直接声称某个 lag 代表固定物理时长。

7.4.2 EP020: 生成模型与概率预测：分布校准、不确定性量化与尾部风险

Template Formula & Rhetorical Function

Function: generative_and_probabilistic_modeling **Register**: technical_specialized

Template Formula:

A probabilistic/ generative model assigns probabilities to possible outcomes, allowing us to quantify uncertainty and evaluate both calibration and sharpness.

Exposition: 用于生成模型（Generative Models）与概率预测论文，阐述为何需要学习完整条件分布而非单一点估计（point estimate）。

Authentic Exemplars in Research Literature

A distributional forecast is arguably more informative when decisions depend on tail risk.

Forecast calibration concerns statistical consistency between forecasts and observations.

Academic Pitfall & Reviewer Perspective

- 将高集中度（sharpness）误认为已经过概率校准（calibrated）。

7.4.3 EP021: 具身智能（Embodied AI）与机器人：空间表征、轨迹跟踪与物理约束

Template Formula & Rhetorical Function

Function: embodied_ai_and_robotics **Register**: technical_specialized

Template Formula:

To bridge the discrepancy between simulation and real- world deployment, the policy enforces [physical/ monotonicity constraints] while adapting to [sensor noise / external perturbations].

Exposition: 用于具身智能、机器人控制与视觉导航写作，准确表达仿真到真实迁移（sim-to-real）、空间方位（spatial orientation）与速度/运动学约束。

Authentic Exemplars in Research Literature

The optimization procedure enforces a non-negativity constraint on the weights while tracking the trajectory.

The sensitivity analysis examines the effect of a small perturbation to the input sensor readings.

Academic Pitfall & Reviewer Perspective

- 把仿真器（simulation）中的理想合成观测（synthetic data）直接等同于真实物理传感器的含噪观测。

7.4.4 EP022: 消融实验（Ablation Study）与基线公平对比（Fair Baseline Comparison）

Template Formula & Rhetorical Function

Function: `ablation_and_experimental_evaluation` **Register**: `formal_academic`

Template Formula:

```
To disentangle the contribution of [Component A] from [Component B], we compare the full model against a conventional baseline under comparable conditions.
```

Exposition: 用于实验与消融分析章节，严谨剥离各模块的独立贡献，避免只挑有利指标汇报（selective reporting）。

Authentic Exemplars in Research Literature

Every proposed method should be compared against a relevant baseline under comparable conditions.

The study design aims to disentangle the effect of training from prior experience.

Academic Pitfall & Reviewer Perspective

- 使用 unparallelled / groundbreaking 等空洞吹嘘词代替具体的消融实验差值与误差棒说明。

7.4.5 EP023: 六级与雅思托福（CET-6 / IELTS / TOEFL）议论文：去模板化“主张—机制—具体实例—让步”结构

Template Formula & Rhetorical Function

Function: `cet6_ielts_toefl_essay_writing` **Register**: `general_academic`

Template Formula:

```
While [Concession: competing view/ cost], [Main Claim: policy/ technology] can [Verb: facilitate/mitigate] [Outcome] by [Concrete Mechanism], as illustrated by [Specific Real-World Case].
```

Exposition: 专为六级、雅思 Task 2 与托福独立写作设计：告别陈腐套话，用一句话完成让步对比，紧接着给出具体作用机制与可信实例。

Authentic Exemplars in Research Literature

While expanding public transport requires substantial infrastructure investment, subsidising bus fares can facilitate access to education for low-income students by reducing daily commuting costs.

Asynchronous online courses allow working adults to study at their own pace; nevertheless, they cannot wholly replace face-to-face seminars that stimulate spontaneous debate.

Academic Pitfall & Reviewer Perspective

- 只写夸张口号（如 *Online education is a double-edged sword and plays a vital role*）而不展开具体原因机制与实例。

7.4.6 EP024: 学术演讲（Tech Talk）、研究陈述（SoP）与审稿回复（Rebuttal）的自然衔接

Template Formula & Rhetorical Function

Function: `oral_presentation_sop_and_rebuttal` **Register**: `spoken_academic`

Template Formula:

```
In response to the reviewer's question regarding [X], we have clarified [Y] and added an experiment on [Z]; as this slide/ table shows, the conclusion remains robust.
```

Exposition: 用于国际会议口头报告、博士/教职申请研究陈述（Research Statement）及同行评审回复，做到礼貌、直接、有理有据。

Authentic Exemplars in Research Literature

I would like to begin by explaining the sampling procedure, and I will allow time for questions at the end.

The revision addresses the limitation explicitly and incorporates feedback from the pilot evaluation.

Academic Pitfall & Reviewer Perspective

- 在 Rebuttal 中情绪化反驳（hostile remarks）或回避审稿人指出的核心假设缺陷。

Part III

Discourse Logic & Argumentation Engine / 篇章逻辑与论证引擎

Chapter 8

The Information Flow Paradigm / 篇章信息流与主位推进模型

优秀的学术段落不是由离散句子的简单拼盘，而是一条精密咬合的认知齿轮链条。读者在理解学术技术论述时，依赖每一句话句首对前文概念的自然锚定。本章系统阐述主位推进（Theme-Rheme Progression）的底层规律，确立以已知信息引导新信息的根本法则。

8.1 The Given-to-New Fundamental Law (L001) / 旧信息在前、新信息在后原则

句子包含两个核心信息层级：主位（Theme，通常位于句首，代表已知信息 Given）与述位（Rheme，位于句中和句末，代表新信息 New）。连贯段落的核心秘诀在于：后一句的主位必须承接前一句的述位或已知概念，形成链式信息流。

8.2 Thematic Progression Architectures (L002 & L003) / 两大主流主位推进架构

1. **恒定主位推进（Constant Theme, L002）**：多句话保持相同主位实体（如同一种算法架构），分别在述位抛出其在速度、精度、内存与鲁棒性等多个维度的评价。
2. **线性链式推进（Linear Rheme-to-Theme, L003）**：第一句句尾的新概念 Y，立即作为第二句句首的已知主位锚点，进而引出新概念 Z，形成 $X \rightarrow Y \Rightarrow Y \rightarrow Z$ 的因果推导闭环。

8.3 Anaphoric Anchoring & Cohesive Scaffolding (L004 & L005) / 壳名词指代与语义钩子

严禁使用裸露的代词（如单字的 *This shows that...*）。应当使用具有概括总结功能的“壳名词（Shell Nouns）”，如 *This discrepancy, This trade-off, Such an anomaly*，既指代明确，又赋予前文定性评价。

Chapter 9

Toulmin Micro-Argumentation Architecture / 图尔敏微论证架构与科学思辨

科学论证不是情绪宣泄，而是基于证据的严密辩护。现代学术话语体系广泛采用图尔敏论证模型（Toulmin Model of Argumentation）来组织段落内的微论证，确保每个学术主张均有实证基准与逻辑保证的支撑。

9.1 The Six-Constituent Toulmin Formalism (L006) / 图尔敏六要素论证模型

- **主张 (Claim)**: 本段的核心学术论点或结论。
- **数据/事实 (Data)**: 实验指标、统计观测数据或实证记录。
- **保证 (Warrant)**: 连接数据与主张的科学定律、理论桥梁或算法先验。
- **支撑 (Backing)**: 证明保证本身成立的底层数学基础或前人基石文献。
- **反驳/例外 (Rebuttal)**: 坦诚承认主张在某些极端边界条件下可能受限。
- **限制词 (Qualifier)**: 审慎限定主张成立的概率范围（typically, substantially, in finite samples）。

9.2 Mechanistic Causality vs. Spurious Correlation (L007) / 机制因果与伪相关辨析

在实验分析段落中，严禁仅凭统计相关性直接断言因果关系。必须交代物理媒介或内部特征流的传导过程，彻底杜绝逻辑假飞跃。

9.3 Concession, Pivot, and Structural Refutation (L008) / 让步、转折与结构化反驳

在学术商榷与审稿回复中，标准的绅士论辩采用“承认合理 → 聚焦边界 → 扭转权重”三步舞姿：首先诚恳承认竞品或审稿人顾虑的合理性，继而指出其隐含假设的脆弱性，最后证明本文方案在关键工况下的决定性收益。

Chapter 10

Zero AI-ese Discourse Surgery & 30 Academic Logic Rules (L001–L030) / 去 AI 化篇章外科重构与 30 条逻辑法则

随着大语言模型的普及，审稿人对充斥着 *delve into, paramount, testaments, tapestry, paves the way* 等浮夸套话的 AI-ese 风格表现出极度反感。本章设立去 AI 化篇章重构手术准则，并完整解析 30 条学术逻辑与论证核心法则（L001–L030）。

10.1 The Zero AI-ese Surgical Protocol (L029) / 去 AI 化外科重构准则

坚决割除大模型生成的“四大浮夸病灶”：

Symptom 1: **空洞修辞帽**: 诸如 *In the ever-evolving landscape of...*，一律删除，直奔领域实体。

Symptom 2: **过度吹捧**: 诸如 *groundbreaking, unprecedented, game-changer*，替换为具体的量化突破指标。

Symptom 3: **散装总结**: 诸如 *In conclusion, it is worth noting that...*，替换为紧凑的因果或对比收束。

Symptom 4: **动词贫乏**: 诸如大量滥用 *plays a crucial role in*，替换为具体动作及物动词（underpins, mitigates, regulates）。

10.2 The Complete 30 Academic Logic Rules Encyclopedia (L001–L030) / 30 条学术逻辑法则全集

10.2.1 L001: Given-to-New Information Flow / 旧信息在前、新信息在后原则（Given-to-New Principle）

Core Logic Principle & Structural Formula

Category: `Information Flow (信息流推进)`

Core Principle: 英语段落连贯的第一动力不是连接词，而是每一句的句首（主位 Theme）承接上一句句末已经出现过的概念（旧信息 Given），再在句末（述位 Rheme）抛出本句真正要推进的核心增量（新信息 New）。

Structural Formula:

`Sentence A: [Subject X] → [Verb] → [New Concept Y]. | Sentence B: [This/Such Y or Pronoun] → [Verb] → [New Concept Z].`

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Autoregressive transformers suffer from severe error accumulation over long horizons. A sudden distribution shift in non-stationary sensor streams further amplifies this error accumulation.

[Idiomatic Reconstruction]:

Autoregressive transformers suffer from severe error accumulation over long horizons. This cumulative drift accelerates sharply when non-stationary sensor streams undergo sudden distribution shifts.

Surgical Analysis: 修改前第二句以长新名词 A sudden distribution shift... 开头，旧信息 this error accumulation 被甩在句末；修改后第二句用 This cumulative drift 紧扣上一句句尾的 error accumulation，再引出新信息 distribution shifts，段落瞬间丝滑。

Academic Trap & Reviewer Perspective

Common Pitfall: 每一句都突兀地以一个全新概念作长主语，而把指代上一句的词塞在句尾，导致读者每读一句都要重新建立思维锚点。

10.2.2 L002: Constant Theme Progression / 平行主位推进模式（Constant Theme Progression）

Core Logic Principle & Structural Formula

Category: `Information Flow (信息流推进)`

Core Principle: 当一段话旨在从多个互补维度刻画同一个核心模型、算法或社会现象时，连续 2–3 句保持相同或同指代的主语（如 Our framework / The temporal encoder / It），让谓语动词承担信息递进。

Structural Formula:

```
S1: [Theme T] + [Action 1 (Architecture)]. | S2: [T / It] + [Action 2 (Optimization)]. | S3: [By combining both, T] + [Action 3 (Empirical Outcome)].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

The encoder extracts multi-scale patches first. Then frequency spectrum filtering is performed by a Fourier block. Finally, high-resolution predictions are generated by the decoder.

[Idiomatic Reconstruction]:

Our architecture processes non-stationary sequences in three stages. It first partitions the raw series into multi-scale temporal patches, then filters high-frequency noise in the Fourier domain, and finally decodes these refined representations into multi-horizon forecasts.

Surgical Analysis: 修改前主语在 encoder、filtering、predictions 之间无序跳转且混用被动语态；修改后以 Our architecture / It 为统一主位，用三个并列强动词 partitions → filters → decodes 清晰串联数据流。

Academic Trap & Reviewer Perspective

Common Pitfall: 为了所谓“句式多变”强行每句换主语、频繁在主动与被动之间横跳，导致读者搞不清到底是谁在执行模块动作。

10.2.3 L003: Linear Rheme-to-Theme Chain / 延续链式推进模式（Linear Chain Progression）

Core Logic Principle & Structural Formula

Category: `Information Flow（信息流推进）`

Core Principle: 阐述因果机制、算法推导或仿真到真实迁移（Sim-to-Real）链条时，严格采用 A → B，B → C，C → D 的咬合结构，上一句的宾语转化为下一句的主语。

Structural Formula:

`S1: A induces B. → S2: This B distorts C. → S3: Once C degrades, D collapses.`

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Unmodeled friction in simulated joints causes real-world grasping failure. Domain randomization often fails because state transitions diverge.

[Idiomatic Reconstruction]:

Unmodeled joint friction in simulators creates a systematic discrepancy between synthetic and physical state transitions. This transition mismatch misleads visuomotor policies during contact-rich phases, ultimately causing real-world grasping failures.

Surgical Analysis: 通过 synthetic and physical state transitions → This transition mismatch → contact-rich phases → grasping failures 形成严密的四环机械传导链。

Academic Trap & Reviewer Perspective

Common Pitfall: 中间跳过关键传导环节（如直接从 A 跳到 D），或者链条顺序颠倒。

10.2.4 L004: Shell Noun / Summary Noun Anaphora / 概念浓缩指代词（This/Such + Summary Noun）

Core Logic Principle & Structural Formula

Category: Cohesion & Reference（指代与衔接）

Core Principle: 严禁单独使用裸露的 This / That / It 指代上一整句话！必须在指示代词后紧跟一个精准概括上一句本质的抽象范畴名词（如 This discrepancy / Such degradation / These computational constraints / This trade-off）。

Structural Formula:

```
Sentence 1: [Full empirical observation or limitation]. → Sentence 2: This/Such [Summary Noun: bottleneck / discrepancy / vulnerability / shift] + [Active Verb: motivates / necessitates / limits / explains]...
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Diffusion models require hundreds of iterative denoising steps at inference time. This prevents them from running on onboard robot controllers.

[Idiomatic Reconstruction]:

Diffusion policies require hundreds of iterative denoising steps at inference time. This computational latency prevents them from running at closed-loop control frequencies on onboard robot hardware.

Surgical Analysis: 用 This computational latency 将上一句“数百步去噪迭代”精准定性为“计算延迟”，既消除了指代歧义，又完成了因果定性。

Academic Trap & Reviewer Perspective

Common Pitfall: 句首单写’This makes the model slow’ 或’Which causes error’，审稿人和雅思考官无法判断 This 究竟指上一句的主语、宾语还是整件事。

10.2.5 L005: Semantic Hooks over Mechanical Connectors / 去机械连接词化：用语义钩子替代 First/Second/Moreover

Core Logic Principle & Structural Formula

Category: Cohesion & Reference（指代与衔接）

Core Principle: 母语学者与高分作文极少每句句首都贴 Firstly, Secondly, Moreover, What is more, In addition 等膏药。应使用语义承接状语（Beyond accuracy gains, / Alongside temporal drift, / At the hardware level,）实现自然递进。

Structural Formula:

```
Instead of 'Moreover, X does Y', use: 'Beyond [Summary of Prior Point], [Subject X] also [Verb Y].'
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

First, our model reduces MSE by 15%. Second, our model uses 40% less memory. Moreover, our model generalizes to unseen sensors.

[Idiomatic Reconstruction]:

Our model reduces forecasting MSE by 15.4% across six benchmarks. Beyond accuracy gains, its sparse attention mechanism cuts peak GPU memory by 40%, enabling zero-shot deployment on unseen sensor arrays.

Surgical Analysis: 用 Beyond accuracy gains 承上启下，用 enabling... 自然引出第三层泛化收益，彻底摆脱机械序数词。

Academic Trap & Reviewer Perspective

Common Pitfall: 一段 5 句话连续用 First, Second, Besides, Moreover，而句子内部毫无词汇呼应，形成典型的中式“八股排比腔”或 AI 列表腔。

10.2.6 L006: Claim-Mechanism-Evidence-Warrant-Qualification / 图尔敏五步微论证模型（Toulmin Micro-Argumentation）

Core Logic Principle & Structural Formula

Category: `Argumentation (论证结构)`

Core Principle: 任何一个独立成段的学术或议论主张，都必须具备完整的五要素：(1) 主张（Claim）、(2) 机理解释（Mechanism/Reason）、(3) 实证数据或具象案例（Evidence）、(4) 逻辑桥梁（Warrant：为什么该证据能证明该主张）、(5) 边界限定（Qualification/Hedging）。

Structural Formula:

`[S1: Claim] → [S2: Underlying Mechanism] → [S3: Concrete Metric/ Instance] → [S4: Warrant/Implication] → [S5: Boundary/Scope Qualification].`

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Patch-level tokenization is much better than point-wise tokenization. It captures local information very well and improves time series forecasting significantly in all cases.

[Idiomatic Reconstruction]:

Patch-level tokenization outperforms point-wise representations in long-horizon time series forecasting [Claim]. Whereas single time steps lack semantic context and amplify high-frequency noise, contiguous sub-series patches preserve local slope and periodicity [Mechanism]. On the ETTm2 benchmark, grouping 16 consecutive steps reduces 720-step MSE by 18.7% compared with point-wise attention [Evidence]. This gain confirms that local structural aggregation mitigates attention dis-

persion over long lookback windows [Warrant], although excessively large patches may blur abrupt transient anomalies [Qualification].

Surgical Analysis: 五句话各司其职：立论 → 对比机理 → ETTm2 实证数据 → 理论印证 → 诚实指出过大 Patch 的边界局限，无懈可击。

Academic Trap & Reviewer Perspective

Common Pitfall: 只有 Claim + 换句话重复一遍 Claim + 空洞夸张形容词，中间既无传导机制也无具象证据。

10.2.7 L007: Causal Mechanism vs. Spurious Correlation / 因果机制链 vs. 表面相关性切分（Causality vs. Correlation）

Core Logic Principle & Structural Formula

Category: `Argumentation（论证结构）`

Core Principle: 在论文实验分析与雅思/托福议论文中，观察到 A 与 B 同时变化时，绝不能直接断言 A causes B；必须区分“直接因果（drives / induces）”、“中介传导（mediates / channels）”与“混杂相关（coincides with / correlates with）”。

Structural Formula:

```
Observation: A correlates with B. → Confounder check: Rather than A directly causing B, confounding factor C drives both → Controlled verification.
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Increasing model parameters makes the robot grasp better in the real world.

[Idiomatic Reconstruction]:

While scaling backbone parameters correlates with higher success rates in simulation, real-world grasping gains stem primarily from diverse tactile pre-training rather than parameter count alone.

Surgical Analysis: 精准用 correlates with 界定表面伴随关系，再用 stem primarily from... rather than... 点明真实因果驱动力。

Academic Trap & Reviewer Perspective

Common Pitfall: 将伴随发生的数据现象直接写成绝对因果，或用 make / let 等口语词粗糙表达因果。

10.2.8 L008: Concession, Pivot, and Structural Refutation / 先让步、再釜底抽薪式反驳（Concession-Pivot-Refutation）

Core Logic Principle & Structural Formula

Category: Argumentation（论证结构）

Core Principle: 无论是论文引言批评前人方法、Rebuttal 回复审稿人，还是六级/雅思大作文回应反方观点，最高级的说服力都来自“先承认对方在特定受限条件下确实有效（Concession），再指出该条件在真实/复杂场景下不成立（Pivot & Refutation）”。

Structural Formula:

```
Admittedly / While [Prior Method / Opposing View] achieves [Specific Merit] under [Strict Assumption A], this assumption breaks down when [Real-World Condition B], resulting in [Concrete Failure].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

MLP-based forecasters are too simple and cannot model complex dependencies, so our Transformer model is necessary.

[Idiomatic Reconstruction]:

While lightweight MLP forecasters achieve competitive accuracy on stationary sequences with stable periodicity, their fixed temporal weights cannot adapt dynamically when industrial sensor streams undergo abrupt regime shifts.

Surgical Analysis: 先肯定 MLP 在平稳周期序列上的竞争力，再精准指出其“固定时间权重无法适应突发工况漂移”的结构性软肋，有理有据。

Academic Trap & Reviewer Perspective

Common Pitfall: 一上来全盘否定前人工作（Previous works are totally wrong/useless），显得极不客观且容易激怒审稿人。

10.2.9 L009: Taxonomic Synthesis vs. Chronological Laundry List / 分类矩阵综述法：拒绝 A did X, B did Y 流水账（Taxonomy over Laundry List）

Core Logic Principle & Structural Formula

Category: Paragraph Architecture（段落架构）

Core Principle: 撰写 Related Work 或文献背景段时，严禁以人名作主语逐句罗列（Smith (2022) proposed X. Lee (2023) proposed Y. Wang (2024) used Z.）。必须以“方法流派/技术范式”为主语进行聚类，把文献引用退居句尾括号或作同位语例证。

Structural Formula:

```
Existing approaches to [Problem] fall into two primary paradigms: [Paradigm
A] and [Paradigm B]. [Paradigm A] methods (e.g., [1, 2]) rely on..., whereas
[Paradigm B] architectures [3, 4] leverage... However, neither paradigm resolves
[Core Bottleneck].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Wu et al. proposed Autoformer in 2021. Zhou et al. proposed FEDformer in 2022. Nie et al. proposed PatchTST in 2023. They all improved time series forecasting.

[Idiomatic Reconstruction]:

Recent deep forecasters mitigate long-range attention complexity either by decomposing seasonal-trend components in the frequency domain (e.g., Autoformer, FEDformer) or by aggregating local sub-series into channel-independent patches (e.g., PatchTST). Yet both paradigms treat cross-variable correlations as static over time, overlooking transient inter-sensor coupling during system faults.

Surgical Analysis: 两句话完成流派二分法归纳 + 代表作锚定 + 共同盲区提炼，学术密度提升数倍。

Academic Trap & Reviewer Perspective

Common Pitfall: 连续 6 句话全部是 [Author] et al. proposed/introduced/developed...，只有罗列没有批判性归类。

10.2.10 L010: The Metric-Baseline-Condition Precision Triad / 定量实证三元组：指标 + 基线 + 边界条件（Metric-Baseline-Condition Triad）

Core Logic Principle & Structural Formula

Category: Precision & Grounding（精确度与实证落地）

Core Principle: 任何宣称“更好、更快、更鲁棒、影响深远”的句子，必须同时交代三个维度：(1) 具体评估指标与幅度（Metric & Magnitude）、(2) 对比参照系（Baseline）、(3) 评测工况或数据集条件（Condition）。

Structural Formula:

```
[Method/ Policy] improves/ reduces [Metric] by [Exact %/ Value] compared with
[Strongest Baseline] under [Specific Dataset/Horizon/Noise Regime].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Our generative policy achieves much better success rates and runs much faster than previous methods.

[Idiomatic Reconstruction]:
Our consistency-distilled policy increases bimanual peg-insertion success rates from 64% to 89% while cutting inference latency from 180 ms to 12 ms relative to a 50-step DDPM baseline.

Surgical Analysis: 用 64%→89%（指标）、180ms→12ms（延迟）、relative to a 50-step DDPM baseline（基线）和 bimanual peg-insertion（任务条件）彻底消灭空洞形容词。

Academic Trap & Reviewer Perspective

Common Pitfall: 只写 significantly improves performance 或 achieves state-of-the-art results，属于典型的“不精确 + 空洞夸张”。

10.2.11 L011: Subject Compression and End-Weight Balance / 段落内主语瘦身与尾重平衡（Subject Compression & End-Weight）

Core Logic Principle & Structural Formula

Category: `Sentence & Paragraph Rhythm（节奏与尾重）`

Core Principle: 在多句段落中，严禁出现超过 10–12 个单词的臃肿主语从句或连环 of 短语作主语（头重脚轻）。把已知的短核心名词放在主语位置（≤6 词），将复杂的条件、机制和结果顺延到谓语动词之后的宾语或状语从句中。

Structural Formula:
`Bad: [The phenomenon that A with B under C fails to D] is common. → Good: [Short Subject A] frequently fails to [Verb D] when [Condition B and C occur].`

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:
The phenomenon that autoregressive language models hallucinate unverified citations when prompted with out-of-distribution academic queries is a serious obstacle.

[Idiomatic Reconstruction]:
Autoregressive language models frequently hallucinate unverified citations when prompted with out-of-distribution academic queries, posing a serious obstacle to automated literature review.

Surgical Analysis: 主语从 17 个词压缩为 3 个词（Autoregressive language models），强动词 hallucinate 提前，句尾用分词结果状语自然收束。

Academic Trap & Reviewer Perspective

Common Pitfall: 受中文“……的现象/问题/事实表明……”影响，写出 The fact that... is... 或 The problem of the degradation of... is... 的畸形句。

10.2.12 L012: One Paragraph, One Controlling Idea / 一段一核：单一控制中心法则（Single Controlling Idea Rule）

Core Logic Principle & Structural Formula

Category: Paragraph Architecture (段落架构)

Core Principle: 一个自然段只能回答一个核心问题（例如：这一段只讲“为什么频域滤波能抑制传感器高频噪声”，就绝不要在同一段末尾顺手塞进“此外我们还开源了代码并降低了显存”）。

Structural Formula:

```
Topic Sentence (Controlling Idea X) → Mechanism of X → Evidence for X → Boundary/Synthesis of X. (Move Concept Y to Next Paragraph).
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Our spatial-temporal attention block captures cross-sensor correlations accurately. Also, training takes only two hours on a single GPU, and the public dataset we used has some missing timestamps that we interpolated linearly.

[Idiomatic Reconstruction]:

Our spatial-temporal attention block captures dynamic cross-sensor correlations by weighting neighbor nodes according to instantaneous phase lag. This phase-aware aggregation prevents faulty sensors from corrupting adjacent channels during localized hardware spikes, reducing multi-sensor MAE by 11.8%.

Surgical Analysis: 砍掉无关的训练耗时与数据清洗杂音，全段聚焦于“相移感知图注意力如何隔离故障传感器并提升多传感器预测精度”。

Academic Trap & Reviewer Perspective

Common Pitfall: 一段话里既讲精度、又讲速度、还顺带抱怨数据集质量，导致段落像杂货铺一样失焦。

10.2.13 L013: Deductive Zoom-In vs. Inductive Zoom-Out / 演绎推进（Zoom-In）与归纳提炼（Zoom-Out）双模切换

Core Logic Principle & Structural Formula

Category: Paragraph Architecture (段落架构)

Core Principle: 写方法论与议论文主论点段宜用演绎法（Zoom-In：首句抛出概括性规律，随后逐句拆解机理与实例）；写实验分析与看图小作文局部段宜用归纳法（Zoom-Out：先报告具体观测对比，末句提炼背后的理论规律）。

Structural Formula:

```
Zoom-In: [General Claim] → [Specific Mechanism] → [Concrete Data]. | Zoom-Out: [Data Observation A vs. B] → [Pattern Across Horizons] → [Theoretical Takeaway].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

At horizon 96, our MSE is 0.182. At horizon 336, our MSE is 0.245. At horizon 720, our MSE is 0.310, while Informer reaches 0.612.

[Idiomatic Reconstruction]:

As the prediction horizon extends from 96 to 720 steps, our model's MSE rises modestly from 0.182 to 0.310, whereas Informer degrades sharply to 0.612. This widening performance gap demonstrates that decoupling trend projection from seasonal attention effectively suppresses long-range error compounding.

Surgical Analysis: 第一句整合多步长数据趋势对比，第二句用 This widening performance gap demonstrates that... 完成归纳升华，点明“趋势与季节解耦抑制长程误差累积”的本质原因。

Academic Trap & Reviewer Perspective

Common Pitfall: 在实验分析段只罗列完数据就戛然而止，缺少最后一句 Zoom-Out 的机理升华（Why this pattern matters）。

10.2.14 L014: Intuition-First Equation Integration / 公式与散文无缝咬合：先给直觉动机，后推符号定义（Intuition-Before-Math）

Core Logic Principle & Structural Formula

Category: Methodology Writing（方法章节逻辑）

Core Principle: 在 Methodology 段落中引入数学公式时，严禁突兀甩出'The equation is defined as follows:'。必须遵循三步曲：(1) 物理/算法直觉（为什么要算这个量）→ (2) 数学定义（把公式作为句子的语法宾语或同位语自然嵌入）→ (3) 关键符号的物理含义与边界。

Structural Formula:

```
To [Intuitive Goal], we formulate [Concept] as: [Equation], where [Symbol A] denotes [Meaning] and [Symbol B] constrains [Boundary].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

We use a gating weight. The formula is w = sigmoid(Wx + b). In this formula, x is the feature and w is the weight.

[Idiomatic Reconstruction]:

To suppress high-frequency sensor jitter before temporal aggregation, we compute a channel-wise attenuation gate $g_t = \sigma(W_g h_t + b_g)$*, where* $h_t \in \mathbb{R}^d$ *denotes the instantaneous hidden state and* $g_t \in (0,1)^d$ *dynamically scales down noisy channels.*

Surgical Analysis: 一句话之内完成“目的状语（抑制高频抖动）+ 核心主谓宾（计算通道衰减门）+ 公式同位语 + where 定语从句解释符号维度与物理作用”，紧凑专业。

Academic Trap & Reviewer Perspective

Common Pitfall: 公式前后全是碎片句（Here is the formula: ... In the formula, x is input, y is output.），破坏段落连贯。

10.2.15 L015: Pattern Interpretation over Table Recitation / 图表叙事铁律：解读规律而非朗读表格（Interpret Patterns, Never Read Tables Aloud）

Core Logic Principle & Structural Formula

Category: `Experimental Writing（实验章节逻辑）`

Core Principle: 读者自己会看表格数字。实验段落的任务不是把 Table 1 的每一行数字抄进正文，而是回答三个问题：(1) 跨数据集的整体胜率与极值在哪？(2) 哪类基线在什么条件下崩盘了？(3) 这种差异验证了我们在引言/方法中提出的什么假设？

Structural Formula:
`[Table Reference + Overall Dominance] → [Contrast between Baseline Families under Hard Regime] → [Mechanistic Explanation linking back to Method].`

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:
Table 1 shows the results. On dataset A, our model gets 91%. On dataset B, our model gets 88%. Our model is the best on all datasets.

[Idiomatic Reconstruction]:
Table 1 shows that our model achieves the lowest forecasting error on 26 out of 30 benchmark settings. While point-wise Transformers degrade rapidly on volatile electricity load datasets (e.g., ECL), our multi-scale spectral mixer reduces MSE by an average of 16.4%, confirming that frequency-domain denoising stabilizes long-horizon representations.

Surgical Analysis: 既有全局概览（26/30 最优），又有困难场景（波动剧烈的电力负载 ECL）下的流派对比，最后回扣方法论核心假设。

Academic Trap & Reviewer Perspective

Common Pitfall: 把表格每一行数字机械抄进正文，只报数不提炼规律与原因。

10.2.16 L016: Counterfactual Ablation Attribution / 消融实验归因逻辑：移除模块 → 暴露失效模式 → 反证模块必要性（Ablation Attribution）

Core Logic Principle & Structural Formula

Category: `Experimental Writing (实验章节逻辑)`

Core Principle: 消融实验段落（Ablation Study）的核心逻辑是“反事实归因”：不仅要报告拿掉模块 A 后指标下降了多少，更要指出拿掉 A 后模型具体在什么子任务/子场景上出现了什么退化，从而锁定模块 A 的不可替代性。

Structural Formula:

```
Removing [Module A] (w/ o A) causes [Metric] to drop by [X%], with the sharpest degradation occurring under [Specific Hard Condition]. This drop indicates that without [Module A], the model fails to [Specific Function].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

When we remove the kinematics constraint loss, the success rate drops by 14%. This shows that the kinematics constraint loss is very important for our robot.

[Idiomatic Reconstruction]:

Removing the kinematic feasibility loss (w/o $\mathcal{L}_{kin}$) reduces real-world drawer-opening success from 86% to 72%. Inspection of failed trajectories reveals that without $\mathcal{L}_{kin}$, the diffusion policy frequently generates wrist orientations that exceed joint torque limits during the pull phase.

Surgical Analysis: 不仅给出 86%→72% 的量化回落，还通过轨迹检查揭示了“缺少运动学约束导致手腕姿态超出关节扭矩极限”的具体物理死因。

Academic Trap & Reviewer Perspective

Common Pitfall: 只写 Without module A, accuracy drops by 3%, which proves module A is very important and effective（同义反复，未解释为什么降）。

10.2.17 L017: Constructive Boundary Framing in Limitations / 局限性（Limitations）的无损诚实写法：界定适用边界而非自毁贡献

Core Logic Principle & Structural Formula

Category: `Discussion & Hedging (边界与审慎表达)`

Core Principle: 顶会强制要求写 Limitations。高水平写法是：明确指出当前方法在什么极端假设违背下（如极端稀疏采样、毫秒级超高速碰撞）性能会受限，并立即给出造成该边界的客观算力/物理原因及可行的下一步演进路径。

Structural Formula:

```
While our framework performs robustly under [Standard Regime], its accuracy/
efficiency is currently bounded when [Extreme Condition]. This limitation arises
because [Objective Mechanism], suggesting a promising direction in [Concrete
Future Fix].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Our model has some limitations. It is too slow when there are many variables and does not work well on irregular time series.

[Idiomatic Reconstruction]:

While our cross-variate graph module scales linearly with sequence length, its quadratic complexity with respect to channel count ($O(C^2)$) increases memory overhead when $C > 1,000$. Incorporating low-rank channel projection or sparse k-NN graph routing offers a direct path to scaling toward ultra-high-dimensional sensor arrays.

Surgical Analysis: 把“慢”精确化为通道维度的 $O(C^2)$ 复杂度边界（$C > 1,000$），并顺势给出低秩投影或稀疏 k-NN 路由的建设性解法。

Academic Trap & Reviewer Perspective

Common Pitfall: 要么写成虚假自夸（Our only limitation is that our model is too general），要么写成致命自毁（Our model fails completely on real data）。

10.2.18 L018: Spoken Rhythm, Visual Anchoring, and Rhetorical Signposting / 口头学术演讲的听觉信息流：短分句、视觉锚点与设问引导（Oral Signposting）

Core Logic Principle & Structural Formula

Category: Presentation & Speech（学术演讲与汇报）

Core Principle: 听众无法像读论文那样倒回去重读长定语从句。把书面段转化为演讲稿时，必须：(1) 砍掉多重嵌套从句，拆为 10–15 词的清晰短句；(2) 先给视觉坐标（Look at the red curve in the upper-right panel）再讲结论；(3) 用轻量设问（Why does this spike happen? Because...）引导注意力。

Structural Formula:

```
[Visual Anchor: Turning to the right panel of Slide 4,] → [Short Rhetorical
Question: why does the baseline diverge after step 50?] → [Crisp Causal Answer:
The answer lies in compound orientation drift.]
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

As illustrated in Figure 4, the trajectory generated by the baseline model without temporal consistency regularization exhibits substantial oscillation around the target waypoint.

[Idiomatic Reconstruction]:
Take a look at the dashed blue trajectory in Figure 4. Notice how violently the robot arm oscillates near the target waypoint. Why does this happen? Without temporal consistency regularization, adjacent action chunks conflict with one another.

Surgical Analysis: 用 Take a look at... → Notice how... → Why does this happen? → Without X, Y conflicts... 营造极具现场掌控力的母语级 Tech Talk 节奏。

Academic Trap & Reviewer Perspective

Common Pitfall: 拿着书面论文原封不动照念长达 40 个词的复合句，台下听众在前 10 秒就丢失了主谓关系。

10.2.19 L019: Evidence-First Polite Rebuttal Protocol / 顶会 Rebuttal 黄金四步：致谢认同 → 结论前置 → 新增证据/表格 → 修订位置锚定

Core Logic Principle & Structural Formula

Category: `Rebuttal & Peer Review (审稿回复逻辑)`

Core Principle: 回复审稿人尖锐质疑时，绝不能情绪化辩解或绕圈子。第一句简短感谢并重申问题核心；第二句直接给出明确结论（Direct Answer）；第三至四句呈上新增实验数据或理论澄清（New Empirical Evidence）；末句指明已在修订稿哪一节哪一表更新。

Structural Formula:

```
We thank Reviewer X for raising this insightful question regarding [Topic].
[Direct Answer: Our method maintains 84% success even under 30% occlusion].
Specifically, our new ablation in Table R1 shows that [Data]. We have
incorporated this analysis into Section 4.3 (highlighted in blue).
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:
You misunderstood our paper. We already explained this in Section 3. Our model is actually very robust to noise.

[Idiomatic Reconstruction]:
We thank the reviewer for highlighting the question of sensor noise robustness. To verify performance under severe corruption, we evaluated our policy under Gaussian observation noise ($\sigma \in [0.05, 0.20]$). As shown in Table R2 below, our model degrades by only 3.2% at $\sigma = 0.20$, compared with a 21.5% drop for the Diffusion Policy baseline. We have added these results to Appendix D.

Surgical Analysis: 礼貌专业，直接用新增噪声区间实验数据（3.2% vs. 21.5% 跌幅对比）让审稿人心服口服。

Academic Trap & Reviewer Perspective

Common Pitfall: 用大段文字教育审稿人“你没看懂我们的论文”，或者只口头承诺“我们以后会做实验”而不给真实数据。

10.2.20 L020: Counterfactual Consequence Chain / 反事实假设论证法（Counterfactual Reasoning in Exam Essays）

Core Logic Principle & Structural Formula

Category: Exam Essays（六级/雅思/托福论证）

Core Principle: 在六级、雅思 Task 2 和托福独立写作中，当正面说理写满两句感觉词穷时，立即引入一句“反事实推演（Absent this policy / Without such regulation / Were universities to abandon X）”：指出如果缺失该条件，会引发什么具体负面连锁反应，从而反向锁死中心论点。

Structural Formula:

```
[Positive Mechanism S1-      S2]. Conversely, without [Core Factor X], [System/
Individuals] would inevitably face [Concrete Negative Outcome Y], as seen when
[Real-World Scenario Z].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Independent critical thinking is very important for college students in the AI era. Students should think independently so that they can have a bright future.

[Idiomatic Reconstruction]:

Training students to verify algorithmic outputs against primary sources cultivates rigorous analytical judgment. Without this verification habit, undergraduates risk accepting plausible-sounding AI hallucinations as factual premises, eroding their capacity to diagnose complex engineering failures.

Surgical Analysis: 第一句正面阐述“核查一手文献培养分析判断力”，第二句用 Without this verification habit... 反推不核查带来的具体后果（把逼真的 AI 幻觉当成事实前提），论证立体深刻。

Academic Trap & Reviewer Perspective

Common Pitfall: 正面车轱辘话来回重复三遍（X is important. Therefore we must do X because X is very beneficial to society.）。

10.2.21 L021: Concrete Single-Sentence Micro-Case Illustration / 具象微观场景举例法：用一句话精确案例终结空洞模板（Micro-Case Grounding）

Core Logic Principle & Structural Formula

Category: Exam Essays（六级/雅思/托福论证）

Core Principle: 六级/雅思/托福作文严禁编造虚假的调查数据（如 *A recent survey by Harvard shows 87.3% of people...*），也严禁写空洞口号（*For example, technology helps doctors a lot*）。最高分做法是用一句 'For instance, when [Specific Actor] implements [Specific Tool/Policy] in [Specific Scenario], [Concrete Measurable Effect]' 构建具象行业微观切片。

Structural Formula:

```
For instance, when [municipal grid operators / radiology departments / logistics warehouses] deploy [specific technology/ policy], they can [concrete operational improvement], thereby [link back to paragraph claim].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

For example, according to a recent survey, 92% of factories that use robots become very rich and successful.

[Idiomatic Reconstruction]:

For instance, when automotive assembly plants deploy collaborative robots for repetitive welding and heavy chassis lifting, human technicians can transition toward quality inspection and fault diagnostics, reducing workplace musculoskeletal injuries while raising line throughput.

Surgical Analysis: 不编造假百分比，而是通过“汽车总装厂部署协作机器人承担重复焊接与底盘搬运 → 技工转向质检与故障诊断 → 减少工伤并提升产线吞吐量”展现母语级具体说服力。

Academic Trap & Reviewer Perspective

Common Pitfall: 举例要么假大空（For example, Jack studied hard and succeeded），要么编造一眼假的统计数字。

10.2.22 L022: Eliminating Tautology and Circular Explanations / 杜绝循环论证与同义反复（Eliminating Circular Reasoning / Tautology）

Core Logic Principle & Structural Formula

Category: Logical Fallacy Prevention（逻辑谬误防范）

Core Principle: 原因分句与结果分句不能只是换个近义词把同一件事说两遍（如 *The model generalizes well because it has strong generalization ability*）。because / since / as 之后必须提供处于更底层的 ** 结构、物理或统计机制 **。

Structural Formula:
`Claim: [High- Level Property A]. → Reason: Because [Low- Level Structural/ Statistical Mechanism B], NOT [Synonym of Property A].`

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:
Our policy is more adaptable to new objects because it possesses superior cross-object adaptability.

[Idiomatic Reconstruction]:
Our policy adapts reliably to unseen object geometries because it conditions grasp proposals on local surface normals rather than global category templates.

Surgical Analysis: 将“具有跨物体适应性”替换为底层几何机制“以局部表面法向量而非全局类别模板作为抓取条件”，真正回答了 Why。

Academic Trap & Reviewer Perspective

Common Pitfall: 经常写出“因为模型更鲁棒，所以它在噪声下表现更好”这种零信息增量的同义反复句。

10.2.23 L023: Moving Beyond False Dichotomies to Pareto Trade-offs / 破除二元对立谬误：展现多维权衡（Transcending False Dichotomies）

Core Logic Principle & Structural Formula

Category: `Logical Fallacy Prevention (逻辑谬误防范)`

Core Principle: 无论是讨论模型设计（精度 vs. 速度、生成多样性 vs. 保真度）还是社会议题（发展 AI vs. 保护就业、基础理论 vs. 工程应用），避免非黑即白的极端断言；用兼容性或帕累托改进框架（reconcile / bridge / balance X and Y by...）构建高阶论点。

Structural Formula:
`Rather than treating [Goal A] and [Goal B] as mutually exclusive, [Approach/ Policy] reconciles both by [Mechanism C].`

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:
Universities should stop teaching theoretical courses and only teach practical coding skills.

[Idiomatic Reconstruction]:
Rather than treating mathematical foundations and engineering practice as mutually exclusive, computer science curricula can integrate both by embedding convex optimization and probability theory directly into hands-on machine learning studio projects.

Surgical Analysis: 用 Rather than treating A and B as mutually exclusive, ... reconciles/integrates both by... 展现成熟严密的学术思辨。

Academic Trap & Reviewer Perspective

Common Pitfall: 非此即彼，为了夸大一端而把另一端贬得一无是处。

10.2.24 L024: Non-Stationary Drift and Multi-Scale Decomposition Logic / 时间序列非平稳性与多尺度解耦论证规范（Time Series Non-Stationarity Logic）

Core Logic Principle & Structural Formula

Category: `Domain Discourse (时间序列专属论证)`

Core Principle: 论述时间序列问题时，必须清晰区分两类矛盾：(1) 统计分布漂移（Distribution Shift：均值/方差随时间突变）与局部时序语义（Intrinsic Temporal Dependency）之间的矛盾；(2) 短期高频波动（High-frequency Fluctuations）与长期宏观趋势（Long-term Trend）的混叠干扰。

Structural Formula:

```
Naive normalization removes [Non- stationary shift] at the expense of [Intrinsic scale information]. → Our reversible/  decoupled module disentangles [Component A] from [Component B], restoring [Target Property].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Real-world time series are very complicated and change all the time, so normal normalization makes predictions bad.

[Idiomatic Reconstruction]:

While instance normalization mitigates non-stationary mean and variance shifts across lookback windows, stripping these statistics discards amplitude cues essential for forecasting peak electricity loads. Our adaptive de-stationary projection restores horizon-specific scale statistics during decoding, preserving both stationarity in attention computation and physical amplitude fidelity in the final output.

Surgical Analysis: 精准切中时间序列领域最核心的“去平稳化归一化抹除物理幅值信息 vs. 注意力层需要平稳分布”的张力。

Academic Trap & Reviewer Perspective

Common Pitfall: 笼统说 time series data is very complex and hard to predict，说不清究竟是协变量偏移、频率混叠还是采样不规则。

10.2.25 L025: Embodied Perception-Action Loop and Sim-to-Real Logic / 具身智能感知-决策-控制闭环与 Sim-to-Real 论证规范（Embodied AI Closed-Loop Logic）

Core Logic Principle & Structural Formula

Category: Domain Discourse（具身智能专属论证）

Core Principle: 论述具身智能（Embodied AI）与机器人策略时，严禁将其等同于静态图像识别。必须突出 ** 闭环反馈（Closed-loop feedback）**、** 分布外状态累积偏移（Covariate shift in sequential decision-making）**、** 接触动力学不确定性（Contact-rich dynamics）** 以及 ** 机载实时推理延迟约束（Onboard real-time latency）**。

Structural Formula:

```
Unlike static perception where errors remain isolated, closed- loop manipulation compounds small action deviations into out-  of-  distribution states. → To maintain stability under contact uncertainty, our policy [Mechanism].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

The robot sees the image from the camera and predicts the action to pick up the cup accurately.

[Idiomatic Reconstruction]:

Unlike static visual recognition where prediction errors remain isolated across samples, closed-loop robotic manipulation compounds minor trajectory deviations into unfamiliar contact states. To prevent this covariate drift during precision assembly, our visuomotor policy conditions short-horizon action chunks on real-time wrist force-torque feedback.

Surgical Analysis: 鲜明对比静态视觉与闭环操作中的误差累积（covariate drift），并引出视觉-力觉多模态短视野动作块控制机制。

Academic Trap & Reviewer Perspective

Common Pitfall: 把机器人控制写得像普通分类任务（The robot looks at the image and outputs the right class of movement）。

10.2.26 L026: Fidelity, Mode Coverage, and Sampling Efficiency Trilemma / 生成模型三元权衡论证规范：生成质量、模式覆盖度与采样效率（Generative Trilemma Logic）

Core Logic Principle & Structural Formula

Category: `Domain Discourse（生成模型专属论证）`

Core Principle: 分析生成模型（Diffusion, Flow Matching, Autoregressive, GAN, VAE）时，段落逻辑应围绕生成三难困境（Trilemma）展开：(1) 样本保真度（Sample Fidelity / Temporal Realism）、(2) 分布模式覆盖度/多样性（Mode Coverage / Diversity，防止 Mode Collapse）、(3) 推理采样步数与速度（Inference NFE / Sampling Latency）。

Structural Formula:

```
While [Model Family A] excels at [Fidelity/    Diversity], it incurs [High NFE / Mode Collapse]. → By reformulating the generation trajectory via [Rectified Flow / Consistency Distillation], our approach preserves [Property 1] while reducing sampling steps from [N] to [k].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Diffusion models generate very realistic data, but they are too slow. Our model makes them fast and still good.

[Idiomatic Reconstruction]:

While denoising diffusion models capture multi-modal trajectory distributions without mode collapse, their reliance on tens of iterative solver steps precludes real-time control. By straightening the probability flow ODE via reflow distillation, our generator preserves multi-modal path diversity while compressing inference to a single forward pass.

Surgical Analysis: 精准使用 multi-modal trajectory distributions、without mode collapse、iterative solver steps、straightening the probability flow ODE 等领域高阶表达。

Academic Trap & Reviewer Perspective

Common Pitfall: 只说 the generated samples look very good and realistic，缺乏分布覆盖度与采样效率维度的严谨坐标。

10.2.27 L027: Inter-Paragraph Handshake and Bridge Clauses / 段尾留钩与段首咬合：跨段落无缝过渡法则（Paragraph-to-Paragraph Handshake）

Core Logic Principle & Structural Formula

Category: `Paragraph Transitions (段间衔接)`

Core Principle: 整篇小论文或大作文不是几个孤立段落的拼盘。上一段的末句应为下一段埋下伏笔，或者下一段的首句用前置状语短语（Having established [Paragraph N's core], we now turn to [Paragraph N+1's core] / While [Paragraph N's factor] accounts for X, [Paragraph N+1's factor] governs Y）完成段间握手。

Structural Formula:

```
End of Para A: ...resolving [Issue X] in isolation is insufficient without addressing [Issue Y]. → Start of Para B: To address this [Issue Y], we introduce [Module B]...
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

[End of Para 1] ...thus extracting periodic features. [Start of Para 2] Next, we use a contrastive loss function to train the network.

[Idiomatic Reconstruction]:

[End of Para 1] ...thus extracting multi-period representations from raw sensor streams. [Start of Para 2] Having isolated these periodic representations, we next align them across heterogeneous operating regimes via a regime-aware contrastive objective.

Surgical Analysis: 用 Having isolated these periodic representations, we next... 将上一段的输出直接作为下一段损失函数设计的输入，全篇浑然一体。

Academic Trap & Reviewer Perspective

Common Pitfall: 段落之间毫无过渡，上一段讲完频域特征，下一段突然毫无征兆地讲损失函数，像看幻灯片跳页。

10.2.28 L028: Diagnosing and Bridging Missing Logical Premises / 诊断与修复逻辑跳跃（Bridging Logical Leaps / Missing Premises）

Core Logic Principle & Structural Formula

Category: `Logical Fallacy Prevention (逻辑谬误防范)`

Core Principle: 作者脑子里清楚 A → B → C，但写出来只写了 A，紧接着就 wrote Therefore, C。修改时必须站在审稿人/读者视角，把隐含的中间桥接前提 B（Missing Premise）显式补写出来。

Structural Formula:

```
Premise A: [Observation]. → Missing Bridge B: [Because A implies Property B
under Condition K], → Conclusion C: [Therefore, Design/Policy C is necessary].
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

Industrial time series contain missing values at irregular intervals. Therefore, we use a continuous-time neural ODE encoder.

[Idiomatic Reconstruction]:

Industrial sensor streams frequently suffer from irregular missing timestamps due to packet loss. Because standard discrete-step RNNs assume uniform sampling intervals and distort time-delta dynamics when inputs are missing, we model latent state evolution between arbitrary observation times using a continuous-time neural ODE.

Surgical Analysis: 补全了关键中间前提："离散步长 RNN 默认等间隔采样，在缺失值下会扭曲时间间隔动力学"，从而让引入连续时间神经 ODE 变得顺理成章。

Academic Trap & Reviewer Perspective

Common Pitfall: 滥用 Therefore / Thus / Hence 强行连接两句根本没有直接推出关系的话。

10.2.29 L029: Stripping LLM Rhetorical Padding at the Paragraph Level / 篇章级去 AI 味手术：砍掉空洞修辞外壳，直奔命题内核（Zero AI-ese Discourse Surgery）

Core Logic Principle & Structural Formula

Category: `Style & Anti-AI-ese (去 AI 化文风)`

Core Principle: 大模型生成的段落有三大典型臭味：(1) 句首喜欢戴高帽（In the rapidly evolving landscape of... / It serves as a testament to...）；(2) 段中喜欢滥用排比与抽象隐喻（not only... but also..., tapestry, beacon, realm, paradigm shift, delve into, underscore）；(3) 段尾喜欢升华口号（paving the way for a brighter future）。修改时必须一刀砍掉所有修辞外壳，首句第一词直接落到具体研究对象或议题实体上。

Structural Formula:

```
Delete: 'In the realm of X, it is undeniable that Y plays a pivotal role in
spearheading Z.' → Write: '[Concrete Entity Y] governs [Specific Mechanism Z]
by [Measurable Action].'
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:

In the rapidly evolving landscape of artificial intelligence, time series forecasting stands as a cornerstone that plays a pivotal role in spearheading industrial transformation, serving as a testament to data-driven decision-making.

[Idiomatic Reconstruction]:
Accurate multi-horizon time series forecasting enables power grid operators to balance renewable generation against fluctuating consumer demand.

Surgical Analysis: 砍掉 32 个词的 AI 废话外壳，用 16 个词的紧凑主谓宾直接锁定“多步时间序列预测帮电网调度员平衡新能源发电与负荷波动”。

Academic Trap & Reviewer Perspective

Common Pitfall: 误以为辞藻越华丽、排比越工整、抽象大词越多，英语就越高级。

10.2.30 L030: Seven-Point Post-Drafting Paragraph Audit / 成段后七维自检清单（The 7-Point Paragraph Audit Checklist）

Core Logic Principle & Structural Formula

Category: `Quality Assurance (段落自检闭环)`

Core Principle: 写完或改完任何一个英语段落，必须依次用七把尺子过筛：(1) 首句是否一眼看清本段唯一核心主张？(2) 每句主语是否控制在 6–8 词以内（无头重脚轻）？(3) 句与句之间是否遵循“旧信息在前、新信息在后”？(4) 所有的 This/Such 后是否都跟了精准汇总名词？(5) 是否彻底清除了口语词与 AI 套话？(6) 所有的程度主张是否都有具体指标、基线或微观案例支撑？(7) 主动与被动语态的切换是否服务于信息焦点？

Structural Formula:

```
Audit(Para) = Unity(1) + ShortSubject(2) + GivenToNew(3) + SummaryNoun(4) +
ZeroAI_ZeroColloquial(5) + ConcreteMetricOrCase(6) + PurposefulVoice(7).
```

Defective Draft vs. Idiomatic Reconstruction

[Defective Draft]:
Checking only spelling and grammar while leaving a 22-word subject, bare 'This', and unsubstantiated 'very good performance' untouched.

[Idiomatic Reconstruction]:
Applying the 7-point audit systematically: compressing the 22-word subject to 4 words, replacing bare 'This' with 'This phase lag', and replacing 'very good' with 'a 14.6% MSE reduction on Weather'.

Surgical Analysis: 将抽象的“改作文/改论文”转化为 7 个可逐项打勾的工程化核验动作。

Academic Trap & Reviewer Perspective

Common Pitfall: 写完只检查拼写和单复数，不检查段落骨架、信息流与论证闭环。

Part IV

The 48 Core Paragraph Blueprints Encyclopedia / 48 大段落构建蓝图全书

Chapter 11

Track 1: Research Context & Problem Articulation (BP01–BP08) / 研究背景、问题界定与贡献提炼

本章系统拆解该赛道的 8 大核心段落构建蓝图。每一项蓝图均涵盖明确的用户交际意图、适用场景、逐句动作规范（Move Structure）、句际信息咬合公式、病段与发表级定稿对照、成分剖析表格以及跨学科/文体迁移指南。

11.0.1 BP01: 确立研究背景与真实工程价值（拒绝空喊口号）

Communicative Intent: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Applicable Context: 论文 Introduction 首段，或开题报告背景首段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。
- **S2.** S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。
- **S3.** S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。
- **S4.** S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。

Logic Chaining & Cohesion Formula:

Given-to-New 链：Core Task (S1) → Practical Deployment (S2) → Real-world Disturbances (S3) → Consequence of Failure (S4). 绑定逻辑规则：L001, L011, L029.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In the rapidly developing modern world, time series forecasting plays a pivotal role in many fields. The fact that industrial sensors often have noise and distribution changes makes forecasting very hard and bad for decision making.

Diagnostic Autopsy: (1) 首句 AI 套话（In the rapidly developing... plays a pivotal role）；(2) 第二句严重头重脚轻（The fact that... 11 词从句作主语）且用词口语化（very hard and bad）。

[Publication-Grade Model Paragraph]:

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [任务锚定 Topic Anchor]	*Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.* **Constituent Analysis**: 对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [应用场景展开 Concrete Scenarios]	*In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.* **Constituent Analysis**: 对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [苛刻工况约束 Operational Bottleneck]	*However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.* **Constituent Analysis**: 对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [研究必要性收束 Imperative Hook]	*Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.* **Constituent Analysis**: 对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

Paper Intro 首段直奔领域实体；若改用于学术演讲开场，可将 S1-S2 压缩为一句设问 + 一张故障代价对比图；若用于六级/雅思科技类首段，可将工业传感器替换为更通用的公共基础设施场景。

11.0.2 BP02: 总结现有主流范式及其隐含假设（欲抑先扬铺垫段）

Communicative Intent: 我这一段想客观概括前人主流模型（如 Transformer / Diffusion / MLP）取得了什么进展，但点出它们共同依赖的一个脆弱前提假设。

Applicable Context: Introduction 第二段（Literature Context & Hidden Assumption）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。
- **S2.** S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。
- **S3.** S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。
- **S4.** S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。

Logic Chaining & Cohesion Formula:

先让步肯定（While/Recent paradigms...）→ 后釜底抽薪揭示隐含前提（Yet these architectures implicitly assume...）。绑定逻辑规则：L008, L009.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Many researchers have studied this problem. Wu et al. used Autoformer. Nie et al. used PatchTST. They are very good, but they all think the data distribution does not change, which is wrong in real life.

Diagnostic Autopsy: (1) 流水账罗列人名（A used X, B used Y）；(2) 口语化严重（very good, think, which is wrong in real life）。

[Publication-Grade Model Paragraph]:

Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition. These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments. Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window. When external shocks alter underlying

system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [流派归类 Taxonomic Opening]	*Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition.* **Constituent Analysis**: 对应步骤：S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [代表性机制展开 Mechanism Recap]	*These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments.* **Constituent Analysis**: 对应步骤：S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [转折揭示隐含假设 Pivot to Hidden Assumption]	*Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window.* **Constituent Analysis**: 对应步骤：S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [假设破裂后果 Breakdown Under Stress]	*When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.* **Constituent Analysis**: 对应步骤：S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

适用于任何领域的 Intro 第二段：用 implicitly assume that... 一针见血点出前人工作的理论边界。

11.0.3 BP03: 尖锐指出核心技术瓶颈与失效机理（Gap Identification）

Communicative Intent: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Applicable Context: Introduction 第二/三段核心痛点段（Problem Diagnosis）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。
- **S2.** S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。
- **S3.** S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。
- **S4.** S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。

Logic Chaining & Cohesion Formula:

因果链式推进（A induces B → This B compounds into C → Naive fix D sacrifices E）。绑定逻辑规则：L003, L004, L024/L025.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

The problem of error accumulation in long-term forecasting and the losing of scale information when using RevIN normalization is very serious and hard to solve by simple ways.

Diagnostic Autopsy: (1) 长达 21 词的巨型名词短语作主语（头重脚轻）；(2) 谓语表语极度空洞（is very serious and hard to solve by simple ways）。

[Publication-Grade Model Paragraph]:

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [核心痛点立论 Bottleneck Claim]	*Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.* **Constituent Analysis**: 对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [第一层失效机理 Primary Failure Mechanism]	*While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.* **Constituent Analysis**: 对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [第二层耦合困境 Secondary Dilemma]	*Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.* **Constituent Analysis**: 对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [凝练核心研究问题 Research Question Formulation]	*Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.* **Constituent Analysis**: 对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

通过 While [Fix A]..., it erases...; Conversely, [Fix B] exposes... 构建极具张力的“两难困境（Dilemma）”，为亮出你的新方法做最完美的铺垫。

11.0.4 BP04: 亮出本文核心洞察与破局方法总览（Key Insight & Method Overview）

Communicative Intent: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。

Applicable Context: Introduction 第三段（Our Approach & Key Insight）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [方法登场与定位 Method Introduction]: To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal].

S2. S2 [核心直觉洞察 Key Insight]: Our key insight is that [Physical/Statistical Principle], 点破破局的底层支点。

S3. S3 [模块协同主干 Module Synergy]: 用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。

S4. S4 [闭环收益总结 Closed-Loop Benefit]: 说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。

Logic Chaining & Cohesion Formula:

Problem Anchor → Key Insight → Dual-Module Mechanism → Pareto Resolution. 绑定逻辑规则：L002, L023.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In this paper, we propose a novel and groundbreaking model called DyTS-Net. It has three modules: Module A, Module B, and Module C. They work together to delve into the data and get state-of-the-art performance.

Diagnostic Autopsy: (1) 典型 AI 自吹词（novel and groundbreaking, delve into, state-of-the-art）; (2) 机械报菜名（Module A, B, C），完全没讲清为什么这套组合能解决上一段的痛点。

[Publication-Grade Model Paragraph]:

To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch. Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [方法登场与定位 Method Introduction]	*To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.* **Constituent Analysis**: 对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [核心直觉洞察 Key Insight]	*Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.* **Constituent Analysis**: 对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [模块协同主干 Module Synergy]	*Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.* **Constituent Analysis**: 对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [闭环收益总结 Closed-Loop Benefit]	*Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.* **Constituent Analysis**: 对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

先给 Key Insight（核心直觉），再讲模块分工，审稿人一眼就能看懂你的创新不是拼积木，而是有严密的理论动机。

11.0.5 BP05: 挑战与机制一对一映射段（Challenge-to-Mechanism Mapping）

Communicative Intent: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。

Applicable Context: Introduction 第四段或 Methodology 开篇总览段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges.

S2. S2 [难点 1 + 靶向设计 1]：First, because [Challenge 1], we design [Mechanism 1] to [Specific Function].

S3. S3 [难点 2 + 靶向设计 2]：Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function].

S4. S4 [整体计算/泛化特性 System Property]：Together, these components enable [Model] to operate at [Efficiency/Robustness Metric].

Logic Chaining & Cohesion Formula:

Challenge 1 → Mechanism 1; Challenge 2 → Mechanism 2 → Joint System Property. 绑定逻辑规则：L003, L011.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

There are two challenges. The first challenge is that noise is big. We use a filter. The second challenge is that speed is slow. We use distillation.

Diagnostic Autopsy: 全部是散装幼稚短句（is big, is slow, We use...），毫无学术机制深度。

[Publication-Grade Model Paragraph]:

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [落地难点总起 Implementation Hurdle]	*Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.* **Constituent Analysis**: 对应步骤：S1 [落地难点总起 Implementation Hurdle]: Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [难点 1 + 靶向设计 1]	*First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.* **Constituent Analysis**: 对应步骤：S2 [难点 1 + 靶向设计 1]: First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [难点 2 + 靶向设计 2]	*Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.* **Constituent Analysis**: 对应步骤：S3 [难点 2 + 靶向设计 2]: Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [整体计算/泛化特性 System Property]	*Together, these designs sustain 50 Hz reactive control under partial visual occlusion.* **Constituent Analysis**: 对应步骤：S4 [整体计算/泛化特性 System Property]: Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

每个机制都以 because / to prevent 引导的真实难点作为前置状语，彻底杜绝“为了加模块而加模块”的凑工作量嫌疑。

11.0.6 BP06: 实验证言与性能突破预览段（Quantitative Empirical Preview）

Communicative Intent: 我这一段想在 Introduction 结尾前，用三句话硬核总结我们在哪些基准数据集/真实硬件上做了实验，取得了多大幅度的领先。

Applicable Context: Introduction 倒数第二段（Empirical Validation Preview）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [评测广度 Scope of Evaluation]：We evaluate [Model] across [N] public benchmarks and [Real-world Hardware/Scenario], spanning [Tasks].

S2. S2 [核心精度突破 Primary Accuracy Gain]：实验主指标对比（必须包含具体百分比、最强基线名称及困难场景表现）。

S3. S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]：报告计算速度、显存占用或零样本跨场景泛化收益。

Logic Chaining & Cohesion Formula:

Evaluation Scope → Accuracy Gain over Strongest Baselines → Efficiency/Generalization Gain. 绑定逻辑规则：L010, L015.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

We did comprehensive experiments on many datasets. The results show that our model beats all baselines by a large margin and is very fast.

Diagnostic Autopsy: 全段零数字（many datasets, a large margin, very fast），没有任何实质信息。

[Publication-Grade Model Paragraph]:

We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors. Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions. Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [评测广度 Scope of Evaluation]	*We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors.* **Constituent Analysis**: 对应步骤：S1 [评测广度 Scope of Evaluation]：We evaluate [Model] across [N] public benchmarks and [Real-world Hardware/Scenario], spanning [Tasks]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [核心精度突破 Primary Accuracy Gain]	*Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions.* **Constituent Analysis**: 对应步骤：S2 [核心精度突破 Primary Accuracy Gain]：实验主指标对比（必须包含具体百分比、最强基线名称及困难场景表现）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]	*Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.* **Constituent Analysis**: 对应步骤：S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]：报告计算速度、显存占用或零样本跨场景泛化收益。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

用 8 个基准、1,400 个传感器、12.8%/17.4% MSE 降幅、45% FLOPs 削减支撑论点，零 AI 形容词却极具震撼力。

11.0.7 BP07: 凝练论文三点核心贡献（Contributions Summary）

Communicative Intent: 我这一段想在 Introduction 最末尾清晰列出本文的三点核心贡献（理论/视角贡献 + 核心算法/架构贡献 + 实验与开源实证贡献）。

Applicable Context: Introduction 末段（Summary of Contributions）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [总起引导句 Lead-in]：In summary, our main contributions are threefold:
- **S2.** S2 [贡献 1：新视角/问题形式化 Conceptual/Formulation Contribution]：用强动词名物化或动词原形开头（We identify and formalize... / We establish...），点明打破了什么传统局限。
- **S3.** S3 [贡献 2：核心方法与模块 Algorithmic/Architectural Contribution]：We propose [Model/Mechanism], which integrates [A] and [B] to achieve [Theoretical/Functional Guarantee].

S4. S4 [贡献 3：实证与系统突破 Empirical & Practical Contribution]：We demonstrate through extensive benchmarks and physical deployment that [Model] reduces [Metric] by [X%] while [Efficiency Gain].

Logic Chaining & Cohesion Formula:

Parallel Grammatical Structure（严格平行结构：全部用 We + 强及物动词 identify / propose / demonstrate 开头）。绑定逻辑规则：L002, L010.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Our contributions are: 1. A new model is proposed. 2. Solving the non-stationary problem. 3. Good results on datasets.

Diagnostic Autopsy: 语法结构完全不平行（被动句 + 动名词短语 + 名词短语），且内容空泛。

[Publication-Grade Model Paragraph]:

In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability. (2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker. (3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [总起引导句 Lead-in]	*In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability.* **Constituent Analysis**: 对应步骤：S1 [总起引导句 Lead-in]：In summary, our main contributions are threefold: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [贡献 1	*(2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker.* **Constituent Analysis**: 对应步骤：S2 [贡献 1：新视角/问题形式化 Conceptual/Formulation Contribution]：用强动词名物化或动词原形开头（We identify and formalize... / We establish...），点明打破了什么传统局限。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [贡献 2	*(3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.* **Constituent Analysis**: 对应步骤：S3 [贡献 2：核心方法与模块 Algorithmic/Architectural Contribution]：We propose [Model/Mechanism], which integrates [A] and [B] to achieve [Theoretical/Functional Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

三点贡献严格遵循 'We formalize... / We propose... / We demonstrate...' 平行主谓宾结构，分别对应“理论发现 → 方法构建 → 实证突破”。

11.0.8 BP08: 高密度四句式 Abstract 全段构建（Complete 4-Move Abstract）

Communicative Intent: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Applicable Context: 论文 Abstract 全段，或学术演讲摘要（Talk Abstract）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。

S2. S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause].

S3. S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]...

S4. S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result].

Logic Chaining & Cohesion Formula:

Context (S1) → Gap (S2) → Method & Mechanism (S3-S4) → Quantitative Proof (S5). 绑定逻辑规则：L001, L008, L010, L029.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Time series imputation is very important in the era of big data. Existing generative models delve into the data distribution but are too slow and fail when missing rates are high. We propose FlowImp, a novel framework that uses conditional flow matching and frequency guidance. Extensive experiments show that FlowImp achieves state-of-the-art results on all benchmarks.

Diagnostic Autopsy: 包含 in the era of big data, delve into, novel, state-of-the-art 等典型低质套话，且没有任何具体机制因果与量化数字。

[Publication-Grade Model Paragraph]:

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [背景与任务重要性 Context]	*Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.* **Constituent Analysis**: 对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [现有局限与痛点 Gap/Bottleneck]	*While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.* **Constituent Analysis**: 对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3-S4 [本文方法与核心机制 Approach & Mechanism]	*We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.* **Constituent Analysis**: 对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S5 [量化实证结果 Quantitative Results]	*Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.* **Constituent Analysis**: 对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S5 [量化实证结果 Quantitative Results]	*Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.* **Constituent Analysis**: 对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

108 个词的黄金顶会摘要：无一字废话，痛点具体（50% 连续块缺失 + 采样延迟），机制清晰（频谱先验直线传输路径 + 傅里叶相位正则），数据硬核（MAE 降 19.3%，提速 24 倍）。

Chapter 12

Track 2: Critical Literature Synthesis & Theoretical Framework (BP09–BP16) / 文献分类综述、范式定位与理论框架

本章系统拆解该赛道的 8 大核心段落构建蓝图。每一项蓝图均涵盖明确的用户交际意图、适用场景、逐句动作规范（Move Structure）、句际信息咬合公式、病段与发表级定稿对照、成分剖析表格以及跨学科/文体迁移指南。

12.0.1 BP09: 相关工作分类综述段（拒绝 A did X, B did Y 流水账）

Communicative Intent: 我这一段想写 Related Work 的一个子方向：把过去三年的 6–8 篇代表性论文按技术路线分成两类进行归纳，并指出它们的共同盲区。

Applicable Context: Related Work 章节各小节首段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [分类总纲 Taxonomic Thesis]：Existing methods for [Task] can be categorized into two main families: [Family A] and [Family B].
- **S2.** S2 [流派 A 归纳与代表作 Family A Synthesis]：[Family A] approaches (e.g., [Models]) focus on [Mechanism A], effectively addressing [Sub-problem 1].
- **S3.** S3 [流派 B 归纳与对比 Family B Synthesis]：By contrast, [Family B] architectures (e.g., [Models]) reformulate the problem via [Mechanism B] to reduce [Cost/Complexity].
- **S4.** S4 [共同盲区提炼 Shared Blind Spot]：Despite their respective strengths, both families overlook [Critical Factor], which degrades performance under [Target Scenario].

Logic Chaining & Cohesion Formula:

Two-Branch Taxonomy → Family A → Contrastive Family B → Shared Limitation. 绑定逻辑规则：L009, L008.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In recent years, many works studied time series foundation models. TimeGPT trained a transformer on large data. Moment used masked pre-training. Chronos quantized time series into tokens. Lag-Llama used lag features. However, our model is different from them.

Diagnostic Autopsy: 典型的“报菜名流水账”（连续四句主语分别是四个模型名），末句 our model is different 毫无营养。

[Publication-Grade Model Paragraph]:

Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization. Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains. Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives. Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [分类总纲 Taxonomic Thesis]	*Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization.* **Constituent Analysis**: 对应步骤：S1 [分类总纲 Taxonomic Thesis]：Existing methods for [Task] can be categorized into two main families: [Family A] and [Family B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [流派 A 归纳与代表作 Family A Synthesis]	*Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains.* **Constituent Analysis**: 对应步骤：S2 [流派 A 归纳与代表作 Family A Synthesis]：[Family A] approaches (e.g., [Models]) focus on [Mechanism A], effectively addressing [Sub-problem 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [流派 B 归纳与对比 Family B Synthesis]	*Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives.* **Constituent Analysis**: 对应步骤：S3 [流派 B 归纳与对比 Family B Synthesis]：By contrast, [Family B] architectures (e.g., [Models]) reformulate the problem via [Mechanism B] to reduce [Cost/Complexity]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [共同盲区提炼 Shared Blind Spot]	*Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.* **Constituent Analysis**: 对应步骤：S4 [共同盲区提炼 Shared Blind Spot]：Despite their respective strengths, both families overlook [Critical Factor], which degrades performance under [Target Scenario]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

把散乱的文献提炼为“连续 Patch 重构 vs. 离散词表量化”两大阵营，最后一句切中“单通道独立预训练忽略多变量物理守恒约束”的共同软肋。

12.0.2 BP10: 对比现有流派优劣并精准定位本文差异（Contrastive Positioning）

Communicative Intent: 我这一段想在 Related Work 小节末尾，把我们的方法与最接近的 2 个竞争方案做精准对比，防止审稿人说我们只是对某篇旧文的微调（Incremental）。

Applicable Context: Related Work 小节末段，或 Rebuttal 创新性辨析段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [锁定最相近工作 Closest Prior Work]：Among prior studies, [Closest Method A] and [Closest Method B] are most closely related to our approach.
- **S2.** S2 [剖析其局限性 Structural Difference 1]：While [Method A] also employs [Shared Concept], it relies on [Limitation 1], restricting its applicability to [Narrow Setting].
- **S3.** S3 [剖析本文本质突破 Core Departure]：In contrast, our framework introduces [Core Innovation], which fundamentally eliminates [Bottleneck] by [Mechanism].
- **S4.** S4 [实证或理论优势锁定 Empirical/Theoretical Edge]：Consequently, our approach achieves [Capability] without requiring [Expensive Requirement of Prior Work].

Logic Chaining & Cohesion Formula:

Acknowledge Overlap → Pinpoint Structural Divergence → Explain Functional Gain. 绑定逻辑规则：L008, L019.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Some works also use frequency domain like us, such as FEDformer and TimesNet. But our method is much better and more novel because we use a new module.

Diagnostic Autopsy: 回避了具体技术异同，只喊口号说 much better and more novel，审稿人最反感这种敷衍对比。

[Publication-Grade Model Paragraph]:

Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach. While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes. In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases.

> Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [锁定最相近工作 Closest Prior Work]	*Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach.* **Constituent Analysis**: 对应步骤：S1 [锁定最相近工作 Closest Prior Work]：Among prior studies, [Closest Method A] and [Closest Method B] are most closely related to our approach. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [剖析其局限性 Structural Difference 1]	*While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes.* **Constituent Analysis**: 对应步骤：S2 [剖析其局限性 Structural Difference 1]：While [Method A] also employs [Shared Concept], it relies on [Limitation 1], restricting its applicability to [Narrow Setting]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [剖析本文本质突破 Core Departure]	*In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases.* **Constituent Analysis**: 对应步骤：S3 [剖析本文本质突破 Core Departure]：In contrast, our framework introduces [Core Innovation], which fundamentally eliminates [Bottleneck] by [Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [实证或理论优势锁定 Empirical/ Theoretical Edge]	*Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.* **Constituent Analysis**: 对应步骤：S4 [实证或理论优势锁定 Empirical/Theoretical Edge]：Consequently, our approach achieves [Capability] without requiring [Expensive Requirement of Prior Work]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

坦然承认都用了傅里叶变换（Acknowledge overlap），随即精准对比“全局静态 top-k 选频 vs. 滑动子区间自适应小波带宽调节”，彻底打消 incremental 质疑。

12.0.3 BP11: 问题形式化定义与符号约定段（Problem Formulation & Notation）

Communicative Intent: 我这一段想在 Methodology 开篇（Section 3.1 Problem Formulation），严谨、简洁地定义输入张量、输出目标、约束条件与优化目标。

Applicable Context: Methodology 第 3.1 节 Problem Formulation

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [任务总体形式化 Task Definition]: Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning].

S2. S2 [补充条件或模态输入 Auxiliary/Condition Inputs]: At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime].

S3. S3 [预测/控制目标 Objective]: Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$ over horizon H (or action trajectory $a_{t:t+K}$).

S4. S4 [核心分布/物理约束 Core Constraint]: Crucially, unlike standard stationary settings where $P(\mathbf{X})$ remains fixed, we address the regime where [Mathematical Constraint/Shift] varies across time.

Logic Chaining & Cohesion Formula:

Input Space → Auxiliary Conditions → Target Mapping → Crucial Distributional Constraint. 绑定逻辑规则: L014, L011.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In this section, we define the problem. The input is X. X has length L and C channels. The output is Y. Y has length H. We want to make the error between predicted Y and true Y as small as possible.

Diagnostic Autopsy: 全是碎片化的幼稚简单句（The input is X. X has...），缺乏数学严谨性与核心科学问题界定。

[Publication-Grade Model Paragraph]:

Consider a multivariate time series $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ recording C correlated sensors over lookback horizon L. Given an binary observation mask $\mathbf{M} \in \{0, 1\}^{L \times C}$ indicating irregular sensor dropouts, our objective is to learn a generative mapping $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ over future horizon H. Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where $\mathbb{E}[x_t]$ and $\mathrm{Cov}(x_t)$ evolve continuously between the lookback and forecasting windows.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [任务总体形式化 Task Definition]	*Consider a multivariate time series* $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ *recording* C *correlated sensors over lookback horizon* L. **Constituent Analysis**: 对应步骤：S1 [任务总体形式化 Task Definition]：Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [补充条件或模态输入 Auxiliary/Condition Inputs]	*Given an binary observation mask* $\mathbf{M} \in \{0,1\}^{L \times C}$ *indicating irregular sensor dropouts, our objective is to learn a generative mapping* $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ *over future horizon* H. **Constituent Analysis**: 对应步骤：S2 [补充条件或模态输入 Auxiliary/Condition Inputs]：At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [预测/控制目标 Objective]	*Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where* $\mathbb{E}[x_t]$ *and* $Cov(x_t)$ *evolve continuously between the lookback and forecasting windows.* **Constituent Analysis**: 对应步骤：S3 [预测/控制目标 Objective]：Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$ over horizon H (or action trajectory $a_{t:t+K}$). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

四句话把输入维度、缺失掩码、条件生成分布与非平稳矩漂移约束交代得滴水不漏。

12.0.4 BP12: 整体模型架构数据流俯瞰段（Framework Overview & Pipeline Walk-through）

Communicative Intent: 我这一段想配合论文里的 Figure 2（模型总架构图），用一段话把输入张量从左到右流经各个模块直到输出预测结果的完整全貌讲清楚。

Applicable Context: Methodology 第 3.2 节 Architecture Overview

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]：Figure 2 illustrates the overall architecture of [Model], which processes [Input] through three sequential stages: [Stage 1], [Stage 2], and [Stage 3].

S2. S2 [阶段 1 数据流 Stage 1 Flow]：Given raw input $\mathbf{X}$, the [Module 1] first transforms [Raw Input] into [Intermediate Representation $\mathbf{Z}^{(1)}$] to [Functional Purpose 1].

S3. S3 [阶段 2 数据流 Stage 2 Flow]：Next, these [Summary Noun: representations $\mathbf{Z}^{(1)}$] are fed into [Module 2], which [Mechanism 2] to produce [Refined State $\mathbf{Z}^{(2)}$].

S4. S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]：Finally, the [Module 3] integrates $\mathbf{Z}^{(2)}$ with [Prior/Scale Cue] to output the final [Prediction/Action Trajectory].

Logic Chaining & Cohesion Formula:
严格的链式数据流（$\mathbf{X} \rightarrow \mathbf{Z}^{(1)} \rightarrow \mathbf{Z}^{(2)} \rightarrow \hat{\mathbf{Y}}$），每一句句首承接上一句的输出张量。绑定逻辑规则：L001, L003, L004.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Figure 2 shows our model. There is a Patch Encoder on the left. In the middle, there is a Spectral Block and a Graph Block. On the right, a Linear Decoder outputs the result.

Diagnostic Autopsy: 写成看图说话的静态空间方位描述（on the left, in the middle, on the right），没有体现张量变换的数据流动与功能因果。

[Publication-Grade Model Paragraph]:

Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages. Given an input sequence $\mathbf{X}$, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches $\mathbf{Z}_0$. These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph. Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output $\hat{\mathbf{Y}}$.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]	*Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages.* **Constituent Analysis**: 对应步骤：S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]：Figure 2 illustrates the overall architecture of [Model], which processes [Input] through three sequential stages: [Stage 1], [Stage 2], and [Stage 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [阶段 1 数据流 Stage 1 Flow]	*Given an input sequence* $\mathbf{X}$*, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches* $\mathbf{Z}_0$*.* **Constituent Analysis**: 对应步骤：S2 [阶段 1 数据流 Stage 1 Flow]：Given raw input **X**, the [Module 1] first transforms [Raw Input] into [Intermediate Representation $\mathbf{Z}^{(1)}$] to [Functional Purpose 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [阶段 2 数据流 Stage 2 Flow]	*These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph.* **Constituent Analysis**: 对应步骤：S3 [阶段 2 数据流 Stage 2 Flow]：Next, these [Summary Noun: representations $\mathbf{Z}^{(1)}$] are fed into [Module 2], which [Mechanism 2] to produce [Refined State $\mathbf{Z}^{(2)}$]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]	*Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output* $\hat{\mathbf{Y}}$*.* **Constituent Analysis**: 对应步骤：S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]: Finally, the [Module 3] integrates $\mathbf{Z}^{(2)}$ with [Prior/Scale Cue] to output the final [Prediction/Action Trajectory]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

不讲“左边有什么右边有什么”，而是沿着张量 $\mathbf{X} \rightarrow \mathbf{Z}_0 \rightarrow$ refined representations $\rightarrow \hat{\mathbf{Y}}$ 的变换链条，每一步都带出明确的目的状语。

12.0.5 BP13: 核心子模块：直觉动机与数学设计无缝衔接段（Intuition-before-Math）

Communicative Intent: 我这一段想详细介绍我们提出的某个核心创新模块：先用两句话讲清楚为什么直接用传统算子会出问题，再顺理成章推导出我们的新公式。

Applicable Context: Methodology 核心子章节（Section 3.3 / 3.4）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [传统算子缺陷 Motivation]: Standard [Conventional Operator, e.g., softmax self-attention] computes [Operation], which suffers from [Pathology] when applied to [Domain Condition].

S2. S2 [物理/统计直觉 Physical Intuition]: To overcome this pathology, we observe that [Key Domain Property] can be effectively regularized by [Intuitive Operation].

S3. S3 [数学公式定义 Mathematical Formulation]: Formally, given input representation $H \in \mathbb{R}^{N\times d}$, we formulate the [Module Name] as: [Equation], where [Symbol Explanations].

S4. S4 [理论/计算性质 Guarantee]: By construction, this formulation guarantees [Desirable Property, e.g., $O(N)$ complexity / translation invariance].

Logic Chaining & Cohesion Formula:

Pathology of Standard Operator → Physical Intuition → Equation Integration → Theoretical Guarantee. 绑定逻辑规则: L014, L003.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In this subsection, we introduce the Phase-Aware Attention. The formula is Attention(Q, K, V) = softmax(QK^T / sqrt(d) + M)V. Here M is a phase matrix. It makes the model work better.

Diagnostic Autopsy: 直接空降公式，读者完全不知道矩阵 M 是从哪冒出来的、为什么要加在 softmax 里面。

[Publication-Grade Model Paragraph]:

Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps. To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies. Formally, given query and key projections $Q, K \in \mathbb{R}^{N\times d}$, we modulate the attention logits via a phase-alignment mask: $\mathcal{A}(Q,K,V) = \mathrm{softmax}\big(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1\big)V$, where $\Delta\Phi_{Q,K} \in [-\pi,\pi]^{N\times N}$ denotes the instantaneous Hilbert phase difference matrix and $\lambda > 0$ controls phase selectivity. Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [传统算子缺陷 Motivation]	*Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps.* **Constituent Analysis**: 对应步骤：S1 [传统算子缺陷 Motivation]：Standard [Conventional Operator, e.g., softmax self-attention] computes [Operation], which suffers from [Pathology] when applied to [Domain Condition]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [物理/统计直觉 Physical Intuition]	*To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies.* **Constituent Analysis**: 对应步骤：S2 [物理/统计直觉 Physical Intuition]：To overcome this pathology, we observe that [Key Domain Property] can be effectively regularized by [Intuitive Operation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [数学公式定义 Mathematical Formulation]	*Formally, given query and key projections* $Q, K \in \mathbb{R}^{N\times d}$*, we modulate the attention logits via a phase-alignment mask:* $\mathcal{A}(Q,K,V) = softmax(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1)V$*, where* $\Delta\Phi_{Q,K} \in [-\pi,\pi]^{N\times N}$ *denotes the instantaneous Hilbert phase difference matrix and* $\lambda > 0$ *controls phase selectivity.* **Constituent Analysis**: 对应步骤：S3 [数学公式定义 Mathematical Formulation]：Formally, given input representation $H \in \mathbb{R}^{N\times d}$, we formulate the [Module Name] as: [Equation], where [Symbol Explanations]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [理论/计算性质 Guarantee]	*Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.* **Constituent Analysis**: 对应步骤：S4 [理论/计算性质 Guarantee]：By construction, this formulation guarantees [Desirable Property, e.g., $O(N)$ complexity / translation invariance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

从传统点积注意力的缺陷（只看幅值对齐易受噪声尖峰误导）自然推导出希尔伯特相位差惩罚项 $-\lambda\|\Delta\Phi_{Q,K}\|_1$，公式嵌入如行云流水。

12.0.6 BP14: 时间序列专属：多尺度频域-时域耦合机制阐述段（Time Series Mechanism）

Communicative Intent: 我这一段想专门阐述时间序列模型中“如何在频域捕获全局周期性，同时在时域保留局部突发异常（Transient Anomalies）”的双分支耦合机制。

Applicable Context: 时间序列论文 Methodology 章节或开题报告

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [单域局限性 Single-Domain Limitation]：Modeling non-stationary time series solely in the time domain struggles to capture long-range periodicities, whereas pure Fourier projection blurs localized transient anomalies.

S2. S2 [频域分支设计 Frequency Branch]：To capture global seasonal structure, our spectral branch applies real-valued FFT $\mathcal{F}(\cdot)$ along the temporal dimension and reweights dominant harmonic components.

S3. S3 [时域分支设计 Temporal Branch]：In parallel, a multi-scale depthwise convolution branch operates directly on local patches to preserve sharp step changes and localized spikes.

S4. S4 [自适应门控融合 Adaptive Fusion]：An uncertainty-aware gate then fuses both branches step-by-step, assigning higher weight to the temporal branch whenever local residual variance exceeds a threshold.

Logic Chaining & Cohesion Formula:

Time-Frequency Dilemma → Spectral Global Branch → Temporal Local Branch → Uncertainty-Gated Fusion.

绑定逻辑规则：L023, L024.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

We use both FFT and CNN in our model. FFT gets global features and CNN gets local features. Then we add them together to get final features.

Diagnostic Autopsy: 描述过于粗浅（gets global features, gets local features, add them together），像本科生课程作业报告。

[Publication-Grade Model Paragraph]:

Modeling telemetry streams solely in the time domain struggles to span multi-day periodicities, whereas pure Fourier projection spreads localized transient spikes across the entire spectrum. To reconcile both scales, our dual-domain block processes input patches along two complementary pathways. The spectral pathway applies real-valued RFFT to isolate top-k harmonic bases governing macro-periodicity, while the parallel temporal pathway employs dilated depthwise convolutions to retain sharp step changes and localized anomalies. An energy-ratio gate then fuses both representations at each time step, dynamically shifting weight toward the temporal pathway whenever local high-frequency residual energy spikes.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [单域局限性 Single-Domain Limitation]	*Modeling telemetry streams solely in the time domain struggles to span multi-day periodicities, whereas pure Fourier projection spreads localized transient spikes across the entire spectrum.* **Constituent Analysis**: 对应步骤：S1 [单域局限性 Single-Domain Limitation]: Modeling non-stationary time series solely in the time domain struggles to capture long-range periodicities, whereas pure Fourier projection blurs localized transient anomalies. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [频域分支设计 Frequency Branch]	*To reconcile both scales, our dual-domain block processes input patches along two complementary pathways.* **Constituent Analysis**: 对应步骤：S2 [频域分支设计 Frequency Branch]: To capture global seasonal structure, our spectral branch applies real-valued FFT $\mathcal{F}(\cdot)$ along the temporal dimension and reweights dominant harmonic components. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [时域分支设计 Temporal Branch]	*The spectral pathway applies real-valued RFFT to isolate top-k harmonic bases governing macro-periodicity, while the parallel temporal pathway employs dilated depthwise convolutions to retain sharp step changes and localized anomalies.* **Constituent Analysis**: 对应步骤：S3 [时域分支设计 Temporal Branch]: In parallel, a multi-scale depthwise convolution branch operates directly on local patches to preserve sharp step changes and localized spikes. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [自适应门控融合 Adaptive Fusion]	*An energy-ratio gate then fuses both representations at each time step, dynamically shifting weight toward the temporal pathway whenever local high-frequency residual energy spikes.* **Constituent Analysis**: 对应步骤：S4 [自适应门控融合 Adaptive Fusion]: An uncertainty-aware gate then fuses both branches step-by-step, assigning higher weight to the temporal branch whenever local residual variance exceeds a threshold. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

把“FFT + CNN 相加”升华为“频域谐波基提取 + 时域空洞深度卷积保留瞬态尖峰 + 能量比自适应门控融合”。

12.0.7 BP15: 生成模型专属：条件控制与采样去噪轨迹阐述段（Generative Modeling Mechanism）

Communicative Intent: 我这一段想阐述生成模型（Diffusion / Flow Matching / Autoregressive）如何在满足外部物理约束或条件观测的前提下，生成高质量且多样化的序列/轨迹。

Applicable Context: 生成模型 / 时序生成 / 轨迹生成论文 Methodology 章节

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]: Applying post-hoc classifier guidance to pre-trained diffusion samplers often pushes intermediate noisy states off the data manifold, inducing high-frequency artifacts.
- **S2.** S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]: To enforce hard boundary constraints without manifold deviation, we embed [Condition/Constraint] directly into the velocity field / denoising kernel.
- **S3.** S3 [具体轨迹更新方程 Trajectory Update Formulation]: At solver step $\tau \in [0, 1]$, the conditional velocity field $v_\theta(x_\tau, \tau \mid c)$ projects the unconstrained flow onto the tangent space of feasible trajectories.
- **S4.** S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]: This tangent projection guarantees exact boundary satisfaction at $\tau = 1$ while permitting straight 4-step Euler integration.

Logic Chaining & Cohesion Formula:

Guidance Artifact Problem → Intrinsic Velocity Conditioning → Tangent Projection Step → Fast Exact Convergence. 绑定逻辑规则：L026, L014.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Normal diffusion guidance makes the generated time series look unnatural when we add constraints. So we put the constraint inside the flow matching model, and it generates realistic samples in only 4 steps.

Diagnostic Autopsy: 缺乏生成模型核心术语（data manifold, velocity field, tangent projection, Euler integration），解释过于口语化。

[Publication-Grade Model Paragraph]:

Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations. To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field. At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator. This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]	*Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations.* **Constituent Analysis**: 对应步骤：S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]: Applying post-hoc classifier guidance to pre-trained diffusion samplers often pushes intermediate noisy states off the data manifold, inducing high-frequency artifacts. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]	*To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field.* **Constituent Analysis**: 对应步骤：S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]: To enforce hard boundary constraints without manifold deviation, we embed [Condition/Constraint] directly into the velocity field / denoising kernel. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [具体轨迹更新方程 Trajectory Update Formulation]	*At flow time $\tau \in [0,1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator.* **Constituent Analysis**: 对应步骤：S3 [具体轨迹更新方程 Trajectory Update Formulation]: At solver step $\tau \in [0,1]$, the conditional velocity field $v_\theta(x_\tau, \tau \mid c)$ projects the unconstrained flow onto the tangent space of feasible trajectories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]	*This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.* **Constituent Analysis**: 对应步骤：S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]: This tangent projection guarantees exact boundary satisfaction at $\tau = 1$ while permitting straight 4-step Euler integration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

地道运用 off the data manifold, conditional flow-matching velocity field, null space projection, straight transport paths 等生成模型顶会核心表达。

12.0.8 BP16: 具身智能专属：多模态感知-动作闭环与 Sim-to-Real 阐述段（Embodied AI Policy）

Communicative Intent: 我这一段想阐述具身智能机器人策略（Embodied Policy）如何融合视觉、本体感觉（Proprioception）或触觉，并克服仿真到真实硬件的动力学差距（Sim-to-Real Gap）。

Applicable Context: 具身智能 / 机器人操作（Robotic Manipulation）论文 Methodology 章节

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [单模态开环局限 Open-Loop Visual Limitation]：Relying solely on RGB-D point clouds leaves manipulation policies vulnerable to self-occlusion by the robot gripper during fine-grained contact.

S2. S2 [多模态时空对齐 Multi-Modal Alignment]：To maintain state observability during contact, our encoder aligns asynchronous 30 Hz visual tokens with 500 Hz joint proprioception and wrist force-torque histories.

S3. S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]：Rather than predicting single-step torques prone to jitter, the policy outputs overlapping K-step SE(3) end-effector action chunks smoothed via exponential temporal ensembling.

S4. S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]：During physical deployment, a lightweight history adapter infers latent friction and payload parameters online, bridging the sim-to-real dynamics gap without policy fine-tuning.

Logic Chaining & Cohesion Formula:

Visual Occlusion Bottleneck → Asynchronous Multimodal Alignment → Action Chunking → Online Sim-to-Real System Identification. 绑定逻辑规则：L025, L003.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

When the robot hand blocks the camera, the robot cannot see the object clearly. So we add force sensors and joint angles. Also, we predict multiple steps of actions and use an adapter to transfer from simulation to the real robot.

Diagnostic Autopsy: 全是大白话（blocks the camera, cannot see clearly, add force sensors），没有体现异步采样率对齐、SE(3) 动作块与在线隐空间动力学辨识。

[Publication-Grade Model Paragraph]:

Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object. To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings. Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling. At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [单模态开环局限 Open-Loop Visual Limitation]	*Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object.* **Constituent Analysis**: 对应步骤：S1 [单模态开环局限 Open-Loop Visual Limitation]：Relying solely on RGB-D point clouds leaves manipulation policies vulnerable to self-occlusion by the robot gripper during fine-grained contact. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [多模态时空对齐 Multi-Modal Alignment]	*To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings.* **Constituent Analysis**: 对应步骤：S2 [多模态时空对齐 Multi-Modal Alignment]：To maintain state observability during contact, our encoder aligns asynchronous 30 Hz visual tokens with 500 Hz joint proprioception and wrist force-torque histories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]	*Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling.* **Constituent Analysis**: 对应步骤：S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]：Rather than predicting single-step torques prone to jitter, the policy outputs overlapping K-step SE(3) end-effector action chunks smoothed via exponential temporal ensembling. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]	*At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.* **Constituent Analysis**: 对应步骤：S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]：During physical deployment, a lightweight history adapter infers latent friction and payload parameters online, bridging the sim-to-real dynamics gap without policy fine-tuning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

精准涵盖 self-occlusion, contact-rich manipulation, 30 Hz visual vs. 500 Hz proprioception, SE(3) action chunks, exponential temporal ensembling, online system identification。

Chapter 13

Track 3: Methodological Rigor & Experimental Justification (BP17–BP24) / 实验协议、基准对比、消融与局限性

本章系统拆解该赛道的 8 大核心段落构建蓝图。每一项蓝图均涵盖明确的用户交际意图、适用场景、逐句动作规范（Move Structure）、句际信息咬合公式、病段与发表级定稿对照、成分剖析表格以及跨学科/文体迁移指南。

13.0.1 BP17: 实验设置、数据集特性与强基线选型说明段（Experimental Setup & Baselines）

Communicative Intent: 我这一段想在 Experiments 开篇说明：我们为什么选这几类数据集和这几类基线模型，证明我们的实验设计是全面、公平且直击痛点的。

Applicable Context: Experiments 第 4.1 节 Experimental Setup

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [数据集选型逻辑 Dataset Selection Rationale]: To evaluate performance across diverse non-stationary regimes, we conduct experiments on [N] benchmarks spanning [Domain 1], [Domain 2], and [Domain 3].
- **S2.** S2 [极端/困难特性突出 Challenging Dataset Properties]: Notably, [Dataset A] exhibits [Hard Property 1, e.g., strong covariate shift], whereas [Dataset B] tests scalability across [Hard Property 2, e.g., 862 correlated channels].
- **S3.** S3 [基线流派全覆盖 Baseline Family Coverage]: We compare our method against [K] competitive baselines grouped into three families: [Family 1], [Family 2], and [Family 3].
- **S4.** S4 [公平评测协议 Fair Evaluation Protocol]: All models are evaluated under identical lookback windows ($L = 96$), prediction horizons ($H \in \{96, 192, 336, 720\}$), and hyperparameter search budgets.

Logic Chaining & Cohesion Formula:

Dataset Diversity → Challenging Regime Highlights → Categorized Strong Baselines → Strict Fairness Protocol. 绑定逻辑规则：L009, L010.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

We use ETT, Weather, Electricity, and Traffic datasets. We compare with Autoformer, PatchTST, iTransformer, and DLinear. We use MSE and MAE as metrics, and the lookback window is 96.

Diagnostic Autopsy: 只是把数据集名字和基线名字干巴巴列了一遍，没有说明选这些数据集和基线背后的科学验证意图。

[Publication-Grade Model Paragraph]:

To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic). Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability. We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer). All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [数据集选型逻辑 Dataset Selection Rationale]	*To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic).* **Constituent Analysis**: 对应步骤：S1 [数据集选型逻辑 Dataset Selection Rationale]：To evaluate performance across diverse non-stationary regimes, we conduct experiments on [N] benchmarks spanning [Domain 1], [Domain 2], and [Domain 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [极端/困难特性突出 Challenging Dataset Properties]	*Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability.* **Constituent Analysis**: 对应步骤：S2 [极端/困难特性突出 Challenging Dataset Properties]：Notably, [Dataset A] exhibits [Hard Property 1, e.g., strong covariate shift], whereas [Dataset B] tests scalability across [Hard Property 2, e.g., 862 correlated channels]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [基线流派全覆盖 Baseline Family Coverage]	*We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer).* **Constituent Analysis**: 对应步骤：S3 [基线流派全覆盖 Baseline Family Coverage]：We compare our method against [K] competitive baselines grouped into three families: [Family 1], [Family 2], and [Family 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [公平评测协议 Fair Evaluation Protocol]	*All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.* **Constituent Analysis**: 对应步骤：S4 [公平评测协议 Fair Evaluation Protocol]：All models are evaluated under identical lookback windows ($L = 96$), prediction horizons ($H \in \{96, 192, 336, 720\}$), and hyperparameter search budgets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

让审稿人立刻感受到实验设计的严密性：每个数据集测什么极限能力、每个基线代表什么流派、评测协议如何保证绝对公平。

13.0.2 BP18: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）

Communicative Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Applicable Context: Experiments 第 4.2 节 Main Results

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [全局统治力概述 Overall Dominance]: Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations.

S2. S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]: Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical.

S3. S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]: Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism].

S4. S4 [回扣本文核心机制 Mechanistic Confirmation]: By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$.

Logic Chaining & Cohesion Formula:

Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则: L015, L013, L010.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

As shown in Table 1, our model gets the best MSE on ETTm1, ETTm2, Weather, and Traffic. On Weather dataset, our MSE is 0.165, which is 12% lower than PatchTST. On Traffic, our MSE is 0.382, which beats iTransformer. This proves our model is the best.

Diagnostic Autopsy: 典型的“报账式朗读表格”，只报数字不分析机理，末句 This proves our model is the best 毫无信息量。

[Publication-Grade Model Paragraph]:

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [全局统治力概述 Overall Dominance]	*Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.* **Constituent Analysis**: 对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]	*Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.* **Constituent Analysis**: 对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]	*Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.* **Constituent Analysis**: 对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [回扣本文核心机制 Mechanistic Confirmation]	*By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.* **Constituent Analysis**: 对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

把冷冰冰的表格数字还原为生动的算法博弈：为什么在 Traffic 上赢 PatchTST（需要跨传感器空间传播），为什么在 $H = 720$ Weather 上赢 iTransformer（抑制阵风过拟合与长程漂移）。

13.0.3 BP19: 消融实验逐组件归因论证段（Counterfactual Ablation Study）

Communicative Intent: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Applicable Context: Experiments 第 4.3 节 Ablation Study

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [消融实验总括 Ablation Setup]: To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C].

S2. S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]: Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A].

S3. S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]: Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B].

S4. S4 [协同效应总结 Synergistic Conclusion]: Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning.

Logic Chaining & Cohesion Formula:

Ablation Overview → w/o Module A (Specific Failure) → w/o Module B (Specific Failure) → Cross-Module Synergy. 绑定逻辑规则: L016, L007.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Table 2 shows the ablation study. Without Module A, MSE increases from 0.180 to 0.215. Without Module B, MSE increases to 0.204. Without Loss C, MSE increases to 0.195. Thus, every part is necessary.

Diagnostic Autopsy: 流水账报数，没有区分“拿掉 A 坏在什么场景、拿掉 B 坏在什么场景”，也没有解释模块间的协同关系。

[Publication-Grade Model Paragraph]:

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [消融实验总括 Ablation Setup]	*To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.* **Constituent Analysis**: 对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]	*Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.* **Constituent Analysis**: 对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]	*Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.* **Constituent Analysis**: 对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [协同效应总结 Synergistic Conclusion]	*Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.* **Constituent Analysis**: 对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

精彩绝伦的消融论证：拿掉 Scale-ODE 坏在“换季温度基线漂移（+19.4%）”，拿掉 Phase-Attn 坏在“321 通道电网高频负荷尖峰串扰（+13.8%）”，最后点出两者的互锁增益。

13.0.4 BP20: 超参数敏感性与极端噪声鲁棒性分析段（Sensitivity & Robustness Analysis）

Communicative Intent: 我这一段想回应审稿人最常问的：你的模型是不是对超参数很敏感？在缺失值或传感器强噪声下会不会突然崩盘？

Applicable Context: Experiments 第 4.4 节 Sensitivity & Robustness

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [分析目的 Objective]: We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4.

S2. S2 [超参数平坦区间 Plateau Stability]: Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau.

S3. S3 [极端边缘工况压力测试 Stress Test under Corruption]: Under synthetic sensor dropouts up to [40%] and Gaussian noise ($\sigma = 0.3$), baseline accuracy plummets by over [35%], whereas our model degrades by less than [6.5%].

S4. S4 [鲁棒性物理归因 Robustness Mechanism]: This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation.

Logic Chaining & Cohesion Formula:

Goal → Wide Hyperparameter Plateau → Stress-Test Comparison → Mechanistic Explanation. 绑定逻辑规则: L010, L019.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Figure 4 shows hyperparameter sensitivity. When lambda changes from 0.1 to 1.0, the result does not change much. Also, when we add noise, our model is still very stable.

Diagnostic Autopsy: 模糊口语（does not change much, still very stable），缺乏量化波动范围与抗噪机理。

[Publication-Grade Model Paragraph]:

Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption. Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$. Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), pointwise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%. This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [分析目的 Objective]	*Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption.* **Constituent Analysis**: 对应步骤：S1 [分析目的 Objective]：We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [超参数平坦区间 Plateau Stability]	*Varying the phase regularization weight λ across* $[0.05, 0.50]$ *alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around* $\lambda = 0.15$. **Constituent Analysis**: 对应步骤：S2 [超参数平坦区间 Plateau Stability]：Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [极端边缘工况压力测试 Stress Test under Corruption]	*Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%.* **Constituent Analysis**: 对应步骤：S3 [极端边缘工况压力测试 Stress Test under Corruption]：Under synthetic sensor dropouts up to [40%] and Gaussian noise (σ = 0.3), baseline accuracy plummets by over [35%], whereas our model degrades by less than [6.5%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [鲁棒性物理归因 Robustness Mechanism]	*This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.* **Constituent Analysis**: 对应步骤：S4 [鲁棒性物理归因 Robustness Mechanism]: This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

给出 < 1.4% 的超参平坦区间与 $4.8% *vs.* $34.2% 的强噪声衰减对比，彻底封堵审稿人对调参难度的质疑。

13.0.5 BP21: 极端/反常情况与失败案例坦诚剖析段（Failure Case & Outlier Analysis）

Communicative Intent: 我这一段想通过可视化案例（Case Study / Qualitative Analysis），既展示模型在复杂跳变下的精准跟踪，又坦诚剖析一个模型预测失准的极端案例。

Applicable Context: Experiments 第 4.5 节 Qualitative & Failure Case Analysis

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [定性对比切入 Qualitative Setup]: Figure 5 visualizes predicted trajectories during an abrupt regime shift on [Dataset/Task].

S2. S2 [成功案例亮点 Success Highlight]: While [Baseline] lags behind the sudden step change by [K steps] and overshoots, our model rapidly locks onto the new amplitude regime within [M steps].

S3. S3 [失败案例坦诚剖析 Honest Failure Mode]: Nevertheless, inspecting the bottom-right failure case reveals that when an external manual override occurs without historical precursors, all purely endogenous forecasters—including ours—exhibit a transient lag.

S4. S4 [改进启示 Constructive Takeaway]: Incorporating exogenous event logs or textual maintenance records is necessary to anticipate such unscheduled interventions.

Logic Chaining & Cohesion Formula:

Visual Setup → Success under Shift → Honest Failure Case under Unobserved Exogenous Shock → Actionable Insight. 绑定逻辑规则：L017, L018.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Figure 5 shows that our prediction curve is very close to the ground truth curve, while baseline curves are far away. Sometimes our model also makes small errors when the data changes too suddenly.

Diagnostic Autopsy: 只说 very close / far away，没有指出具体的滞后（phase lag）、超调（overshoot）以及外生干预（exogenous shock）的物理本质。

[Publication-Grade Model Paragraph]:

Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark. Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error. Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response. Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [定性对比切入 Qualitative Setup]	*Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark.* **Constituent Analysis**: 对应步骤：S1 [定性对比切入 Qualitative Setup]：Figure 5 visualizes predicted trajectories during an abrupt regime shift on [Dataset/Task]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [成功案例亮点 Success Highlight]	*Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error.* **Constituent Analysis**: 对应步骤：S2 [成功案例亮点 Success Highlight]：While [Baseline] lags behind the sudden step change by [K steps] and overshoots, our model rapidly locks onto the new amplitude regime within [M steps]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [失败案例坦诚剖析 Honest Failure Mode]	*Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response.* **Constituent Analysis**: 对应步骤：S3 [失败案例坦诚剖析 Honest Failure Mode]：Nevertheless, inspecting the bottom-right failure case reveals that when an external manual override occurs without historical precursors, all purely endogenous forecasters—including ours—exhibit a transient lag. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [改进启示 Constructive Takeaway]	*Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.* **Constituent Analysis**: 对应步骤：S4 [改进启示 Constructive Takeaway]：Incorporating exogenous event logs or textual maintenance records is necessary to anticipate such unscheduled interventions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

内行看门道：用 phase lag（相位滞后）、post-recovery overshoot（恢复后超调）和 zero-precursor manual intervention（无先兆人工干预）展现极高的专业素养。

13.0.6 BP22: 计算开销、推理延迟与精度帕累托权衡段（Efficiency-Accuracy Pareto Analysis）

Communicative Intent: 我这一段想说明我们的模型不仅准，而且算得快、显存省，在“精度-延迟/显存”帕累托前沿（Pareto Frontier）上显著优于现有大模型。

Applicable Context: Experiments 第 4.6 节 Computational Efficiency

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [硬件与测试基准 Hardware Setup]: Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU.

S2. S2 [复杂度理论与实测对比 Complexity & Runtime]: By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch by [X%] and peak VRAM by [Y%].

S3. S3 [实时闭环部署可行性 Real-Time Viability]: At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control deadline.

S4. S4 [帕累托结论 Pareto Takeaway]: Consequently, [Our Model] achieves superior accuracy at a fraction of the computational footprint required by heavyweight foundation models.

Logic Chaining & Cohesion Formula:

Hardware Benchmark → Complexity Reduction & Memory Gain → Real-Time Latency Margin → Pareto Conclusion. 绑定逻辑规则：L010, L023.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Our model is very fast and uses little memory. It only takes 4.2 ms to run, so it can be used in real-time applications easily.

Diagnostic Autopsy: 缺少复杂度来源分析（为什么快）、参照硬件与实时控制硬截止时间（Deadline）对比。

[Publication-Grade Model Paragraph]:

Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module. By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer. At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [硬件与测试基准 Hardware Setup]	*Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module.* **Constituent Analysis**: 对应步骤：S1 [硬件与测试基准 Hardware Setup]：Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [复杂度理论与实测对比 Complexity & Runtime]	*By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer.* **Constituent Analysis**: 对应步骤：S2 [复杂度理论与实测对比 Complexity & Runtime]：By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch by [X%] and peak VRAM by [Y%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [实时闭环部署可行性 Real-Time Viability]	*At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.* **Constituent Analysis**: 对应步骤：S3 [实时闭环部署可行性 Real-Time Viability]：At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control deadline. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

将理论复杂度（$O(L^2) \rightarrow O(L \log L)$）、机载芯片（Orin AGX）、推理耗时（6.8 ms vs. 20 ms 控制周期预算）与任务成功率完美缝合。

13.0.7 BP23: 局限性（Limitations）客观陈述与未来扩展段（Constructive Limitations）

Communicative Intent: 我这一段想写论文末尾或 NeurIPS/ICLR Checklist 要求的 Limitations 段落：诚实指出当前方法的两个客观边界，但绝不贬损论文的核心贡献。

Applicable Context: Discussion / Limitations 章节

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [边界 1：极端高维或极度稀疏场景 Boundary 1]：Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries.

S2. S2 [边界 1 机理分析与解法 Mechanism & Solution 1]：First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy.

S3. S3 [边界 2：特定假设限制 Boundary 2]：Second, our continuous scale tracker assumes piece-wise differentiable drift and may under-estimate instantaneous discontinuous jumps caused by hardware resets.

S4. S4 [未来演进方向 Future Extension]：Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work.

Logic Chaining & Cohesion Formula:

Two Objective Boundaries → [Boundary 1 + Technical Remedy] → [Boundary 2 + Stochastic Extension]. 绑定逻辑规则：L017.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Our paper has some limitations. We did not test on very big datasets because we do not have enough GPUs. In the future, we will try more datasets and make the model better.

Diagnostic Autopsy: 写成了“因为没卡所以没做完实验”的借口，不仅毫无学术价值，反而会诱发审稿人拒稿。

[Publication-Grade Model Paragraph]:

Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting. First, while our spectral branch scales quasi-linearly with sequence length, computing pair-wise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy. Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration. Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [边界 1	*Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting.* **Constituent Analysis**: 对应步骤: S1 [边界 1: 极端高维或极度稀疏场景 Boundary 1]: Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [边界 1 机理分析与解法 Mechanism & Solution 1]	*First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy.* **Constituent Analysis**: 对应步骤: S2 [边界 1 机理分析与解法 Mechanism & Solution 1]: First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [边界 2	*Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration.* **Constituent Analysis**: 对应步骤: S3 [边界 2: 特定假设限制 Boundary 2]: Second, our continuous scale tracker assumes piece-wise differentiable drift and may underestimate instantaneous discontinuous jumps caused by hardware resets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [未来演进方向 Future Extension]	*Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.* **Constituent Analysis**: 对应步骤: S4 [未来演进方向 Future Extension]: Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

极具学者风范的 Limitations：精准指出 $O(C^2)$ 通道复杂度与分段平滑假设的数学边界，并分别给出“锚点低秩图稀疏化”与“跳跃扩散随机微分方程”的高阶解法。

13.0.8 BP24: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）

Communicative Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Applicable Context: Conclusion 章节全段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [核心科学问题闭合 Scientific Closure]: In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain].

S2. S2 [方法论精髓回顾 Methodological Synthesis]: By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty].

S3. S3 [实证结果高度概括 Empirical Recap]: Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups.

S4. S4 [跨领域迁移展望 Broader Outlook]: Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments.

Logic Chaining & Cohesion Formula:

Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则: L013, L023, L029.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In conclusion, we proposed DyTS-Net in this paper. It has a spectral module and an ODE module. Experiments show that it achieves state-of-the-art performance on many datasets. We hope our work will inspire more researchers in the future.

Diagnostic Autopsy: 机械重复摘要且句式干瘪，末句 We hope our work will inspire... 属于典型的中式客套收尾。

[Publication-Grade Model Paragraph]:

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [核心科学问题闭合 Scientific Closure]	*In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.* **Constituent Analysis**: 对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [方法论精髓回顾 Methodological Synthesis]	*By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.* **Constituent Analysis**: 对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [实证结果高度概括 Empirical Recap]	*Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.* **Constituent Analysis**: 对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [跨领域迁移展望 Broader Outlook]	*Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.* **Constituent Analysis**: 对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

首句点明“解决了归一化平稳性与物理幅值保真度之间的长期张力”，末句自然延展到信息物理系统的异常检测与闭环预测控制，余音绕梁。

Chapter 14

Track 4: Data Diagnostics, Pattern Reporting & Visual Cross-Reference (BP25–BP32) / 数据图表汇报、趋势对比与多图印证

本章系统拆解该赛道的 8 大核心段落构建蓝图。每一项蓝图均涵盖明确的用户交际意图、适用场景、逐句动作规范（Move Structure）、句际信息咬合公式、病段与发表级定稿对照、成分剖析表格以及跨学科/文体迁移指南。

14.0.1 BP25: 图表小作文首段：题目改写与总体趋势概览（Paraphrase + Macro Overview）

Communicative Intent: 我这一段想写雅思 Task 1 或数据报告的第一段：用同义句法改写图表标题，并用两句话高度凝练全图最核心的 2 个宏观趋势（极值、反超拐点或主次格局），绝不堆砌零碎数字。

Applicable Context: IELTS Academic Task 1 首段（Introduction + Overview）或实验趋势总览段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [图表命题同义改写 Prompt Paraphrase]：The line graph / bar chart compares [Variable/ Metric] across [Categories/Regions] over a [N]-year period from [Year A] to [Year B].
- **S2.** S2 [宏观趋势 1：总体走向与反超 Overview Trend 1]：Overall, while [Category A and B] experienced a steady upward trajectory, [Category C] declined markedly, leading to a reversal in dominance around [Mid-Year].
- **S3.** S3 [宏观趋势 2：极值或差距特征 Overview Trend 2]：Notably, [Category A] registered the most dramatic growth throughout the period, ultimately overtaking all other sectors to become the primary contributor.

Logic Chaining & Cohesion Formula:

Paraphrase → Overall Trajectory Contrast → Dominant Crossover/ Extremum (Zero raw numbers in Overview!). 绑定逻辑规则：L013, L015.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

The chart shows the energy consumption from 2000 to 2025. We can see that solar energy went up a lot from 5% to 35%, and coal went down from 60% to 25%. Wind also went up.

Diagnostic Autopsy: (1) 词汇低幼（went up a lot, went down）；(2) 把具体百分比过早塞进了 Overview，导致后面细节段无话可写。

[Publication-Grade Model Paragraph]:

The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025. Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily. Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [图表命题同义改写 Prompt Paraphrase]	*The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025.* **Constituent Analysis**: 对应步骤：S1 [图表命题同义改写 Prompt Paraphrase]：The line graph / bar chart compares [Variable/Metric] across [Categories/Regions] over a [N]-year period from [Year A] to [Year B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [宏观趋势 1	*Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily.* **Constituent Analysis**: 对应步骤：S2 [宏观趋势 1：总体走向与反超 Overview Trend 1]：Overall, while [Category A and B] experienced a steady upward trajectory, [Category C] declined markedly, leading to a reversal in dominance around [Mid-Year]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [宏观趋势 2	*Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.* **Constituent Analysis**: 对应步骤：S3 [宏观趋势 2：极值或差距特征 Overview Trend 2]：Notably, [Category A] registered the most dramatic growth throughout the period, ultimately overtaking all other sectors to become the primary contributor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

不露一个具体数字，却精准抓住了“可再生能源上扬 vs. 煤炭萎缩”的总基调以及“2018 年主次反转（crossover）”的决定性拐点，稳拿 Task Achievement 8+。

14.0.2 BP26: 动态时间序列趋势分段描述（暴涨、缓降、平台期与拐点）

Communicative Intent: 我这一段想写动态折线图（或模型训练收敛曲线）的主体细节段：把一条经历“初期缓慢爬升 → 中期爆发增长 → 后期高位平台期”的曲线，以及另一条“冲高回落”的曲线对比写活。

Applicable Context: IELTS Task 1 动态图主体段 / 论文训练动力学与收敛曲线分析段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [起始点位对比 Initial Standing]：At the beginning of the period in [2000], [Entity A] stood at a modest [12%],less than a quarter of [Entity B]’s initial figure ([54%]).

S2. S2 [多阶段演进：缓升转暴涨 Phased Acceleration]：After rising marginally over the firstfive years, [Entity A] surged nearly threefold between [2005] and [2018], peaking at [48%] before stabilizing on a high plateau.

S3. S3 [反向曲线对比：冲高回落 Counter-Trajectory]：Conversely, [Entity B] climbed briefly to [61%] in [2004] but subsequently entered a prolonged decline, dropping by more than half to [24%] by [2025].

S4. S4 [终局交叉总结 Terminal Comparison]：Consequently, the initial [42]-percentage-point gap between the two categories had completely reversed by the final year.

Logic Chaining & Cohesion Formula:

Initial Disparity → Multi-Phase Surge & Plateau (Entity A) → Peak-then-Decline (Entity B) → Gap Reversal.

绑定逻辑规则：L010, L013.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In 2000, A was 12% and B was 54%. Then in 2005, A increased to 15%. In 2018, A increased to 48%. In 2025, A was 47%. B increased to 61% in 2004, and then decreased to 24% in 2025.

Diagnostic Autopsy: 典型的“流水账报年份数字”（In 2000... In 2005... In 2018...），句式单一且毫无趋势洞察。

[Publication-Grade Model Paragraph]:

In 2000, battery-electric vehicles accounted for a mere 4% of municipal fleet purchases, trailing diesel buses at 62%. Following five years of marginal fluctuation, electric adoption accelerated sharply after 2006, quadrupling to 38% by 2018 before leveling off at approximately 41% through 2025. By contrast, diesel procurement peaked briefly at 67% in 2004 but subsequently underwent a sustained contraction, plunging to 19% by the end of the survey. This divergent trajectory reversed the initial 58-percentage-point deficit into a more than twofold lead for electric fleets.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [起始点位对比 Initial Standing]	*In 2000, battery-electric vehicles accounted for a mere 4% of municipal fleet purchases, trailing diesel buses at 62%.* **Constituent Analysis**: 对应步骤：S1 [起始点位对比 Initial Standing]：At the beginning of the period in [2000], [Entity A] stood at a modest [12%],less than a quarter of [Entity B]'s initial figure ([54%]). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [多阶段演进	*Following five years of marginal fluctuation, electric adoption accelerated sharply after 2006, quadrupling to 38% by 2018 before leveling off at approximately 41% through 2025.* **Constituent Analysis**: 对应步骤：S2 [多阶段演进：缓升转暴涨 Phased Acceleration]：After rising marginally over the firstfive years, [Entity A] surged nearly threefold between [2005] and [2018], peaking at [48%] before stabilizing on a high plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [反向曲线对比	*By contrast, diesel procurement peaked briefly at 67% in 2004 but subsequently underwent a sustained contraction, plunging to 19% by the end of the survey.* **Constituent Analysis**: 对应步骤：S3 [反向曲线对比：冲高回落 Counter-Trajectory]：Conversely, [Entity B] climbed briefly to [61%] in [2004] but subsequently entered a prolonged decline, dropping by more than half to [24%] by [2025]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [终局交叉总结 Terminal Comparison]	*This divergent trajectory reversed the initial 58-percentage-point deficit into a more than twofold lead for electric fleets.* **Constituent Analysis**: 对应步骤：S4 [终局交叉总结 Terminal Comparison]：Consequently, the initial [42]-percentage-point gap between the two categories had completely reversed by the final year. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

灵活交替使用 accounted for a mere 4%, trailing..., Following five years of marginal fluctuation, quadrupling to..., leveling off at..., underwent a sustained contraction，句式富于韵律且零 AI 套话。

14.0.3 BP27: 静态多组数据横向对比与倍差/占比凸显段（Cross-Group Proportions）

Communicative Intent: 我这一段想写静态柱状图或饼图的主体段：横向对比不同国家/不同模型在不同开销维度上的占比差异，灵活运用倍数（double/triple）、分数（nearly a third）与排序结构。

Applicable Context: IELTS Task 1 静态图表段 / 论文开销拆解（Breakdown）段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [主导项锁定 Dominant Category]：Regarding [Group 1], [Category A] claims the largest share at [46%], nearly double the proportion allocated to [Category B] ([24%]).

S2. S2 [组间横向反差 Cross-Group Contrast]：A markedly different allocation pattern emerges in [Group 2], where [Category C] dominates at [52%], while [Category A] shrinks to just [15%].

S3. S3 [次要项合并归纳 Minor Categories Grouping]：Meanwhile, the remaining categories—[Category D and E]—collectively account for less than [12%] across both groups.

S4. S4 [结构性差异提炼 Structural Takeaway]：These figures indicate that whereas [Group 1] prioritizes [Function A], [Group 2] concentrates resources heavily on [Function C].

Logic Chaining & Cohesion Formula:

Dominant Item with Multiplier → Sharp Cross-Group Contrast → Pooled Minor Items → Structural Synthesis.

绑定逻辑规则：L009, L010.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In Group 1, A is 46%, B is 24%, C is 18%, D is 7%, and E is 5%. In Group 2, C is 52%, B is 22%, A is 15%, D is 6%, and E is 5%.

Diagnostic Autopsy: 把文字写成了 CSV 表格转储，完全没有归并次要项，也没有提炼倍差与结构反差。

[Publication-Grade Model Paragraph]:

Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%). A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocessing dominates energy usage at 51%, while attention overhead contracts to just 16%. Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption. This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [主导项锁定 Dominant Category]	*Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%).* **Constituent Analysis**: 对应步骤：S1 [主导项锁定 Dominant Category]：Regarding [Group 1], [Category A] claims the largest share at [46%], nearly double the proportion allocated to [Category B] ([24%]). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [组间横向反差 Cross-Group Contrast]	*A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocessing dominates energy usage at 51%, while attention overhead contracts to just 16%.* **Constituent Analysis**: 对应步骤：S2 [组间横向反差 Cross-Group Contrast]：A markedly different allocation pattern emerges in [Group 2], where [Category C] dominates at [52%], while [Category A] shrinks to just [15%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [次要项合并归纳 Minor Categories Grouping]	*Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption.* **Constituent Analysis**: 对应步骤：S3 [次要项合并归纳 Minor Categories Grouping]：Meanwhile, the remaining categories—[Category D and E]—collectively account for less than [12%] across both groups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [结构性差异提炼 Structural Takeaway]	*This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.* **Constituent Analysis**: 对应步骤：S4 [结构性差异提炼 Structural Takeaway]：These figures indicate that whereas [Group 1] prioritizes [Function A], [Group 2] concentrates resources heavily on [Function C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

第一句用倍数（48%, exactly double... 24%），第二句抓跨组反转（51% vs. 16%），第三句用 collectively account for under 11% 打包次要项，末句点破深层规律。

14.0.4 BP28: 双图关联与互补/因果印证段（Synthesizing Dual Charts）

Communicative Intent: 我这一段想写双图题（如一张图是算力投入/研发经费变化，另一张图是模型误差/碳排放强度变化）的综合关联段，阐明两张图之间的内在协同或背离关系。

Applicable Context: IELTS Task 1 双图组合题 / 论文多指标联合分析段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [双图联动锚定 Cross-Chart Link]: Synthesizing both charts reveals a clear inverse / positive relationship between [Chart 1 Variable] and [Chart 2 Variable] up to [Threshold Point].

S2. S2 [区间 1 协同变化 Phase 1 Co-movement]: Specifically, as [Variable X in Chart 1] expanded from [A] to [B] during [Period 1], [Variable Y in Chart 2] dropped correspondingly by [C%].

S3. S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]: However, beyond [Threshold Point], although [Variable X] continued to double, [Variable Y] plateaued around [Value], indicating diminishing marginal returns.

S4. S4 [例外个体点拨 Exception Highlight]: An exception to this pattern is [Entity Z], which achieved the lowest [Variable Y] despite maintaining moderate [Variable X].

Logic Chaining & Cohesion Formula:

Global Correlation → Co-movement Phase → Diminishing Returns / Divergence → Notable Outlier. 绑定逻辑规则：L007, L013.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Chart 1 shows R&D spending and Chart 2 shows defect rates. When R&D spending goes up, defect rates go down. Country Z spends little money but has low defect rates.

Diagnostic Autopsy: 极其简陋，未能刻画“初期强相关 → 超过阈值后边际收益递减 → 特例分析”的层次感。

[Publication-Grade Model Paragraph]:

Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold. Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [双图联动锚定 Cross-Chart Link]	*Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold.* **Constituent Analysis**: 对应步骤：S1 [双图联动锚定 Cross-Chart Link]：Synthesizing both charts reveals a clear inverse / positive relationship between [Chart 1 Variable] and [Chart 2 Variable] up to [Threshold Point]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [区间 1 协同变化 Phase 1 Co-movement]	*Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8% to 0.9% (Figure 2).* **Constituent Analysis**: 对应步骤：S2 [区间 1 协同变化 Phase 1 Co-movement]：Specifically, as [Variable X in Chart 1] expanded from [A] to [B] during [Period 1], [Variable Y in Chart 2] dropped correspondingly by [C%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]	*Beyond $40 million, however, further capital expenditure yielded diminishing returns: doubling investment to $80 million reduced defects by only an additional 0.1 percentage points.* **Constituent Analysis**: 对应步骤：S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]：However, beyond [Threshold Point], although [Variable X] continued to double, [Variable Y] plateaued around [Value], indicating diminishing marginal returns. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [例外个体点拨 Exception Highlight]	*Notably, Plant D defied this expenditure curve, achieving a 0.7% defect rate with just $22 million through closed-loop sensor calibration.* **Constituent Analysis**: 对应步骤：S4 [例外个体点拨 Exception Highlight]：An exception to this pattern is [Entity Z], which achieved the lowest [Variable Y] despite maintaining moderate [Variable X]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

将两张图的数据有机编织在一起，精准点出 $40M 拐点后的边际收益递减（diminishing returns）与 Plant D 高效特例。

14.0.5 BP29: 流程图/算法流水线与闭环工序衔接段（Process & Closed-Loop Lifecycle）

Communicative Intent: 我这一段想写雅思 Task 1 工艺/自然循环流程图（或论文算法数据预处理与闭环训练流水线）：把 5–6 个串联、分叉与回环步骤写得层次分明，绝不机械连用 Then, Then, Then。

Applicable Context: IELTS Task 1 流程图（Process Diagram）/ 论文系统工程 Pipeline 描述段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [流程总括与起止界定 Lifecycle Overview]：The diagram illustrates the closed-loop pipeline for [Process Name], comprising [N] main stages from initial [Input Stage] to continuous [Feedback/Output Stage].

S2. S2 [前段采集与预处理 Initial Ingestion & Filtering]：The process begins when raw [Material/Data] is collected via [Source] and passed through [Filter/Chamber] to remove [Impurities/Outliers].

S3. S3 [中段并行分流 Parallel Branching]：Once purified, the stream bifurcates into two parallel pathways: [Fraction A] undergoes [Process A], whereas [Fraction B] is routed to [Process B].

S4. S4 [末端合成与闭环回流 Synthesis & Closed-Loop Recycling]：The resulting outputs are subsequently merged in [Unit C], with residual by-products / high-error trajectories recycled back into the initial buffer for iterative refinement.

Logic Chaining & Cohesion Formula:

Pipeline Overview → Ingestion & Filtering → Parallel Bifurcation → Terminal Merger & Feedback Loop. 绑定逻辑规则：L003, L005.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

First, raw data is collected. Then it is cleaned. Then it goes into two branches. Branch 1 does A, and Branch 2 does B. Then they are combined. Finally, bad samples go back to the beginning.

Diagnostic Autopsy: 连续四个 Then，词汇贫乏（cleaned, goes into, does A, bad samples go back）。

[Publication-Grade Model Paragraph]:

The diagram illustrates the closed-loop data curation pipeline for embodied robot learning, spanning four primary stages from teleoperated collection to failure-driven retraining. The cycle begins when raw multimodal demonstrations are recorded via VR teleoperation and filtered to discard kinematic singularities and dropped frames. Once validated, the trajectory stream bifurcates into two parallel pathways: visual frames undergo background augmentation, while proprioceptive states are normalized into local gripper coordinates. These processed streams are subsequently synchronized to train the visuomotor policy, whose real-world rollout failures are logged and fed back into the teleoperation buffer to close the active-learning loop.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [流程总括与起止界定 Lifecycle Overview]	*The diagram illustrates the closed-loop data curation pipeline for embodied robot learning, spanning four primary stages from teleoperated collection to failure-driven retraining.* **Constituent Analysis**: 对应步骤：S1 [流程总括与起止界定 Lifecycle Overview]: The diagram illustrates the closed-loop pipeline for [Process Name], comprising [N] main stages from initial [Input Stage] to continuous [Feedback/Output Stage]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [前段采集与预处理 Initial Ingestion & Filtering]	*The cycle begins when raw multimodal demonstrations are recorded via VR teleoperation and filtered to discard kinematic singularities and dropped frames.* **Constituent Analysis**: 对应步骤：S2 [前段采集与预处理 Initial Ingestion & Filtering]: The process begins when raw [Material/Data] is collected via [Source] and passed through [Filter/Chamber] to remove [Impurities/Outliers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [中段并行分流 Parallel Branching]	*Once validated, the trajectory stream bifurcates into two parallel pathways: visual frames undergo background augmentation, while proprioceptive states are normalized into local gripper coordinates.* **Constituent Analysis**: 对应步骤：S3 [中段并行分流 Parallel Branching]: Once purified, the stream bifurcates into two parallel pathways: [Fraction A] undergoes [Process A], whereas [Fraction B] is routed to [Process B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [末端合成与闭环回流 Synthesis & Closed-Loop Recycling]	*These processed streams are subsequently synchronized to train the visuomotor policy, whose real-world rollout failures are logged and fed back into the teleoperation buffer to close the active-learning loop.* **Constituent Analysis**: 对应步骤：S4 [末端合成与闭环回流 Synthesis & Closed-Loop Recycling]: The resulting outputs are subsequently merged in [Unit C], with residual by-products / high-error trajectories recycled back into the initial buffer for iterative refinement. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

用 The cycle begins when... → Once validated, the stream bifurcates into... → These processed streams are subsequently synchronized..., whose failures are fed back... 完美呈现分支与闭环。

14.0.6 BP30: 历史实测与未来预测跨时态衔接段（Historical vs. Projected Trends）

Communicative Intent: 我这一段想处理同时包含过去历史数据（如 2010–2025）和未来预测数据（如 2026–2040）的图表段落，严格区分过去实测时态与未来审慎预测表达。

Applicable Context: IELTS Task 1 含未来预测年份的图表 / 产业算力与能耗预测报告

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [历史实测轨迹 Historical Trajectory (Past Tense)]：Between [2010] and [2025], global data-center electricity demand rose steadily from [200 TWh] to [460 TWh], driven by cloud migration.

S2. S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]：From [2026] onward, consumption is projected to accelerate sharply, with forecasts indicating a surge to [1,050 TWh] by [2035].

S3. S3 [结构占比重塑 Structural Shift in Projections]：Within this projected expansion, generative AI training and inference—which accounted for less than [10%] in [2023]—are anticipated to represent over [45%] of total load by [2040].

S4. S4 [条件式审慎收尾 Conditional Hedging]：Nevertheless, energy-efficient hardware adoption is expected to moderate the growth slope during the final five years.

Logic Chaining & Cohesion Formula:

Historical Past (rose/accounted for) → Pivot Year → Future Projection Verbs (is projected/anticipated/expected to) → Moderating Factor. 绑定逻辑规则：L010, L017.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

From 2010 to 2025, demand increased from 200 to 460. In 2035, it will increase to 1050. AI was 10% in 2023 and will be 45% in 2040.

Diagnostic Autopsy: 未来年份全部生硬使用 will be / will increase，违反学术图表报告对未来预测需使用审慎预测动词（is projected / anticipated / forecast to）的规范。

[Publication-Grade Model Paragraph]:

Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh. From 2026 onward, total demand is projected to climb at a substantially steeper rate, with baseline forecasts pointing to 1,050 TWh by 2035. Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040. However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [历史实测轨迹 Historical Trajectory (Past Tense)]	*Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh.* **Constituent Analysis**: 对应步骤：S1 [历史实测轨迹 Historical Trajectory (Past Tense)]：Between [2010] and [2025], global data-center electricity demand rose steadily from [200 TWh] to [460 TWh], driven by cloud migration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]	*From 2026 onward, total demand is projected to climb at a substantially steeper rate, with baseline forecasts pointing to 1,050 TWh by 2035.* **Constituent Analysis**: 对应步骤：S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]：From [2026] onward, consumption is projected to accelerate sharply, with forecasts indicating a surge to [1,050 TWh] by [2035]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [结构占比重塑 Structural Shift in Projections]	*Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040.* **Constituent Analysis**: 对应步骤：S3 [结构占比重塑 Structural Shift in Projections]：Within this projected expansion, generative AI training and inference—which accounted for less than [10%] in [2023]—are anticipated to represent over [45%] of total load by [2040]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [条件式审慎收尾 Conditional Hedging]	*However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.* **Constituent Analysis**: 对应步骤：S4 [条件式审慎收尾 Conditional Hedging]：Nevertheless, energy-efficient hardware adoption is expected to moderate the growth slope during the final five years. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

过去实测用 more than doubled / represented，未来预测交替使用 is projected to climb / with forecasts pointing to / are anticipated to account for / is expected to temper，时态与语气滴水不漏。

14.0.7 BP31: 托福综合写作 / 学术争议正反交锋段（Point-by-Point Reading vs. Lecture Synthesis）

Communicative Intent: 我这一段想写托福综合写作（Integrated Writing）或文献争议交锋段：先用 1 句话精准概括阅读材料（或正方文献）的分论点，再用 2–3 句话详细呈现教授讲座（或反方实证）是如何釜底抽薪推翻该论点的。

Applicable Context: TOEFL Integrated Writing 主体段 / 论文理论争议辨析段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [正方/阅读论点概括 Reading Claim & Rationale]: The reading passage contends that [Claim A], arguing that [Reading's Supporting Mechanism].

S2. S2 [反方/听力直接反驳 Lecture Direct Refutation]: The lecturer directly challenges this assertion by pointing out that [Core Flaw / Neglected Factor in Reading's Premise].

S3. S3 [反方具体反证展开 Empirical Counter-Evidence]: Specifically, she explains that when [Real-World Condition], [Mechanism B occurs instead], which [Negates the Claimed Benefit].

S4. S4 [结论锁定 Refutation Closure]: Consequently, rather than achieving [Reading's Goal], the proposed measure actually exacerbates [Negative Outcome].

Logic Chaining & Cohesion Formula:

Reading Claim → Lecturer's Direct Challenge → Specific Counter-Mechanism → Reversed Outcome. 绑定逻辑规则: L008, L019.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

The reading says that replacing all delivery trucks with autonomous drones will save energy. The professor disagrees. She says drones use a lot of battery to stay in the air and can only carry one package at a time.

Diagnostic Autopsy: 句式碎片化（The reading says... The professor disagrees. She says...），缺乏紧凑的逻辑因果嵌套。

[Publication-Grade Model Paragraph]:

The reading passage contends that replacing urban delivery vans with autonomous quadcopter drones will substantially reduce last-mile transportation energy consumption, citing the lighter weight of aerial vehicles. The lecturer directly challenges this premise by examining energy efficiency per kilogram of delivered freight. Specifically, she points out that unlike ground vehicles supported passively by road surfaces, multi-rotor drones must expend continuous battery power simply to counteract gravity and headwinds while carrying only a single parcel per trip. Consequently, servicing a dense apartment block via individual drone flights consumes nearly three times more energy per package than a single consolidated van route.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [正方/阅读论点概括 Reading Claim & Rationale]	*The reading passage contends that replacing urban delivery vans with autonomous quadcopter drones will substantially reduce last-mile transportation energy consumption, citing the lighter weight of aerial vehicles.* **Constituent Analysis**: 对应步骤：S1 [正方/阅读论点概括 Reading Claim & Rationale]：The reading passage contends that [Claim A], arguing that [Reading's Supporting Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [反方/听力直接反驳 Lecture Direct Refutation]	*The lecturer directly challenges this premise by examining energy efficiency per kilogram of delivered freight.* **Constituent Analysis**: 对应步骤：S2 [反方/听力直接反驳 Lecture Direct Refutation]：The lecturer directly challenges this assertion by pointing out that [Core Flaw / Neglected Factor in Reading's Premise]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [反方具体反证展开 Empirical Counter-Evidence]	*Specifically, she points out that unlike ground vehicles supported passively by road surfaces, multi-rotor drones must expend continuous battery power simply to counteract gravity and headwinds while carrying only a single parcel per trip.* **Constituent Analysis**: 对应步骤：S3 [反方具体反证展开 Empirical Counter-Evidence]：Specifically, she explains that when [Real-World Condition], [Mechanism B occurs instead], which [Negates the Claimed Benefit]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [结论锁定 Refutation Closure]	*Consequently, servicing a dense apartment block via individual drone flights consumes nearly three times more energy per package than a single consolidated van route.* **Constituent Analysis**: 对应步骤：S4 [结论锁定 Refutation Closure]：Consequently, rather than achieving [Reading's Goal], the proposed measure actually exacerbates [Negative Outcome]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

用 contends that..., citing... 紧凑概括阅读，再用 directly challenges this premise by examining... → Specifically, she points out that unlike... → Consequently... 形成极具压迫感的反驳闭环。

14.0.8 BP32: 异常数据点（Outliers）与局部背离现象精准批注段（Outlier & Anomaly Annotation）

Communicative Intent: 我这一段想在图表或实验结果段中，专门挑出违背总体规律的 1–2 个异常点（Outliers）或局部反常尖峰进行精准描述与成因剖析。

Applicable Context: IELTS Task 1 特例段 / 论文实验反常现象（Anomalous Observation）剖析段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [总体规律背景下的特例锚定 Outlier Identification]: Although [General Trend holds across most groups/steps], a notable departure from this pattern occurs in [Specific Point/Group].

S2. S2 [反常幅度量化 Quantitative Deviation]: Whereas peer categories cluster tightly within $[A, B]$, [Outlier Entity] spikes/drops to [Extreme Value], deviating by more than [X%] from the group median.

S3. S3 [底层物理/统计成因剖析 Root Cause Analysis]: Closer inspection indicates that this anomaly stems from [Specific Structural/External Factor] rather than random variance.

S4. S4 [方法论启示 Methodological Implication]: Accounting for such localized departures is critical when calibrating [Downstream Model/Policy].

Logic Chaining & Cohesion Formula:

General Rule vs. Notable Departure → Quantitative Deviation → Root Cause Attribution → Calibration Take-away. 绑定逻辑规则：L007, L013.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Most datasets follow this rule, except Exchange dataset. On Exchange dataset, deep models are worse than simple linear models. This is because financial data has strong random walk noise.

Diagnostic Autopsy: 表达平淡（follow this rule, except, worse than），缺乏量化的偏离描述与深层统计定性。

[Publication-Grade Model Paragraph]:

Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset. Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate. Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks. This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [总体规律背景下的特例锚定 Outlier Identification]	*Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset.* **Constituent Analysis**: 对应步骤：S1 [总体规律背景下的特例锚定 Outlier Identification]：Although [General Trend holds across most groups/steps], a notable departure from this pattern occurs in [Specific Point/Group]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [反常幅度量化 Quantitative Deviation]	*Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate.* **Constituent Analysis**: 对应步骤：S2 [反常幅度量化 Quantitative Deviation]：Whereas peer categories cluster tightly within $[A, B]$, [Outlier Entity] spikes/drops to [Extreme Value], deviating by more than [X%] from the group median. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [底层物理/统计成因剖析 Root Cause Analysis]	*Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks.* **Constituent Analysis**: 对应步骤：S3 [底层物理/统计成因剖析 Root Cause Analysis]：Closer inspection indicates that this anomaly stems from [Specific Structural/External Factor] rather than random variance. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [方法论启示 Methodological Implication]	*This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.* **Constituent Analysis**: 对应步骤：S4 [方法论启示 Methodological Implication]：Accounting for such localized departures is critical when calibrating [Downstream Model/Policy]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

将“Exchange 数据集上深度模型不如线性模型”这一经典反常现象，剖析为“日频汇率收益近乎鞅过程（near-martingale）、弱周期性导致高容量注意力层过拟合瞬态市场冲击”，深刻且地道。

Chapter 15

Track 5: Discussion, Speculation Boundaries & Critical Nuance (BP33–BP40) / 议论破题、正方立论、双层因果与让步反驳

本章系统拆解该赛道的 8 大核心段落构建蓝图。每一项蓝图均涵盖明确的用户交际意图、适用场景、逐句动作规范（Move Structure）、句际信息咬合公式、病段与发表级定稿对照、成分剖析表格以及跨学科/文体迁移指南。

15.0.1 BP33: 议论文破题首段：具象趋势锚定 + 争议张力 + 明确中心论点（Zero-Template Hook & Thesis）

Communicative Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉'With the rapid development of science and technology, more and more people...' 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Applicable Context: CET-6 / IELTS Task 2 / TOEFL Independent Essay 首段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。
- **S2.** S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality].
- **S3.** S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2].

Logic Chaining & Cohesion Formula:

Concrete Phenomenon (No Cliché) → Balanced Tension → Conditional & Specific Thesis Statement. 绑定逻辑规则：L023, L029.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

With the rapid development of science and technology, artificial intelligence is becoming more and more popular in universities. Some people think it is good, while others think it is bad. In my opinion, every coin has two sides and we should use it properly.

Diagnostic Autopsy: 汇集了中国考生三大扣分重灾区：(1) With the rapid development of...；(2) more and more popular / good vs. bad；(3) Every coin has two sides（零立场的废话中心论点）。

[Publication-Grade Model Paragraph]:

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [具象现象锚定 Concrete Trend Anchor]	*Large language models and automated code assistants have rapidly become routine study aids across university campuses.* **Constituent Analysis**: 对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [核心争议张力 Core Tension / Debate]	*While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.* **Constituent Analysis**: 对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality].	句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [带条件的中心论点 Nuanced Thesis Statement]	*In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.* **Constituent Analysis**: 对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2].	句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

第一句直接点出大语言模型与代码助手进入大学校园；第二句精准提炼正反双方核心交锋；第三句给出清晰、可操作且带明确理由的中心论点（先掌握基础分析能力再在进阶课程引入）。

15.0.2 BP34: 核心正方分论点段：主张 → 机制推演 → 具象行业微案例 → 扣题闭环（PEEL 满分段）

Communicative Intent: 我这一段想写大作文的第一个主体段（Body Paragraph 1）：论证某个核心优势或必要性，做到有理有据、有具象场景支撑，绝不车轱辘话来回说。

Applicable Context: CET-6 / IELTS Task 2 / TOEFL 议论文核心主体段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [分论点主题句 Topic Claim]：一句话清晰亮出本段唯一控制中心（[Policy/Factor X] strengthens [Outcome Y] by [Core Mechanism]）。

S2. S2 [因果机制推演 Causal Elaboration]：Explain HOW and WHY：阐明如果没有这层干扰或有了这项机制，认知/经济/技术流程会发生什么具体改变。

S3. S3 [具象微观案例 Concrete Micro-Case]：For instance, when [Specific Group/Industry] [Specific Action], [Measurable/Observable Consequence]（绝不编造假百分比，用具体场景细节说话）。

S4. S4 [反事实对比或扣题升华 Warrant / Link Back]：Thus, [Summary Noun] shifts human effort from [Low-value Routine] toward [High-value Judgment].

Logic Chaining & Cohesion Formula:

Topic Claim → Causal Mechanism → Concrete Micro-Case (L021) → Warrant Closure. 绑定逻辑规则：L006, L012, L021.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

First, AI can improve working efficiency a lot. People have many boring tasks every day. If AI does these tasks, people can save time and do better things. For example, my friend uses AI to write emails and saves two hours every day. So AI is very helpful.

Diagnostic Autopsy: (1) 口语词泛滥（a lot, boring tasks, better things, my friend）；(2) 个人化琐碎举例（my friend saves two hours）不符合学术议论文语域。

[Publication-Grade Model Paragraph]:

Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment. In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formu-

lating a hypothesis. When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation. Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [分论点主题句 Topic Claim]	*Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment.* **Constituent Analysis**: 对应步骤：S1 [分论点主题句 Topic Claim]：一句话清晰亮出本段唯一控制中心（[Policy/Factor X] strengthens [Outcome Y] by [Core Mechanism]）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [因果机制推演 Causal Elaboration]	*In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis.* **Constituent Analysis**: 对应步骤：S2 [因果机制推演 Causal Elaboration]：Explain HOW and WHY：阐明如果没有这层干扰或有了这项机制，认知/经济/技术流程会发生什么具体改变。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [具象微观案例 Concrete Micro-Case]	*When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation.* **Constituent Analysis**: 对应步骤：S3 [具象微观案例 Concrete Micro-Case]：For instance, when [Specific Group/Industry] [Specific Action], [Measurable/Observable Consequence]（绝不编造假百分比，用具体场景细节说话）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [反事实对比或扣题升华 Warrant / Link Back]	*Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.* **Constituent Analysis**: 对应步骤：S4 [反事实对比或扣题升华 Warrant / Link Back]：Thus, [Summary Noun] shifts human effort from [Low-value Routine] toward [High-value Judgment]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

从抽象机制（腾出认知带宽用于高阶诊断判断）自然过渡到具体行业切片（医院放射科部署自动筛查算法初筛正常胸片 → 医生聚焦疑难病理与问诊），说服力极强。

15.0.3 BP35: 深层原因剖析段：表层诱因与结构性根源双层挖掘（Dual-Layer Causal Diagnosis）

Communicative Intent: 我这一段想在问题分析类作文（Why is this happening?）中，剖析某个社会/科研痛点背后的深层原因（既指出直接表层诱因，又挖出底层的考核激励或结构性根源）。

Applicable Context: CET-6 / IELTS Task 2 原因分析段，或学术评论文章

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [原因总起句 Dual-Cause Thesis]：The widespread emergence of [Problem X] stems from both immediate technological pressures and deeper institutional incentives.

S2. S2 [第一层：直接操作/技术诱因 Immediate Driver]：At the operational level, [Factor A] lowers the cost of [Behavior], tempting individuals/organizations to [Short-term Action].

S3. S3 [第二层：深层制度/激励根源 Structural Root Cause]：More fundamentally, existing evaluation systems reward [Superficial Metric] rather than [Long-term Quality], creating powerful pressure to prioritize speed over rigor.

S4. S4 [双因叠加后果 Compounding Effect]：When effortless generation meets quantity-driven evaluation, [Problem X] becomes systemic rather than isolated.

Logic Chaining & Cohesion Formula:

Dual-Cause Opening → Immediate Operational Driver → Deeper Institutional Root → Systemic Compounding. 绑定逻辑规则：L007, L005.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Why do many students copy AI answers directly? First, AI is very easy and fast to use. Second, students are lazy and have too much homework. So they just copy and paste.

Diagnostic Autopsy: 把复杂社会教育现象简单归结为“学生懒（students are lazy）”，缺乏操作成本与评价体系两个层面的结构化深度。

[Publication-Grade Model Paragraph]:

The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design. At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading. More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique. When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [原因总起句 Dual-Cause Thesis]	*The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design.* **Constituent Analysis**: 对应步骤：S1 [原因总起句 Dual-Cause Thesis]：The widespread emergence of [Problem X] stems from both immediate technological pressures and deeper institutional incentives. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [第一层	*At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading.* **Constituent Analysis**: 对应步骤：S2 [第一层：直接操作/技术诱因 Immediate Driver]：At the operational level, [Factor A] lowers the cost of [Behavior], tempting individuals/organizations to [Short-term Action]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [第二层	*More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique.* **Constituent Analysis**: 对应步骤：S3 [第二层：深层制度/激励根源 Structural Root Cause]：More fundamentally, existing evaluation systems reward [Superficial Metric] rather than [Long-term Quality], creating powerful pressure to prioritize speed over rigor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [双因叠加后果 Compounding Effect]	*When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.* **Constituent Analysis**: 对应步骤：S4 [双因叠加后果 Compounding Effect]：When effortless generation meets quantity-driven evaluation, [Problem X] becomes systemic rather than isolated. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

用 At an immediate level... 与 More fundamentally... 剥洋葱般剖析“极低生成摩擦成本”与“奖励表面流利度的公式化命题考核”，末句 When A meets B, C inevitably crowds out D 极具思辨穿透力。

15.0.4 BP36: 让步-反驳段：优雅承认反方合理成分并强力扭转权重（Concession & Refutation）

Communicative Intent: 我这一段想在议论文中回应反方观点（Some people argue that...）：先大度承认反方担忧在短期或局部确实存在，再用结构性证据或制度设计证明该问题完全可控，从而锁死我方立场。

Applicable Context: CET-6 / IELTS Task 2 / TOEFL 让步反驳段（Body Paragraph 2 or 3）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [呈现反方最有力的担忧 Articulate Opposing Concern]：Admittedly, critics raise a valid concern that [Opposing Argument, e.g., automation displaces entry-level clerical and manufacturing workers].

S2. S2 [承认局部真实性 Acknowledge Partial Validity]：In the short term, rapid deployment of [Technology/Policy] can indeed create transitional friction for workers whose routines are easily codified.

S3. S3 [釜底抽薪式转折 Structural Pivot]：However, viewing [X] purely as a zero-sum threat overlooks how productivity gains generate entirely new categories of skilled employment.

S4. S4 [历史/现实反证与制度补救 Evidence & Mitigation]：Just as spreadsheet software reduced manual bookkeeping while expanding demand for financial analysts, pairing automation with targeted vocational retraining allows displaced workers to transition into system supervision and maintenance roles.

Logic Chaining & Cohesion Formula:

Admittedly (Valid Concern) → Short-term Reality → However (Zero-Sum Fallacy Pivot) → Historical Analogy & Policy Mitigation. 绑定逻辑规则：L008, L023.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Some people think robots will take all our jobs and make everyone poor. This is wrong. Robots will create new jobs, and people can learn new skills to find better jobs easily.

Diagnostic Autopsy: 生硬武断（This is wrong, find better jobs easily），既没有共情反方的合理短期阵痛，也没有给出可信的历史类比或转型机制。

[Publication-Grade Model Paragraph]:

Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers. During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules. However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles. Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [呈现反方最有力的担忧 Articulate Opposing Concern]	*Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers.* **Constituent Analysis**: 对应步骤：S1 [呈现反方最有力的担忧 Articulate Opposing Concern]：Admittedly, critics raise a valid concern that [Opposing Argument, e.g., automation displaces entry-level clerical and manufacturing workers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [承认局部真实性 Acknowledge Partial Validity]	*During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules.* **Constituent Analysis**: 对应步骤：S2 [承认局部真实性 Acknowledge Partial Validity]：In the short term, rapid deployment of [Technology/Policy] can indeed create transitional friction for workers whose routines are easily codified. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [釜底抽薪式转折 Structural Pivot]	*However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles.* **Constituent Analysis**: 对应步骤：S3 [釜底抽薪式转折 Structural Pivot]：However, viewing [X] purely as a zero-sum threat overlooks how productivity gains generate entirely new categories of skilled employment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [历史/现实反证与制度补救 Evidence & Mitigation]	*Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.* **Constituent Analysis**: 对应步骤：S4 [历史/现实反证与制度补救 Evidence & Mitigation]：Just as spreadsheet software reduced manual bookkeeping while expanding demand for financial analysts, pairing automation with targeted vocational retraining allows displaced workers to transition into system supervision and maintenance roles. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

先用 Admittedly... does create localized employment friction 展现客观严谨，再用 However, treating X as a fixed-pie zero-sum game overlooks... + 电子表格软件的历史类比彻底瓦解反方悲观论。

15.0.5 BP37: 问题诊断与靶向解决方案段：拒绝“政府应该立法”万金油（Targeted Problem-Solution）

Communicative Intent: 我这一段想写 Problem-Solution 类作文的解决方案段：针对前文诊断出的两个具体根源，提出两项可操作、有明确执行主体和传导机制的靶向对策，拒绝只喊’The government should make laws and raise awareness’。

Applicable Context: CET-6 / IELTS Task 2 / 政策与工程治理方案段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [对策总起句 Targeted Solution Thesis]：Mitigating [Problem X] effectively requires coordinated intervention across both [Level A: Technical/Curriculum Design] and [Level B: Institutional Governance].

S2. S2 [对策 1：操作层可验证改革 Actionable Measure 1]：First, [Specific Actor 1, e.g., university departments] should replace [Vulnerable Practice] with [Verifiable Mechanism, e.g., oral defense and in-class iterative drafting], ensuring that [Specific Effect 1].

S3. S3 [对策 2：制度/标准层激励对齐 Actionable Measure 2]：Concurrently, [Specific Actor 2, e.g., regulatory agencies / industry consortia] must establish [Concrete Standard, e.g., mandatory provenance watermarking and audit logs] to penalize [Misconduct].

S4. S4 [协同闭环效果 Combined Outcome]：Together, aligning pedagogical verification with technical traceability curbs misuse without stifling legitimate innovation.

Logic Chaining & Cohesion Formula:

Dual-Level Intervention → Actor 1 + Concrete Mechanism → Actor 2 + Concrete Standard → Balanced Closure. 绑定逻辑规则：L002, L010.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

To solve this problem, first, the government should pass strict laws to punish bad people. Second, schools and media should educate the public and raise their awareness. Only in this way can we build a harmonious society.

Diagnostic Autopsy: 典型的中式六级万金油套话（pass strict laws, raise awareness, Only in this way can we...），放之四海而皆空，考官已产生严重审美疲劳。

[Publication-Grade Model Paragraph]:

Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria. First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review. Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions

and replication studies. Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [对策总起句 Targeted Solution Thesis]	*Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria.* **Constituent Analysis**: 对应步骤：S1 [对策总起句 Targeted Solution Thesis]：Mitigating [Problem X] effectively requires coordinated intervention across both [Level A: Technical/Curriculum Design] and [Level B: Institutional Governance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [对策 1	*First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review.* **Constituent Analysis**: 对应步骤：S2 [对策 1：操作层可验证改革 Actionable Measure 1]：First, [Specific Actor 1, e.g., university departments] should replace [Vulnerable Practice] with [Verifiable Mechanism, e.g., oral defense and in-class iterative drafting], ensuring that [Specific Effect 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [对策 2	*Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies.* **Constituent Analysis**: 对应步骤：S3 [对策 2：制度/标准层激励对齐 Actionable Measure 2]：Concurrently, [Specific Actor 2, e.g., regulatory agencies / industry consortia] must establish [Concrete Standard, e.g., mandatory provenance watermarking and audit logs] to penalize [Misconduct]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [协同闭环效果 Combined Outcome]	*Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.* **Constituent Analysis**: 对应步骤：S4 [协同闭环效果 Combined Outcome]：Together, aligning pedagogical verification with technical traceability curbs misuse without stifling legitimate innovation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

每一项对策都有明确的动作主体（期刊编委会 / 高校聘任委员会）、具体抓手（强制提交可执行代码容器与原始传感器日志 / 考核权重向开源与复现倾斜）及因果闭环。

15.0.6 BP38: 利弊权衡段：证明结构性长期收益远超短期可管可控成本（Outweigh / Trade-off Analysis）

Communicative Intent: 我这一段想回应'Do the advantages outweigh the disadvantages?' 题型：把弊端定性为"暂时的、可通过技术/规则收敛的局部成本"，把优势定性为"不可替代的结构性长期突破"。

Applicable Context: IELTS Task 2 (Advantages vs. Disadvantages) / 论文设计权衡（Trade-off）段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [权衡定性主题句 Asymmetry Thesis]: Weighing both dimensions reveals that the long-term structural benefits of [X] far outweigh its manageable transitional costs.

S2. S2 [将弊端限定为可收敛成本 Bounding the Drawback]: While [Drawback A, e.g., initial capital outlay / training overhead] poses a tangible hurdle, this cost is bounded and diminishes rapidly as [Mitigating Factor, e.g., hardware amortization / open-source standardization] matures.

S3. S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]: By contrast, the core advantage—[Benefit B, e.g., eliminating human exposure to hazardous environments / discovering nonlinear scientific laws]—delivers compounding value that manual methods cannot replicate at any cost.

S4. S4 [权衡结论 Trade-off Verdict]: Therefore, addressing the friction through targeted safeguards is far more rational than forgoing the transformative gain.

Logic Chaining & Cohesion Formula:

Asymmetry Thesis → Bound the Drawback (Temporary & Mitigable) → Highlight Compounding Advantage (Irreplaceable) → Rational Verdict. 绑定逻辑规则：L008, L023.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Every coin has two sides. Although open-source AI has some security risks, it also has many advantages like helping researchers and small companies. So I think the advantages are bigger than the disadvantages.

Diagnostic Autopsy: 只把好坏两边并列摆出来，然后生硬宣布 advantages are bigger than disadvantages，缺乏"为什么 A 大于 B"的权衡标尺（可逆成本 vs. 不可替代收益）。

[Publication-Grade Model Paragraph]:

Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks. While 公开 releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters. By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable

architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics. Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [权衡定性主题句 Asymmetry Thesis]	*Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks.* **Constituent Analysis**: 对应步骤：S1 [权衡定性主题句 Asymmetry Thesis]：Weighing both dimensions reveals that the long-term structural benefits of [X] far outweigh its manageable transitional costs. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [将弊端限定为可收敛成本 Bounding the Drawback]	*While 公开 releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters.* **Constituent Analysis**: 对应步骤：S2 [将弊端限定为可收敛成本 Bounding the Drawback]：While [Drawback A, e.g., initial capital outlay / training overhead] poses a tangible hurdle, this cost is bounded and diminishes rapidly as [Mitigating Factor, e.g., hardware amortization / open-source standardization] matures. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]	*By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics.* **Constituent Analysis**: 对应步骤：S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]：By contrast, the core advantage—[Benefit B, e.g., eliminating human exposure to hazardous environments / discovering non-linear scientific laws]—delivers compounding value that manual methods cannot replicate at any cost. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [权衡结论 Trade-off Verdict]	*Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.* **Constituent Analysis**: 对应步骤：S4 [权衡结论 Trade-off Verdict]：Therefore, addressing the friction through targeted safeguards is far more rational than forgoing the transformative gain. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

提供了一套无敌的权衡逻辑标尺：弊端是可被纵深防御缓解的（mitigable via defense-in-depth），而优势（科学可复现性与透明审计）是黑盒 API 无法替代的（irreplaceable）。

15.0.7 BP39: 反事实推理段：如果不采取某项原则，系统将如何一步步滑向失控（Counterfactual Escalation）

Communicative Intent: 我这一段想在议论文或科研立项书中，通过严密的“反事实推演（If we fail to... / Absent X...）”强化紧迫感与说服力。

Applicable Context: CET-6 / IELTS Task 2 / TOEFL 深化论证段，或论文 Motivation 强化段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [正面基准命题 Positive Baseline]：Maintaining rigorous [Principle/Standard X] serves as the primary safeguard against [Systemic Risk Y].
- **S2.** S2 [反事实假设切入 Counterfactual Premise]：Absent this safeguard—if [Actors] were to rely exclusively on [Unchecked Alternative]—[Immediate Vulnerability] would quickly emerge.
- **S3.** S3 [连锁恶化传导 Cascading Deterioration]：Once [Immediate Vulnerability] takes hold, [Secondary Consequence] compounds across cycles, eventually eroding [Core Foundation].
- **S4.** S4 [实证印证或结论锁定 Empirical Echo]：This degradation dynamic mirrors [Real-world/Technical Precedent], underscoring why [Principle X] cannot be compromised for short-term convenience.

Logic Chaining & Cohesion Formula:

Positive Safeguard → Counterfactual Premise (Absent X...) → Cascading Breakdown → Precedent Confirmation. 绑定逻辑规则：L020, L003.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Curating clean human data is very important. If we do not use clean human data and only use AI-generated data to train new AI models, the models will become worse and worse and finally collapse.

Diagnostic Autopsy: 虽然有如果……就会……的意思，但表达过于口语化（worse and worse），缺少“尾部稀缺分布丢失 → 方差坍缩”的机制链条。

[Publication-Grade Model Paragraph]:

Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models. Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations. In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse. This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [正面基准命题 Positive Baseline]	*Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models.* **Constituent Analysis**: 对应步骤：S1 [正面基准命题 Positive Baseline]：Maintaining rigorous [Principle/Standard X] serves as the primary safeguard against [Systemic Risk Y]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [反事实假设切入 Counterfactual Premise]	*Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations.* **Constituent Analysis**: 对应步骤：S2 [反事实假设切入 Counterfactual Premise]：Absent this safeguard—if [Actors] were to rely exclusively on [Unchecked Alternative]—[Immediate Vulnerability] would quickly emerge. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [连锁恶化传导 Cascading Deterioration]	*In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse.* **Constituent Analysis**: 对应步骤：S3 [连锁恶化传导 Cascading Deterioration]：Once [Immediate Vulnerability] takes hold, [Secondary Consequence] compounds across cycles, eventually eroding [Core Foundation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [实证印证或结论锁定 Empirical Echo]	*This degenerative feedback loop —known empirically as model autophagy — underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.* **Constituent Analysis**: 对应步骤：S4 [实证印证或结论锁定 Empirical Echo]：This degradation dynamic mirrors [Real-world/Technical Precedent], underscoring why [Principle X] cannot be compromised for short-term convenience. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

正面立论 → Absent this human data anchor (反事实切入) → 尾部稀疏事件被截断导致分布向高密度模式坍缩（机理传导）→ 点名模型自噬（model autophagy）实证结论。

15.0.8 BP40: 议论文收尾段：跳出简单重复，完成核心权衡整合与边界升华（Synthesis Conclusion）

Communicative Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Applicable Context: CET-6 / IELTS Task 2 / TOEFL 结尾段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward.

S2. S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk].

S3. S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed.

Logic Chaining & Cohesion Formula:

Balanced Restatment → Dual-Pillar Synthesis → Grounded Governance Principle (Zero Slogans!). 绑定逻辑规则：L023, L029.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

In conclusion, AI has both advantages and disadvantages. We should make full use of its advantages and avoid its disadvantages. Let us work hand in hand to welcome a bright future of AI!

Diagnostic Autopsy: 结尾段最忌讳的三句废话全中：既有好处也有坏处 + 扬长避短 + 让我们携手共创美好未来（口号感叹句）。

[Publication-Grade Model Paragraph]:

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [重申核心权衡与立场 Restate Nuanced Thesis]	*In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.* **Constituent Analysis**: 对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [整合全文两大支柱机制 Synthesize Dual Pillars]	*By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.* **Constituent Analysis**: 对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [理性边界展望（可选）Grounded Forward Look]	*Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.* **Constituent Analysis**: 对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

第一句让步重申立场，第二句合并“透明审计轨迹 + 靶向技能培训”两大抓手，第三句用 not on replacing..., but on elevating... 理性收尾，格调高雅。

Chapter 16

Track 6: Conclusion, Impact Articulation & High-Stakes Genres (BP41–BP48) / 学术演讲、顶会 Rebuttal 与个人陈述 SoP

本章系统拆解该赛道的 8 大核心段落构建蓝图。每一项蓝图均涵盖明确的用户交际意图、适用场景、逐句动作规范（Move Structure）、句际信息咬合公式、病段与发表级定稿对照、成分剖析表格以及跨学科/文体迁移指南。

16.0.1 BP41: 学术演讲开场 60 秒 Hook：用直观痛点或反直觉现象抓住全场（Oral Opening Hook）

Communicative Intent: 我这一段想作为顶会 Oral Presentation、组会汇报或博士面试 PPT 第一页的开场白：在 45–60 秒内不用任何晦涩公式，让全场听众立刻听懂“我们为什么要做这件事、现有方法到底卡在哪”。

Applicable Context: Tech Talk / Conference Oral / 组会开场逐字稿（Slide 1–2）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [场景直觉切入 Concrete Scenario Hook]：Good morning everyone. Imagine deploying a humanoid robot to assemble flexible cables on a moving factory line—or forecasting grid load when a storm hits.

S2. S2 [反直觉痛点设问 Counter-intuitive Breakdown]：Standard deep models perform remarkably well when conditions stay calm, yet why do they fail abruptly the moment contact dynamics or weather patterns shift?

S3. S3 [一针见血揭晓答案 Crisp Root Cause]：The culprit is a hidden stationarity assumption baked into standard attention layers.

S4. S4 [亮出今日主旨 Talk Roadmap]：In this talk, I will show how our framework, [Model Name], breaks this bottleneck and achieves [Key Metric Gain] in real-time deployment.

Logic Chaining & Cohesion Formula:

Concrete Visual Scenario → Rhetorical Question → Crisp Culprit Identification → Talk Promise. 绑定逻辑规则：L018, L029.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Hello everyone, my name is XXX. Today the title of my presentation is 'Adaptive De-stationary Spectral-Spatial Transformer for Multivariate Time Series Forecasting'. First I will talk about background, second methodology, third experiments, and finally conclusion.

Diagnostic Autopsy: 最催眠的开场方式：念一遍长标题 + 报一遍八股目录（First background, second methodology...），台下听众前 30 秒就开始看手机。

[Publication-Grade Model Paragraph]:

Good morning everyone. Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines. Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges needed to predict peak loads. In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [场景直觉切入 Concrete Scenario Hook]	*Good morning everyone.* **Constituent Analysis**: 对应步骤：S1 [场景直觉切入 Concrete Scenario Hook]: Good morning everyone. Imagine deploying a humanoid robot to assemble flexible cables on a moving factory line—or forecasting grid load when a storm hits. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [反直觉痛点设问 Counter-intuitive Breakdown]	*Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines.* **Constituent Analysis**: 对应步骤：S2 [反直觉痛点设问 Counter-intuitive Breakdown]: Standard deep models perform remarkably well when conditions stay calm, yet why do they fail abruptly the moment contact dynamics or weather patterns shift? | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [一针见血揭晓答案 Crisp Root Cause]	*Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges needed to predict peak loads.* **Constituent Analysis**: 对应步骤：S3 [一针见血揭晓答案 Crisp Root Cause]: The culprit is a hidden stationarity assumption baked into standard attention layers. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [亮出今日主旨 Talk Roadmap]	*In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.* **Constituent Analysis**: 对应步骤：S4 [亮出今日主旨 Talk Roadmap]: In this talk, I will show how our framework, [Model Name], breaks this bottleneck and achieves [Key Metric Gain] in real-time deployment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

用一个风暴突袭风电场的具象画面 + 一个直击灵魂的设问（why do their predictions lag right when the storm hits?）瞬间抓住全场注意力。

16.0.2 BP42: 演讲幻灯片与复杂图表视觉引导段（Visual Signposting in Tech Talks）

Communicative Intent: 我这一段想在演讲到架构图或核心实验曲线（Slide 4 / Slide 7）时，用最自然的口语引导听众的眼睛先看哪、再看哪，瞬间读懂图表背后的胜出逻辑。

Applicable Context: 学术演讲图表/架构讲解逐字稿（Visual Walkthrough）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [视觉坐标锁定 Spatial Pointer]：Let's look at the comparison in the right panel of Slide 5, where the horizontal axis tracks prediction horizon and the vertical axis shows MSE.

S2. S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]：Notice the dashed orange curve representing the standard Transformer baseline: past 336 steps, error shoots up almost vertically.

S3. S3 [引导观察我方稳定曲线 Guide to Our Curve]：Now compare that with our solid blue curve at the bottom, which stays flat even out to 720 steps.

S4. S4 [一句话点破视觉差距原因 One-Sentence Takeaway]：That gap comes directly from our neural-ODE scale tracker preventing long-range drift.

Logic Chaining & Cohesion Formula:

Spatial Coordinate → Notice Baseline Failure → Compare Our Curve → One-Sentence Mechanistic Takeaway.

绑定逻辑规则：L018, L015.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Here is Figure 3. You can see many lines here. Our model is the blue line and it is the lowest, which means our model is the best on all horizons.

Diagnostic Autopsy: 没有给听众任何坐标引导（You can see many lines here），听众还在找坐标轴是什么意思时你已经翻页了。

[Publication-Grade Model Paragraph]:

Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours. Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours. Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours. That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [视觉坐标锁定 Spatial Pointer]	*Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours.* **Constituent Analysis**: 对应步骤：S1 [视觉坐标锁定 Spatial Pointer]：Let's look at the comparison in the right panel of Slide 5, where the horizontal axis tracks prediction horizon and the vertical axis shows MSE. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]	*Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours.* **Constituent Analysis**: 对应步骤：S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]：Notice the dashed orange curve representing the standard Transformer baseline: past 336 steps, error shoots up almost vertically. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [引导观察我方稳定曲线 Guide to Our Curve]	*Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours.* **Constituent Analysis**: 对应步骤：S3 [引导观察我方稳定曲线 Guide to Our Curve]：Now compare that with our solid blue curve at the bottom, which stays flat even out to 720 steps. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [一句话点破视觉差距原因 One-Sentence Takeaway]	*That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.* **Constituent Analysis**: 对应步骤：S4 [一句话点破视觉差距原因 One-Sentence Takeaway]：That gap comes directly from our neural-ODE scale tracker preventing long-range drift. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

像激光笔一样精准引导听众视线：先定坐标轴 → 再看橙色虚线在 336 小时后飙升 → 再对比底部蓝色实线的平稳 → 最后点出右侧 17% 差距的来源。

16.0.3 BP43: 演讲中化繁为简讲解核心公式与算法直觉（Explaining Math Orally）

Communicative Intent: 我这一段想在 PPT 上出现核心损失函数或注意力公式时，用 4 句通俗、精准的口语把公式背后的物理/几何直觉讲透，而不是在台上干巴巴念希腊字母。

Applicable Context: 学术演讲方法详解页（Explaining Equations on Slides）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [先讲直觉目标 Intuition First]：Before looking at the math at the bottom of the slide, here is the simple physical intuition we want to capture.

S2. S2 [日常/物理类比 Physical Analogy]：When two sensors monitor the same mechanical shaft, genuine vibrations share a consistent phase lag, whereas random electrical glitches fire out of phase.

S3. S3 [指向公式核心项 Point to Key Term]：That is exactly what the red penalty term $\|\Delta\Phi\|_1$ inside the softmax does: it measures phase mismatch between channels.

S4. S4 [一句话总结效果 Operational Effect]：Whenever two channels fall out of phase, this term suppresses their attention weight automatically.

Logic Chaining & Cohesion Formula:

Intuition Before Symbols → Physical Reality → Pointer to Highlighted Math Term → Operational Effect. 绑定逻辑规则：L014, L018.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

The formula on this slide is Q times K transpose divided by square root of d minus lambda times delta phi L1 norm, and then softmax times V.

Diagnostic Autopsy: 在台上把公式符号逐字念一遍（Q times K transpose divided by...）是学术演讲的大忌，听众听得云里雾里。

[Publication-Grade Model Paragraph]:

Before looking at the equation in the middle box, here is the physical intuition behind it. When two sensors sit on the same turbine shaft, genuine mechanical vibrations travel with a consistent phase lag, whereas random sensor glitches fire completely out of phase. That physical rule is captured by the highlighted red term, $\|\Delta\Phi_{Q,K}\|_1$, right inside the softmax. Whenever two channels show inconsistent phase lag, this penalty automatically zeroes out their attention weight.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [先讲直觉目标 Intuition First]	*Before looking at the equation in the middle box, here is the physical intuition behind it.* **Constituent Analysis**: 对应步骤：S1 [先讲直觉目标 Intuition First]：Before looking at the math at the bottom of the slide, here is the simple physical intuition we want to capture. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [日常/物理类比 Physical Analogy]	*When two sensors sit on the same turbine shaft, genuine mechanical vibrations travel with a consistent phase lag, whereas random sensor glitches fire completely out of phase.* **Constituent Analysis**: 对应步骤：S2 [日常/物理类比 Physical Analogy]：When two sensors monitor the same mechanical shaft, genuine vibrations share a consistent phase lag, whereas random electrical glitches fire out of phase. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [指向公式核心项 Point to Key Term]	*That physical rule is captured by the highlighted red term, $\|\Delta\Phi_{Q,K}\|_1$, right inside the softmax.* **Constituent Analysis**: 对应步骤：S3 [指向公式核心项 Point to Key Term]：That is exactly what the red penalty term $\|\Delta\Phi\|_1$ inside the softmax does: it measures phase mismatch between channels. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [一句话总结效果 Operational Effect]	*Whenever two channels show inconsistent phase lag, this penalty automatically zeroes out their attention weight.* **Constituent Analysis**: 对应步骤：S4 [一句话总结效果 Operational Effect]：Whenever two channels fall out of phase, this term suppresses their attention weight automatically. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

先讲涡轮机轴承上传感器真实振动与随机电噪声的相位差直觉，再用手指着标红的 $\|\Delta\Phi_{Q,K}\|_1$ 项一句话点破其作用，台下外行内行都能瞬间秒懂。

16.0.4 BP44: 演讲总结页（Take-home Message）与问答（Q&A）从容应答段

Communicative Intent: 我这一段想用于演讲最后一页总结（Take-home Messages）以及回答台下教授/审稿人犀利提问时的标准应答模板。

Applicable Context: 学术演讲收尾页 & Q&A 现场问答

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [凝练核心带走信息 Core Take-Home Message]：To wrap up, if there is one takeaway from our work today, it is that [One-Sentence Core Principle].

S2. S2 [成果三点速览 Rapid Recap]：By combining [A] and [B], [Model] delivers [Metric 1], [Metric 2], and [Open-Source/Practical Asset].

S3. S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]：(For Q&A) That is a great question regarding [Topic]—the short answer is [Direct Conclusion].

S4. S4 [Q&A 展开与诚实边界 Q&A Elaboration]：In our current experiments, we observed [Empirical Fact], and extending this to [Questioner's Scenario] can be handled by [Concrete Mechanism].

Logic Chaining & Cohesion Formula:

Single Takeaway → Quantitative Triad → Q&A Direct Answer → Empirical Grounding. 绑定逻辑规则：L018, L019.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

That's all for my presentation. Thank you for listening. Do you have any questions? (When asked a hard question): Well, maybe we didn't test that, I think maybe it can work in the future.

Diagnostic Autopsy: 收尾平淡（That's all），回答问题时犹豫闪烁（Well, maybe... I think maybe...），显得对自己的工作极不自信。

[Publication-Grade Model Paragraph]:

To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics. Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen. — [Q&A Response]: Thank you for that sharp question on irregular missing timestamps. The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [凝练核心带走信息 Core Take-Home Message]	*To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics.* **Constituent Analysis**: 对应步骤：S1 [凝练核心带走信息 Core Take-Home Message]：To wrap up, if there is one takeaway from our work today, it is that [One-Sentence Core Principle]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [成果三点速览 Rapid Recap]	*Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen.* **Constituent Analysis**: 对应步骤：S2 [成果三点速览 Rapid Recap]：By combining [A] and [B], [Model] delivers [Metric 1], [Metric 2], and [Open-Source/Practical Asset]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]	*Thank you for that sharp question on irregular missing timestamps.* **Constituent Analysis**: 对应步骤：S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]：(For Q&A) That is a great question regarding [Topic]—the short answer is [Direct Conclusion]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [Q&A 展开与诚实边界 Q&A Elaboration]	*The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.* **Constituent Analysis**: 对应步骤: S4 [Q&A 展开与诚实边界 Q&A Elaboration]: In our current experiments, we observed [Empirical Fact], and extending this to [Questioner's Scenario] can be handled by [Concrete Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

总结页用一句话提炼灵魂 takeaway；问答环节用 The short answer is... 结论前置 + 备份页实测数据（backup slide）从容控场。

16.0.5 BP45: 顶会 Rebuttal 全局总括与新增实验汇总段（Author Rebuttal General Response）

Communicative Intent: 我这一段想写 ICLR / NeurIPS / ICML / CVPR 审稿回复（Rebuttal）最顶部的致全体审稿人总括段（Global Response）：在 150 词内用最提气的语气感谢审稿人肯定，并一眼亮出我们在 Rebuttal 期间赶做出的所有硬核新增实验。

Applicable Context: 顶会 Author Rebuttal 置顶总括段（Official Global Response）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [致谢并提炼审稿人共同好评 Gratitude & Shared Praise]: We thank all reviewers for their constructive feedback and for recognizing our work's [Quote Reviewer 1 Praise] (R1), [Quote Reviewer 2 Praise] (R2), and [Quote Reviewer 3 Praise] (R3).

S2. S2 [概览新增实证工作 Overview of New Experiments]: In response to your valuable suggestions, we have conducted [N] sets of new experiments and theoretical analyses, summarized below and highlighted in blue in the revised PDF:

S3. S3 [分条列出硬核新增实验 Bullet Summary of Additions]: (1) Added comparisons against recent baselines [A, B] on [Dataset] (Table R1, addressing R1, R3); (2) Evaluated real-world robustness under [Harsh Condition] (Appendix D, R2); (3) Provided wall-clock latency and memory profiling across sequence lengths up to $L = 4,096$ (Appendix E, R3).

S4. S4 [诚挚收束 Closing Courtesies]: We address each specific question in the individual responses below and welcome any further discussion.

Logic Chaining & Cohesion Formula:

Synthesize Reviewer Praise → Announce Concrete Revisions → Numbered Evidence Bullets mapped to Reviewers → Courteous Hook. 绑定逻辑规则: L019, L010.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Thank you to all reviewers for your time and comments. We have carefully read your reviews and answered all your questions below. We hope you can raise your scores.

Diagnostic Autopsy: 空洞客套且直接讨要分数（hope you can raise your scores），没有在第一眼展示任何新增实证工作量。

[Publication-Grade Model Paragraph]:

We thank all reviewers for their constructive feedback and for recognizing our framework's 'principled formulation of the stationarity-amplitude trade-off' (R1, R3), 'strong empirical gains across eight benchmarks' (R2), and 'clear ablation design' (R4). To address your valuable suggestions, we conducted four sets of new experiments during the rebuttal period (incorporated in blue in the revised manuscript): (1) Benchmarked against two latest 2025 foundation models, Chronos and MOIRAI, across all horizons (Table R1, addressing R1, R3); (2) Added real-robot grasping evaluations under dynamic lighting and 25% camera occlusion (Section 4.5, R2); and (3) Included full GPU memory and wall-clock latency scaling curves up to $L = 4,096$ (Appendix E, R4). Below we address each reviewer's specific questions in detail.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [致谢并提炼审稿人共同好评 Gratitude & Shared Praise]	*We thank all reviewers for their constructive feedback and for recognizing our framework's 'principled formulation of the stationarity-amplitude trade-off' (R1, R3), 'strong empirical gains across eight benchmarks' (R2), and 'clear ablation design' (R4).* **Constituent Analysis**: 对应步骤：S1 [致谢并提炼审稿人共同好评 Gratitude & Shared Praise]：We thank all reviewers for their constructive feedback and for recognizing our work's [Quote Reviewer 1 Praise] (R1), [Quote Reviewer 2 Praise] (R2), and [Quote Reviewer 3 Praise] (R3). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [概览新增实证工作 Overview of New Experiments]	*To address your valuable suggestions, we conducted four sets of new experiments during the rebuttal period (incorporated in blue in the revised manuscript): (1) Benchmarked against two latest 2025 foundation models, Chronos and MOIRAI, across all horizons (Table R1, addressing R1, R3); (2) Added real-robot grasping evaluations under dynamic lighting and 25% camera occlusion (Section 4.5, R2); and (3) Included full GPU memory and wall-clock latency scaling curves up to $L = 4,096$ (Appendix E, R4).* **Constituent Analysis**: 对应步骤：S2 [概览新增实证工作 Overview of New Experiments]：In response to your valuable suggestions, we have conducted [N] sets of new experiments and theoretical analyses, summarized below and highlighted in blue in the revised PDF: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [分条列出硬核新增实验 Bullet Summary of Additions]	*Below we address each reviewer's specific questions in detail.* **Constituent Analysis**: 对应步骤：S3 [分条列出硬核新增实验 Bullet Summary of Additions]：(1) Added comparisons against recent baselines [A, B] on [Dataset] (Table R1, addressing R1, R3); (2) Evaluated real-world robustness under [Harsh Condition] (Appendix D, R2); (3) Provided wall-clock latency and memory profiling across sequence lengths up to $L = 4,096$ (Appendix E, R3). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

把审稿人的原话好评巧妙嵌入首句（给 AC 留下极佳第一印象），紧接着用编号清单亮出 Rebuttal 期间实打实补做的 3 组硬核实验，翻盘率拉满。

16.0.6 BP46: 顶会 Rebuttal 针对“创新性存疑（Limited Novelty / Incremental）”的有力澄清段

Communicative Intent: 我这一段想回复审稿人最致命的质疑：'The method seems like a simple combination of existing Module A and Module B (limited novelty).' 既不顶撞审稿人，又能用理论 + 消融数据证明“绝对不是简单拼凑”。

Applicable Context: 顶会 Rebuttal 核心争议攻防段（Novelty Defense）

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [礼貌接招并定位误解核心 Courteous Reframing]：We appreciate Reviewer X's comment on architectural design and would like to clarify the fundamental distinction between our coupled formulation and a naive combination of [A] and [B].

S2. S2 [点破简单拼接为何会失败 Why Naive Combination Fails]：Directly stacking standard [A] and [B] fails on [Problem] because [Theoretical Conflict / Gradient Interference].

S3. S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]：Our core contribution lies not in using [A] or [B] in isolation, but in deriving the [Specific Coupling Mechanism C], which resolves [Conflict] by [Mathematical Guarantee].

S4. S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline ('Naive A+B') in Table R2: our coupled formulation outperforms 'Naive A+B' by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking.

Logic Chaining & Cohesion Formula:

Clarify Distinction → Explain Why Naive Stacking Fails → Highlight Core Coupling Mechanism → Prove with 'Naive A+B' Ablation Data. 绑定逻辑规则：L019, L016.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

We disagree that our paper lacks novelty. Although A and B existed before, nobody used them together on time series forecasting. Also our results are very good, so our paper is novel.

Diagnostic Autopsy: 承认了"只是把现成的 A 和 B 第一次拿到这个任务上拼起来（nobody used them together）"，等于坐实了审稿人的 A+B 拼凑指控！

[Publication-Grade Model Paragraph]:

We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs. Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory. Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq. 7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention. To verify this empirically, we implemented a direct 'Naive FFT + ODE' concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over 'Naive FFT + ODE', confirming that the gain stems from our mathematical coupling rather than module stacking.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [礼貌接招并定位误解核心 Courteous Reframing]	*We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs.* **Constituent Analysis**: 对应步骤：S1 [礼貌接招并定位误解核心 Courteous Reframing]：We appreciate Reviewer X's comment on architectural design and would like to clarify the fundamental distinction between our coupled formulation and a naive combination of [A] and [B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [点破简单拼接为何会失败 Why Naive Combination Fails]	*Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory.* **Constituent Analysis**: 对应步骤：S2 [点破简单拼接为何会失败 Why Naive Combination Fails]：Directly stacking standard [A] and [B] fails on [Problem] because [Theoretical Conflict / Gradient Interference]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]	*Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq.* **Constituent Analysis**: 对应步骤：S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]：Our core contribution lies not in using [A] or [B] in isolation, but in deriving the [Specific Coupling Mechanism C], which resolves [Conflict] by [Mathematical Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]	*7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention.* **Constituent Analysis**: 对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline ('Naive A+B') in Table R2: our coupled formulation outperforms 'Naive A+B' by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]	*To verify this empirically, we implemented a direct 'Naive FFT + ODE' concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over 'Naive FFT + ODE', confirming that the gain stems from our mathematical coupling rather than module stacking.* **Constituent Analysis**: 对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline ('Naive A+B') in Table R2: our coupled formulation outperforms 'Naive A+B' by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

教科书级的 Novelty 防御战：先讲直接拼凑 A+B 为什么在数学上会冲突，再讲我们的双向耦合算子如何破局，最后直接掏出 'Naive A+B' 对比实验（赢 15.2%）一剑封喉。

16.0.7 BP47: 顶会 Rebuttal 针对“缺少最新基线或算力不公平（Missing Baselines / Fairness）”的补全段

Communicative Intent: 我这一段想回复审稿人要求补测最新基线模型，或者质疑我们的提升是不是因为参数量更大/训练更久。

Applicable Context: 顶会 Rebuttal 实验补全与公平性自证段

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

S1. S1 [直接响应并呈上新表 Direct Fulfilment]：Following your suggestion, we have evaluated [Requested Baselines X and Y] under identical splits and added a parameter-matched variant of our model in Table R3.

S2. S2 [对比最新基线结果 Performance vs. New Baselines]：Across all four horizons on [Datasets], our method outperforms [New Baseline X] by an average of [9.8%] in MSE and [New Baseline Y] by [12.1%].

S3. S3 [打消参数量/算力质疑 Parameter & FLOPs Parity Proof]：Regarding model capacity, even when we scale down our hidden dimension from $d = 256$ to $d = 128$ ('Ours-Small', using 38% fewer parameters and 52% fewer FLOPs than [Baseline X]), 'Ours-Small' still outperforms [Baseline X] by [7.4%].

S4. S4 [结论锁定 Conclusion]：These results confirm that our improvements derive from inductive bias rather than increased parameter count.

Logic Chaining & Cohesion Formula:

Direct Fulfilment → Margin over New Baselines → Parameter-Reduced Variant ('Ours-Small') Proof → Inductive Bias Takeaway. 绑定逻辑规则：L019, L010.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

We added the baseline X you asked for. Our model is 0.182 and X is 0.205, so we are still better. Also our model does not have many parameters.

Diagnostic Autopsy: 过于单薄，特别是回应参数量质疑时只口头说 does not have many parameters，远不如构造一个缩小版 'Ours-Small' 依然吊打对手有说服力。

[Publication-Grade Model Paragraph]:

Following your suggestion, we benchmarked against the two recent baselines (TimeMixer and FreTS) and evaluated a capacity-reduced variant of our architecture ('DyTS-Lite') in Table R3. Across all horizons on Weather and ECL, standard DyTS-Net reduces MSE by 9.6% over TimeMixer and 13.4% over FreTS. Furthermore, to ensure strict capacity fairness, 'DyTS-Lite' shrinks the hidden dimension to $d = 64$—requiring 42% fewer parameters and 55% fewer FLOPs than TimeMixer—yet still outperforms TimeMixer by 6.8% in average MSE. This confirms that our gains stem from structural-scale disentanglement rather than model size.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [直接响应并呈上新表 Direct Fulfilment]	*Following your suggestion, we benchmarked against the two recent baselines (TimeMixer and FreTS) and evaluated a capacity-reduced variant of our architecture ('DyTS-Lite') in Table R3.* **Constituent Analysis**: 对应步骤：S1 [直接响应并呈上新表 Direct Fulfilment]：Following your suggestion, we have evaluated [Requested Baselines X and Y] under identical splits and added a parameter-matched variant of our model in Table R3. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [对比最新基线结果 Performance vs. New Baselines]	*Across all horizons on Weather and ECL, standard DyTS-Net reduces MSE by 9.6% over TimeMixer and 13.4% over FreTS.* **Constituent Analysis**: 对应步骤：S2 [对比最新基线结果 Performance vs. New Baselines]：Across all four horizons on [Datasets], our method outperforms [New Baseline X] by an average of [9.8%] in MSE and [New Baseline Y] by [12.1%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [打消参数量/算力质疑 Parameter & FLOPs Parity Proof]	*Furthermore, to ensure strict capacity fairness, 'DyTS-Lite' shrinks the hidden dimension to $d = 64$—requiring 42% fewer parameters and 55% fewer FLOPs than TimeMixer—yet still outperforms TimeMixer by 6.8% in average MSE.* **Constituent Analysis**: 对应步骤：S3 [打消参数量/算力质疑 Parameter & FLOPs Parity Proof]：Regarding model capacity, even when we scale down our hidden dimension from $d = 256$ to $d = 128$ ('Ours-Small', using 38% fewer parameters and 52% fewer FLOPs than [Baseline X]), 'Ours-Small' still outperforms [Baseline X] by [7.4%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [结论锁定 Conclusion]	*This confirms that our gains stem from structural-scale disentanglement rather than model size.* **Constituent Analysis**: 对应步骤：S4 [结论锁定 Conclusion]：These results confirm that our improvements derive from inductive bias rather than increased parameter count. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

既补齐了审稿人点名的最新基线，又主动用小号模型（少 42% 参数、少 55% 计算量却依然反超 6.8%）彻底封死“靠堆参数取胜”的质疑。

16.0.8 BP48: 科研个人陈述（SoP / Research Narrative）：从时间序列/生成建模迈向具身智能的闭环叙事段

Communicative Intent: 我这一段想写申请博士/博后/访问学者或个人学术主页的 Research Statement / SoP 核心段：把我已经做过的深度学习/时间序列/生成模型工作，与我未来想探索的具身智能（Embodied AI）无缝串联成一条充满内驱力的科研主线。

Applicable Context: Statement of Purpose (SoP) / Research Statement / 学术自我介绍

Move Structure & Cohesion Logic Engine

Sentence-by-Sentence Move Blueprint:

- **S1.** S1 [过往研究主线凝练 Past Research Thread]：My prior research has centered on modeling non-stationary dynamics and uncertainty in high-dimensional sequential data.
- **S2.** S2 [代表作攻克过程与硬核成果 Concrete Achievement]：In my recent work on [Time Series / Generative Modeling], I observed that [Core Bottleneck], which led me to formulate [Method]—reducing long-horizon drift by [X%] across industrial telemetry benchmarks.
- **S3.** S3 [向具身智能跃迁的科学顿悟 Scientific Pivot to Embodied AI]：While passive sequence forecasting predicts how an environment evolves, physical decision-making requires closing the loop: actions actively alter future state distributions.
- **S4.** S4 [未来研究愿景 Future Research Agenda]：Building on my foundation in continuous-time generative modeling, my doctoral/future research aims to develop sample-efficient visuomotor world models that enable robots to adapt online to non-stationary contact dynamics.

Logic Chaining & Cohesion Formula:

Past Core Thread → Concrete Method & Metric → Intellectual Bridge (Passive Forecasting → Active Closed-Loop Control) → Future Embodied AI Agenda. 绑定逻辑规则：L003, L025, L029.

Defective Draft vs. Zero AI-ese Model Paragraph

[Defective Draft]:

Since I was a child, I have been fascinated by artificial intelligence and robots. In my master's degree, I did some research on time series forecasting and published a paper. In the future, I want to study embodied AI because it is a very hot topic and can help humans a lot.

Diagnostic Autopsy: 包含 SoP 两大禁忌：(1) 童年情怀套话（Since I was a child, I have been fascinated by...）；(2) 盲目追热点（because it is a very hot topic），没讲清你过去的时序/生成建模背景赋予了你做具身智能什么独特优势。

[Publication-Grade Model Paragraph]:

My research focuses on learning reliable representations and generative trajectories from non-stationary sequential data. While investigating multivariate industrial telemetry, I found that conventional attention mechanisms collapse when underlying system dynamics shift—motivating me to design a neural-ODE coupled spectral forecaster that reduced long-range drift by 17.4%. Yet passive sequence forecasting only observes how a system evolves; intelligent physical agents must actively intervene within that loop, where every motor command alters future state distributions. Bridging my background in non-stationary time-series modeling and fast flow-matching samplers, my future research aims to build adaptive visuomotor world models that allow embodied robots to generalize across unfamiliar contact regimes in real time.

Sentence-by-Sentence Microscopic Breakdown

Move & Sentence	Authentic Text & Constituent / Flow Annotation
S1 [过往研究主线凝练 Past Research Thread]	*My research focuses on learning reliable representations and generative trajectories from non-stationary sequential data.* **Constituent Analysis**: 对应步骤：S1 [过往研究主线凝练 Past Research Thread]: My prior research has centered on modeling non-stationary dynamics and uncertainty in high-dimensional sequential data. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S2 [代表作攻克过程与硬核成果 Concrete Achievement]	*While investigating multivariate industrial telemetry, I found that conventional attention mechanisms collapse when underlying system dynamics shift—motivating me to design a neural-ODE coupled spectral forecaster that reduced long-range drift by 17.4%.* **Constituent Analysis**: 对应步骤：S2 [代表作攻克过程与硬核成果 Concrete Achievement]: In my recent work on [Time Series / Generative Modeling], I observed that [Core Bottleneck], which led me to formulate [Method]—reducing long-horizon drift by [X%] across industrial telemetry benchmarks. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S3 [向具身智能跃迁的科学顿悟 Scientific Pivot to Embodied AI]	*Yet passive sequence forecasting only observes how a system evolves; intelligent physical agents must actively intervene within that loop, where every motor command alters future state distributions.* **Constituent Analysis**: 对应步骤：S3 [向具身智能跃迁的科学顿悟 Scientific Pivot to Embodied AI]: While passive sequence forecasting predicts how an environment evolves, physical decision-making requires closing the loop: actions actively alter future state distributions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
S4 [未来研究愿景 Future Research Agenda]	*Bridging my background in non-stationary time-series modeling and fast flow-matching samplers, my future research aims to build adaptive visuomotor world models that allow embodied robots to generalize across unfamiliar contact regimes in real time.* **Constituent Analysis**: 对应步骤：S4 [未来研究愿景 Future Research Agenda]: Building on my foundation in continuous-time generative modeling, my doctoral/future research aims to develop sample-efficient visuomotor world models that enable robots to adapt online to non-stationary contact dynamics. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Cross-Genre Adaptation & Field Guidance

没有一句童年套话！将“过去的非平稳时间序列与快速流匹配生成模型”通过“从被动观测序列演进到主动闭环物理交互”的深刻科学洞察，自然推演出“构建自适应具身智能世界模型”的未来蓝图。

Part V

Diagnostic Revision Clinic & 60 Surgical Cases

深度病段诊所与母语级重构图谱

Chapter 17

Diagnostic Reconstructive Clinic

学术论文写作的本质是“思维的外显化”与“信息的精细化编码”。在初学者与跨语种学者的初稿中，往往充斥着结构松散、逻辑跳跃、套话连篇等典型病态段落。本章建立系统化的学术语言病理诊断与重构诊所，涵盖 5 大核心病灶分类、60 例跨工科、理科、医科、经管与社科的真实临床案例（RC01–RC60），贯彻“七维病灶透视”与“四步重构手术”法，帮助读者形成敏锐的语病自检与母语级升华能力。

The Five-Tier Defect Taxonomy / 五大核心语病分类

Category I: **头重脚轻长主语 + 散装短句堆砌（Top-Heavy Subject & Choppy Flow）**（Cases RC01–RC12）：主语包含冗长名词短语或从句，谓语动词极度延后，句法失去平衡；随后又退化为初中水平散碎短句，缺乏严谨从属复合结构。

Category II: **口语化严重 + 缺失比较基准与维度（Colloquialism & Imprecision）**（Cases RC13–RC24）：大量使用日常交际口语动词（be, make, get, think, very good, pretty fast），缺乏精准学术 reporting verbs；对比描述中缺少基准线与量化维度。

Category III: **机械连接词膏药 + 旧新信息流断裂与逻辑跳跃（Mechanical Connectors & Logical Leap）**（Cases RC25–RC36）：过度依赖 *First, Second, Moreover, Furthermore* 作为机械粘合剂，句子之间缺乏语义钩子，前后两句话题突变，新信息强行插入。

Category IV: **重度 AI 套话 + 空洞夸张无具体指标（Heavy AI-ese & Hollow Hyperbole）**（Cases RC37–RC48）：充斥 *In the rapidly evolving era of..., delve into, novel and groundbreaking, paves the way* 等无意义修辞帽，无任何具体指标、参数与物理场景支撑。

Category V: **过度名词化僵尸句 + 被动语态滥用（Over-Nominalization & Passive Abuse）**（Cases RC49–RC60）：盲目堆叠抽象名词短语（The implementation of the execution of the evaluation was conducted by...），全段连续采用冷冰冰的被动语态，掩盖真正动作行为体。

17.1 Category I: Top-Heavy Subject & Choppy Flow / 头重脚轻与散句堆砌

17.1.1 Case RC01: 非平稳时间序列预测与分布漂移消除 (Non-Stationary Time Series Forecasting)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Linked Architecture: Blueprint `BP01` | Logic Rule `L001`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In the rapidly developing modern world, time series forecasting plays a pivotal role in many fields. The fact that industrial sensors often have noise and distribution changes makes forecasting very hard and bad for decision making.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Constituent Alignment & Rhetorical Moves

- **S1 [任务锚定 Topic Anchor]:** *Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.*
  对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [应用场景展开 Concrete Scenarios]:** *In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.*
  对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [苛刻工况约束 Operational Bottleneck]:** *However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.*
  对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [研究必要性收束 Imperative Hook]:** *Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.*
  对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.2 Case RC02: 非平稳时间序列预测与分布漂移消除 (Non-Stationary Time Series Forecasting)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想客观概括前人主流模型（如 Transformer / Diffusion / MLP）取得了什么进展，但点出它们共同依赖的一个脆弱前提假设。
Linked Architecture: Blueprint `BP02` | Logic Rule `L002`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Many researchers have studied this problem. Wu et al. used Autoformer. Nie et al. used PatchTST. They are very good, but they all think the data distribution does not change, which is wrong in real life.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:
Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition. These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments. Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window. When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.

Constituent Alignment & Rhetorical Moves

- **S1 [流派归类 Taxonomic Opening]:** *Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition.*
  对应步骤：S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表性机制展开 Mechanism Recap]:** *These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments.*
  对应步骤：S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [转折揭示隐含假设 Pivot to Hidden Assumption]:** *Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window.*
  对应步骤：S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [假设破裂后果 Breakdown Under Stress]:** *When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.*
  对应步骤：S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.3 Case RC03: 非平稳时间序列预测与分布漂移消除 (Non-Stationary Time Series Forecasting)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。
Linked Architecture: Blueprint `BP03` | Logic Rule `L003`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
The problem of error accumulation in long-term forecasting and the losing of scale information when using RevIN normalization is very serious and hard to solve by simple ways.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Constituent Alignment & Rhetorical Moves

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*
  对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*
  对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.4 Case RC04: 非平稳时间序列预测与分布漂移消除 (Non-Stationary Time Series Forecasting)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。
Linked Architecture: Blueprint `BP04` | Logic Rule `L004`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
In this paper, we propose a novel and groundbreaking model called DyTS-Net. It has three modules: Module A, Module B, and Module C. They work together to delve into the data and get state-of-the-art performance.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:
To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.

Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Constituent Alignment & Rhetorical Moves

- **S1 [方法登场与定位 Method Introduction]:** *To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.*
  对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心直觉洞察 Key Insight]:** *Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.*
  对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [模块协同主干 Module Synergy]:** *Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.*
  对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [闭环收益总结 Closed-Loop Benefit]:** *Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.*
  对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.5 Case RC05: 非平稳时间序列预测与分布漂移消除 (Non-Stationary Time Series Forecasting)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。
Linked Architecture: Blueprint `BP05` | Logic Rule `L005`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

There are two challenges. The first challenge is that noise is big. We use a filter. The second challenge is that speed is slow. We use distillation.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Constituent Alignment & Rhetorical Moves

- **S1 [落地难点总起 Implementation Hurdle]:** *Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.*
  对应步骤：S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [难点 1 + 靶向设计 1]:** *First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.*
  对应步骤：S2 [难点 1 + 靶向设计 1]: First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [难点 2 + 靶向设计 2]:** *Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.*

对应步骤：S3 [难点 2 + 靶向设计 2]：Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [整体计算/泛化特性 System Property]:** *Together, these designs sustain 50 Hz reactive control under partial visual occlusion.*
  对应步骤：S4 [整体计算/泛化特性 System Property]：Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.6 Case RC06: 非平稳时间序列预测与分布漂移消除 (Non-Stationary Time Series Forecasting)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Introduction 结尾前，用三句话硬核总结我们在哪些基准数据集/真实硬件上做了实验，取得了多大幅度的领先。

Linked Architecture: Blueprint `BP06` | Logic Rule `L006`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

We did comprehensive experiments on many datasets. The results show that our model beats all baselines by a large margin and is very fast.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors. Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions. Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.

Constituent Alignment & Rhetorical Moves

- **S1 [评测广度 Scope of Evaluation]:** *We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors.*
  对应步骤：S1 [评测广度 Scope of Evaluation]：We evaluate [Model] across [N] public benchmarks and [Real-world Hardware/Scenario], spanning [Tasks]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心精度突破 Primary Accuracy Gain]:** *Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions.*
  对应步骤：S2 [核心精度突破 Primary Accuracy Gain]：实验主指标对比（必须包含具体百分比、最强基线名称及困难场景表现）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]:** *Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.*
  对应步骤：S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]：报告计算速度、显存占用或零样本跨场景泛化收益。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.7 Case RC07: 基于扩散/流匹配的缺失时序插补与生成 (Diffusion & Flow-Matching Time Series Imputation)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Introduction 最末尾清晰列出本文的三点核心贡献（理论/视角贡献 + 核心算法/架构贡献 + 实验与开源实证贡献）。
Linked Architecture: Blueprint `BP07` | Logic Rule `L007`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Our contributions are: 1. A new model is proposed. 2. Solving the non-stationary problem. 3. Good results on datasets.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability. (2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker. (3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.

Constituent Alignment & Rhetorical Moves

- **S1 [总起引导句 Lead-in]:** *In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability.*
  对应步骤：S1 [总起引导句 Lead-in]：In summary, our main contributions are threefold: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [贡献 1:** *(2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker.*
  对应步骤：S2 [贡献 1：新视角/问题形式化 Conceptual/Formulation Contribution]：用强动词名物化或动词原形开头（We identify and formalize... / We establish...），点明打破了什么传统局限。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [贡献 2:** *(3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.*
  对应步骤：S3 [贡献 2：核心方法与模块 Algorithmic/Architectural Contribution]：We propose [Model/Mechanism], which integrates [A] and [B] to achieve [Theoretical/Functional Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.8 Case RC08: 基于扩散/流匹配的缺失时序插补与生成 (Diffusion & Flow-Matching Time Series Imputation)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Linked Architecture: Blueprint `BP08` | Logic Rule `L008`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Time series imputation is very important in the era of big data. Existing generative models delve into the data distribution but are too slow and fail when missing rates are high. We propose FlowImp, a novel framework that uses conditional flow matching and frequency guidance. Extensive experiments show that FlowImp achieves state-of-the-art results on all benchmarks.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Constituent Alignment & Rhetorical Moves

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*

对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.9 Case RC09: 基于扩散/流匹配的缺失时序插补与生成 (Diffusion & Flow-Matching Time Series Imputation)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写 Related Work 的一个子方向：把过去三年的 6–8 篇代表性论文按技术路线分成两类进行归纳，并指出它们的共同盲区。

Linked Architecture: Blueprint `BP09` | Logic Rule `L009`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In recent years, many works studied time series foundation models. TimeGPT trained a transformer on large data. Moment used masked pre-training. Chronos quantized time series into tokens. Lag-Llama used lag features. However, our model is different from them.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization. Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains. Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives. Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.

Constituent Alignment & Rhetorical Moves

- **S1 [分类总纲 Taxonomic Thesis]:** *Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization.*
  对应步骤：S1 [分类总纲 Taxonomic Thesis]：Existing methods for [Task] can be categorized into two main families: [Family A] and [Family B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [流派 A 归纳与代表作 Family A Synthesis]:** *Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains.*
  对应步骤：S2 [流派 A 归纳与代表作 Family A Synthesis]：[Family A] approaches (e.g., [Models]) focus on [Mechanism A], effectively addressing [Sub-problem 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [流派 B 归纳与对比 Family B Synthesis]:** *Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives.*
  对应步骤：S3 [流派 B 归纳与对比 Family B Synthesis]：By contrast, [Family B] architectures (e.g., [Models]) reformulate the problem via [Mechanism B] to reduce [Cost/Complexity]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [共同盲区提炼 Shared Blind Spot]:** *Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.*
  对应步骤：S4 [共同盲区提炼 Shared Blind Spot]：Despite their respective strengths, both families overlook [Critical Factor], which degrades performance under [Target Scenario]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.10 Case RC10: 基于扩散/流匹配的缺失时序插补与生成 (Diffusion & Flow-Matching Time Series Imputation)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Related Work 小节末尾，把我们的方法与最接近的 2 个竞争方案做精准对比，防止审稿人说我们只是对某篇旧文的微调（Incremental）。
Linked Architecture: Blueprint `BP10` | Logic Rule `L010`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Some works also use frequency domain like us, such as FEDformer and TimesNet. But our method is much better and more novel because we use a new module.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach. While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes. In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases. Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.

Constituent Alignment & Rhetorical Moves

- **S1 [锁定最相近工作 Closest Prior Work]:** *Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach.*
  对应步骤：S1 [锁定最相近工作 Closest Prior Work]：Among prior studies, [Closest Method A] and [Closest Method B] are most closely related to our approach. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [剖析其局限性 Structural Difference 1]:** *While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes.*
  对应步骤：S2 [剖析其局限性 Structural Difference 1]：While [Method A] also employs [Shared Concept], it relies on [Limitation 1], restricting its applicability to [Narrow Setting]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [剖析本文本质突破 Core Departure]:** *In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases.*
  对应步骤：S3 [剖析本文本质突破 Core Departure]：In contrast, our framework introduces [Core Innovation], which fundamentally eliminates [Bottleneck] by [Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证或理论优势锁定 Empirical/Theoretical Edge]:** *Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.*
  对应步骤：S4 [实证或理论优势锁定 Empirical/Theoretical Edge]：Consequently, our approach achieves [Capability] without requiring [Expensive Requirement of Prior Work]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.11 Case RC11: 基于扩散/流匹配的缺失时序插补与生成 (Diffusion & Flow-Matching Time Series Imputation)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Methodology 开篇（Section 3.1 Problem Formulation），严谨、简洁地定义输入张量、输出目标、约束条件与优化目标。

Linked Architecture: Blueprint `BP11` | Logic Rule `L011`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In this section, we define the problem. The input is X. X has length L and C channels. The output is Y. Y has length H. We want to make the error between predicted Y and true Y as small as possible.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Consider a multivariate time series $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ recording C correlated sensors over lookback horizon L. Given an binary observation mask $\mathbf{M} \in \{0, 1\}^{L \times C}$ indicating irregular sensor dropouts, our objective is to learn a generative mapping $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ over future horizon H. Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where $\mathbb{E}[x_t]$ and $\mathrm{Cov}(x_t)$ evolve continuously between the lookback and forecasting windows.

Constituent Alignment & Rhetorical Moves

- **S1 [任务总体形式化 Task Definition]:** *Consider a multivariate time series* $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ *recording C correlated sensors over lookback horizon L.*
  对应步骤: S1 [任务总体形式化 Task Definition]: Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [补充条件或模态输入 Auxiliary/Condition Inputs]:** *Given an binary observation mask* $\mathbf{M} \in \{0,1\}^{L \times C}$ *indicating irregular sensor dropouts, our objective is to learn a generative mapping* $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ *over future horizon* H.
  对应步骤：S2 [补充条件或模态输入 Auxiliary/Condition Inputs]：At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [预测/控制目标 Objective]:** *Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where* $\mathbb{E}[x_t]$ *and* $Cov(x_t)$ *evolve continuously between the lookback and forecasting windows.*
  对应步骤：S3 [预测/控制目标 Objective]：Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$ over horizon H (or action trajectory $a_{t:t+K}$). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.1.12 Case RC12: 基于扩散/流匹配的缺失时序插补与生成 (Diffusion & Flow-Matching Time Series Imputation)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想配合论文里的 Figure 2（模型总架构图），用一段话把输入张量从左到右流经各个模块直到输出预测结果的完整全貌讲清楚。
Linked Architecture: Blueprint `BP12` | Logic Rule `L012`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Figure 2 shows our model. There is a Patch Encoder on the left. In the middle, there is a Spectral Block and a Graph Block. On the right, a Linear Decoder outputs the result.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages. Given an input sequence $\mathbf{X}$, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches $\mathbf{Z}_0$. These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph. Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output $\hat{\mathbf{Y}}$.

Constituent Alignment & Rhetorical Moves

- **S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]:** *Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages.*
  对应步骤：S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]：Figure 2 illustrates the overall architecture of [Model], which processes [Input] through three sequential stages: [Stage 1], [Stage 2], and [Stage 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [阶段 1 数据流 Stage 1 Flow]:** *Given an input sequence* $\mathbf{X}$*, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches* $\mathbf{Z}_0$.
  对应步骤：S2 [阶段 1 数据流 Stage 1 Flow]：Given raw input $\mathbf{X}$, the [Module 1] first transforms [Raw Input] into [Intermediate Representation $\mathbf{Z}^{(1)}$] to [Functional Purpose 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [阶段 2 数据流 Stage 2 Flow]:** *These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph.*
  对应步骤：S3 [阶段 2 数据流 Stage 2 Flow]：Next, these [Summary Noun: representations $\mathbf{Z}^{(1)}$] are fed into [Module 2], which [Mechanism 2] to produce [Refined State $\mathbf{Z}^{(2)}$]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]:** *Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output* $\hat{\mathbf{Y}}$.
  对应步骤：S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]：Finally, the [Module 3] integrates $\mathbf{Z}^{(2)}$ with [Prior/Scale Cue] to output the final [Prediction/Action Trajectory]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2 Category II: Colloquialism & Imprecision / 口语化表达与缺乏量化基准

17.2.1 Case RC13: 具身智能视觉运动策略与仿真到真实迁移 (Embodied AI Visuomotor Policy & Sim-to-Real)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想详细介绍我们提出的某个核心创新模块：先用两句话讲清楚为什么直接用传统算子会出问题，再顺理成章推导出我们的新公式。

Linked Architecture: Blueprint `BP13` | Logic Rule `L013`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In this subsection, we introduce the Phase-Aware Attention. The formula is Attention(Q, K, V) = softmax(QK^T / sqrt(d) + M)V. Here M is a phase matrix. It makes the model work better.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps. To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies. Formally, given query and key projections $Q, K \in \mathbb{R}^{N\times d}$, we modulate the attention logits via a phase-alignment mask: $\mathcal{A}(Q,K,V) = \mathrm{softmax}\big(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1\big)V$, where $\Delta\Phi_{Q,K} \in [-\pi,\pi]^{N\times N}$ denotes the instantaneous Hilbert phase difference matrix and $\lambda > 0$ controls phase selectivity. Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.

Constituent Alignment & Rhetorical Moves

- **S1 [传统算子缺陷 Motivation]:** *Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps.*
  对应步骤：S1 [传统算子缺陷 Motivation]：Standard [Conventional Operator, e.g., softmax self-attention] computes [Operation], which suffers from [Pathology] when applied to [Domain Condition]. | 句首锚定

旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [物理/统计直觉 Physical Intuition]:** *To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies.*
  对应步骤：S2 [物理/统计直觉 Physical Intuition]：To overcome this pathology, we observe that [Key Domain Property] can be effectively regularized by [Intuitive Operation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [数学公式定义 Mathematical Formulation]:** *Formally, given query and key projections* $Q, K \in \mathbb{R}^{N\times d}$*, we modulate the attention logits via a phase-alignment mask:* $\mathcal{A}(Q,K,V) = softmax\big(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1\big)V$*, where* $\Delta\Phi_{Q,K} \in [-\pi,\pi]^{N\times N}$ *denotes the instantaneous Hilbert phase difference matrix and* $\lambda > 0$ *controls phase selectivity.*
  对应步骤：S3 [数学公式定义 Mathematical Formulation]：Formally, given input representation $H \in \mathbb{R}^{N\times d}$, we formulate the [Module Name] as: [Equation], where [Symbol Explanations]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [理论/计算性质 Guarantee]:** *Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.*
  对应步骤：S4 [理论/计算性质 Guarantee]：By construction, this formulation guarantees [Desirable Property, e.g., $O(N)$ complexity / translation invariance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.2 Case RC14: 具身智能视觉运动策略与仿真到真实迁移 (Embodied AI Visuomotor Policy & Sim-to-Real)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想专门阐述时间序列模型中“如何在频域捕获全局周期性，同时在时域保留局部突发异常（Transient Anomalies）”的双分支耦合机制。

Linked Architecture: Blueprint `BP14` | Logic Rule `L014`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

We use both FFT and CNN in our model. FFT gets global features and CNN gets local features. Then we add them together to get final features.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Modeling telemetry streams solely in the time domain struggles to span multi-day periodicities, whereas pure Fourier projection spreads localized transient spikes across the entire spectrum. To reconcile both scales, our dual-domain block processes input patches along two complementary pathways. The spectral pathway applies real-valued RFFT to isolate top-k harmonic bases governing macro-periodicity, while the parallel temporal pathway employs dilated depthwise convolutions to retain sharp step changes and localized anomalies. An energy-ratio gate then fuses both representations at each time step, dynamically shifting weight toward the temporal pathway whenever local high-frequency residual energy spikes.

Constituent Alignment & Rhetorical Moves

- **S1 [单域局限性 Single-Domain Limitation]:** *Modeling telemetry streams solely in the time domain struggles to span multi-day periodicities, whereas pure Fourier projection spreads localized transient spikes across the entire spectrum.*
  对应步骤：S1 [单域局限性 Single-Domain Limitation]：Modeling non-stationary time series solely in the time domain struggles to capture long-range periodicities, whereas pure Fourier projection blurs localized transient anomalies. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [频域分支设计 Frequency Branch]:** *To reconcile both scales, our dual-domain block processes input patches along two complementary pathways.*
  对应步骤：S2 [频域分支设计 Frequency Branch]：To capture global seasonal structure, our spectral branch applies real-valued FFT $\mathcal{F}(\cdot)$ along the temporal dimension and reweights dominant harmonic components. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [时域分支设计 Temporal Branch]:** *The spectral pathway applies real-valued RFFT to isolate top-k harmonic bases governing macro-periodicity, while the parallel temporal pathway employs dilated depthwise convolutions to retain sharp step changes and localized anomalies.*
  对应步骤：S3 [时域分支设计 Temporal Branch]：In parallel, a multi-scale depthwise convolution branch operates directly on local patches to preserve sharp step changes and localized spikes. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [自适应门控融合 Adaptive Fusion]:** *An energy-ratio gate then fuses both representations at each time step, dynamically shifting weight toward the temporal pathway whenever local high-frequency residual energy spikes.*
  对应步骤：S4 [自适应门控融合 Adaptive Fusion]：An uncertainty-aware gate then fuses both branches step-by-step, assigning higher weight to the temporal branch whenever local residual variance exceeds a threshold. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.3 Case RC15: 具身智能视觉运动策略与仿真到真实迁移 (Embodied AI Visuomotor Policy & Sim-to-Real)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想阐述生成模型（Diffusion / Flow Matching / Autoregressive）如何在满足外部物理约束或条件观测的前提下，生成高质量且多样化的序列/轨迹。

Linked Architecture: Blueprint `BP15` | Logic Rule `L015`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Normal diffusion guidance makes the generated time series look unnatural when we add constraints. So we put the constraint inside the flow matching model, and it generates realistic samples in only 4 steps.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations. To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field. At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear

constraint operator. This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.

Constituent Alignment & Rhetorical Moves

- **S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]:** *Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations.*
  对应步骤：S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]：Applying post-hoc classifier guidance to pre-trained diffusion samplers often pushes intermediate noisy states off the data manifold, inducing high-frequency artifacts. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]:** *To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field.*
  对应步骤: S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]: To enforce hard boundary constraints without manifold deviation, we embed [Condition/Constraint] directly into the velocity field / denoising kernel. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具体轨迹更新方程 Trajectory Update Formulation]:** *At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator.*
  对应步骤：S3 [具体轨迹更新方程 Trajectory Update Formulation]：At solver step $\tau \in [0, 1]$, the conditional velocity field $v_\theta(x_\tau, \tau \mid c)$ projects the unconstrained flow onto the tangent space of feasible trajectories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]:** *This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.*
  对应步骤：S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]：This tangent projection guarantees exact boundary satisfaction at $\tau = 1$ while permitting straight 4-step Euler integration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.4 Case RC16: 具身智能视觉运动策略与仿真到真实迁移 (Embodied AI Visuomotor Policy & Sim-to-Real)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想阐述具身智能机器人策略（Embodied Policy）如何融合视觉、本体感觉（Proprioception）或触觉，并克服仿真到真实硬件的动力学差距（Sim-to-Real Gap）。

Linked Architecture: Blueprint `BP16` | Logic Rule `L016`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

When the robot hand blocks the camera, the robot cannot see the object clearly. So we add force sensors and joint angles. Also, we predict multiple steps of actions and use an adapter to transfer from simulation to the real robot.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object. To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings. Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling. At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.

Constituent Alignment & Rhetorical Moves

- **S1 [单模态开环局限 Open-Loop Visual Limitation]:** *Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object.*

对应步骤：S1 [单模态开环局限 Open-Loop Visual Limitation]：Relying solely on RGB-D point clouds leaves manipulation policies vulnerable to self-occlusion by the robot gripper during fine-grained contact. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [多模态时空对齐 Multi-Modal Alignment]:** *To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings.*
  对应步骤：S2 [多模态时空对齐 Multi-Modal Alignment]：To maintain state observability during contact, our encoder aligns asynchronous 30 Hz visual tokens with 500 Hz joint proprioception and wrist force-torque histories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]:** *Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling.*
  对应步骤：S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]：Rather than predicting single-step torques prone to jitter, the policy outputs overlapping K-step SE(3) end-effector action chunks smoothed via exponential temporal ensembling. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]:** *At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.*
  对应步骤：S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]：During physical deployment, a lightweight history adapter infers latent friction and payload parameters online, bridging the sim-to-real dynamics gap without policy fine-tuning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.5 Case RC17: 具身智能视觉运动策略与仿真到真实迁移 (Embodied AI Visuomotor Policy & Sim-to-Real)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Experiments 开篇说明：我们为什么选这几类数据集和这几类基线模型，证明我们的实验设计是全面、公平且直击痛点的。
Linked Architecture: Blueprint `BP17` | Logic Rule `L017`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
We use ETT, Weather, Electricity, and Traffic datasets. We compare with Autoformer, PatchTST, iTransformer, and DLinear. We use MSE and MAE as metrics, and the lookback window is 96.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic). Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability. We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer). All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.

Constituent Alignment & Rhetorical Moves

- **S1 [数据集选型逻辑 Dataset Selection Rationale]:** *To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic).*
  对应步骤：S1 [数据集选型逻辑 Dataset Selection Rationale]：To evaluate performance across diverse non-stationary regimes, we conduct experiments on [N] benchmarks spanning [Domain 1], [Domain 2], and [Domain 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [极端/困难特性突出 Challenging Dataset Properties]:** *Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability.*
  对应步骤：S2 [极端/困难特性突出 Challenging Dataset Properties]：Notably, [Dataset A] exhibits [Hard Property 1, e.g., strong covariate shift], whereas [Dataset B] tests scalability across [Hard Property 2, e.g., 862 correlated channels]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [基线流派全覆盖 Baseline Family Coverage]:** *We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers*

(PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer).
对应步骤：S3 [基线流派全覆盖 Baseline Family Coverage]：We compare our method against [K] competitive baselines grouped into three families: [Family 1], [Family 2], and [Family 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [公平评测协议 Fair Evaluation Protocol]:** *All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.*
对应步骤：S4 [公平评测协议 Fair Evaluation Protocol]：All models are evaluated under identical lookback windows ($L = 96$), prediction horizons ($H \in \{96, 192, 336, 720\}$), and hyperparameter search budgets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.6 Case RC18: 具身智能视觉运动策略与仿真到真实迁移 (Embodied AI Visuomotor Policy & Sim-to-Real)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。
Linked Architecture: Blueprint `BP18` | Logic Rule `L018`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
As shown in Table 1, our model gets the best MSE on ETTm1, ETTm2, Weather, and Traffic. On Weather dataset, our MSE is 0.165, which is 12% lower than PatchTST. On Traffic, our MSE is 0.382, which beats iTransformer. This proves our model is the best.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Constituent Alignment & Rhetorical Moves

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*
  对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.*
  对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.*
  对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*
  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.7 Case RC19: 多模态灵巧操作与力触觉闭环反馈 (Multimodal Robotic Manipulation & Tactile Feedback)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Linked Architecture: Blueprint `BP19` | Logic Rule `L019`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Table 2 shows the ablation study. Without Module A, MSE increases from 0.180 to 0.215. Without Module B, MSE increases to 0.204. Without Loss C, MSE increases to 0.195. Thus, every part is necessary.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Constituent Alignment & Rhetorical Moves

- **S1 [消融实验总括 Ablation Setup]:** *To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.*
  对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and

[Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]:** *Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.*
  对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]:** *Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.*
  对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同效应总结 Synergistic Conclusion]:** *Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.*
  对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.8 Case RC20: 多模态灵巧操作与力触觉闭环反馈 (Multimodal Robotic Manipulation & Tactile Feedback)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想回应审稿人最常问的：你的模型是不是对超参数很敏感？在缺失值或传感器强噪声下会不会突然崩盘？

Linked Architecture: Blueprint BP20 | Logic Rule L020

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Figure 4 shows hyperparameter sensitivity. When lambda changes from 0.1 to 1.0, the result does not change much. Also, when we add noise, our model is still very stable.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption. Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$. Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%. This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.

Constituent Alignment & Rhetorical Moves

- **S1 [分析目的 Objective]:** *Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption.*
  对应步骤：S1 [分析目的 Objective]：We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [超参数平坦区间 Plateau Stability]:** *Varying the phase regularization weight λ across* $[0.05, 0.50]$ *alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around* $\lambda = 0.15$.
  对应步骤：S2 [超参数平坦区间 Plateau Stability]：Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [极端边缘工况压力测试 Stress Test under Corruption]:** *Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%.*
  对应步骤：S3 [极端边缘工况压力测试 Stress Test under Corruption]：Under synthetic sensor dropouts up to [40%] and Gaussian noise ($\sigma = 0.3$), baseline accuracy plummets by over [35%], whereas our model degrades by less than [6.5%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [鲁棒性物理归因 Robustness Mechanism]:** *This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.*
  对应步骤：S4 [鲁棒性物理归因 Robustness Mechanism]：This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.9 Case RC21: 多模态灵巧操作与力触觉闭环反馈 (Multimodal Robotic Manipulation & Tactile Feedback)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想通过可视化案例（Case Study / Qualitative Analysis），既展示模型在复杂跳变下的精准跟踪，又坦诚剖析一个模型预测失准的极端案例。
Linked Architecture: Blueprint `BP21` | Logic Rule `L021`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Figure 5 shows that our prediction curve is very close to the ground truth curve, while baseline curves are far away. Sometimes our model also makes small errors when the data changes too suddenly.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:
Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark. Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error. Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response. Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.

Constituent Alignment & Rhetorical Moves

- **S1 [定性对比切入 Qualitative Setup]:** *Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark.*
  对应步骤：S1 [定性对比切入 Qualitative Setup]：Figure 5 visualizes predicted trajectories during an abrupt regime shift on [Dataset/Task]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [成功案例亮点 Success Highlight]:** *Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error.*
  对应步骤：S2 [成功案例亮点 Success Highlight]：While [Baseline] lags behind the sudden step change by [K steps] and overshoots, our model rapidly locks onto the new amplitude regime within [M steps]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [失败案例坦诚剖析 Honest Failure Mode]:** *Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response.*
  对应步骤：S3 [失败案例坦诚剖析 Honest Failure Mode]：Nevertheless, inspecting the bottom-right failure case reveals that when an external manual override occurs without historical precursors, all purely endogenous forecasters—including ours—exhibit a transient lag. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [改进启示 Constructive Takeaway]:** *Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.*
  对应步骤：S4 [改进启示 Constructive Takeaway]：Incorporating exogenous event logs or textual maintenance records is necessary to anticipate such unscheduled interventions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.10 Case RC22: 多模态灵巧操作与力触觉闭环反馈 (Multimodal Robotic Manipulation & Tactile Feedback)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想说明我们的模型不仅准，而且算得快、显存省，在“精度-延迟/显存”帕累托前沿（Pareto Frontier）上显著优于现有大模型。
Linked Architecture: Blueprint `BP22` | Logic Rule `L022`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Our model is very fast and uses little memory. It only takes 4.2 ms to run, so it can be used in real-time applications easily.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module. By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer. At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.

Constituent Alignment & Rhetorical Moves

- **S1 [硬件与测试基准 Hardware Setup]:** *Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module.*
  对应步骤：S1 [硬件与测试基准 Hardware Setup]：Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [复杂度理论与实测对比 Complexity & Runtime]:** *By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer.*
  对应步骤：S2 [复杂度理论与实测对比 Complexity & Runtime]：By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch by [X%] and peak VRAM by [Y%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实时闭环部署可行性 Real-Time Viability]:** *At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.*
  对应步骤：S3 [实时闭环部署可行性 Real-Time Viability]：At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control deadline. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.11 Case RC23: 多模态灵巧操作与力触觉闭环反馈 (Multimodal Robotic Manipulation & Tactile Feedback)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写论文末尾或 NeurIPS/ICLR Checklist 要求的 Limitations 段落：诚实指出当前方法的两个客观边界，但绝不贬损论文的核心贡献。

Linked Architecture: Blueprint `BP23` | Logic Rule `L023`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Our paper has some limitations. We did not test on very big datasets because we do not have enough GPUs. In the future, we will try more datasets and make the model better.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting. First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy. Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration. Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.

Constituent Alignment & Rhetorical Moves

- **S1 [边界 1:** *Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting.*
  对应步骤：S1 [边界 1：极端高维或极度稀疏场景 Boundary 1]: Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [边界 1 机理分析与解法 Mechanism & Solution 1]:** *First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy.*
  对应步骤：S2 [边界 1 机理分析与解法 Mechanism & Solution 1]：First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [边界 2:** *Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration.*
  对应步骤：S3 [边界 2：特定假设限制 Boundary 2]：Second, our continuous scale tracker assumes piecewise differentiable drift and may under-estimate instantaneous discontinuous jumps caused by hardware resets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [未来演进方向 Future Extension]:** *Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.*
  对应步骤：S4 [未来演进方向 Future Extension]：Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.2.12 Case RC24: 多模态灵巧操作与力触觉闭环反馈 (Multimodal Robotic Manipulation & Tactile Feedback)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。
Linked Architecture: Blueprint BP24 | Logic Rule L024

Defective Paragraph vs. Publication Revision

[Defective Draft]:
In conclusion, we proposed DyTS-Net in this paper. It has a spectral module and an ODE module. Experiments show that it achieves state-of-the-art performance on many datasets. We hope our work will inspire more researchers in the future.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Constituent Alignment & Rhetorical Moves

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*
  对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3 Category III: Mechanical Connectors & Logical Leap / 机械连接词与逻辑跳跃

17.3.1 Case RC25: 长序列 Transformer 稀疏注意力与分块表征 (Long-Horizon Transformer Efficiency & Patching)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写雅思 Task 1 或数据报告的第一段：用同义句法改写图表标题，并用两句话高度凝练全图最核心的 2 个宏观趋势（极值、反超拐点或主次格局），绝不堆砌零碎数字。

Linked Architecture: Blueprint BP25 | Logic Rule L025

Defective Paragraph vs. Publication Revision

[Defective Draft]:

The chart shows the energy consumption from 2000 to 2025. We can see that solar energy went up a lot from 5% to 35%, and coal went down from 60% to 25%. Wind also went up.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025. Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance

contracted steadily. Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.

Constituent Alignment & Rhetorical Moves

- **S1 [图表命题同义改写 Prompt Paraphrase]:** *The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025.*
  对应步骤：S1 [图表命题同义改写 Prompt Paraphrase]：The line graph / bar chart compares [Variable/Metric] across [Categories/Regions] over a [N]-year period from [Year A] to [Year B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [宏观趋势 1:** *Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily.*
  对应步骤：S2 [宏观趋势 1：总体走向与反超 Overview Trend 1]：Overall, while [Category A and B] experienced a steady upward trajectory, [Category C] declined markedly, leading to a reversal in dominance around [Mid-Year]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [宏观趋势 2:** *Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.*
  对应步骤：S3 [宏观趋势 2：极值或差距特征 Overview Trend 2]：Notably, [Category A] registered the most dramatic growth throughout the period, ultimately overtaking all other sectors to become the primary contributor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.2 Case RC26: 长序列 Transformer 稀疏注意力与分块表征 (Long-Horizon Transformer Efficiency & Patching)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写动态折线图（或模型训练收敛曲线）的主体细节段：把一条经历“初期缓慢爬升 → 中期爆发增长 → 后期高位平台期”的曲线，以及另一条“冲高回落”的曲线对比写活。

Linked Architecture: Blueprint `BP26` | Logic Rule `L026`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In 2000, A was 12% and B was 54%. Then in 2005, A increased to 15%. In 2018, A increased to 48%. In 2025, A was 47%. B increased to 61% in 2004, and then decreased to 24% in 2025.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

In 2000, battery-electric vehicles accounted for a mere 4% of municipal fleet purchases, trailing diesel buses at 62%. Following five years of marginal fluctuation, electric adoption accelerated sharply after 2006, quadrupling to 38% by 2018 before leveling off at approximately 41% through 2025. By contrast, diesel procurement peaked briefly at 67% in 2004 but subsequently underwent a sustained contraction, plunging to 19% by the end of the survey. This divergent trajectory reversed the initial 58-percentage-point deficit into a more than twofold lead for electric fleets.

Constituent Alignment & Rhetorical Moves

- **S1 [起始点位对比 Initial Standing]:** *In 2000, battery-electric vehicles accounted for a mere 4% of municipal fleet purchases, trailing diesel buses at 62%.*
  对应步骤：S1 [起始点位对比 Initial Standing]：At the beginning of the period in [2000], [Entity A] stood at a modest [12%],less than a quarter of [Entity B]'s initial figure ([54%]). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [多阶段演进:** *Following five years of marginal fluctuation, electric adoption accelerated sharply after 2006, quadrupling to 38% by 2018 before leveling off at approximately 41% through 2025.*
  对应步骤：S2 [多阶段演进：缓升转暴涨 Phased Acceleration]：After rising marginally over the firstfive years, [Entity A] surged nearly threefold between [2005] and [2018], peaking at [48%] before stabilizing on a high plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [反向曲线对比:** *By contrast, diesel procurement peaked briefly at 67% in 2004 but subsequently underwent a sustained contraction, plunging to 19% by the end of the survey.*
  对应步骤：S3 [反向曲线对比：冲高回落 Counter-Trajectory]：Conversely, [Entity B] climbed briefly to [61%] in [2004] but subsequently entered a prolonged decline, dropping by more than half to [24%] by [2025]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [终局交叉总结 Terminal Comparison]:** *This divergent trajectory reversed the initial 58-percentage-point deficit into a more than twofold lead for electric fleets.*

对应步骤：S4 [终局交叉总结 Terminal Comparison]：Consequently, the initial [42]-percentage-point gap between the two categories had completely reversed by the final year. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.3 Case RC27: 长序列 Transformer 稀疏注意力与分块表征 (Long-Horizon Transformer Efficiency & Patching)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写静态柱状图或饼图的主体段：横向对比不同国家/不同模型在不同开销维度上的占比差异，灵活运用倍数（double/triple）、分数（nearly a third）与排序结构。

Linked Architecture: Blueprint BP27 | Logic Rule L027

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In Group 1, A is 46%, B is 24%, C is 18%, D is 7%, and E is 5%. In Group 2, C is 52%, B is 22%, A is 15%, D is 6%, and E is 5%.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%). A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocessing dominates energy usage at 51%, while attention overhead contracts to just 16%. Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption. This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.

Constituent Alignment & Rhetorical Moves

- **S1 [主导项锁定 Dominant Category]:** *Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%).*
  对应步骤：S1 [主导项锁定 Dominant Category]：Regarding [Group 1], [Category A] claims the largest share at [46%], nearly double the proportion allocated to [Category B] ([24%]). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [组间横向反差 Cross-Group Contrast]:** *A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocessing dominates energy usage at 51%, while attention overhead contracts to just 16%.*
  对应步骤：S2 [组间横向反差 Cross-Group Contrast]：A markedly different allocation pattern emerges in [Group 2], where [Category C] dominates at [52%], while [Category A] shrinks to just [15%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [次要项合并归纳 Minor Categories Grouping]:** *Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption.*
  对应步骤：S3 [次要项合并归纳 Minor Categories Grouping]：Meanwhile, the remaining categories—[Category D and E]—collectively account for less than [12%] across both groups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [结构性差异提炼 Structural Takeaway]:** *This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.*
  对应步骤：S4 [结构性差异提炼 Structural Takeaway]：These figures indicate that whereas [Group 1] prioritizes [Function A], [Group 2] concentrates resources heavily on [Function C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.4 Case RC28: 长序列 Transformer 稀疏注意力与分块表征 (Long-Horizon Transformer Efficiency & Patching)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写双图题（如一张图是算力投入/研发经费变化，另一张图是模型误差/碳排放强度变化）的综合关联段，阐明两张图之间的内在协同或背离关系。
Linked Architecture: Blueprint `BP28` | Logic Rule `L028`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Chart 1 shows R&D spending and Chart 2 shows defect rates. When R&D spending goes up, defect rates go down. Country Z spends little money but has low defect rates.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold. Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8

Constituent Alignment & Rhetorical Moves

- **S1 [双图联动锚定 Cross-Chart Link]:** *Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold.*
  对应步骤：S1 [双图联动锚定 Cross-Chart Link]：Synthesizing both charts reveals a clear inverse / positive relationship between [Chart 1 Variable] and [Chart 2 Variable] up to [Threshold Point]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [区间 1 协同变化 Phase 1 Co-movement]:** *Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8% to 0.9% (Figure 2).*
  对应步骤：S2 [区间 1 协同变化 Phase 1 Co-movement]：Specifically, as [Variable X in Chart 1] expanded from [A] to [B] during [Period 1], [Variable Y in Chart 2] dropped correspondingly by [C%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]:** *Beyond $40 million, however, further capital expenditure yielded diminishing returns: doubling investment to $80 million reduced defects by only an additional 0.1 percentage points.*
  对应步骤：S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]：However, beyond [Threshold Point], although [Variable X] continued to double, [Variable Y] plateaued around [Value], indicating diminishing marginal returns. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [例外个体点拨 Exception Highlight]:** *Notably, Plant D defied this expenditure curve, achieving a 0.7% defect rate with just $22 million through closed-loop sensor calibration.*
  对应步骤：S4 [例外个体点拨 Exception Highlight]：An exception to this pattern is [Entity Z], which achieved the lowest [Variable Y] despite maintaining moderate [Variable X]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.5 Case RC29: 长序列 Transformer 稀疏注意力与分块表征 (Long-Horizon Transformer Efficiency & Patching)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写雅思 Task 1 工艺/自然循环流程图（或论文算法数据预处理与闭环训练流水线）：把 5–6 个串联、分叉与回环步骤写得层次分明，绝不机械连用 Then, Then, Then。

Linked Architecture: Blueprint `BP29` | Logic Rule `L029`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

First, raw data is collected. Then it is cleaned. Then it goes into two branches. Branch 1 does A, and Branch 2 does B. Then they are combined. Finally, bad samples go back to the beginning.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 ‘This/Such + 汇总名词 ‘ 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

The diagram illustrates the closed-loop data curation pipeline for embodied robot learning, spanning four primary stages from teleoperated collection to failure-driven retraining. The cycle begins when raw multimodal demonstrations are recorded via VR teleoperation and filtered to discard kinematic singularities and dropped frames. Once validated, the trajectory stream bifurcates into two parallel pathways: visual frames undergo background augmentation, while proprioceptive states are normalized into local gripper coordinates. These processed streams are subsequently synchronized to train the visuomotor policy, whose real-world rollout failures are logged and fed back into the teleoperation buffer to close the active-learning loop.

Constituent Alignment & Rhetorical Moves

- **S1 [流程总括与起止界定 Lifecycle Overview]:** *The diagram illustrates the closed-loop data curation pipeline for embodied robot learning, spanning four primary stages from teleoperated collection to failure-driven retraining.*
  对应步骤：S1 [流程总括与起止界定 Lifecycle Overview]：The diagram illustrates the closed-loop pipeline for [Process Name], comprising [N] main stages from initial [Input Stage] to continuous [Feedback/Output Stage]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [前段采集与预处理 Initial Ingestion & Filtering]:** *The cycle begins when raw multimodal demonstrations are recorded via VR teleoperation and filtered to discard kinematic singularities and dropped frames.*
  对应步骤：S2 [前段采集与预处理 Initial Ingestion & Filtering]：The process begins when raw [Material/Data] is collected via [Source] and passed through [Filter/Chamber] to remove [Impurities/Outliers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [中段并行分流 Parallel Branching]:** *Once validated, the trajectory stream bifurcates into two parallel pathways: visual frames undergo background augmentation, while proprioceptive states are normalized into local gripper coordinates.*
  对应步骤：S3 [中段并行分流 Parallel Branching]：Once purified, the stream bifurcates into two parallel pathways: [Fraction A] undergoes [Process A], whereas [Fraction B] is routed to [Process B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [末端合成与闭环回流 Synthesis & Closed-Loop Recycling]:** *These processed streams are subsequently synchronized to train the visuomotor policy, whose real-world rollout failures are logged and fed back into the teleoperation buffer to close the active-learning loop.*
  对应步骤：S4 [末端合成与闭环回流 Synthesis & Closed-Loop Recycling]：The resulting outputs are subsequently merged in [Unit C], with residual by-products / high-error trajectories recycled back into the initial buffer for iterative refinement. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.6 Case RC30: 长序列 Transformer 稀疏注意力与分块表征 (Long-Horizon Transformer Efficiency & Patching)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想处理同时包含过去历史数据（如 2010–2025）和未来预测数据（如 2026–2040）的图表段落，严格区分过去实测时态与未来审慎预测表达。
Linked Architecture: Blueprint BP30 | Logic Rule L030

Defective Paragraph vs. Publication Revision

[Defective Draft]:

From 2010 to 2025, demand increased from 200 to 460. In 2035, it will increase to 1050. AI was 10% in 2023 and will be 45% in 2040.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh. From 2026 onward, total demand is projected to climb at a substantially steeper rate, with baseline forecasts pointing to 1,050 TWh by 2035. Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040. However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.

Constituent Alignment & Rhetorical Moves

- **S1 [历史实测轨迹 Historical Trajectory (Past Tense)]:** *Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh.*
  对应步骤：S1 [历史实测轨迹 Historical Trajectory (Past Tense)]：Between [2010] and [2025], global data-center electricity demand rose steadily from [200 TWh] to [460 TWh], driven by cloud migration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]:** *From 2026 onward, total demand is projected to climb at a substantially steeper rate, with baseline forecasts pointing to 1,050 TWh by 2035.*
  对应步骤：S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]：From [2026] onward, consumption is projected to accelerate sharply, with forecasts indicating a surge to [1,050 TWh] by [2035]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [结构占比重塑 Structural Shift in Projections]:** *Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040.*

对应步骤：S3 [结构占比重塑 Structural Shift in Projections]：Within this projected expansion, generative AI training and inference—which accounted for less than [10%] in [2023]—are anticipated to represent over [45%] of total load by [2040]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [条件式审慎收尾 Conditional Hedging]:** *However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.*
  对应步骤：S4 [条件式审慎收尾 Conditional Hedging]：Nevertheless, energy-efficient hardware adoption is expected to moderate the growth slope during the final five years. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.7 Case RC31: 顶会口头汇报（Tech Talk）与图表引导 (Academic Tech Talk & Conference Oral Delivery)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写托福综合写作（Integrated Writing）或文献争议交锋段：先用 1 句话精准概括阅读材料（或正方文献）的分论点，再用 2–3 句话详细呈现教授讲座（或反方实证）是如何釜底抽薪推翻该论点的。

Linked Architecture: Blueprint `BP31` | Logic Rule `L001`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

The reading says that replacing all delivery trucks with autonomous drones will save energy. The professor disagrees. She says drones use a lot of battery to stay in the air and can only carry one package at a time.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

The reading passage contends that replacing urban delivery vans with autonomous quadcopter drones will substantially reduce last-mile transportation energy consumption, citing the lighter weight of aerial vehicles. The lecturer directly challenges this premise by examining energy efficiency per

kilogram of delivered freight. Specifically, she points out that unlike ground vehicles supported passively by road surfaces, multi-rotor drones must expend continuous battery power simply to counteract gravity and headwinds while carrying only a single parcel per trip. Consequently, servicing a dense apartment block via individual drone flights consumes nearly three times more energy per package than a single consolidated van route.

Constituent Alignment & Rhetorical Moves

- **S1 [正方/阅读论点概括 Reading Claim & Rationale]:** *The reading passage contends that replacing urban delivery vans with autonomous quadcopter drones will substantially reduce last-mile transportation energy consumption, citing the lighter weight of aerial vehicles.*
  对应步骤：S1 [正方/阅读论点概括 Reading Claim & Rationale]：The reading passage contends that [Claim A], arguing that [Reading's Supporting Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反方/听力直接反驳 Lecture Direct Refutation]:** *The lecturer directly challenges this premise by examining energy efficiency per kilogram of delivered freight.*
  对应步骤：S2 [反方/听力直接反驳 Lecture Direct Refutation]：The lecturer directly challenges this assertion by pointing out that [Core Flaw / Neglected Factor in Reading's Premise]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [反方具体反证展开 Empirical Counter-Evidence]:** *Specifically, she points out that unlike ground vehicles supported passively by road surfaces, multi-rotor drones must expend continuous battery power simply to counteract gravity and headwinds while carrying only a single parcel per trip.*
  对应步骤：S3 [反方具体反证展开 Empirical Counter-Evidence]：Specifically, she explains that when [Real-World Condition], [Mechanism B occurs instead], which [Negates the Claimed Benefit]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [结论锁定 Refutation Closure]:** *Consequently, servicing a dense apartment block via individual drone flights consumes nearly three times more energy per package than a single consolidated van route.*
  对应步骤：S4 [结论锁定 Refutation Closure]：Consequently, rather than achieving [Reading's Goal], the proposed measure actually exacerbates [Negative Outcome]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.8 Case RC32: 顶会口头汇报（Tech Talk）与图表引导 (Academic Tech Talk & Conference Oral Delivery)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在图表或实验结果段中，专门挑出违背总体规律的 1–2 个异常点（Outliers）或局部反常尖峰进行精准描述与成因剖析。

Linked Architecture: Blueprint `BP32` | Logic Rule `L002`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Most datasets follow this rule, except Exchange dataset. On Exchange dataset, deep models are worse than simple linear models. This is because financial data has strong random walk noise.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset. Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate. Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks. This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.

Constituent Alignment & Rhetorical Moves

- **S1 [总体规律背景下的特例锚定 Outlier Identification]:** *Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset.*
  对应步骤：S1 [总体规律背景下的特例锚定 Outlier Identification]：Although [General Trend holds across most groups/steps], a notable departure from this pattern occurs in [Specific Point/Group]. | 句首

锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [反常幅度量化 Quantitative Deviation]:** *Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate.*
  对应步骤：S2 [反常幅度量化 Quantitative Deviation]：Whereas peer categories cluster tightly within $[A, B]$, [Outlier Entity] spikes/drops to [Extreme Value], deviating by more than [X%] from the group median. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [底层物理/统计成因剖析 Root Cause Analysis]:** *Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks.*
  对应步骤：S3 [底层物理/统计成因剖析 Root Cause Analysis]：Closer inspection indicates that this anomaly stems from [Specific Structural/External Factor] rather than random variance. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [方法论启示 Methodological Implication]:** *This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.*
  对应步骤：S4 [方法论启示 Methodological Implication]：Accounting for such localized departures is critical when calibrating [Downstream Model/Policy]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.9 Case RC33: 顶会口头汇报（Tech Talk）与图表引导 (Academic Tech Talk & Conference Oral Delivery)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉'With the rapid development of science and technology, more and more people...' 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Linked Architecture: Blueprint `BP33` | Logic Rule `L003`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

With the rapid development of science and technology, artificial intelligence is becoming more and more popular in universities. Some people think it is good, while others think it is bad. In my opinion, every coin has two sides and we should use it properly.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Constituent Alignment & Rhetorical Moves

- **S1 [具象现象锚定 Concrete Trend Anchor]:** *Large language models and automated code assistants have rapidly become routine study aids across university campuses.*
  对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心争议张力 Core Tension / Debate]:** *While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.*
  对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [带条件的中心论点 Nuanced Thesis Statement]:** *In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.*
  对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.10 Case RC34: 顶会口头汇报（Tech Talk）与图表引导 (Academic Tech Talk & Conference Oral Delivery)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写大作文的第一个主体段（Body Paragraph 1）：论证某个核心优势或必要性，做到有理有据、有具象场景支撑，绝不车轱辘话来回说。

Linked Architecture: Blueprint BP34 | Logic Rule L004

Defective Paragraph vs. Publication Revision

[Defective Draft]:

First, AI can improve working efficiency a lot. People have many boring tasks every day. If AI does these tasks, people can save time and do better things. For example, my friend uses AI to write emails and saves two hours every day. So AI is very helpful.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment. In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis. When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation. Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.

Constituent Alignment & Rhetorical Moves

- **S1 [分论点主题句 Topic Claim]:** *Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment.*
  对应步骤：S1 [分论点主题句 Topic Claim]：一句话清晰亮出本段唯一控制中心（[Policy/Factor X] strengthens [Outcome Y] by [Core Mechanism]）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [因果机制推演 Causal Elaboration]:** *In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis.*
  对应步骤：S2 [因果机制推演 Causal Elaboration]：Explain HOW and WHY：阐明如果没有这层干扰或有了这项机制，认知/经济/技术流程会发生什么具体改变。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具象微观案例 Concrete Micro-Case]:** *When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation.*
  对应步骤：S3 [具象微观案例 Concrete Micro-Case]：For instance, when [Specific Group/Industry] [Specific Action], [Measurable/Observable Consequence]（绝不编造假百分比，用具体场景细节说话）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [反事实对比或扣题升华 Warrant / Link Back]:** *Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.*
  对应步骤：S4 [反事实对比或扣题升华 Warrant / Link Back]：Thus, [Summary Noun] shifts human effort from [Low-value Routine] toward [High-value Judgment]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.11 Case RC35: 顶会口头汇报（Tech Talk）与图表引导 (Academic Tech Talk & Conference Oral Delivery)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在问题分析类作文（Why is this happening?）中，剖析某个社会/科研痛点背后的深层原因（既指出直接表层诱因，又挖出底层的考核激励或结构性根源）。

Linked Architecture: Blueprint `BP35` | Logic Rule `L005`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Why do many students copy AI answers directly? First, AI is very easy and fast to use. Second, students are lazy and have too much homework. So they just copy and paste.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design. At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading. More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique. When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.

Constituent Alignment & Rhetorical Moves

- **S1 [原因总起句 Dual-Cause Thesis]:** *The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design.*
  对应步骤：S1 [原因总起句 Dual-Cause Thesis]：The widespread emergence of [Problem X] stems from both immediate technological pressures and deeper institutional incentives. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层:** *At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading.*
  对应步骤：S2 [第一层：直接操作/技术诱因 Immediate Driver]：At the operational level, [Factor A] lowers the cost of [Behavior], tempting individuals/organizations to [Short-term Action]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [第二层:** *More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique.*
  对应步骤：S3 [第二层：深层制度/激励根源 Structural Root Cause]：More fundamentally, existing evaluation systems reward [Superficial Metric] rather than [Long-term Quality], creating powerful pressure to prioritize speed over rigor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [双因叠加后果 Compounding Effect]:** *When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.*
  对应步骤：S4 [双因叠加后果 Compounding Effect]：When effortless generation meets quantity-driven evaluation, [Problem X] becomes systemic rather than isolated. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.3.12 Case RC36: 顶会口头汇报（Tech Talk）与图表引导 (Academic Tech Talk & Conference Oral Delivery)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在议论文中回应反方观点（Some people argue that...）：先大度承认反方担忧在短期或局部确实存在，再用结构性证据或制度设计证明该问题完全可控，从而锁死我方立场。

Linked Architecture: Blueprint `BP36` | Logic Rule `L006`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Some people think robots will take all our jobs and make everyone poor. This is wrong. Robots will create new jobs, and people can learn new skills to find better jobs easily.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers. During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules. However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles. Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.

Constituent Alignment & Rhetorical Moves

- **S1 [呈现反方最有力的担忧 Articulate Opposing Concern]:** *Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers.*
  对应步骤：S1 [呈现反方最有力的担忧 Articulate Opposing Concern]：Admittedly, critics raise a valid concern that [Opposing Argument, e.g., automation displaces entry-level clerical and manufacturing

workers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [承认局部真实性 Acknowledge Partial Validity]:** *During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules.*
  对应步骤：S2 [承认局部真实性 Acknowledge Partial Validity]：In the short term, rapid deployment of [Technology/Policy] can indeed create transitional friction for workers whose routines are easily codified. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [釜底抽薪式转折 Structural Pivot]:** *However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles.*
  对应步骤：S3 [釜底抽薪式转折 Structural Pivot]：However, viewing [X] purely as a zero-sum threat overlooks how productivity gains generate entirely new categories of skilled employment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [历史/现实反证与制度补救 Evidence & Mitigation]:** *Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.*
  对应步骤：S4 [历史/现实反证与制度补救 Evidence & Mitigation]：Just as spreadsheet software reduced manual bookkeeping while expanding demand for financial analysts, pairing automation with targeted vocational retraining allows displaced workers to transition into system supervision and maintenance roles. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4 Category IV: Heavy AI-ese & Hollow Hyperbole / 重度 AI 套话与空洞夸张

17.4.1 Case RC37: 顶会审稿回复（Rebuttal）与消融防御 (Conference Author Rebuttal & Reviewer Defense)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写 Problem-Solution 类作文的解决方案段：针对前文诊断出的两个具体根源，提出两项可操作、有明确执行主体和传导机制的靶向对策，拒绝只喊'The government should make laws and raise awareness'。

Linked Architecture: Blueprint `BP37` | Logic Rule `L007`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

To solve this problem, first, the government should pass strict laws to punish bad people. Second, schools and media should educate the public and raise their awareness. Only in this way can we build a harmonious society.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria. First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review. Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies. Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.

Constituent Alignment & Rhetorical Moves

- **S1 [对策总起句 Targeted Solution Thesis]:** *Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria.*
  对应步骤：S1 [对策总起句 Targeted Solution Thesis]：Mitigating [Problem X] effectively requires coordinated intervention across both [Level A: Technical/Curriculum Design] and [Level B: Institutional Governance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对策 1:** *First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review.*
  对应步骤：S2 [对策 1：操作层可验证改革 Actionable Measure 1]：First, [Specific Actor 1, e.g., university departments] should replace [Vulnerable Practice] with [Verifiable Mechanism, e.g., oral defense and in-class iterative drafting], ensuring that [Specific Effect 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [对策 2:** *Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies.*
  对应步骤：S3 [对策 2：制度/标准层激励对齐 Actionable Measure 2]：Concurrently, [Specific Actor 2, e.g., regulatory agencies / industry consortia] must establish [Concrete Standard, e.g., mandatory provenance watermarking and audit logs] to penalize [Misconduct]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同闭环效果 Combined Outcome]:** *Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.*
  对应步骤：S4 [协同闭环效果 Combined Outcome]：Together, aligning pedagogical verification with technical traceability curbs misuse without stifling legitimate innovation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.2 Case RC38: 顶会审稿回复（Rebuttal）与消融防御 (Conference Author Rebuttal & Reviewer Defense)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想回应'Do the advantages outweigh the disadvantages?' 题型：把弊端定性为"暂时的、可通过技术/规则收敛的局部成本"，把优势定性为"不可替代的结构性长期突破"。

Linked Architecture: Blueprint `BP38` | Logic Rule `L008`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Every coin has two sides. Although open-source AI has some security risks, it also has many advantages like helping researchers and small companies. So I think the advantages are bigger than the disadvantages.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks. While 公开 releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters. By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics. Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.

Constituent Alignment & Rhetorical Moves

- **S1 [权衡定性主题句 Asymmetry Thesis]:** *Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks.*
  对应步骤：S1 [权衡定性主题句 Asymmetry Thesis]：Weighing both dimensions reveals that the long-term structural benefits of [X] far outweigh its manageable transitional costs. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [将弊端限定为可收敛成本 Bounding the Drawback]:** *While* 公开 *releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters.*
  对应步骤：S2 [将弊端限定为可收敛成本 Bounding the Drawback]：While [Drawback A, e.g., initial capital outlay / training overhead] poses a tangible hurdle, this cost is bounded and diminishes rapidly as [Mitigating Factor, e.g., hardware amortization / open-source standardization] matures. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]:** *By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics.*
  对应步骤：S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]：By contrast, the core advantage—[Benefit B, e.g., eliminating human exposure to hazardous environments / discovering non-linear scientific laws]—delivers compounding value that manual methods cannot replicate at any cost. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [权衡结论 Trade-off Verdict]:** *Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.*

对应步骤：S4 [权衡结论 Trade-off Verdict]：Therefore, addressing the friction through targeted safeguards is far more rational than forgoing the transformative gain. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.3 Case RC39: 顶会审稿回复（Rebuttal）与消融防御 (Conference Author Rebuttal & Reviewer Defense)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在议论文或科研立项书中，通过严密的“反事实推演（If we fail to... / Absent X...）”强化紧迫感与说服力。

Linked Architecture: Blueprint BP39 | Logic Rule L009

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Curating clean human data is very important. If we do not use clean human data and only use AI-generated data to train new AI models, the models will become worse and worse and finally collapse.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 ‘This/Such + 汇总名词 ‘ 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models. Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations. In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse. This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.

Constituent Alignment & Rhetorical Moves

- **S1 [正面基准命题 Positive Baseline]:** *Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models.*
  对应步骤：S1 [正面基准命题 Positive Baseline]：Maintaining rigorous [Principle/Standard X] serves as the primary safeguard against [Systemic Risk Y]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反事实假设切入 Counterfactual Premise]:** *Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations.*
  对应步骤：S2 [反事实假设切入 Counterfactual Premise]：Absent this safeguard—if [Actors] were to rely exclusively on [Unchecked Alternative]—[Immediate Vulnerability] would quickly emerge. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [连锁恶化传导 Cascading Deterioration]:** *In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse.*
  对应步骤：S3 [连锁恶化传导 Cascading Deterioration]：Once [Immediate Vulnerability] takes hold, [Secondary Consequence] compounds across cycles, eventually eroding [Core Foundation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证印证或结论锁定 Empirical Echo]:** *This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.*
  对应步骤：S4 [实证印证或结论锁定 Empirical Echo]：This degradation dynamic mirrors [Real-world/Technical Precedent], underscoring why [Principle X] cannot be compromised for short-term convenience. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.4 Case RC40: 顶会审稿回复（Rebuttal）与消融防御 (Conference Author Rebuttal & Reviewer Defense)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Linked Architecture: Blueprint `BP40` | Logic Rule `L010`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In conclusion, AI has both advantages and disadvantages. We should make full use of its advantages and avoid its disadvantages. Let us work hand in hand to welcome a bright future of AI!

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Constituent Alignment & Rhetorical Moves

- **S1 [重申核心权衡与立场 Restate Nuanced Thesis]:** *In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.*
  对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [整合全文两大支柱机制 Synthesize Dual Pillars]:** *By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.*
  对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [理性边界展望（可选）Grounded Forward Look]:** *Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.*
  对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.5 Case RC41: 顶会审稿回复（Rebuttal）与消融防御 (Conference Author Rebuttal & Reviewer Defense)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想作为顶会 Oral Presentation、组会汇报或博士面试 PPT 第一页的开场白：在 45–60 秒内不用任何晦涩公式，让全场听众立刻听懂“我们为什么要做这件事、现有方法到底卡在哪”。

Linked Architecture: Blueprint `BP41` | Logic Rule `L011`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Hello everyone, my name is XXX. Today the title of my presentation is 'Adaptive De-stationary Spectral-Spatial Transformer for Multivariate Time Series Forecasting'. First I will talk about background, second methodology, third experiments, and finally conclusion.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Good morning everyone. Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines. Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges

needed to predict peak loads. In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.

Constituent Alignment & Rhetorical Moves

- **S1 [场景直觉切入 Concrete Scenario Hook]:** *Good morning everyone.*
  对应步骤：S1 [场景直觉切入 Concrete Scenario Hook]：Good morning everyone. Imagine deploying a humanoid robot to assemble flexible cables on a moving factory line—or forecasting grid load when a storm hits. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反直觉痛点设问 Counter-intuitive Breakdown]:** *Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines.*
  对应步骤：S2 [反直觉痛点设问 Counter-intuitive Breakdown]：Standard deep models perform remarkably well when conditions stay calm, yet why do they fail abruptly the moment contact dynamics or weather patterns shift? | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [一针见血揭晓答案 Crisp Root Cause]:** *Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges needed to predict peak loads.*
  对应步骤：S3 [一针见血揭晓答案 Crisp Root Cause]：The culprit is a hidden stationarity assumption baked into standard attention layers. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [亮出今日主旨 Talk Roadmap]:** *In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.*
  对应步骤：S4 [亮出今日主旨 Talk Roadmap]：In this talk, I will show how our framework, [Model Name], breaks this bottleneck and achieves [Key Metric Gain] in real-time deployment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.6 Case RC42: 顶会审稿回复（Rebuttal）与消融防御 (Conference Author Rebuttal & Reviewer Defense)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在演讲到架构图或核心实验曲线（Slide 4 / Slide 7）时，用最自然的口语引导听众的眼睛先看哪、再看哪，瞬间读懂图表背后的胜出逻辑。
Linked Architecture: Blueprint `BP42` | Logic Rule `L012`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Here is Figure 3. You can see many lines here. Our model is the blue line and it is the lowest, which means our model is the best on all horizons.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours. Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours. Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours. That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.

Constituent Alignment & Rhetorical Moves

- **S1 [视觉坐标锁定 Spatial Pointer]:** *Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours.*
  对应步骤：S1 [视觉坐标锁定 Spatial Pointer]：Let's look at the comparison in the right panel of Slide 5, where the horizontal axis tracks prediction horizon and the vertical axis shows MSE. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]:** *Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours.*
  对应步骤：S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]：Notice the dashed orange curve representing the standard Transformer baseline: past 336 steps, error shoots up almost vertically. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [引导观察我方稳定曲线 Guide to Our Curve]:** *Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours.*
  对应步骤：S3 [引导观察我方稳定曲线 Guide to Our Curve]：Now compare that with our solid blue curve at the bottom, which stays flat even out to 720 steps. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [一句话点破视觉差距原因 One-Sentence Takeaway]:** *That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.*
  对应步骤：S4 [一句话点破视觉差距原因 One-Sentence Takeaway]：That gap comes directly from our neural-ODE scale tracker preventing long-range drift. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.7 Case RC43: 雅思小作文动态/静态图表与实证数据综述 (IELTS Task 1 & Technical Data Synthesis)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 PPT 上出现核心损失函数或注意力公式时，用 4 句通俗、精准的口语把公式背后的物理/几何直觉讲透，而不是在台上干巴巴念希腊字母。
Linked Architecture: Blueprint `BP43` | Logic Rule `L013`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
The formula on this slide is Q times K transpose divided by square root of d minus lambda times delta phi L1 norm, and then softmax times V.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:
Before looking at the equation in the middle box, here is the physical intuition behind it. When two sensors sit on the same turbine shaft, genuine mechanical vibrations travel with a consistent phase lag, whereas random sensor glitches fire completely out of phase. That physical rule is captured by the highlighted red term, $\|\Delta\Phi_{Q,K}\|_1$, right inside the softmax. Whenever two channels show inconsistent phase lag, this penalty automatically zeroes out their attention weight.

Constituent Alignment & Rhetorical Moves

- **S1 [先讲直觉目标 Intuition First]:** *Before looking at the equation in the middle box, here is the physical intuition behind it.*
  对应步骤：S1 [先讲直觉目标 Intuition First]：Before looking at the math at the bottom of the slide, here is the simple physical intuition we want to capture. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [日常/物理类比 Physical Analogy]:** *When two sensors sit on the same turbine shaft, genuine mechanical vibrations travel with a consistent phase lag, whereas random sensor glitches fire completely out of phase.*
  对应步骤：S2 [日常/物理类比 Physical Analogy]：When two sensors monitor the same mechanical shaft, genuine vibrations share a consistent phase lag, whereas random electrical glitches fire out of phase. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [指向公式核心项 Point to Key Term]:** *That physical rule is captured by the highlighted red term, $\|\Delta\Phi_{Q,K}\|_1$, right inside the softmax.*
  对应步骤：S3 [指向公式核心项 Point to Key Term]：That is exactly what the red penalty term $\|\Delta\Phi\|_1$ inside the softmax does: it measures phase mismatch between channels. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [一句话总结效果 Operational Effect]:** *Whenever two channels show inconsistent phase lag, this penalty automatically zeroes out their attention weight.*
  对应步骤：S4 [一句话总结效果 Operational Effect]：Whenever two channels fall out of phase, this term suppresses their attention weight automatically. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.8 Case RC44: 雅思小作文动态/静态图表与实证数据综述 (IELTS Task 1 & Technical Data Synthesis)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想用于演讲最后一页总结（Take-home Messages）以及回答台下教授/审稿人犀利提问时的标准应答模板。
Linked Architecture: Blueprint `BP44` | Logic Rule `L014`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
That's all for my presentation. Thank you for listening. Do you have any questions? (When asked a hard question): Well, maybe we didn't test that, I think maybe it can work in the future.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics. Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen. — [Q&A Response]: Thank you for that sharp question on irregular missing timestamps. The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.

Constituent Alignment & Rhetorical Moves

- **S1 [凝练核心带走信息 Core Take-Home Message]:** *To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics.*
  对应步骤：S1 [凝练核心带走信息 Core Take-Home Message]：To wrap up, if there is one takeaway from our work today, it is that [One-Sentence Core Principle]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [成果三点速览 Rapid Recap]:** *Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen.*
  对应步骤：S2 [成果三点速览 Rapid Recap]：By combining [A] and [B], [Model] delivers [Metric 1], [Metric 2], and [Open-Source/Practical Asset]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]:** *Thank you for that sharp question on irregular missing timestamps.*
  对应步骤：S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]：(For Q&A) That is a great question regarding [Topic]—the short answer is [Direct Conclusion]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [Q&A 展开与诚实边界 Q&A Elaboration]:** *The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.*
  对应步骤：S4 [Q&A 展开与诚实边界 Q&A Elaboration]：In our current experiments, we observed [Empirical Fact], and extending this to [Questioner's Scenario] can be handled by [Concrete Mechanism].

| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.9 Case RC45: 雅思小作文动态/静态图表与实证数据综述 (IELTS Task 1 & Technical Data Synthesis)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写 ICLR / NeurIPS / ICML / CVPR 审稿回复（Rebuttal）最顶部的致全体审稿人总括段（Global Response）：在 150 词内用最提气的语气感谢审稿人肯定，并一眼亮出我们在 Rebuttal 期间赶做出的所有硬核新增实验。

Linked Architecture: Blueprint `BP45` | Logic Rule `L015`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Thank you to all reviewers for your time and comments. We have carefully read your reviews and answered all your questions below. We hope you can raise your scores.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

We thank all reviewers for their constructive feedback and for recognizing our framework's 'principled formulation of the stationarity-amplitude trade-off' (R1, R3), 'strong empirical gains across eight benchmarks' (R2), and 'clear ablation design' (R4). To address your valuable suggestions, we conducted four sets of new experiments during the rebuttal period (incorporated in blue in the revised manuscript): (1) Benchmarked against two latest 2025 foundation models, Chronos and MOIRAI, across all horizons (Table R1, addressing R1, R3); (2) Added real-robot grasping evaluations under dynamic lighting and 25% camera occlusion (Section 4.5, R2); and (3) Included full GPU memory and wall-clock latency scaling curves up to $L = 4,096$ (Appendix E, R4). Below we address each reviewer's specific questions in detail.

Constituent Alignment & Rhetorical Moves

- **S1 [致谢并提炼审稿人共同好评 Gratitude & Shared Praise]:** *We thank all reviewers for their constructive feedback and for recognizing our framework's 'principled formulation of the stationarity-amplitude trade-off' (R1, R3), 'strong empirical gains across eight benchmarks' (R2), and 'clear ablation design' (R4).*
  对应步骤: S1 [致谢并提炼审稿人共同好评 Gratitude & Shared Praise]: We thank all reviewers for their constructive feedback and for recognizing our work's [Quote Reviewer 1 Praise] (R1), [Quote Reviewer 2 Praise] (R2), and [Quote Reviewer 3 Praise] (R3). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [概览新增实证工作 Overview of New Experiments]:** *To address your valuable suggestions, we conducted four sets of new experiments during the rebuttal period (incorporated in blue in the revised manuscript): (1) Benchmarked against two latest 2025 foundation models, Chronos and MOIRAI, across all horizons (Table R1, addressing R1, R3); (2) Added real-robot grasping evaluations under dynamic lighting and 25% camera occlusion (Section 4.5, R2); and (3) Included full GPU memory and wall-clock latency scaling curves up to $L = 4,096$ (Appendix E, R4).*
  对应步骤: S2 [概览新增实证工作 Overview of New Experiments]: In response to your valuable suggestions, we have conducted [N] sets of new experiments and theoretical analyses, summarized below and highlighted in blue in the revised PDF: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [分条列出硬核新增实验 Bullet Summary of Additions]:** *Below we address each reviewer's specific questions in detail.*
  对应步骤: S3 [分条列出硬核新增实验 Bullet Summary of Additions]: (1) Added comparisons against recent baselines [A, B] on [Dataset] (Table R1, addressing R1, R3); (2) Evaluated real-world robustness under [Harsh Condition] (Appendix D, R2); (3) Provided wall-clock latency and memory profiling across sequence lengths up to $L = 4,096$ (Appendix E, R3). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.10 Case RC46: 雅思小作文动态/静态图表与实证数据综述 (IELTS Task 1 & Technical Data Synthesis)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想回复审稿人最致命的质疑: 'The method seems like a simple combination of existing Module A and Module B (limited novelty).' 既不顶撞审稿人，又能用理论 + 消融数据证明"绝对不是简单拼凑"。

Linked Architecture: Blueprint `BP46` | Logic Rule `L016`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

We disagree that our paper lacks novelty. Although A and B existed before, nobody used them together on time series forecasting. Also our results are very good, so our paper is novel.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs. Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory. Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq. 7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention. To verify this empirically, we implemented a direct 'Naive FFT + ODE' concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over 'Naive FFT + ODE', confirming that the gain stems from our mathematical coupling rather than module stacking.

Constituent Alignment & Rhetorical Moves

- **S1 [礼貌接招并定位误解核心 Courteous Reframing]:** *We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs.*
  对应步骤：S1 [礼貌接招并定位误解核心 Courteous Reframing]：We appreciate Reviewer X's comment on architectural design and would like to clarify the fundamental distinction between our coupled formulation and a naive combination of [A] and [B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [点破简单拼接为何会失败 Why Naive Combination Fails]:** *Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory.*
  对应步骤：S2 [点破简单拼接为何会失败 Why Naive Combination Fails]：Directly stacking standard [A] and [B] fails on [Problem] because [Theoretical Conflict / Gradient Interference]. | 句首锚定旧信

息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]:** *Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq.*
  对应步骤：S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]：Our core contribution lies not in using [A] or [B] in isolation, but in deriving the [Specific Coupling Mechanism C], which resolves [Conflict] by [Mathematical Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]:** *7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention.*
  对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline ('Naive A+B') in Table R2: our coupled formulation outperforms 'Naive A+B' by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]:** *To verify this empirically, we implemented a direct 'Naive FFT + ODE' concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over 'Naive FFT + ODE', confirming that the gain stems from our mathematical coupling rather than module stacking.*
  对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline ('Naive A+B') in Table R2: our coupled formulation outperforms 'Naive A+B' by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.11 Case RC47: 雅思小作文动态/静态图表与实证数据综述 (IELTS Task 1 & Technical Data Synthesis)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想回复审稿人要求补测最新基线模型，或者质疑我们的提升是不是因为参数量更大/训练更久。

Linked Architecture: Blueprint `BP47` | Logic Rule `L017`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

We added the baseline X you asked for. Our model is 0.182 and X is 0.205, so we are still better. Also our model does not have many parameters.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Following your suggestion, we benchmarked against the two recent baselines (TimeMixer and FreTS) and evaluated a capacity-reduced variant of our architecture ('DyTS-Lite') in Table R3. Across all horizons on Weather and ECL, standard DyTS-Net reduces MSE by 9.6% over TimeMixer and 13.4% over FreTS. Furthermore, to ensure strict capacity fairness, 'DyTS-Lite' shrinks the hidden dimension to $d = 64$—requiring 42% fewer parameters and 55% fewer FLOPs than TimeMixer—yet still outperforms TimeMixer by 6.8% in average MSE. This confirms that our gains stem from structural-scale disentanglement rather than model size.

Constituent Alignment & Rhetorical Moves

- **S1 [直接响应并呈上新表 Direct Fulfilment]:** *Following your suggestion, we benchmarked against the two recent baselines (TimeMixer and FreTS) and evaluated a capacity-reduced variant of our architecture ('DyTS-Lite') in Table R3.*
  对应步骤：S1 [直接响应并呈上新表 Direct Fulfilment]：Following your suggestion, we have evaluated [Requested Baselines X and Y] under identical splits and added a parameter-matched variant of our model in Table R3. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比最新基线结果 Performance vs. New Baselines]:** *Across all horizons on Weather and ECL, standard DyTS-Net reduces MSE by 9.6% over TimeMixer and 13.4% over FreTS.*
  对应步骤：S2 [对比最新基线结果 Performance vs. New Baselines]：Across all four horizons on [Datasets], our method outperforms [New Baseline X] by an average of [9.8%] in MSE and [New Baseline Y] by [12.1%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [打消参数量/算力质疑 Parameter & FLOPs Parity Proof]:** *Furthermore, to ensure strict capacity fairness, 'DyTS-Lite' shrinks the hidden dimension to $d = 64$—requiring 42% fewer parameters and 55% fewer FLOPs than TimeMixer—yet still outperforms TimeMixer by 6.8% in average MSE.*
  对应步骤：S3 [打消参数量/算力质疑 Parameter & FLOPs Parity Proof]：Regarding model capacity, even when we scale down our hidden dimension from $d = 256$ to $d = 128$ ('Ours-Small', using 38%

fewer parameters and 52% fewer FLOPs than [Baseline X]), 'Ours-Small' still outperforms [Baseline X] by [7.4%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [结论锁定 Conclusion]:** *This confirms that our gains stem from structural-scale disentanglement rather than model size.*
  对应步骤: S4 [结论锁定 Conclusion]: These results confirm that our improvements derive from inductive bias rather than increased parameter count. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.4.12 Case RC48: 雅思小作文动态/静态图表与实证数据综述 (IELTS Task 1 & Technical Data Synthesis)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写申请博士/博后/访问学者或个人学术主页的 Research Statement / SoP 核心段：把我已经做过的深度学习/时间序列/生成模型工作，与我未来想探索的具身智能（Embodied AI）无缝串联成一条充满内驱力的科研主线。

Linked Architecture: Blueprint `BP48` | Logic Rule `L018`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Since I was a child, I have been fascinated by artificial intelligence and robots. In my master's degree, I did some research on time series forecasting and published a paper. In the future, I want to study embodied AI because it is a very hot topic and can help humans a lot.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】: 删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】: 将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】: 在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】: 补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

My research focuses on learning reliable representations and generative trajectories from non-stationary sequential data. While investigating multivariate industrial telemetry, I found that conventional attention mechanisms collapse when underlying system dynamics shift—motivating me to design a neural-ODE coupled spectral forecaster that reduced long-range drift by 17.4%. Yet passive

sequence forecasting only observes how a system evolves; intelligent physical agents must actively intervene within that loop, where every motor command alters future state distributions. Bridging my background in non-stationary time-series modeling and fast flow-matching samplers, my future research aims to build adaptive visuomotor world models that allow embodied robots to generalize across unfamiliar contact regimes in real time.

Constituent Alignment & Rhetorical Moves

- **S1 [过往研究主线凝练 Past Research Thread]:** *My research focuses on learning reliable representations and generative trajectories from non-stationary sequential data.*
  对应步骤：S1 [过往研究主线凝练 Past Research Thread]：My prior research has centered on modeling non-stationary dynamics and uncertainty in high-dimensional sequential data. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表作攻克过程与硬核成果 Concrete Achievement]:** *While investigating multivariate industrial telemetry, I found that conventional attention mechanisms collapse when underlying system dynamics shift—motivating me to design a neural-ODE coupled spectral forecaster that reduced long-range drift by 17.4%.*
  对应步骤：S2 [代表作攻克过程与硬核成果 Concrete Achievement]：In my recent work on [Time Series / Generative Modeling], I observed that [Core Bottleneck], which led me to formulate [Method]—reducing long-horizon drift by [X%] across industrial telemetry benchmarks. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [向具身智能跃迁的科学顿悟 Scientific Pivot to Embodied AI]:** *Yet passive sequence forecasting only observes how a system evolves; intelligent physical agents must actively intervene within that loop, where every motor command alters future state distributions.*
  对应步骤：S3 [向具身智能跃迁的科学顿悟 Scientific Pivot to Embodied AI]：While passive sequence forecasting predicts how an environment evolves, physical decision-making requires closing the loop: actions actively alter future state distributions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [未来研究愿景 Future Research Agenda]:** *Bridging my background in non-stationary time-series modeling and fast flow-matching samplers, my future research aims to build adaptive visuomotor world models that allow embodied robots to generalize across unfamiliar contact regimes in real time.*
  对应步骤：S4 [未来研究愿景 Future Research Agenda]：Building on my foundation in continuous-time generative modeling, my doctoral/future research aims to develop sample-efficient visuomotor world models that enable robots to adapt online to non-stationary contact dynamics. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5 Category V: Over-Nominalization & Passive Abuse / 过度名词化与被动语态滥用

17.5.1 Case RC49: 六级/雅思大作文：AI 自动化、教育与科研伦理 (CET-6 & IELTS Task 2 Technology & Society Essay)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Linked Architecture: Blueprint `BP01` | Logic Rule `L019`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In the rapidly developing modern world, time series forecasting plays a pivotal role in many fields. The fact that industrial sensors often have noise and distribution changes makes forecasting very hard and bad for decision making.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Constituent Alignment & Rhetorical Moves

- **S1 [任务锚定 Topic Anchor]:** *Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.*
  对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [应用场景展开 Concrete Scenarios]:** *In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.*

对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [苛刻工况约束 Operational Bottleneck]:** *However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.*
  对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [研究必要性收束 Imperative Hook]:** *Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.*
  对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.2 Case RC50: 六级/雅思大作文：AI 自动化、教育与科研伦理 (CET-6 & IELTS Task 2 Technology & Society Essay)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想客观概括前人主流模型（如 Transformer / Diffusion / MLP）取得了什么进展，但点出它们共同依赖的一个脆弱前提假设。

Linked Architecture: Blueprint `BP02` | Logic Rule `L020`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Many researchers have studied this problem. Wu et al. used Autoformer. Nie et al. used PatchTST. They are very good, but they all think the data distribution does not change, which is wrong in real life.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition. These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments. Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window. When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.

Constituent Alignment & Rhetorical Moves

- **S1 [流派归类 Taxonomic Opening]:** *Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition.*
  对应步骤：S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表性机制展开 Mechanism Recap]:** *These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments.*
  对应步骤：S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [转折揭示隐含假设 Pivot to Hidden Assumption]:** *Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window.*
  对应步骤：S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [假设破裂后果 Breakdown Under Stress]:** *When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.*
  对应步骤：S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.3 Case RC51: 六级/雅思大作文：AI 自动化、教育与科研伦理 (CET-6 & IELTS Task 2 Technology & Society Essay)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Linked Architecture: Blueprint `BP03` | Logic Rule `L021`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

The problem of error accumulation in long-term forecasting and the losing of scale information when using RevIN normalization is very serious and hard to solve by simple ways.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Constituent Alignment & Rhetorical Moves

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*
  对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*
  对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.4 Case RC52: 六级/雅思大作文：AI 自动化、教育与科研伦理 (CET-6 & IELTS Task 2 Technology & Society Essay)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。
Linked Architecture: Blueprint `BP04` | Logic Rule `L022`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
In this paper, we propose a novel and groundbreaking model called DyTS-Net. It has three modules: Module A, Module B, and Module C. They work together to delve into the data and get state-of-the-art performance.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch. Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Constituent Alignment & Rhetorical Moves

- **S1 [方法登场与定位 Method Introduction]:** *To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.*
  对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心直觉洞察 Key Insight]:** *Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.*
  对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [模块协同主干 Module Synergy]:** *Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.*
  对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [闭环收益总结 Closed-Loop Benefit]:** *Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.*
  对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.5 Case RC53: 六级/雅思大作文：AI 自动化、教育与科研伦理 (CET-6 & IELTS Task 2 Technology & Society Essay)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。

Linked Architecture: Blueprint `BP05` | Logic Rule `L023`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

There are two challenges. The first challenge is that noise is big. We use a filter. The second challenge is that speed is slow. We use distillation.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Constituent Alignment & Rhetorical Moves

- **S1 [落地难点总起 Implementation Hurdle]:** *Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.*
  对应步骤：S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [难点 1 + 靶向设计 1]:** *First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.*
  对应步骤：S2 [难点 1 + 靶向设计 1]: First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [难点 2 + 靶向设计 2]:** *Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.*
  对应步骤：S3 [难点 2 + 靶向设计 2]: Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [整体计算/泛化特性 System Property]:** *Together, these designs sustain 50 Hz reactive control under partial visual occlusion.*
  对应步骤：S4 [整体计算/泛化特性 System Property]: Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.6 Case RC54: 六级/雅思大作文：AI 自动化、教育与科研伦理 (CET-6 & IELTS Task 2 Technology & Society Essay)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Introduction 结尾前，用三句话硬核总结我们在哪些基准数据集/真实硬件上做了实验，取得了多大幅度的领先。
Linked Architecture: Blueprint `BP06` | Logic Rule `L024`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
We did comprehensive experiments on many datasets. The results show that our model beats all baselines by a large margin and is very fast.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors. Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions. Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.

Constituent Alignment & Rhetorical Moves

- **S1 [评测广度 Scope of Evaluation]:** *We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors.*
  对应步骤：S1 [评测广度 Scope of Evaluation]：We evaluate [Model] across [N] public benchmarks and [Real-world Hardware/Scenario], spanning [Tasks]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心精度突破 Primary Accuracy Gain]:** *Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions.*
  对应步骤：S2 [核心精度突破 Primary Accuracy Gain]：实验主指标对比（必须包含具体百分比、最强基线名称及困难场景表现）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]:** *Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.*
  对应步骤：S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]：报告计算速度、显存占用或零样本跨场景泛化收益。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.7 Case RC55: 托福独立与综合写作：理论基础与工程实践权衡 (TOEFL Independent & Integrated Academic Writing)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Introduction 最末尾清晰列出本文的三点核心贡献（理论/视角贡献 + 核心算法/架构贡献 + 实验与开源实证贡献）。

Linked Architecture: Blueprint `BP07` | Logic Rule `L025`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Our contributions are: 1. A new model is proposed. 2. Solving the non-stationary problem. 3. Good results on datasets.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability. (2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker. (3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.

Constituent Alignment & Rhetorical Moves

- **S1 [总起引导句 Lead-in]:** *In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability.*
  对应步骤：S1 [总起引导句 Lead-in]：In summary, our main contributions are threefold: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [贡献 1:** *(2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker.*
  对应步骤：S2 [贡献 1：新视角/问题形式化 Conceptual/Formulation Contribution]：用强动词名物化或动词原形开头（We identify and formalize... / We establish...），点明打破了什么传统局限。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [贡献 2:** *(3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.*

对应步骤：S3 [贡献 2：核心方法与模块 Algorithmic/Architectural Contribution]：We propose [Model/Mechanism], which integrates [A] and [B] to achieve [Theoretical/Functional Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.8 Case RC56: 托福独立与综合写作：理论基础与工程实践权衡 (TOEFL Independent & Integrated Academic Writing)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Linked Architecture: Blueprint `BP08` | Logic Rule `L026`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

Time series imputation is very important in the era of big data. Existing generative models delve into the data distribution but are too slow and fail when missing rates are high. We propose FlowImp, a novel framework that uses conditional flow matching and frequency guidance. Extensive experiments show that FlowImp achieves state-of-the-art results on all benchmarks.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under

50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Constituent Alignment & Rhetorical Moves

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*
  对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.9 Case RC57: 托福独立与综合写作：理论基础与工程实践权衡 (TOEFL Independent & Integrated Academic Writing)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想写 Related Work 的一个子方向：把过去三年的 6–8 篇代表性论文按技术路线分成两类进行归纳，并指出它们的共同盲区。

Linked Architecture: Blueprint `BP09` | Logic Rule `L027`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In recent years, many works studied time series foundation models. TimeGPT trained a transformer on large data. Moment used masked pre-training. Chronos quantized time series into tokens. Lag-Llama used lag features. However, our model is different from them.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization. Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains. Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives. Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.

Constituent Alignment & Rhetorical Moves

- **S1 [分类总纲 Taxonomic Thesis]:** *Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization.*
  对应步骤：S1 [分类总纲 Taxonomic Thesis]：Existing methods for [Task] can be categorized into two main families: [Family A] and [Family B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [流派 A 归纳与代表作 Family A Synthesis]:** *Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains.*
  对应步骤：S2 [流派 A 归纳与代表作 Family A Synthesis]：[Family A] approaches (e.g., [Models]) focus on [Mechanism A], effectively addressing [Sub-problem 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [流派 B 归纳与对比 Family B Synthesis]:** *Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives.*
  对应步骤：S3 [流派 B 归纳与对比 Family B Synthesis]：By contrast, [Family B] architectures (e.g., [Models]) reformulate the problem via [Mechanism B] to reduce [Cost/Complexity]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [共同盲区提炼 Shared Blind Spot]:** *Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.*
  对应步骤：S4 [共同盲区提炼 Shared Blind Spot]：Despite their respective strengths, both families overlook [Critical Factor], which degrades performance under [Target Scenario]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.10 Case RC58: 托福独立与综合写作：理论基础与工程实践权衡 (TOEFL Independent & Integrated Academic Writing)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Related Work 小节末尾，把我们的方法与最接近的 2 个竞争方案做精准对比，防止审稿人说我们只是对某篇旧文的微调（Incremental）。
Linked Architecture: Blueprint `BP10` | Logic Rule `L028`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Some works also use frequency domain like us, such as FEDformer and TimesNet. But our method is much better and more novel because we use a new module.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach. While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes. In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases. Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.

Constituent Alignment & Rhetorical Moves

- **S1 [锁定最相近工作 Closest Prior Work]:** *Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach.*
  对应步骤：S1 [锁定最相近工作 Closest Prior Work]：Among prior studies, [Closest Method A] and [Closest Method B] are most closely related to our approach. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [剖析其局限性 Structural Difference 1]:** *While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes.*
  对应步骤：S2 [剖析其局限性 Structural Difference 1]：While [Method A] also employs [Shared Concept], it relies on [Limitation 1], restricting its applicability to [Narrow Setting]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [剖析本文本质突破 Core Departure]:** *In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases.*
  对应步骤：S3 [剖析本文本质突破 Core Departure]：In contrast, our framework introduces [Core Innovation], which fundamentally eliminates [Bottleneck] by [Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证或理论优势锁定 Empirical/Theoretical Edge]:** *Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.*
  对应步骤：S4 [实证或理论优势锁定 Empirical/Theoretical Edge]：Consequently, our approach achieves [Capability] without requiring [Expensive Requirement of Prior Work]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.11 Case RC59: 托福独立与综合写作：理论基础与工程实践权衡 (TOEFL Independent & Integrated Academic Writing)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想在 Methodology 开篇（Section 3.1 Problem Formulation），严谨、简洁地定义输入张量、输出目标、约束条件与优化目标。

Linked Architecture: Blueprint `BP11` | Logic Rule `L029`

Defective Paragraph vs. Publication Revision

[Defective Draft]:

In this section, we define the problem. The input is X. X has length L and C channels. The output is Y. Y has length H. We want to make the error between predicted Y and true Y as small as possible.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Consider a multivariate time series $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ recording C correlated sensors over lookback horizon L. Given an binary observation mask $\mathbf{M} \in \{0, 1\}^{L \times C}$ indicating irregular sensor dropouts, our objective is to learn a generative mapping $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ over future horizon H. Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where $\mathbb{E}[x_t]$ and $\mathrm{Cov}(x_t)$ evolve continuously between the lookback and forecasting windows.

Constituent Alignment & Rhetorical Moves

- **S1 [任务总体形式化 Task Definition]:** *Consider a multivariate time series* $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ *recording C correlated sensors over lookback horizon L.*
  对应步骤: S1 [任务总体形式化 Task Definition]: Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [补充条件或模态输入 Auxiliary/Condition Inputs]:** *Given an binary observation mask* $\mathbf{M} \in \{0,1\}^{L\times C}$ *indicating irregular sensor dropouts, our objective is to learn a generative mapping* $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ *over future horizon* H.
  对应步骤：S2 [补充条件或模态输入 Auxiliary/Condition Inputs]：At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [预测/控制目标 Objective]:** *Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where* $\mathbb{E}[x_t]$ *and* $Cov(x_t)$ *evolve continuously between the lookback and forecasting windows.*
  对应步骤：S3 [预测/控制目标 Objective]：Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H\times C}$ over horizon H (or action trajectory $a_{t:t+K}$). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

17.5.12 Case RC60: 托福独立与综合写作：理论基础与工程实践权衡 (TOEFL Independent & Integrated Academic Writing)

Clinical Profile & Diagnostic Focus

Communicative Goal: 我这一段想配合论文里的 Figure 2（模型总架构图），用一段话把输入张量从左到右流经各个模块直到输出预测结果的完整全貌讲清楚。
Linked Architecture: Blueprint `BP12` | Logic Rule `L030`

Defective Paragraph vs. Publication Revision

[Defective Draft]:
Figure 2 shows our model. There is a Patch Encoder on the left. In the middle, there is a Spectral Block and a Graph Block. On the right, a Linear Decoder outputs the result.

[4-Step Reconstructive Surgery]:

Step 1: 第一步【砍壳提主干】：删去句首 AI 修辞帽与口语词，将核心实体（≤5 词）提作全句主语。

Step 2: 第二步【强动词替换】：将 is very bad / makes it good 替换为精准及物动词（degrades / stabilizes / decouples / reduces）。

Step 3: 第三步【咬合信息流】：在第二、三句句首使用 'This/Such + 汇总名词 ' 承接上一句宾语，形成 Given-to-New 锁链。

Step 4: 第四步【注入实证锚点】：补全具体评测指标、对比基线或微观工程场景，彻底消除空洞夸张。

[Publication-Ready Revision]:

Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages. Given an input sequence $\mathbf{X}$, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches $\mathbf{Z}_0$. These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph. Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output $\hat{\mathbf{Y}}$.

Constituent Alignment & Rhetorical Moves

- **S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]:** *Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages.*
  对应步骤：S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]：Figure 2 illustrates the overall architecture of [Model], which processes [Input] through three sequential stages: [Stage 1], [Stage 2], and [Stage 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [阶段 1 数据流 Stage 1 Flow]:** *Given an input sequence* $\mathbf{X}$*, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches* $\mathbf{Z}_0$.
  对应步骤：S2 [阶段 1 数据流 Stage 1 Flow]：Given raw input $\mathbf{X}$, the [Module 1] first transforms [Raw Input] into [Intermediate Representation $\mathbf{Z}^{(1)}$] to [Functional Purpose 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [阶段 2 数据流 Stage 2 Flow]:** *These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph.*
  对应步骤：S3 [阶段 2 数据流 Stage 2 Flow]：Next, these [Summary Noun: representations $\mathbf{Z}^{(1)}$] are fed into [Module 2], which [Mechanism 2] to produce [Refined State $\mathbf{Z}^{(2)}$]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]:** *Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output* $\hat{\mathbf{Y}}$.
  对应步骤：S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]：Finally, the [Module 3] integrates $\mathbf{Z}^{(2)}$ with [Prior/Scale Cue] to output the final [Prediction/Action Trajectory]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Part VI

Full-Text Anthology
全篇范文与段落蓝图链拆解文库

Chapter 18

Computer Science & Artificial Intelligence / 计算机科学与人工智能

本章精选该学科领域及对应文体的完整学术范文。每一篇范文不仅展示母语级发表标准的正文全貌，更清晰拆解全篇的“段落蓝图链条（Structural Blueprint Chain）”，并对每个自然段逐一剖析其核心交际意图、句际衔接逻辑公式（Cohesion Formula）以及每句话的句法成分与推进机制。

18.1 FT01: 非平稳多变量时间序列预测完整小论文（DyTS-Net: Decoupling Spectral Phase and Continuous Scale）

Composition Architecture & Structural Blueprint Chain

Genre: `AI_Mini_Paper` **Paragraph Count:** 12

Blueprint Chain: `BP08` → `BP01` → `BP02` → `BP03` → `BP04` → `BP07` → `BP11` → `BP12` → `BP13` → `BP18` → `BP19` → `BP24`

Paragraph 1: 高密度四句式 Abstract 全段构建（Complete 4-Move Abstract）(BP08)

Expressive Intent: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Cohesion Formula: Context (S1) → Gap (S2) → Method & Mechanism (S3-S4) → Quantitative Proof (S5). 绑定逻辑规则：L001, L008, L010, L029.

Publication-Grade Authentic Text

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*
  对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 确立研究背景与真实工程价值（拒绝空喊口号）(BP01)

Expressive Intent: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Cohesion Formula: Given-to-New 链：Core Task (S1) → Practical Deployment (S2) → Real-world Disturbances (S3) → Consequence of Failure (S4). 绑定逻辑规则：L001, L011, L029.

Publication-Grade Authentic Text

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务锚定 Topic Anchor]:** *Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.*
  对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [应用场景展开 Concrete Scenarios]:** *In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.*
  对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [苛刻工况约束 Operational Bottleneck]:** *However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.*
  对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [研究必要性收束 Imperative Hook]:** *Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.*
  对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 总结现有主流范式及其隐含假设（欲抑先扬铺垫段）(BP02)

Expressive Intent: 我这一段想客观概括前人主流模型（如 Transformer / Diffusion / MLP）取得了什么进展，但点出它们共同依赖的一个脆弱前提假设。

Cohesion Formula: 先让步肯定（While/Recent paradigms...）→ 后釜底抽薪揭示隐含前提（Yet these architectures implicitly assume...）。绑定逻辑规则：L008, L009.

Publication-Grade Authentic Text

Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition. These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments. Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window. When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [流派归类 Taxonomic Opening]:** *Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition.*
  对应步骤：S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表性机制展开 Mechanism Recap]:** *These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments.*
  对应步骤：S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [转折揭示隐含假设 Pivot to Hidden Assumption]:** *Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window.*
  对应步骤：S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [假设破裂后果 Breakdown Under Stress]:** *When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.*
  对应步骤：S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 尖锐指出核心技术瓶颈与失效机理（Gap Identification）(BP03)

Expressive Intent: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Cohesion Formula: 因果链式推进（A induces B → This B compounds into C → Naive fix D sacrifices E）。绑定逻辑规则：L003, L004, L024/L025.

Publication-Grade Authentic Text

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*
  对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*
  对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 亮出本文核心洞察与破局方法总览（Key Insight & Method Overview）(BP04)

Expressive Intent: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。

Cohesion Formula: Problem Anchor → Key Insight → Dual-Module Mechanism → Pareto Resolution. 绑定逻辑规则：L002, L023.

Publication-Grade Authentic Text

To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch. Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [方法登场与定位 Method Introduction]:** *To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.*
  对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心直觉洞察 Key Insight]:** *Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.*
  对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [模块协同主干 Module Synergy]:** *Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.*
  对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [闭环收益总结 Closed-Loop Benefit]:** *Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.*

对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 凝练论文三点核心贡献（Contributions Summary）(BP07)

Expressive Intent: 我这一段想在 Introduction 最末尾清晰列出本文的三点核心贡献（理论/视角贡献 + 核心算法/架构贡献 + 实验与开源实证贡献）。

Cohesion Formula: Parallel Grammatical Structure（严格平行结构：全部用 We + 强及物动词 identify / propose / demonstrate 开头）。绑定逻辑规则：L002, L010.

Publication-Grade Authentic Text

In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability. (2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker. (3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总起引导句 Lead-in]:** *In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability.*
  对应步骤：S1 [总起引导句 Lead-in]：In summary, our main contributions are threefold: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [贡献 1:** *(2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker.*
  对应步骤：S2 [贡献 1：新视角/问题形式化 Conceptual/Formulation Contribution]：用强动词名物化或动词原形开头（We identify and formalize... / We establish...），点明打破了什么传统局限。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [贡献 2:** *(3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.*
  对应步骤：S3 [贡献 2：核心方法与模块 Algorithmic/Architectural Contribution]：We propose [Model/Mechanism], which integrates [A] and [B] to achieve [Theoretical/Functional Guarantee]. | 句首锚定旧

信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 7: 问题形式化定义与符号约定段（Problem Formulation & Notation）(BP11)

Expressive Intent: 我这一段想在 Methodology 开篇（Section 3.1 Problem Formulation），严谨、简洁地定义输入张量、输出目标、约束条件与优化目标。

Cohesion Formula: Input Space → Auxiliary Conditions → Target Mapping → Crucial Distributional Constraint.
绑定逻辑规则：L014, L011.

Publication-Grade Authentic Text

Consider a multivariate time series $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ recording C correlated sensors over lookback horizon L. Given an binary observation mask $\mathbf{M} \in \{0, 1\}^{L \times C}$ indicating irregular sensor dropouts, our objective is to learn a generative mapping $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ over future horizon H. Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where $\mathbb{E}[x_t]$ and $\mathrm{Cov}(x_t)$ evolve continuously between the lookback and forecasting windows.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务总体形式化 Task Definition]:** *Consider a multivariate time series* $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ *recording* C *correlated sensors over lookback horizon* L.
  对应步骤：S1 [任务总体形式化 Task Definition]: Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [补充条件或模态输入 Auxiliary/Condition Inputs]:** *Given an binary observation mask* $\mathbf{M} \in \{0, 1\}^{L \times C}$ *indicating irregular sensor dropouts, our objective is to learn a generative mapping* $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ *over future horizon* H.
  对应步骤：S2 [补充条件或模态输入 Auxiliary/Condition Inputs]: At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [预测/控制目标 Objective]:** *Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where* $\mathbb{E}[x_t]$ *and* $Cov(x_t)$ *evolve continuously between the lookback and forecasting windows.*
  对应步骤：S3 [预测/控制目标 Objective]: Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$ over horizon H (or action trajectory $a_{t:t+K}$). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 8: 整体模型架构数据流俯瞰段（Framework Overview & Pipeline Walk-through）(BP12)

Expressive Intent: 我这一段想配合论文里的 Figure 2（模型总架构图），用一段话把输入张量从左到右流经各个模块直到输出预测结果的完整全貌讲清楚。

Cohesion Formula: 严格的链式数据流（$\mathbf{X} \rightarrow \mathbf{Z}^{(1)} \rightarrow \mathbf{Z}^{(2)} \rightarrow \hat{\mathbf{Y}}$），每一句句首承接上一句的输出张量。绑定逻辑规则：L001, L003, L004.

Publication-Grade Authentic Text

Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages. Given an input sequence $\mathbf{X}$, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches $\mathbf{Z}_0$. These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph. Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output $\hat{\mathbf{Y}}$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]:** *Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages.*
  对应步骤：S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]: Figure 2 illustrates the overall architecture of [Model], which processes [Input] through three sequential stages: [Stage 1], [Stage 2], and [Stage 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [阶段 1 数据流 Stage 1 Flow]:** *Given an input sequence* $\mathbf{X}$*, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches* $\mathbf{Z}_0$.
  对应步骤：S2 [阶段 1 数据流 Stage 1 Flow]: Given raw input $\mathbf{X}$, the [Module 1] first transforms [Raw Input] into [Intermediate Representation $\mathbf{Z}^{(1)}$] to [Functional Purpose 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [阶段 2 数据流 Stage 2 Flow]:** *These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph.*
  对应步骤：S3 [阶段 2 数据流 Stage 2 Flow]: Next, these [Summary Noun: representations $\mathbf{Z}^{(1)}$] are fed into [Module 2], which [Mechanism 2] to produce [Refined State $\mathbf{Z}^{(2)}$]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]:** *Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output* $\hat{\mathbf{Y}}$.

对应步骤：S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]：Finally, the [Module 3] integrates $\mathbf{Z}^{(2)}$ with [Prior/Scale Cue] to output the final [Prediction/Action Trajectory]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 9: 核心子模块：直觉动机与数学设计无缝衔接段（Intuition-before-Math）(BP13)

Expressive Intent: 我这一段想详细介绍我们提出的某个核心创新模块：先用两句话讲清楚为什么直接用传统算子会出问题，再顺理成章推导出我们的新公式。

Cohesion Formula: Pathology of Standard Operator → Physical Intuition → Equation Integration → Theoretical Guarantee. 绑定逻辑规则：L014, L003.

Publication-Grade Authentic Text

Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps. To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies. Formally, given query and key projections $Q, K \in \mathbb{R}^{N\times d}$, we modulate the attention logits via a phase-alignment mask: $\mathcal{A}(Q,K,V) = \text{softmax}\big(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1\big)V$, where $\Delta\Phi_{Q,K} \in [-\pi,\pi]^{N\times N}$ denotes the instantaneous Hilbert phase difference matrix and $\lambda > 0$ controls phase selectivity. Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [传统算子缺陷 Motivation]:** *Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps.*
  对应步骤：S1 [传统算子缺陷 Motivation]：Standard [Conventional Operator, e.g., softmax self-attention] computes [Operation], which suffers from [Pathology] when applied to [Domain Condition]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [物理/统计直觉 Physical Intuition]:** *To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies.*
  对应步骤：S2 [物理/统计直觉 Physical Intuition]：To overcome this pathology, we observe that [Key Domain Property] can be effectively regularized by [Intuitive Operation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [数学公式定义 Mathematical Formulation]:** *Formally, given query and key projections $Q, K \in \mathbb{R}^{N\times d}$, we modulate the attention logits via a phase-alignment mask: $\mathcal{A}(Q,K,V) =$*

$softmax\left(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1\right)V$, *where* $\Delta\Phi_{Q,K} \in [-\pi,\pi]^{N\times N}$ *denotes the instantaneous Hilbert phase difference matrix and* $\lambda > 0$ *controls phase selectivity.*

对应步骤：S3 [数学公式定义 Mathematical Formulation]：Formally, given input representation $H \in \mathbb{R}^{N\times d}$, we formulate the [Module Name] as: [Equation], where [Symbol Explanations]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [理论/计算性质 Guarantee]:** *Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.*

  对应步骤：S4 [理论/计算性质 Guarantee]：By construction, this formulation guarantees [Desirable Property, e.g., $O(N)$ complexity / translation invariance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 10: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）(BP18)

Expressive Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Cohesion Formula: Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则：L015, L013, L010.

Publication-Grade Authentic Text

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*

  对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.*
  对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.*
  对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*
  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 11: 消融实验逐组件归因论证段（Counterfactual Ablation Study）(BP19)

Expressive Intent: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Cohesion Formula: Ablation Overview → w/o Module A (Specific Failure) → w/o Module B (Specific Failure) → Cross-Module Synergy. 绑定逻辑规则：L016, L007.

Publication-Grade Authentic Text

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load

spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [消融实验总括 Ablation Setup]:** *To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.*
  对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]:** *Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.*
  对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]:** *Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.*
  对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同效应总结 Synergistic Conclusion]:** *Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.*
  对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 12: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）(BP24)

Expressive Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Cohesion Formula: Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则：L013, L023, L029.

Publication-Grade Authentic Text

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*
  对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

18.2 FT02: 基于条件流匹配的块缺失时间序列生成与插补完整小论文（FlowImp: Manifold-Consistent Time Series Imputation）

Composition Architecture & Structural Blueprint Chain

Genre: `AI_Mini_Paper` **Paragraph Count:** 11

Blueprint Chain: `BP08` → `BP01` → `BP09` → `BP10` → `BP11` → `BP15` → `BP17` → `BP18` → `BP20` → `BP23` → `BP24`

Paragraph 1: 高密度四句式 Abstract 全段构建（Complete 4-Move Abstract）(BP08)

Expressive Intent: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Cohesion Formula: Context (S1) → Gap (S2) → Method & Mechanism (S3-S4) → Quantitative Proof (S5). 绑定逻辑规则：L001, L008, L010, L029.

Publication-Grade Authentic Text

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*
  对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 确立研究背景与真实工程价值（拒绝空喊口号）(BP01)

Expressive Intent: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Cohesion Formula: Given-to-New 链：Core Task (S1) → Practical Deployment (S2) → Real-world Disturbances (S3) → Consequence of Failure (S4). 绑定逻辑规则：L001, L011, L029.

Publication-Grade Authentic Text

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务锚定 Topic Anchor]:** *Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.*

对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [应用场景展开 Concrete Scenarios]:** *In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.*
  对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [苛刻工况约束 Operational Bottleneck]:** *However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.*
  对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [研究必要性收束 Imperative Hook]:** *Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.*
  对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 相关工作分类综述段（拒绝 A did X, B did Y 流水账）(BP09)

Expressive Intent: 我这一段想写 Related Work 的一个子方向：把过去三年的 6–8 篇代表性论文按技术路线分成两类进行归纳，并指出它们的共同盲区。

Cohesion Formula: Two-Branch Taxonomy → Family A → Contrastive Family B → Shared Limitation. 绑定逻辑规则：L009, L008.

Publication-Grade Authentic Text

Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization. Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains. Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives. Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分类总纲 Taxonomic Thesis]:** *Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization.*

对应步骤：S1 [分类总纲 Taxonomic Thesis]：Existing methods for [Task] can be categorized into two main families: [Family A] and [Family B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [流派 A 归纳与代表作 Family A Synthesis]:** *Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains.*
  对应步骤：S2 [流派 A 归纳与代表作 Family A Synthesis]：[Family A] approaches (e.g., [Models]) focus on [Mechanism A], effectively addressing [Sub-problem 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [流派 B 归纳与对比 Family B Synthesis]:** *Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives.*
  对应步骤：S3 [流派 B 归纳与对比 Family B Synthesis]：By contrast, [Family B] architectures (e.g., [Models]) reformulate the problem via [Mechanism B] to reduce [Cost/Complexity]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [共同盲区提炼 Shared Blind Spot]:** *Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.*
  对应步骤：S4 [共同盲区提炼 Shared Blind Spot]：Despite their respective strengths, both families overlook [Critical Factor], which degrades performance under [Target Scenario]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 对比现有流派优劣并精准定位本文差异（Contrastive Positioning）(BP10)

Expressive Intent: 我这一段想在 Related Work 小节末尾，把我们的方法与最接近的 2 个竞争方案做精准对比，防止审稿人说我们只是对某篇旧文的微调（Incremental）。

Cohesion Formula: Acknowledge Overlap → Pinpoint Structural Divergence → Explain Functional Gain. 绑定逻辑规则：L008, L019.

Publication-Grade Authentic Text

Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach. While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes. In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases. Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [锁定最相近工作 Closest Prior Work]:** *Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach.*
  对应步骤：S1 [锁定最相近工作 Closest Prior Work]：Among prior studies, [Closest Method A] and [Closest Method B] are most closely related to our approach. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [剖析其局限性 Structural Difference 1]:** *While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes.*
  对应步骤：S2 [剖析其局限性 Structural Difference 1]：While [Method A] also employs [Shared Concept], it relies on [Limitation 1], restricting its applicability to [Narrow Setting]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [剖析本文本质突破 Core Departure]:** *In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases.*
  对应步骤：S3 [剖析本文本质突破 Core Departure]：In contrast, our framework introduces [Core Innovation], which fundamentally eliminates [Bottleneck] by [Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证或理论优势锁定 Empirical/Theoretical Edge]:** *Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.*
  对应步骤：S4 [实证或理论优势锁定 Empirical/Theoretical Edge]：Consequently, our approach achieves [Capability] without requiring [Expensive Requirement of Prior Work]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 问题形式化定义与符号约定段（Problem Formulation & Notation）(BP11)

Expressive Intent: 我这一段想在 Methodology 开篇（Section 3.1 Problem Formulation），严谨、简洁地定义输入张量、输出目标、约束条件与优化目标。

Cohesion Formula: Input Space → Auxiliary Conditions → Target Mapping → Crucial Distributional Constraint.
绑定逻辑规则：L014, L011.

Publication-Grade Authentic Text

Consider a multivariate time series $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ recording C correlated sensors over lookback horizon L. Given an binary observation mask $\mathbf{M} \in \{0, 1\}^{L \times C}$ indicating irregular sensor dropouts, our objective is to learn a generative mapping $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ over future horizon H. Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where $\mathbb{E}[x_t]$ and $\mathrm{Cov}(x_t)$ evolve continuously between the lookback and forecasting windows.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务总体形式化 Task Definition]:** *Consider a multivariate time series* $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ *recording C correlated sensors over lookback horizon L.*
  对应步骤: S1 [任务总体形式化 Task Definition]: Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [补充条件或模态输入 Auxiliary/Condition Inputs]:** *Given an binary observation mask* $\mathbf{M} \in \{0, 1\}^{L \times C}$ *indicating irregular sensor dropouts, our objective is to learn a generative mapping* $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ *over future horizon H.*
  对应步骤: S2 [补充条件或模态输入 Auxiliary/Condition Inputs]: At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [预测/控制目标 Objective]:** *Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where* $\mathbb{E}[x_t]$ *and* $Cov(x_t)$ *evolve continuously between the lookback and forecasting windows.*
  对应步骤: S3 [预测/控制目标 Objective]: Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$ over horizon H (or action trajectory $a_{t:t+K}$). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 生成模型专属：条件控制与采样去噪轨迹阐述段（Generative Modeling Mechanism）(BP15)

Expressive Intent: 我这一段想阐述生成模型（Diffusion / Flow Matching / Autoregressive）如何在满足外部物理约束或条件观测的前提下，生成高质量且多样化的序列/轨迹。

Cohesion Formula: Guidance Artifact Problem → Intrinsic Velocity Conditioning → Tangent Projection Step → Fast Exact Convergence. 绑定逻辑规则：L026, L014.

Publication-Grade Authentic Text

Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations. To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field. At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator. This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]:** *Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations.*
  对应步骤：S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]：Applying post-hoc classifier guidance to pre-trained diffusion samplers often pushes intermediate noisy states off the data manifold, inducing high-frequency artifacts. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]:** *To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field.*
  对应步骤：S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]：To enforce hard boundary constraints without manifold deviation, we embed [Condition/Constraint] directly into the velocity field / denoising kernel. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具体轨迹更新方程 Trajectory Update Formulation]:** *At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator.*
  对应步骤：S3 [具体轨迹更新方程 Trajectory Update Formulation]：At solver step $\tau \in [0, 1]$, the conditional velocity field $v_\theta(x_\tau, \tau \mid c)$ projects the unconstrained flow onto the tangent space of feasible trajectories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]:** *This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.*
  对应步骤：S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]：This tangent projection guarantees exact boundary satisfaction at $\tau = 1$ while permitting straight 4-step Euler integration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 7: 实验设置、数据集特性与强基线选型说明段（Experimental Setup & Baselines）(BP17)

Expressive Intent: 我这一段想在 Experiments 开篇说明：我们为什么选这几类数据集和这几类基线模型，证明我们的实验设计是全面、公平且直击痛点的。

Cohesion Formula: Dataset Diversity → Challenging Regime Highlights → Categorized Strong Baselines → Strict Fairness Protocol. 绑定逻辑规则：L009, L010.

Publication-Grade Authentic Text

To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic). Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability. We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer). All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [数据集选型逻辑 Dataset Selection Rationale]:** *To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic).*
  对应步骤：S1 [数据集选型逻辑 Dataset Selection Rationale]：To evaluate performance across diverse non-stationary regimes, we conduct experiments on [N] benchmarks spanning [Domain 1], [Domain 2], and [Domain 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [极端/困难特性突出 Challenging Dataset Properties]:** *Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability.*
  对应步骤：S2 [极端/困难特性突出 Challenging Dataset Properties]：Notably, [Dataset A] exhibits [Hard Property 1, e.g., strong covariate shift], whereas [Dataset B] tests scalability across [Hard Property 2, e.g., 862 correlated channels]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [基线流派全覆盖 Baseline Family Coverage]:** *We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer).*
  对应步骤：S3 [基线流派全覆盖 Baseline Family Coverage]：We compare our method against [K] competitive baselines grouped into three families: [Family 1], [Family 2], and [Family 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [公平评测协议 Fair Evaluation Protocol]:** *All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.*
  对应步骤：S4 [公平评测协议 Fair Evaluation Protocol]：All models are evaluated under identical lookback windows ($L = 96$), prediction horizons ($H \in \{96, 192, 336, 720\}$), and hyperparameter search

budgets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 8: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）(BP18)

Expressive Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Cohesion Formula: Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则：L015, L013, L010.

Publication-Grade Authentic Text

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*
  对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.*
  对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates*

by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.

对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*

  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 9: 超参数敏感性与极端噪声鲁棒性分析段（Sensitivity & Robustness Analysis）(BP20)

Expressive Intent: 我这一段想回应审稿人最常问的：你的模型是不是对超参数很敏感？在缺失值或传感器强噪声下会不会突然崩盘？

Cohesion Formula: Goal → Wide Hyperparameter Plateau → Stress-Test Comparison → Mechanistic Explanation. 绑定逻辑规则：L010, L019.

Publication-Grade Authentic Text

Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption. Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$. Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), pointwise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%. This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分析目的 Objective]:** *Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption.*

  对应步骤：S1 [分析目的 Objective]：We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [超参数平坦区间 Plateau Stability]:** *Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$.*
  对应步骤：S2 [超参数平坦区间 Plateau Stability]：Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [极端边缘工况压力测试 Stress Test under Corruption]:** *Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%.*
  对应步骤：S3 [极端边缘工况压力测试 Stress Test under Corruption]：Under synthetic sensor dropouts up to [40%] and Gaussian noise ($\sigma = 0.3$), baseline accuracy plummets by over [35%], whereas our model degrades by less than [6.5%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [鲁棒性物理归因 Robustness Mechanism]:** *This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.*
  对应步骤：S4 [鲁棒性物理归因 Robustness Mechanism]：This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 10: 局限性（Limitations）客观陈述与未来扩展段（Constructive Limitations）(BP23)

Expressive Intent: 我这一段想写论文末尾或 NeurIPS/ICLR Checklist 要求的 Limitations 段落：诚实指出当前方法的两个客观边界，但绝不贬损论文的核心贡献。

Cohesion Formula: Two Objective Boundaries → [Boundary 1 + Technical Remedy] → [Boundary 2 + Stochastic Extension]. 绑定逻辑规则：L017.

Publication-Grade Authentic Text

Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting. First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy. Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration. Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [边界 1:** *Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting.*
  对应步骤：S1 [边界 1：极端高维或极度稀疏场景 Boundary 1]：Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [边界 1 机理分析与解法 Mechanism & Solution 1]:** *First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy.*
  对应步骤：S2 [边界 1 机理分析与解法 Mechanism & Solution 1]：First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [边界 2:** *Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration.*
  对应步骤：S3 [边界 2：特定假设限制 Boundary 2]：Second, our continuous scale tracker assumes piecewise differentiable drift and may under-estimate instantaneous discontinuous jumps caused by hardware resets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [未来演进方向 Future Extension]:** *Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.*
  对应步骤：S4 [未来演进方向 Future Extension]：Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 11: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）(BP24)

Expressive Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Cohesion Formula: Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则：L013, L023, L029.

Publication-Grade Authentic Text

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evalu-

ations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*
  对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

18.3 FT03: 时频双域解耦的长序列异常检测与预测完整小论文（WavePatch: Multi-Scale Wavelet-Temporal Mixing）

Composition Architecture & Structural Blueprint Chain

Genre: `AI_Mini_Paper` **Paragraph Count:** 11

Blueprint Chain: `BP08` → `BP02` → `BP03` → `BP05` → `BP06` → `BP12` → `BP14` → `BP18` → `BP19` → `BP21` → `BP24`

Paragraph 1: 高密度四句式 Abstract 全段构建（Complete 4-Move Abstract）(BP08)

Expressive Intent: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Cohesion Formula: Context (S1) → Gap (S2) → Method & Mechanism (S3-S4) → Quantitative Proof (S5). 绑定逻辑规则：L001, L008, L010, L029.

Publication-Grade Authentic Text

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*
  对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 总结现有主流范式及其隐含假设（欲抑先扬铺垫段）(BP02)

Expressive Intent: 我这一段想客观概括前人主流模型（如 Transformer / Diffusion / MLP）取得了什么进展，但点出它们共同依赖的一个脆弱前提假设。

Cohesion Formula: 先让步肯定（While/Recent paradigms...）→ 后釜底抽薪揭示隐含前提（Yet these architectures implicitly assume...）。绑定逻辑规则：L008, L009.

Publication-Grade Authentic Text

Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition. These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments. Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window. When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [流派归类 Taxonomic Opening]:** *Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition.*
  对应步骤：S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表性机制展开 Mechanism Recap]:** *These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments.*
  对应步骤：S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [转折揭示隐含假设 Pivot to Hidden Assumption]:** *Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window.*
  对应步骤：S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [假设破裂后果 Breakdown Under Stress]:** *When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.*
  对应步骤：S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 尖锐指出核心技术瓶颈与失效机理（Gap Identification）(BP03)

Expressive Intent: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Cohesion Formula: 因果链式推进（A induces B → This B compounds into C → Naive fix D sacrifices E）。绑定逻辑规则：L003, L004, L024/L025.

Publication-Grade Authentic Text

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mech-

anism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*
  对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*
  对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 挑战与机制一对一映射段（Challenge-to-Mechanism Mapping）(BP05)

Expressive Intent: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。

Cohesion Formula: Challenge 1 → Mechanism 1; Challenge 2 → Mechanism 2 → Joint System Property. 绑定逻辑规则：L003, L011.

Publication-Grade Authentic Text

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges,

we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [落地难点总起 Implementation Hurdle]:** *Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.*
  对应步骤：S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [难点 1 + 靶向设计 1]:** *First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.*
  对应步骤：S2 [难点 1 + 靶向设计 1]：First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [难点 2 + 靶向设计 2]:** *Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.*
  对应步骤：S3 [难点 2 + 靶向设计 2]：Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [整体计算/泛化特性 System Property]:** *Together, these designs sustain 50 Hz reactive control under partial visual occlusion.*
  对应步骤：S4 [整体计算/泛化特性 System Property]：Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 实验证言与性能突破预览段（Quantitative Empirical Preview）(BP06)

Expressive Intent: 我这一段想在 Introduction 结尾前，用三句话硬核总结我们在哪些基准数据集/真实硬件上做了实验，取得了多大幅度的领先。

Cohesion Formula: Evaluation Scope → Accuracy Gain over Strongest Baselines → Efficiency/Generalization Gain. 绑定逻辑规则：L010, L015.

Publication-Grade Authentic Text

We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors. Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions. Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [评测广度 Scope of Evaluation]:** *We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors.*
  对应步骤：S1 [评测广度 Scope of Evaluation]：We evaluate [Model] across [N] public benchmarks and [Real-world Hardware/Scenario], spanning [Tasks]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心精度突破 Primary Accuracy Gain]:** *Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions.*
  对应步骤：S2 [核心精度突破 Primary Accuracy Gain]：实验主指标对比（必须包含具体百分比、最强基线名称及困难场景表现）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]:** *Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.*
  对应步骤：S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]：报告计算速度、显存占用或零样本跨场景泛化收益。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 整体模型架构数据流俯瞰段（Framework Overview & Pipeline Walk-through）(BP12)

Expressive Intent: 我这一段想配合论文里的 Figure 2（模型总架构图），用一段话把输入张量从左到右流经各个模块直到输出预测结果的完整全貌讲清楚。

Cohesion Formula: 严格的链式数据流（$\mathbf{X} \to \mathbf{Z}^{(1)} \to \mathbf{Z}^{(2)} \to \hat{\mathbf{Y}}$），每一句句首承接上一句的输出张量。绑定逻辑规则：L001, L003, L004.

Publication-Grade Authentic Text

Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages. Given an input sequence $\mathbf{X}$, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches

$\mathbf{Z}_0$. These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph. Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output $\hat{\mathbf{Y}}$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]:** *Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages.*
  对应步骤：S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]：Figure 2 illustrates the overall architecture of [Model], which processes [Input] through three sequential stages: [Stage 1], [Stage 2], and [Stage 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [阶段 1 数据流 Stage 1 Flow]:** *Given an input sequence* $\mathbf{X}$*, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches* $\mathbf{Z}_0$.
  对应步骤：S2 [阶段 1 数据流 Stage 1 Flow]：Given raw input $\mathbf{X}$, the [Module 1] first transforms [Raw Input] into [Intermediate Representation $\mathbf{Z}^{(1)}$] to [Functional Purpose 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [阶段 2 数据流 Stage 2 Flow]:** *These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph.*
  对应步骤：S3 [阶段 2 数据流 Stage 2 Flow]：Next, these [Summary Noun: representations $\mathbf{Z}^{(1)}$] are fed into [Module 2], which [Mechanism 2] to produce [Refined State $\mathbf{Z}^{(2)}$]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]:** *Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output* $\hat{\mathbf{Y}}$.
  对应步骤：S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]：Finally, the [Module 3] integrates $\mathbf{Z}^{(2)}$ with [Prior/Scale Cue] to output the final [Prediction/Action Trajectory]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 7: 时间序列专属：多尺度频域-时域耦合机制阐述段（Time Series Mechanism）(BP14)

Expressive Intent: 我这一段想专门阐述时间序列模型中“如何在频域捕获全局周期性，同时在时域保留局部突发异常（Transient Anomalies）”的双分支耦合机制。

Cohesion Formula: Time-Frequency Dilemma → Spectral Global Branch → Temporal Local Branch → Uncertainty-Gated Fusion. 绑定逻辑规则：L023, L024.

Publication-Grade Authentic Text

Modeling telemetry streams solely in the time domain struggles to span multi-day periodicities, whereas pure Fourier projection spreads localized transient spikes across the entire spectrum. To reconcile both scales, our dual-domain block processes input patches along two complementary pathways. The spectral pathway applies real-valued RFFT to isolate top-k harmonic bases governing macro-periodicity, while the parallel temporal pathway employs dilated depthwise convolutions to retain sharp step changes and localized anomalies. An energy-ratio gate then fuses both representations at each time step, dynamically shifting weight toward the temporal pathway whenever local high-frequency residual energy spikes.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [单域局限性 Single-Domain Limitation]:** *Modeling telemetry streams solely in the time domain struggles to span multi-day periodicities, whereas pure Fourier projection spreads localized transient spikes across the entire spectrum.*
  对应步骤：S1 [单域局限性 Single-Domain Limitation]：Modeling non-stationary time series solely in the time domain struggles to capture long-range periodicities, whereas pure Fourier projection blurs localized transient anomalies. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [频域分支设计 Frequency Branch]:** *To reconcile both scales, our dual-domain block processes input patches along two complementary pathways.*
  对应步骤：S2 [频域分支设计 Frequency Branch]：To capture global seasonal structure, our spectral branch applies real-valued FFT $\mathcal{F}(\cdot)$ along the temporal dimension and reweights dominant harmonic components. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [时域分支设计 Temporal Branch]:** *The spectral pathway applies real-valued RFFT to isolate top-k harmonic bases governing macro-periodicity, while the parallel temporal pathway employs dilated depthwise convolutions to retain sharp step changes and localized anomalies.*
  对应步骤：S3 [时域分支设计 Temporal Branch]：In parallel, a multi-scale depthwise convolution branch operates directly on local patches to preserve sharp step changes and localized spikes. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [自适应门控融合 Adaptive Fusion]:** *An energy-ratio gate then fuses both representations at each time step, dynamically shifting weight toward the temporal pathway whenever local high-frequency residual energy spikes.*
  对应步骤：S4 [自适应门控融合 Adaptive Fusion]：An uncertainty-aware gate then fuses both branches step-by-step, assigning higher weight to the temporal branch whenever local residual variance exceeds a threshold. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 8: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）(BP18)

Expressive Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Cohesion Formula: Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则：L015, L013, L010.

Publication-Grade Authentic Text

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*
  对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.*
  对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.*
  对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%]

at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*
  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 9: 消融实验逐组件归因论证段（Counterfactual Ablation Study）(BP19)

Expressive Intent: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Cohesion Formula: Ablation Overview → w/o Module A (Specific Failure) → w/o Module B (Specific Failure) → Cross-Module Synergy. 绑定逻辑规则：L016, L007.

Publication-Grade Authentic Text

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [消融实验总括 Ablation Setup]:** *To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.*
  对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]:** *Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.*

对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]:** *Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.*
  对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同效应总结 Synergistic Conclusion]:** *Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.*
  对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 10: 极端/反常情况与失败案例坦诚剖析段（Failure Case & Outlier Analysis）(BP21)

Expressive Intent: 我这一段想通过可视化案例（Case Study / Qualitative Analysis），既展示模型在复杂跳变下的精准跟踪，又坦诚剖析一个模型预测失准的极端案例。

Cohesion Formula: Visual Setup → Success under Shift → Honest Failure Case under Unobserved Exogenous Shock → Actionable Insight. 绑定逻辑规则：L017, L018.

Publication-Grade Authentic Text

Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark. Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error. Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response. Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [定性对比切入 Qualitative Setup]:** *Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark.*

对应步骤：S1 [定性对比切入 Qualitative Setup]：Figure 5 visualizes predicted trajectories during an abrupt regime shift on [Dataset/Task]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [成功案例亮点 Success Highlight]:** *Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error.*
  对应步骤：S2 [成功案例亮点 Success Highlight]：While [Baseline] lags behind the sudden step change by [K steps] and overshoots, our model rapidly locks onto the new amplitude regime within [M steps]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [失败案例坦诚剖析 Honest Failure Mode]:** *Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response.*
  对应步骤：S3 [失败案例坦诚剖析 Honest Failure Mode]：Nevertheless, inspecting the bottom-right failure case reveals that when an external manual override occurs without historical precursors, all purely endogenous forecasters—including ours—exhibit a transient lag. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [改进启示 Constructive Takeaway]:** *Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.*
  对应步骤：S4 [改进启示 Constructive Takeaway]：Incorporating exogenous event logs or textual maintenance records is necessary to anticipate such unscheduled interventions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 11: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）(BP24)

Expressive Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Cohesion Formula: Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则：L013, L023, L029.

Publication-Grade Authentic Text

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*
  对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

18.4 FT04: 具身智能视觉-力触觉闭环操作与 Sim-to-Real 自适应完整小论文（TacFlow: Contact-Aware Visuomotor Policy）

Composition Architecture & Structural Blueprint Chain

Genre: `AI_Mini_Paper` **Paragraph Count:** 11

Blueprint Chain: `BP08` → `BP01` → `BP03` → `BP05` → `BP07` → `BP11` → `BP16` → `BP18` → `BP19` → `BP22` → `BP24`

Paragraph 1: 高密度四句式 Abstract 全段构建（Complete 4-Move Abstract）(BP08)

Expressive Intent: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Cohesion Formula: Context (S1) → Gap (S2) → Method & Mechanism (S3-S4) → Quantitative Proof (S5). 绑定逻辑规则：L001, L008, L010, L029.

Publication-Grade Authentic Text

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*
  对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 确立研究背景与真实工程价值（拒绝空喊口号）(BP01)

Expressive Intent: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Cohesion Formula: Given-to-New 链：Core Task (S1) → Practical Deployment (S2) → Real-world Disturbances (S3) → Consequence of Failure (S4). 绑定逻辑规则：L001, L011, L029.

Publication-Grade Authentic Text

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务锚定 Topic Anchor]:** *Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.*

对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [应用场景展开 Concrete Scenarios]:** *In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.*
  对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [苛刻工况约束 Operational Bottleneck]:** *However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.*
  对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [研究必要性收束 Imperative Hook]:** *Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.*
  对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 尖锐指出核心技术瓶颈与失效机理（Gap Identification）(BP03)

Expressive Intent: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Cohesion Formula: 因果链式推进（A induces B → This B compounds into C → Naive fix D sacrifices E）。绑定逻辑规则：L003, L004, L024/L025.

Publication-Grade Authentic Text

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*

对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*
  对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 挑战与机制一对一映射段（Challenge-to-Mechanism Mapping）(BP05)

Expressive Intent: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。

Cohesion Formula: Challenge 1 → Mechanism 1; Challenge 2 → Mechanism 2 → Joint System Property. 绑定逻辑规则：L003, L011.

Publication-Grade Authentic Text

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [落地难点总起 Implementation Hurdle]:** *Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.*
  对应步骤：S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [难点 1 + 靶向设计 1]:** *First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.*
  对应步骤：S2 [难点 1 + 靶向设计 1]：First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [难点 2 + 靶向设计 2]:** *Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.*
  对应步骤：S3 [难点 2 + 靶向设计 2]：Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [整体计算/泛化特性 System Property]:** *Together, these designs sustain 50 Hz reactive control under partial visual occlusion.*
  对应步骤：S4 [整体计算/泛化特性 System Property]：Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 凝练论文三点核心贡献（Contributions Summary）(BP07)

Expressive Intent: 我这一段想在 Introduction 最末尾清晰列出本文的三点核心贡献（理论/视角贡献 + 核心算法/架构贡献 + 实验与开源实证贡献）。

Cohesion Formula: Parallel Grammatical Structure（严格平行结构：全部用 We + 强及物动词 identify / propose / demonstrate 开头）。绑定逻辑规则：L002, L010.

Publication-Grade Authentic Text

In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability. (2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker. (3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总起引导句 Lead-in]:** *In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability.*
  对应步骤：S1 [总起引导句 Lead-in]：In summary, our main contributions are threefold: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [贡献 1:** *(2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker.*
  对应步骤：S2 [贡献 1：新视角/问题形式化 Conceptual/Formulation Contribution]：用强动词名物化或动词原形开头（We identify and formalize... / We establish...），点明打破了什么传统局限。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [贡献 2:** *(3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.*
  对应步骤：S3 [贡献 2：核心方法与模块 Algorithmic/Architectural Contribution]：We propose [Model/Mechanism], which integrates [A] and [B] to achieve [Theoretical/Functional Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 问题形式化定义与符号约定段（Problem Formulation & Notation）(BP11)

Expressive Intent: 我这一段想在 Methodology 开篇（Section 3.1 Problem Formulation），严谨、简洁地定义输入张量、输出目标、约束条件与优化目标。

Cohesion Formula: Input Space → Auxiliary Conditions → Target Mapping → Crucial Distributional Constraint.
绑定逻辑规则：L014, L011.

Publication-Grade Authentic Text

Consider a multivariate time series $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ recording C correlated sensors over lookback horizon L. Given an binary observation mask $\mathbf{M} \in \{0, 1\}^{L \times C}$ indicating irregular sensor dropouts, our objective is to learn a generative mapping $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ over future horizon H. Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where $\mathbb{E}[x_t]$ and $\mathrm{Cov}(x_t)$ evolve continuously between the lookback and forecasting windows.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务总体形式化 Task Definition]:** *Consider a multivariate time series* $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ *recording* C *correlated sensors over lookback horizon* L.
  对应步骤: S1 [任务总体形式化 Task Definition]: Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [补充条件或模态输入 Auxiliary/Condition Inputs]:** *Given an binary observation mask* $\mathbf{M} \in \{0, 1\}^{L \times C}$ *indicating irregular sensor dropouts, our objective is to learn a generative mapping* $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ *over future horizon* H.
  对应步骤: S2 [补充条件或模态输入 Auxiliary/Condition Inputs]: At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [预测/控制目标 Objective]:** *Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where* $\mathbb{E}[x_t]$ *and* $Cov(x_t)$ *evolve continuously between the lookback and forecasting windows.*
  对应步骤: S3 [预测/控制目标 Objective]: Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$ over horizon H (or action trajectory $a_{t:t+K}$). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 7: 具身智能专属：多模态感知-动作闭环与 Sim-to-Real 阐述段（Embodied AI Policy）(BP16)

Expressive Intent: 我这一段想阐述具身智能机器人策略（Embodied Policy）如何融合视觉、本体感觉（Proprioception）或触觉，并克服仿真到真实硬件的动力学差距（Sim-to-Real Gap）。

Cohesion Formula: Visual Occlusion Bottleneck → Asynchronous Multimodal Alignment → Action Chunking → Online Sim-to-Real System Identification. 绑定逻辑规则：L025, L003.

Publication-Grade Authentic Text

Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object. To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings. Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling. At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [单模态开环局限 Open-Loop Visual Limitation]:** *Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object.*
  对应步骤：S1 [单模态开环局限 Open-Loop Visual Limitation]：Relying solely on RGB-D point clouds leaves manipulation policies vulnerable to self-occlusion by the robot gripper during fine-grained contact. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [多模态时空对齐 Multi-Modal Alignment]:** *To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings.*
  对应步骤：S2 [多模态时空对齐 Multi-Modal Alignment]：To maintain state observability during contact, our encoder aligns asynchronous 30 Hz visual tokens with 500 Hz joint proprioception and wrist force-torque histories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]:** *Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling.*
  对应步骤：S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]：Rather than predicting single-step torques prone to jitter, the policy outputs overlapping K-step SE(3) end-effector action chunks smoothed via exponential temporal ensembling. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]:** *At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.*
  对应步骤：S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]：During physical deployment, a lightweight history adapter infers latent friction and payload parameters online, bridging the sim-to-real dynamics gap without policy fine-tuning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 8: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）(BP18)

Expressive Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Cohesion Formula: Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则：L015, L013, L010.

Publication-Grade Authentic Text

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*
  对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.*
  对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.*
  对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*
  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信

息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 9: 消融实验逐组件归因论证段（Counterfactual Ablation Study）(BP19)

Expressive Intent: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Cohesion Formula: Ablation Overview → w/o Module A (Specific Failure) → w/o Module B (Specific Failure) → Cross-Module Synergy. 绑定逻辑规则：L016, L007.

Publication-Grade Authentic Text

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [消融实验总括 Ablation Setup]:** *To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.*
  对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]:** *Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.*
  对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]:** *Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.*

对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [协同效应总结 Synergistic Conclusion]:** *Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.*
  对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 10: 计算开销、推理延迟与精度帕累托权衡段（Efficiency-Accuracy Pareto Analysis）(BP22)

Expressive Intent: 我这一段想说明我们的模型不仅准，而且算得快、显存省，在“精度-延迟/显存”帕累托前沿（Pareto Frontier）上显著优于现有大模型。

Cohesion Formula: Hardware Benchmark → Complexity Reduction & Memory Gain → Real-Time Latency Margin → Pareto Conclusion. 绑定逻辑规则：L010, L023.

Publication-Grade Authentic Text

Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module. By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer. At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [硬件与测试基准 Hardware Setup]:** *Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module.*
  对应步骤：S1 [硬件与测试基准 Hardware Setup]：Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [复杂度理论与实测对比 Complexity & Runtime]:** *By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer.*
  对应步骤：S2 [复杂度理论与实测对比 Complexity & Runtime]：By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch

by [X%] and peak VRAM by [Y%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [实时闭环部署可行性 Real-Time Viability]:** *At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.*
  对应步骤：S3 [实时闭环部署可行性 Real-Time Viability]：At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control deadline. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 11: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）(BP24)

Expressive Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Cohesion Formula: Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则：L013, L023, L029.

Publication-Grade Authentic Text

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*
  对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定

旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

18.5 FT05: 基于整流蒸馏的单步实时机械臂轨迹生成完整小论文（RectiPolicy: Real-Time Action Chunk Generation）

Composition Architecture & Structural Blueprint Chain

Genre: `AI_Mini_Paper` **Paragraph Count:** 11

Blueprint Chain: `BP08` → `BP02` → `BP04` → `BP05` → `BP10` → `BP15` → `BP16` → `BP18` → `BP22` → `BP23` → `BP24`

Paragraph 1: 高密度四句式 Abstract 全段构建（Complete 4-Move Abstract）(BP08)

Expressive Intent: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Cohesion Formula: Context (S1) → Gap (S2) → Method & Mechanism (S3-S4) → Quantitative Proof (S5). 绑定逻辑规则：L001, L008, L010, L029.

Publication-Grade Authentic Text

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*
  对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 总结现有主流范式及其隐含假设（欲抑先扬铺垫段）(BP02)

Expressive Intent: 我这一段想客观概括前人主流模型（如 Transformer / Diffusion / MLP）取得了什么进展，但点出它们共同依赖的一个脆弱前提假设。

Cohesion Formula: 先让步肯定（While/Recent paradigms...）→ 后釜底抽薪揭示隐含前提（Yet these architectures implicitly assume...）。绑定逻辑规则：L008, L009.

Publication-Grade Authentic Text

Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition. These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments. Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window. When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [流派归类 Taxonomic Opening]:** *Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition.*
  对应步骤：S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表性机制展开 Mechanism Recap]:** *These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments.*
  对应步骤：S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [转折揭示隐含假设 Pivot to Hidden Assumption]:** *Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window.*
  对应步骤：S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [假设破裂后果 Breakdown Under Stress]:** *When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.*
  对应步骤：S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 亮出本文核心洞察与破局方法总览（Key Insight & Method Overview）(BP04)

Expressive Intent: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。

Cohesion Formula: Problem Anchor → Key Insight → Dual-Module Mechanism → Pareto Resolution. 绑定逻辑规则：L002, L023.

Publication-Grade Authentic Text

To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concur-

rently models non-stationary mean-variance evolution through a lightweight neural ODE branch. Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [方法登场与定位 Method Introduction]:** *To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.*
  对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心直觉洞察 Key Insight]:** *Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.*
  对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [模块协同主干 Module Synergy]:** *Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.*
  对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [闭环收益总结 Closed-Loop Benefit]:** *Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.*
  对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 挑战与机制一对一映射段（Challenge-to-Mechanism Mapping）(BP05)

Expressive Intent: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。

Cohesion Formula: Challenge 1 → Mechanism 1; Challenge 2 → Mechanism 2 → Joint System Property. 绑定逻辑规则：L003, L011.

Publication-Grade Authentic Text

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [落地难点总起 Implementation Hurdle]:** *Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.*
  对应步骤：S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [难点 1 + 靶向设计 1]:** *First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.*
  对应步骤：S2 [难点 1 + 靶向设计 1]：First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [难点 2 + 靶向设计 2]:** *Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.*
  对应步骤：S3 [难点 2 + 靶向设计 2]：Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [整体计算/泛化特性 System Property]:** *Together, these designs sustain 50 Hz reactive control under partial visual occlusion.*
  对应步骤：S4 [整体计算/泛化特性 System Property]：Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 对比现有流派优劣并精准定位本文差异（Contrastive Positioning）(BP10)

Expressive Intent: 我这一段想在 Related Work 小节末尾，把我们的方法与最接近的 2 个竞争方案做精准对比，防止审稿人说我们只是对某篇旧文的微调（Incremental）。

Cohesion Formula: Acknowledge Overlap → Pinpoint Structural Divergence → Explain Functional Gain. 绑定逻辑规则：L008, L019.

Publication-Grade Authentic Text

Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach. While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes. In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases. Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [锁定最相近工作 Closest Prior Work]:** *Among spectral forecasters, FEDformer and TimesNet are most closely related to our approach.*
  对应步骤：S1 [锁定最相近工作 Closest Prior Work]：Among prior studies, [Closest Method A] and [Closest Method B] are most closely related to our approach. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [剖析其局限性 Structural Difference 1]:** *While both methods transform temporal representations via fast Fourier transforms, they apply static top-k frequency selection globally across the entire lookback window, which blurs localized transient spikes.*
  对应步骤：S2 [剖析其局限性 Structural Difference 1]：While [Method A] also employs [Shared Concept], it relies on [Limitation 1], restricting its applicability to [Narrow Setting]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [剖析本文本质突破 Core Departure]:** *In contrast, our wavelet-coupled attention dynamically adapts filter bandwidths over sliding sub-intervals, isolating non-stationary burst events without corrupting global seasonal bases.*
  对应步骤：S3 [剖析本文本质突破 Core Departure]：In contrast, our framework introduces [Core Innovation], which fundamentally eliminates [Bottleneck] by [Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证或理论优势锁定 Empirical/Theoretical Edge]:** *Consequently, our model captures localized fault transients without requiring pre-defined frequency thresholds.*
  对应步骤：S4 [实证或理论优势锁定 Empirical/Theoretical Edge]：Consequently, our approach achieves [Capability] without requiring [Expensive Requirement of Prior Work]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 生成模型专属：条件控制与采样去噪轨迹阐述段（Generative Modeling Mechanism）(BP15)

Expressive Intent: 我这一段想阐述生成模型（Diffusion / Flow Matching / Autoregressive）如何在满足外部物理约束或条件观测的前提下，生成高质量且多样化的序列/轨迹。

Cohesion Formula: Guidance Artifact Problem → Intrinsic Velocity Conditioning → Tangent Projection Step → Fast Exact Convergence. 绑定逻辑规则：L026, L014.

Publication-Grade Authentic Text

Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations. To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field. At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator. This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]:** *Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations.*
  对应步骤：S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]：Applying post-hoc classifier guidance to pre-trained diffusion samplers often pushes intermediate noisy states off the data manifold, inducing high-frequency artifacts. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]:** *To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field.*
  对应步骤: S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]: To enforce hard boundary constraints without manifold deviation, we embed [Condition/Constraint] directly into the velocity field / denoising kernel. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具体轨迹更新方程 Trajectory Update Formulation]:** *At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator.*
  对应步骤：S3 [具体轨迹更新方程 Trajectory Update Formulation]：At solver step $\tau \in [0, 1]$, the conditional velocity field $v_\theta(x_\tau, \tau \mid c)$ projects the unconstrained flow onto the tangent space of feasible trajectories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]:** *This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.*
  对应步骤：S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]：This tangent projection guarantees exact boundary satisfaction at $\tau = 1$ while permitting straight 4-step Euler integration. | 句首

锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 7: 具身智能专属：多模态感知-动作闭环与 Sim-to-Real 阐述段（Embodied AI Policy）(BP16)

Expressive Intent: 我这一段想阐述具身智能机器人策略（Embodied Policy）如何融合视觉、本体感觉（Proprioception）或触觉，并克服仿真到真实硬件的动力学差距（Sim-to-Real Gap）。

Cohesion Formula: Visual Occlusion Bottleneck → Asynchronous Multimodal Alignment → Action Chunking → Online Sim-to-Real System Identification. 绑定逻辑规则：L025, L003.

Publication-Grade Authentic Text

Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object. To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings. Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling. At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [单模态开环局限 Open-Loop Visual Limitation]:** *Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object.*
  对应步骤：S1 [单模态开环局限 Open-Loop Visual Limitation]：Relying solely on RGB-D point clouds leaves manipulation policies vulnerable to self-occlusion by the robot gripper during fine-grained contact. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [多模态时空对齐 Multi-Modal Alignment]:** *To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings.*
  对应步骤：S2 [多模态时空对齐 Multi-Modal Alignment]：To maintain state observability during contact, our encoder aligns asynchronous 30 Hz visual tokens with 500 Hz joint proprioception and wrist force-torque histories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]:** *Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling.*

对应步骤：S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]: Rather than predicting single-step torques prone to jitter, the policy outputs overlapping K-step SE(3) end-effector action chunks smoothed via exponential temporal ensembling. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]:** *At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.*
  对应步骤：S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]: During physical deployment, a lightweight history adapter infers latent friction and payload parameters online, bridging the sim-to-real dynamics gap without policy fine-tuning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 8: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）(BP18)

Expressive Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Cohesion Formula: Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则：L015, L013, L010.

Publication-Grade Authentic Text

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*
  对应步骤：S1 [全局统治力概述 Overall Dominance]: Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark,*

where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.

对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.*

  对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*

  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 9: 计算开销、推理延迟与精度帕累托权衡段（Efficiency-Accuracy Pareto Analysis）(BP22)

Expressive Intent: 我这一段想说明我们的模型不仅准，而且算得快、显存省，在“精度-延迟/显存”帕累托前沿（Pareto Frontier）上显著优于现有大模型。

Cohesion Formula: Hardware Benchmark → Complexity Reduction & Memory Gain → Real-Time Latency Margin → Pareto Conclusion. 绑定逻辑规则：L010, L023.

Publication-Grade Authentic Text

Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module. By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer. At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [硬件与测试基准 Hardware Setup]:** *Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module.*
  对应步骤：S1 [硬件与测试基准 Hardware Setup]：Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [复杂度理论与实测对比 Complexity & Runtime]:** *By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer.*
  对应步骤：S2 [复杂度理论与实测对比 Complexity & Runtime]：By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch by [X%] and peak VRAM by [Y%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实时闭环部署可行性 Real-Time Viability]:** *At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.*
  对应步骤：S3 [实时闭环部署可行性 Real-Time Viability]：At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control deadline. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 10: 局限性（Limitations）客观陈述与未来扩展段（Constructive Limitations）(BP23)

Expressive Intent: 我这一段想写论文末尾或 NeurIPS/ICLR Checklist 要求的 Limitations 段落：诚实指出当前方法的两个客观边界，但绝不贬损论文的核心贡献。

Cohesion Formula: Two Objective Boundaries → [Boundary 1 + Technical Remedy] → [Boundary 2 + Stochastic Extension]. 绑定逻辑规则：L017.

Publication-Grade Authentic Text

Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting. First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy. Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration. Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [边界 1:** *Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting.*
  对应步骤：S1 [边界 1：极端高维或极度稀疏场景 Boundary 1]：Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [边界 1 机理分析与解法 Mechanism & Solution 1]:** *First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy.*
  对应步骤：S2 [边界 1 机理分析与解法 Mechanism & Solution 1]：First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [边界 2:** *Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration.*
  对应步骤：S3 [边界 2：特定假设限制 Boundary 2]：Second, our continuous scale tracker assumes piecewise differentiable drift and may under-estimate instantaneous discontinuous jumps caused by hardware resets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [未来演进方向 Future Extension]:** *Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.*
  对应步骤：S4 [未来演进方向 Future Extension]：Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 11: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）(BP24)

Expressive Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Cohesion Formula: Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则：L013, L023, L029.

Publication-Grade Authentic Text

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evalu-

ations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*
  对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Chapter 19

Biomedical & Life Sciences / 生物医学与生命科学

本章精选该学科领域及对应文体的完整学术范文。每一篇范文不仅展示母语级发表标准的正文全貌，更清晰拆解全篇的“段落蓝图链条（Structural Blueprint Chain）”，并对每个自然段逐一剖析其核心交际意图、句际衔接逻辑公式（Cohesion Formula）以及每句话的句法成分与推进机制。

19.1 FT06: 面向高维传感器网络的稀疏相位图注意力完整小论文（PhaseGraph: Scalable Spatio-Temporal Forecasting）

Composition Architecture & Structural Blueprint Chain

Genre: `AI_Mini_Paper` **Paragraph Count:** 11

Blueprint Chain: `BP08` → `BP01` → `BP09` → `BP11` → `BP12` → `BP13` → `BP17` → `BP18` → `BP19` → `BP20` → `BP24`

Paragraph 1: 高密度四句式 Abstract 全段构建（Complete 4-Move Abstract）(BP08)

Expressive Intent: 我这一段想写出整篇小论文的完整 Abstract（摘要）：在 150–180 词内一气呵成交代背景、核心痛点、方法机制与量化实验结果。

Cohesion Formula: Context (S1) → Gap (S2) → Method & Mechanism (S3-S4) → Quantitative Proof (S5). 绑定逻辑规则：L001, L008, L010, L029.

Publication-Grade Authentic Text

Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series. While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively. We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps. Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer. Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [背景与任务重要性 Context]:** *Recovering missing sensor observations is essential for reliable downstream monitoring in industrial and clinical time series.*
  对应步骤：S1 [背景与任务重要性 Context]：1 句话锁定任务目标与核心应用场景（≤20 词）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [现有局限与痛点 Gap/Bottleneck]:** *While diffusion-based imputation models generate plausible trajectories under random missingness, they suffer from high sampling latency and lose global periodic coherence when entire sensor blocks drop out consecutively.*
  对应步骤：S2 [现有局限与痛点 Gap/Bottleneck]：While existing [Paradigms] achieve [Merit], they struggle with [Core Bottleneck] because [Underlying Cause]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3-S4 [本文方法与核心机制 Approach & Mechanism]:** *We present FlowImp, a conditional flow-matching framework that reconstructs contiguous missing blocks in three ODE solver steps.*
  对应步骤：S3-S4 [本文方法与核心机制 Approach & Mechanism]：In this work, we present [Model], which [Core Idea]. Specifically, [Module 1]..., while [Module 2]... | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Specifically, FlowImp anchors straight transport paths on observed spectral priors and enforces cross-channel phase consistency via a lightweight Fourier regularizer.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S5 [量化实证结果 Quantitative Results]:** *Across six multivariate benchmarks under 50% block missingness, FlowImp reduces imputation MAE by 19.3% compared with CSDI while accelerating inference by 24x.*
  对应步骤：S5 [量化实证结果 Quantitative Results]：Experiments on [Benchmarks/Hardware] show that [Model] outperforms [Baselines] by [X%] in [Metric] and achieves [Efficiency/Real-world Result]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 确立研究背景与真实工程价值（拒绝空喊口号）(BP01)

Expressive Intent: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Cohesion Formula: Given-to-New 链：Core Task (S1) → Practical Deployment (S2) → Real-world Disturbances (S3) → Consequence of Failure (S4). 绑定逻辑规则：L001, L011, L029.

Publication-Grade Authentic Text

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务锚定 Topic Anchor]:** *Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.*
  对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [应用场景展开 Concrete Scenarios]:** *In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.*
  对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [苛刻工况约束 Operational Bottleneck]:** *However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.*
  对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [研究必要性收束 Imperative Hook]:** *Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.*
  对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 相关工作分类综述段（拒绝 A did X, B did Y 流水账）(BP09)

Expressive Intent: 我这一段想写 Related Work 的一个子方向：把过去三年的 6–8 篇代表性论文按技术路线分成两类进行归纳，并指出它们的共同盲区。

Cohesion Formula: Two-Branch Taxonomy → Family A → Contrastive Family B → Shared Limitation. 绑定逻辑规则：L009, L008.

Publication-Grade Authentic Text

Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization. Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains. Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives. Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分类总纲 Taxonomic Thesis]:** *Pre-trained time series foundation models generally follow either continuous patch reconstruction or discrete vocabulary tokenization.*
  对应步骤：S1 [分类总纲 Taxonomic Thesis]：Existing methods for [Task] can be categorized into two main families: [Family A] and [Family B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [流派 A 归纳与代表作 Family A Synthesis]:** *Continuous reconstruction frameworks (e.g., PatchTST, MOMENT) pre-train encoders via masked patch regression, preserving fine-grained numerical resolution across continuous domains.*
  对应步骤：S2 [流派 A 归纳与代表作 Family A Synthesis]：[Family A] approaches (e.g., [Models]) focus on [Mechanism A], effectively addressing [Sub-problem 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [流派 B 归纳与对比 Family B Synthesis]:** *Conversely, discrete tokenization approaches (e.g., Chronos) quantize scaled observations into fixed bins to leverage cross-entropy language modeling objectives.*
  对应步骤：S3 [流派 B 归纳与对比 Family B Synthesis]：By contrast, [Family B] architectures (e.g., [Models]) reformulate the problem via [Mechanism B] to reduce [Cost/Complexity]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [共同盲区提炼 Shared Blind Spot]:** *Despite their zero-shot transferability, both paradigms treat univariate channels independently during pre-training, ignoring physical cross-variable conservation laws that govern multivariate engineering systems.*
  对应步骤：S4 [共同盲区提炼 Shared Blind Spot]：Despite their respective strengths, both families overlook [Critical Factor], which degrades performance under [Target Scenario]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 问题形式化定义与符号约定段（Problem Formulation & Notation）(BP11)

Expressive Intent: 我这一段想在 Methodology 开篇（Section 3.1 Problem Formulation），严谨、简洁地定义输入张量、输出目标、约束条件与优化目标。

Cohesion Formula: Input Space → Auxiliary Conditions → Target Mapping → Crucial Distributional Constraint.
绑定逻辑规则：L014, L011.

Publication-Grade Authentic Text

Consider a multivariate time series $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ recording C correlated sensors over lookback horizon L. Given an binary observation mask $\mathbf{M} \in \{0,1\}^{L \times C}$ indicating irregular sensor dropouts, our objective is to learn a generative mapping $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ over future horizon H. Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where $\mathbb{E}[x_t]$ and $\mathrm{Cov}(x_t)$ evolve continuously between the lookback and forecasting windows.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务总体形式化 Task Definition]:** *Consider a multivariate time series* $\mathbf{X}_{1:L} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L \times C}$ *recording C correlated sensors over lookback horizon L.*
  对应步骤：S1 [任务总体形式化 Task Definition]：Consider a [System/Dataset] observed over a lookback window of length L, denoted as $\mathbf{X} = [x_1, \ldots, x_L] \in \mathbb{R}^{L \times C}$, where C represents [Meaning]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [补充条件或模态输入 Auxiliary/Condition Inputs]:** *Given an binary observation mask* $\mathbf{M} \in \{0,1\}^{L \times C}$ *indicating irregular sensor dropouts, our objective is to learn a generative mapping* $p_\theta(\mathbf{Y}_{L+1:L+H} \mid \mathbf{X}_{1:L} \odot \mathbf{M})$ *over future horizon H.*
  对应步骤：S2 [补充条件或模态输入 Auxiliary/Condition Inputs]：At each step t, the system also receives [Condition/Observation] $z_t \in \mathcal{Z}$ under [Noise/Missingness/Dynamic Regime]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [预测/控制目标 Objective]:** *Crucially, unlike standard settings that assume time-invariant marginal distributions, we explicitly account for non-stationary regime shifts where* $\mathbb{E}[x_t]$ *and* $Cov(x_t)$ *evolve continuously between the lookback and forecasting windows.*
  对应步骤：S3 [预测/控制目标 Objective]：Our goal is to learn a parameterized mapping $f_\theta(\cdot)$ that predicts future states $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$ over horizon H (or action trajectory $a_{t:t+K}$). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 整体模型架构数据流俯瞰段（Framework Overview & Pipeline Walk-through）(BP12)

Expressive Intent: 我这一段想配合论文里的 Figure 2（模型总架构图），用一段话把输入张量从左到右流经各个模块直到输出预测结果的完整全貌讲清楚。

Cohesion Formula: 严格的链式数据流（$\mathbf{X} \rightarrow \mathbf{Z}^{(1)} \rightarrow \mathbf{Z}^{(2)} \rightarrow \hat{\mathbf{Y}}$），每一句句首承接上一句的输出张量。绑定逻辑规则：L001, L003, L004.

Publication-Grade Authentic Text

Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages. Given an input sequence $\mathbf{X}$, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches $\mathbf{Z}_0$. These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph. Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output $\hat{\mathbf{Y}}$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]:** *Figure 2 illustrates the overall architecture of DyTS-Net, which maps raw sensor streams to multi-horizon forecasts through three sequential stages.*
  对应步骤：S1 [总图锚定与设计哲学 Visual Anchor & Philosophy]: Figure 2 illustrates the overall architecture of [Model], which processes [Input] through three sequential stages: [Stage 1], [Stage 2], and [Stage 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [阶段 1 数据流 Stage 1 Flow]:** *Given an input sequence* $\mathbf{X}$*, the Adaptive De-stationary Tokenizer first separates local trend-scale statistics from normalized sub-series patches* $\mathbf{Z}_0$.
  对应步骤：S2 [阶段 1 数据流 Stage 1 Flow]: Given raw input $\mathbf{X}$, the [Module 1] first transforms [Raw Input] into [Intermediate Representation $\mathbf{Z}^{(1)}$] to [Functional Purpose 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [阶段 2 数据流 Stage 2 Flow]:** *These normalized patches then pass through stacked Spectral-Spatial Mixing blocks, which jointly filter high-frequency channel noise in the Fourier domain and propagate cross-sensor phase dependencies via a sparse directed graph.*
  对应步骤：S3 [阶段 2 数据流 Stage 2 Flow]: Next, these [Summary Noun: representations $\mathbf{Z}^{(1)}$] are fed into [Module 2], which [Mechanism 2] to produce [Refined State $\mathbf{Z}^{(2)}$]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]:** *Finally, the Scale-Guided Decoder fuses the refined structural representations with the projected neural-ODE scale trajectory to reconstruct the multi-horizon output* $\hat{\mathbf{Y}}$.

对应步骤：S4 [阶段 3 解码与闭环输出 Stage 3 Decoding]：Finally, the [Module 3] integrates $\mathbf{Z}^{(2)}$ with [Prior/Scale Cue] to output the final [Prediction/Action Trajectory]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 核心子模块：直觉动机与数学设计无缝衔接段（Intuition-before-Math）(BP13)

Expressive Intent: 我这一段想详细介绍我们提出的某个核心创新模块：先用两句话讲清楚为什么直接用传统算子会出问题，再顺理成章推导出我们的新公式。

Cohesion Formula: Pathology of Standard Operator → Physical Intuition → Equation Integration → Theoretical Guarantee. 绑定逻辑规则：L014, L003.

Publication-Grade Authentic Text

Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps. To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies. Formally, given query and key projections $Q, K \in \mathbb{R}^{N\times d}$, we modulate the attention logits via a phase-alignment mask: $\mathcal{A}(Q,K,V) = \text{softmax}\big(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1\big)V$, where $\Delta\Phi_{Q,K} \in [-\pi,\pi]^{N\times N}$ denotes the instantaneous Hilbert phase difference matrix and $\lambda > 0$ controls phase selectivity. Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [传统算子缺陷 Motivation]:** *Standard dot-product attention measures token similarity purely by instantaneous amplitude alignment, often matching noisy spikes across unrelated time steps.*
  对应步骤：S1 [传统算子缺陷 Motivation]：Standard [Conventional Operator, e.g., softmax self-attention] computes [Operation], which suffers from [Pathology] when applied to [Domain Condition]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [物理/统计直觉 Physical Intuition]:** *To enforce temporal coherence, we observe that genuine physical interactions between industrial sensors exhibit consistent phase lag across dominant frequencies.*
  对应步骤：S2 [物理/统计直觉 Physical Intuition]：To overcome this pathology, we observe that [Key Domain Property] can be effectively regularized by [Intuitive Operation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [数学公式定义 Mathematical Formulation]:** *Formally, given query and key projections $Q, K \in \mathbb{R}^{N\times d}$, we modulate the attention logits via a phase-alignment mask: $\mathcal{A}(Q,K,V) =$*

$softmax\left(\frac{QK^\top}{\sqrt{d}} - \lambda\|\Delta\Phi_{Q,K}\|_1\right)V$, *where* $\Delta\Phi_{Q,K} \in [-\pi, \pi]^{N\times N}$ *denotes the instantaneous Hilbert phase difference matrix and* $\lambda > 0$ *controls phase selectivity.*

对应步骤：S3 [数学公式定义 Mathematical Formulation]：Formally, given input representation $H \in \mathbb{R}^{N\times d}$, we formulate the [Module Name] as: [Equation], where [Symbol Explanations]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [理论/计算性质 Guarantee]:** *Penalizing large phase discrepancies prevents spurious attention weights between out-of-phase noise bursts.*

  对应步骤：S4 [理论/计算性质 Guarantee]：By construction, this formulation guarantees [Desirable Property, e.g., $O(N)$ complexity / translation invariance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 7: 实验设置、数据集特性与强基线选型说明段（Experimental Setup & Baselines）(BP17)

Expressive Intent: 我这一段想在 Experiments 开篇说明：我们为什么选这几类数据集和这几类基线模型，证明我们的实验设计是全面、公平且直击痛点的。

Cohesion Formula: Dataset Diversity → Challenging Regime Highlights → Categorized Strong Baselines → Strict Fairness Protocol. 绑定逻辑规则：L009, L010.

Publication-Grade Authentic Text

To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic). Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability. We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer). All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [数据集选型逻辑 Dataset Selection Rationale]:** *To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic).*

  对应步骤：S1 [数据集选型逻辑 Dataset Selection Rationale]：To evaluate performance across diverse non-stationary regimes, we conduct experiments on [N] benchmarks spanning [Domain 1], [Domain 2], and [Domain 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [极端/困难特性突出 Challenging Dataset Properties]:** *Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability.*
  对应步骤：S2 [极端/困难特性突出 Challenging Dataset Properties]：Notably, [Dataset A] exhibits [Hard Property 1, e.g., strong covariate shift], whereas [Dataset B] tests scalability across [Hard Property 2, e.g., 862 correlated channels]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [基线流派全覆盖 Baseline Family Coverage]:** *We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer).*
  对应步骤：S3 [基线流派全覆盖 Baseline Family Coverage]：We compare our method against [K] competitive baselines grouped into three families: [Family 1], [Family 2], and [Family 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [公平评测协议 Fair Evaluation Protocol]:** *All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.*
  对应步骤：S4 [公平评测协议 Fair Evaluation Protocol]：All models are evaluated under identical lookback windows ($L = 96$), prediction horizons ($H \in \{96, 192, 336, 720\}$), and hyperparameter search budgets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 8: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）(BP18)

Expressive Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Cohesion Formula: Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则：L015, L013, L010.

Publication-Grade Authentic Text

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By

decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*
  对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.*
  对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.*
  对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*
  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 9: 消融实验逐组件归因论证段（Counterfactual Ablation Study）(BP19)

Expressive Intent: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Cohesion Formula: Ablation Overview → w/o Module A (Specific Failure) → w/o Module B (Specific Failure) → Cross-Module Synergy. 绑定逻辑规则：L016, L007.

Publication-Grade Authentic Text

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [消融实验总括 Ablation Setup]:** *To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.*
  对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]:** *Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.*
  对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]:** *Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.*
  对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同效应总结 Synergistic Conclusion]:** *Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.*

对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 10: 超参数敏感性与极端噪声鲁棒性分析段（Sensitivity & Robustness Analysis）(BP20)

Expressive Intent: 我这一段想回应审稿人最常问的：你的模型是不是对超参数很敏感？在缺失值或传感器强噪声下会不会突然崩盘？

Cohesion Formula: Goal → Wide Hyperparameter Plateau → Stress-Test Comparison → Mechanistic Explanation. 绑定逻辑规则：L010, L019.

Publication-Grade Authentic Text

Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption. Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$. Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%. This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分析目的 Objective]:** *Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption.*
  对应步骤：S1 [分析目的 Objective]：We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [超参数平坦区间 Plateau Stability]:** *Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$.*
  对应步骤：S2 [超参数平坦区间 Plateau Stability]：Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [极端边缘工况压力测试 Stress Test under Corruption]:** *Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%.*
  对应步骤：S3 [极端边缘工况压力测试 Stress Test under Corruption]：Under synthetic sensor dropouts up to [40%] and Gaussian noise ($\sigma = 0.3$), baseline accuracy plummets by over [35%], whereas our model

degrades by less than [6.5%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [鲁棒性物理归因 Robustness Mechanism]:** *This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.*
  对应步骤：S4 [鲁棒性物理归因 Robustness Mechanism]：This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 11: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）(BP24)

Expressive Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Cohesion Formula: Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则：L013, L023, L029.

Publication-Grade Authentic Text

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*
  对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定

旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

19.2 FT07: 顶会 5 分钟 Oral 演讲完整逐字稿：非平稳时间序列预测（含开场 Hook、图表引导、公式直觉与 Q&A）

Composition Architecture & Structural Blueprint Chain

Genre: `Academic_Speech` **Paragraph Count:** 6

Blueprint Chain: `BP41` → `BP03` → `BP04` → `BP43` → `BP42` → `BP44`

Paragraph 1: 学术演讲开场 60 秒 Hook：用直观痛点或反直觉现象抓住全场（Oral Opening Hook）(BP41)

Expressive Intent: 我这一段想作为顶会 Oral Presentation、组会汇报或博士面试 PPT 第一页的开场白：在 45–60 秒内不用任何晦涩公式，让全场听众立刻听懂“我们为什么要做这件事、现有方法到底卡在哪”。

Cohesion Formula: Concrete Visual Scenario → Rhetorical Question → Crisp Culprit Identification → Talk Promise. 绑定逻辑规则：L018, L029.

Publication-Grade Authentic Text

Good morning everyone. Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines. Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges needed to predict peak loads. In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [场景直觉切入 Concrete Scenario Hook]:** *Good morning everyone.*
  对应步骤：S1 [场景直觉切入 Concrete Scenario Hook]：Good morning everyone. Imagine deploying a humanoid robot to assemble flexible cables on a moving factory line—or forecasting grid load when a storm hits. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反直觉痛点设问 Counter-intuitive Breakdown]:** *Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines.*
  对应步骤：S2 [反直觉痛点设问 Counter-intuitive Breakdown]：Standard deep models perform remarkably well when conditions stay calm, yet why do they fail abruptly the moment contact dynamics or weather patterns shift? | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [一针见血揭晓答案 Crisp Root Cause]:** *Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The*

culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges needed to predict peak loads.
对应步骤：S3 [一针见血揭晓答案 Crisp Root Cause]：The culprit is a hidden stationarity assumption baked into standard attention layers. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [亮出今日主旨 Talk Roadmap]:** *In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.*
对应步骤：S4 [亮出今日主旨 Talk Roadmap]：In this talk, I will show how our framework, [Model Name], breaks this bottleneck and achieves [Key Metric Gain] in real-time deployment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 尖锐指出核心技术瓶颈与失效机理（Gap Identification）(BP03)

Expressive Intent: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Cohesion Formula: 因果链式推进（A induces B → This B compounds into C → Naive fix D sacrifices E）。绑定逻辑规则：L003, L004, L024/L025.

Publication-Grade Authentic Text

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*
对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*

对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 亮出本文核心洞察与破局方法总览（Key Insight & Method Overview）(BP04)

Expressive Intent: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。

Cohesion Formula: Problem Anchor → Key Insight → Dual-Module Mechanism → Pareto Resolution. 绑定逻辑规则：L002, L023.

Publication-Grade Authentic Text

To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch. Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [方法登场与定位 Method Introduction]:** *To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.*

对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [核心直觉洞察 Key Insight]:** *Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.*
  对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [模块协同主干 Module Synergy]:** *Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.*
  对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [闭环收益总结 Closed-Loop Benefit]:** *Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.*
  对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 演讲中化繁为简讲解核心公式与算法直觉（Explaining Math Orally）(BP43)

Expressive Intent: 我这一段想在 PPT 上出现核心损失函数或注意力公式时，用 4 句通俗、精准的口语把公式背后的物理/几何直觉讲透，而不是在台上干巴巴念希腊字母。

Cohesion Formula: Intuition Before Symbols → Physical Reality → Pointer to Highlighted Math Term → Operational Effect. 绑定逻辑规则：L014, L018.

Publication-Grade Authentic Text

Before looking at the equation in the middle box, here is the physical intuition behind it. When two sensors sit on the same turbine shaft, genuine mechanical vibrations travel with a consistent phase lag, whereas random sensor glitches fire completely out of phase. That physical rule is captured by the highlighted red term, $\|\Delta\Phi_{Q,K}\|_1$, right inside the softmax. Whenever two channels show inconsistent phase lag, this penalty automatically zeroes out their attention weight.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [先讲直觉目标 Intuition First]:** *Before looking at the equation in the middle box, here is the physical intuition behind it.*
  对应步骤：S1 [先讲直觉目标 Intuition First]：Before looking at the math at the bottom of the slide, here is the simple physical intuition we want to capture. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [日常/物理类比 Physical Analogy]:** *When two sensors sit on the same turbine shaft, genuine mechanical vibrations travel with a consistent phase lag, whereas random sensor glitches fire completely out of phase.*
  对应步骤：S2 [日常/物理类比 Physical Analogy]：When two sensors monitor the same mechanical shaft, genuine vibrations share a consistent phase lag, whereas random electrical glitches fire out of phase. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [指向公式核心项 Point to Key Term]:** *That physical rule is captured by the highlighted red term, $\|\Delta\Phi_{Q,K}\|_1$, right inside the softmax.*
  对应步骤：S3 [指向公式核心项 Point to Key Term]：That is exactly what the red penalty term $\|\Delta\Phi\|_1$ inside the softmax does: it measures phase mismatch between channels. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [一句话总结效果 Operational Effect]:** *Whenever two channels show inconsistent phase lag, this penalty automatically zeroes out their attention weight.*
  对应步骤：S4 [一句话总结效果 Operational Effect]：Whenever two channels fall out of phase, this term suppresses their attention weight automatically. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 演讲幻灯片与复杂图表视觉引导段（Visual Signposting in Tech Talks）(BP42)

Expressive Intent: 我这一段想在演讲到架构图或核心实验曲线（Slide 4 / Slide 7）时，用最自然的口语引导听众的眼睛先看哪、再看哪，瞬间读懂图表背后的胜出逻辑。

Cohesion Formula: Spatial Coordinate → Notice Baseline Failure → Compare Our Curve → One-Sentence Mechanistic Takeaway. 绑定逻辑规则：L018, L015.

Publication-Grade Authentic Text

Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours. Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours. Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours. That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [视觉坐标锁定 Spatial Pointer]:** *Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours.*
  对应步骤：S1 [视觉坐标锁定 Spatial Pointer]：Let's look at the comparison in the right panel of Slide 5, where the horizontal axis tracks prediction horizon and the vertical axis shows MSE. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]:** *Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours.*
  对应步骤：S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]：Notice the dashed orange curve representing the standard Transformer baseline: past 336 steps, error shoots up almost vertically. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [引导观察我方稳定曲线 Guide to Our Curve]:** *Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours.*
  对应步骤：S3 [引导观察我方稳定曲线 Guide to Our Curve]：Now compare that with our solid blue curve at the bottom, which stays flat even out to 720 steps. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [一句话点破视觉差距原因 One-Sentence Takeaway]:** *That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.*
  对应步骤：S4 [一句话点破视觉差距原因 One-Sentence Takeaway]：That gap comes directly from our neural-ODE scale tracker preventing long-range drift. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 演讲总结页（Take-home Message）与问答（Q&A）从容应答段 (BP44)

Expressive Intent: 我这一段想用于演讲最后一页总结（Take-home Messages）以及回答台下教授/审稿人犀利提问时的标准应答模板。

Cohesion Formula: Single Takeaway → Quantitative Triad → Q&A Direct Answer → Empirical Grounding. 绑定逻辑规则：L018, L019.

Publication-Grade Authentic Text

To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics. Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen. — [Q&A Response]: Thank you for that sharp question on irregular missing timestamps. The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [凝练核心带走信息 Core Take-Home Message]:** *To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics.*
  对应步骤：S1 [凝练核心带走信息 Core Take-Home Message]：To wrap up, if there is one takeaway from our work today, it is that [One-Sentence Core Principle]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [成果三点速览 Rapid Recap]:** *Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen.*
  对应步骤：S2 [成果三点速览 Rapid Recap]：By combining [A] and [B], [Model] delivers [Metric 1], [Metric 2], and [Open-Source/Practical Asset]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]:** *Thank you for that sharp question on irregular missing timestamps.*
  对应步骤：S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]：(For Q&A) That is a great question regarding [Topic]—the short answer is [Direct Conclusion]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [Q&A 展开与诚实边界 Q&A Elaboration]:** *The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.*
  对应步骤：S4 [Q&A 展开与诚实边界 Q&A Elaboration]：In our current experiments, we observed [Empirical Fact], and extending this to [Questioner's Scenario] can be handled by [Concrete Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

19.3 FT08: 顶会 5 分钟 Spotlight 演讲完整逐字稿：具身智能实时视觉运动控制（含真机视频解说与延迟对比）

Composition Architecture & Structural Blueprint Chain

Genre: `Academic_Speech` **Paragraph Count:** 6

Blueprint Chain: `BP41` → `BP05` → `BP16` → `BP42` → `BP22` → `BP44`

Paragraph 1: 学术演讲开场 60 秒 Hook：用直观痛点或反直觉现象抓住全场（Oral Opening Hook）(BP41)

Expressive Intent: 我这一段想作为顶会 Oral Presentation、组会汇报或博士面试 PPT 第一页的开场白：在 45–60 秒内不用任何晦涩公式，让全场听众立刻听懂“我们为什么要做这件事、现有方法到底卡在哪”。

Cohesion Formula: Concrete Visual Scenario → Rhetorical Question → Crisp Culprit Identification → Talk Promise. 绑定逻辑规则：L018, L029.

Publication-Grade Authentic Text

Good morning everyone. Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines. Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges needed to predict peak loads. In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [场景直觉切入 Concrete Scenario Hook]:** *Good morning everyone.*
  对应步骤：S1 [场景直觉切入 Concrete Scenario Hook]：Good morning everyone. Imagine deploying a humanoid robot to assemble flexible cables on a moving factory line—or forecasting grid load when a storm hits. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反直觉痛点设问 Counter-intuitive Breakdown]:** *Suppose you are operating a regional wind-power grid when a sudden cold front sweeps across the turbines.*
  对应步骤：S2 [反直觉痛点设问 Counter-intuitive Breakdown]：Standard deep models perform remarkably well when conditions stay calm, yet why do they fail abruptly the moment contact dynamics or weather patterns shift? | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [一针见血揭晓答案 Crisp Root Cause]:** *Standard Transformer forecasters look accurate during calm weather, yet why do their predictions lag by hours right when the storm hits? The*

culprit is instance normalization: in trying to stabilize attention weights, it strips away the exact amplitude surges needed to predict peak loads.
对应步骤：S3 [一针见血揭晓答案 Crisp Root Cause]：The culprit is a hidden stationarity assumption baked into standard attention layers. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [亮出今日主旨 Talk Roadmap]:** *In the next five minutes, I will show how our model, DyTS-Net, resolves this trade-off—cutting extreme-weather forecasting error by 17% while running twice as fast.*
对应步骤：S4 [亮出今日主旨 Talk Roadmap]：In this talk, I will show how our framework, [Model Name], breaks this bottleneck and achieves [Key Metric Gain] in real-time deployment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 挑战与机制一对一映射段（Challenge-to-Mechanism Mapping）(BP05)

Expressive Intent: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。

Cohesion Formula: Challenge 1 → Mechanism 1; Challenge 2 → Mechanism 2 → Joint System Property. 绑定逻辑规则：L003, L011.

Publication-Grade Authentic Text

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [落地难点总起 Implementation Hurdle]:** *Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.*
对应步骤：S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [难点 1 + 靶向设计 1]:** *First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.*

对应步骤：S2 [难点 1 + 靶向设计 1]: First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [难点 2 + 靶向设计 2]:** *Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.*
  对应步骤：S3 [难点 2 + 靶向设计 2]: Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [整体计算/泛化特性 System Property]:** *Together, these designs sustain 50 Hz reactive control under partial visual occlusion.*
  对应步骤：S4 [整体计算/泛化特性 System Property]: Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 具身智能专属：多模态感知-动作闭环与 Sim-to-Real 阐述段（Embodied AI Policy）(BP16)

Expressive Intent: 我这一段想阐述具身智能机器人策略（Embodied Policy）如何融合视觉、本体感觉（Proprioception）或触觉，并克服仿真到真实硬件的动力学差距（Sim-to-Real Gap）。

Cohesion Formula: Visual Occlusion Bottleneck → Asynchronous Multimodal Alignment → Action Chunking → Online Sim-to-Real System Identification. 绑定逻辑规则：L025, L003.

Publication-Grade Authentic Text

Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object. To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings. Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling. At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [单模态开环局限 Open-Loop Visual Limitation]:** *Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object.*
  对应步骤：S1 [单模态开环局限 Open-Loop Visual Limitation]: Relying solely on RGB-D point clouds leaves manipulation policies vulnerable to self-occlusion by the robot gripper during fine-grained contact.

| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [多模态时空对齐 Multi-Modal Alignment]:** *To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings.*
  对应步骤：S2 [多模态时空对齐 Multi-Modal Alignment]：To maintain state observability during contact, our encoder aligns asynchronous 30 Hz visual tokens with 500 Hz joint proprioception and wrist force-torque histories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]:** *Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling.*
  对应步骤：S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]：Rather than predicting single-step torques prone to jitter, the policy outputs overlapping K-step SE(3) end-effector action chunks smoothed via exponential temporal ensembling. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]:** *At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.*
  对应步骤：S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]：During physical deployment, a lightweight history adapter infers latent friction and payload parameters online, bridging the sim-to-real dynamics gap without policy fine-tuning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 演讲幻灯片与复杂图表视觉引导段（Visual Signposting in Tech Talks）(BP42)

Expressive Intent: 我这一段想在演讲到架构图或核心实验曲线（Slide 4 / Slide 7）时，用最自然的口语引导听众的眼睛先看哪、再看哪，瞬间读懂图表背后的胜出逻辑。

Cohesion Formula: Spatial Coordinate → Notice Baseline Failure → Compare Our Curve → One-Sentence Mechanistic Takeaway. 绑定逻辑规则：L018, L015.

Publication-Grade Authentic Text

Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours. Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours. Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours. That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [视觉坐标锁定 Spatial Pointer]:** *Let's focus on the right panel of Slide 6, which plots forecasting error as the prediction window stretches from 96 to 720 hours.*
  对应步骤：S1 [视觉坐标锁定 Spatial Pointer]：Let's look at the comparison in the right panel of Slide 5, where the horizontal axis tracks prediction horizon and the vertical axis shows MSE. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]:** *Follow the dashed orange curve for iTransformer—notice how rapidly the error climbs once we pass 336 hours.*
  对应步骤：S2 [引导观察反方崩溃曲线 Guide to Baseline Curve]：Notice the dashed orange curve representing the standard Transformer baseline: past 336 steps, error shoots up almost vertically. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [引导观察我方稳定曲线 Guide to Our Curve]:** *Now compare that against our solid blue curve along the bottom, which remains nearly flat all the way to 720 hours.*
  对应步骤：S3 [引导观察我方稳定曲线 Guide to Our Curve]：Now compare that with our solid blue curve at the bottom, which stays flat even out to 720 steps. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [一句话点破视觉差距原因 One-Sentence Takeaway]:** *That 17% gap at the far right comes directly from decoupling phase attention from continuous scale tracking.*
  对应步骤：S4 [一句话点破视觉差距原因 One-Sentence Takeaway]：That gap comes directly from our neural-ODE scale tracker preventing long-range drift. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 计算开销、推理延迟与精度帕累托权衡段（Efficiency-Accuracy Pareto Analysis）(BP22)

Expressive Intent: 我这一段想说明我们的模型不仅准，而且算得快、显存省，在“精度-延迟/显存”帕累托前沿（Pareto Frontier）上显著优于现有大模型。

Cohesion Formula: Hardware Benchmark → Complexity Reduction & Memory Gain → Real-Time Latency Margin → Pareto Conclusion. 绑定逻辑规则：L010, L023.

Publication-Grade Authentic Text

Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module. By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer. At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [硬件与测试基准 Hardware Setup]:** *Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module.*
  对应步骤：S1 [硬件与测试基准 Hardware Setup]：Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [复杂度理论与实测对比 Complexity & Runtime]:** *By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer.*
  对应步骤：S2 [复杂度理论与实测对比 Complexity & Runtime]：By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch by [X%] and peak VRAM by [Y%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实时闭环部署可行性 Real-Time Viability]:** *At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.*
  对应步骤：S3 [实时闭环部署可行性 Real-Time Viability]：At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control deadline. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 演讲总结页（Take-home Message）与问答（Q&A）从容应答段 (BP44)

Expressive Intent: 我这一段想用于演讲最后一页总结（Take-home Messages）以及回答台下教授/审稿人犀利提问时的标准应答模板。

Cohesion Formula: Single Takeaway → Quantitative Triad → Q&A Direct Answer → Empirical Grounding. 绑定逻辑规则：L018, L019.

Publication-Grade Authentic Text

To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics. Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen. — [Q&A Response]: Thank you for that sharp question on irregular missing timestamps. The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [凝练核心带走信息 Core Take-Home Message]:** *To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics.*
  对应步骤：S1 [凝练核心带走信息 Core Take-Home Message]：To wrap up, if there is one takeaway from our work today, it is that [One-Sentence Core Principle]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [成果三点速览 Rapid Recap]:** *Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen.*
  对应步骤：S2 [成果三点速览 Rapid Recap]：By combining [A] and [B], [Model] delivers [Metric 1], [Metric 2], and [Open-Source/Practical Asset]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]:** *Thank you for that sharp question on irregular missing timestamps.*
  对应步骤：S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]：(For Q&A) That is a great question regarding [Topic]—the short answer is [Direct Conclusion]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [Q&A 展开与诚实边界 Q&A Elaboration]:** *The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.*
  对应步骤：S4 [Q&A 展开与诚实边界 Q&A Elaboration]：In our current experiments, we observed [Empirical Fact], and extending this to [Questioner's Scenario] can be handled by [Concrete Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

19.4 FT09: ICLR/NeurIPS 满分审稿回复全文（Author Rebuttal：含置顶全局回复 + 创新性辩护 + 新增基线与消融表）

Composition Architecture & Structural Blueprint Chain

Genre: `Conference_Rebuttal` **Paragraph Count:** 6

Blueprint Chain: `BP45` → `BP46` → `BP47` → `BP19` → `BP20` → `BP23`

Paragraph 1: 顶会 Rebuttal 全局总括与新增实验汇总段（Author Rebuttal General Response）(BP45)

Expressive Intent: 我这一段想写 ICLR / NeurIPS / ICML / CVPR 审稿回复（Rebuttal）最顶部的致全体审稿人总括段（Global Response）：在 150 词内用最提气的语气感谢审稿人肯定，并一眼亮出我们在 Rebuttal 期间赶做出的所有硬核新增实验。

Cohesion Formula: Synthesize Reviewer Praise → Announce Concrete Revisions → Numbered Evidence Bullets mapped to Reviewers → Courteous Hook. 绑定逻辑规则：L019, L010.

Publication-Grade Authentic Text

We thank all reviewers for their constructive feedback and for recognizing our framework's 'principled formulation of the stationarity-amplitude trade-off' (R1, R3), 'strong empirical gains across eight benchmarks' (R2), and 'clear ablation design' (R4). To address your valuable suggestions, we conducted four sets of new experiments during the rebuttal period (incorporated in blue in the revised manuscript): (1) Benchmarked against two latest 2025 foundation models, Chronos and MOIRAI, across all horizons (Table R1, addressing R1, R3); (2) Added real-robot grasping evaluations under dynamic lighting and 25% camera occlusion (Section 4.5, R2); and (3) Included full GPU memory and wall-clock latency scaling curves up to $L = 4,096$ (Appendix E, R4). Below we address each reviewer's specific questions in detail.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [致谢并提炼审稿人共同好评 Gratitude & Shared Praise]:** *We thank all reviewers for their constructive feedback and for recognizing our framework's 'principled formulation of the stationarity-amplitude trade-off' (R1, R3), 'strong empirical gains across eight benchmarks' (R2), and 'clear ablation design' (R4).*
  对应步骤: S1 [致谢并提炼审稿人共同好评 Gratitude & Shared Praise]: We thank all reviewers for their constructive feedback and for recognizing our work's [Quote Reviewer 1 Praise] (R1), [Quote Reviewer 2 Praise] (R2), and [Quote Reviewer 3 Praise] (R3). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [概览新增实证工作 Overview of New Experiments]:** *To address your valuable suggestions, we conducted four sets of new experiments during the rebuttal period (incorporated in blue in the revised manuscript): (1) Benchmarked against two latest 2025 foundation models,*

Chronos and MOIRAI, across all horizons (Table R1, addressing R1, R3); (2) Added real-robot grasping evaluations under dynamic lighting and 25% camera occlusion (Section 4.5, R2); and (3) Included full GPU memory and wall-clock latency scaling curves up to $L = 4,096$ (Appendix E, R4).

对应步骤：S2 [概览新增实证工作 Overview of New Experiments]：In response to your valuable suggestions, we have conducted [N] sets of new experiments and theoretical analyses, summarized below and highlighted in blue in the revised PDF: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [分条列出硬核新增实验 Bullet Summary of Additions]:** *Below we address each reviewer's specific questions in detail.*

  对应步骤：S3 [分条列出硬核新增实验 Bullet Summary of Additions]：(1) Added comparisons against recent baselines [A, B] on [Dataset] (Table R1, addressing R1, R3); (2) Evaluated real-world robustness under [Harsh Condition] (Appendix D, R2); (3) Provided wall-clock latency and memory profiling across sequence lengths up to $L = 4,096$ (Appendix E, R3). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 顶会 Rebuttal 针对“创新性存疑（Limited Novelty / Incremental）”的有力澄清段 (BP46)

Expressive Intent: 我这一段想回复审稿人最致命的质疑：'The method seems like a simple combination of existing Module A and Module B (limited novelty).' 既不顶撞审稿人，又能用理论 + 消融数据证明“绝对不是简单拼凑”。

Cohesion Formula: Clarify Distinction → Explain Why Naive Stacking Fails → Highlight Core Coupling Mechanism → Prove with 'Naive A+B' Ablation Data. 绑定逻辑规则：L019, L016.

Publication-Grade Authentic Text

We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs. Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory. Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq. 7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention. To verify this empirically, we implemented a direct 'Naive FFT + ODE' concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over 'Naive FFT + ODE', confirming that the gain stems from our mathematical coupling rather than module stacking.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [礼貌接招并定位误解核心 Courteous Reframing]:** *We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs.*
  对应步骤：S1 [礼貌接招并定位误解核心 Courteous Reframing]：We appreciate Reviewer X's comment on architectural design and would like to clarify the fundamental distinction between our coupled formulation and a naive combination of [A] and [B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [点破简单拼接为何会失败 Why Naive Combination Fails]:** *Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory.*
  对应步骤：S2 [点破简单拼接为何会失败 Why Naive Combination Fails]：Directly stacking standard [A] and [B] fails on [Problem] because [Theoretical Conflict / Gradient Interference]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]:** *Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq.*
  对应步骤：S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]：Our core contribution lies not in using [A] or [B] in isolation, but in deriving the [Specific Coupling Mechanism C], which resolves [Conflict] by [Mathematical Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]:** *7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention.*
  对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline ('Naive A+B') in Table R2: our coupled formulation outperforms 'Naive A+B' by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]:** *To verify this empirically, we implemented a direct 'Naive FFT + ODE' concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over 'Naive FFT + ODE', confirming that the gain stems from our mathematical coupling rather than module stacking.*
  对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline ('Naive A+B') in Table R2: our coupled formulation outperforms 'Naive A+B' by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 顶会 Rebuttal 针对“缺少最新基线或算力不公平（Missing Baselines / Fairness）”的补全段 (BP47)

Expressive Intent: 我这一段想回复审稿人要求补测最新基线模型，或者质疑我们的提升是不是因为参数量更大/训练更久。

Cohesion Formula: Direct Fulfilment → Margin over New Baselines → Parameter-Reduced Variant (‘Ours-Small‘) Proof → Inductive Bias Takeaway. 绑定逻辑规则：L019, L010.

Publication-Grade Authentic Text

Following your suggestion, we benchmarked against the two recent baselines (TimeMixer and FreTS) and evaluated a capacity-reduced variant of our architecture (‘DyTS-Lite‘) in Table R3. Across all horizons on Weather and ECL, standard DyTS-Net reduces MSE by 9.6% over TimeMixer and 13.4% over FreTS. Furthermore, to ensure strict capacity fairness, ‘DyTS-Lite‘ shrinks the hidden dimension to $d = 64$—requiring 42% fewer parameters and 55% fewer FLOPs than TimeMixer—yet still outperforms TimeMixer by 6.8% in average MSE. This confirms that our gains stem from structural-scale disentanglement rather than model size.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [直接响应并呈上新表 Direct Fulfilment]:** *Following your suggestion, we benchmarked against the two recent baselines (TimeMixer and FreTS) and evaluated a capacity-reduced variant of our architecture (‘DyTS-Lite‘) in Table R3.*
  对应步骤：S1 [直接响应并呈上新表 Direct Fulfilment]：Following your suggestion, we have evaluated [Requested Baselines X and Y] under identical splits and added a parameter-matched variant of our model in Table R3. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对比最新基线结果 Performance vs. New Baselines]:** *Across all horizons on Weather and ECL, standard DyTS-Net reduces MSE by 9.6% over TimeMixer and 13.4% over FreTS.*
  对应步骤：S2 [对比最新基线结果 Performance vs. New Baselines]：Across all four horizons on [Datasets], our method outperforms [New Baseline X] by an average of [9.8%] in MSE and [New Baseline Y] by [12.1%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [打消参数量/算力质疑 Parameter & FLOPs Parity Proof]:** *Furthermore, to ensure strict capacity fairness, ‘DyTS-Lite‘ shrinks the hidden dimension to $d = 64$—requiring 42% fewer parameters and 55% fewer FLOPs than TimeMixer—yet still outperforms TimeMixer by 6.8% in average MSE.*
  对应步骤：S3 [打消参数量/算力质疑 Parameter & FLOPs Parity Proof]：Regarding model capacity, even when we scale down our hidden dimension from $d = 256$ to $d = 128$ (‘Ours-Small‘, using 38% fewer parameters and 52% fewer FLOPs than [Baseline X]), ‘Ours-Small‘ still outperforms [Baseline X] by [7.4%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [结论锁定 Conclusion]:** *This confirms that our gains stem from structural-scale disentanglement rather than model size.*
  对应步骤: S4 [结论锁定 Conclusion]: These results confirm that our improvements derive from inductive bias rather than increased parameter count. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 消融实验逐组件归因论证段（Counterfactual Ablation Study）(BP19)

Expressive Intent: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Cohesion Formula: Ablation Overview → w/o Module A (Specific Failure) → w/o Module B (Specific Failure) → Cross-Module Synergy. 绑定逻辑规则：L016, L007.

Publication-Grade Authentic Text

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [消融实验总括 Ablation Setup]:** *To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.*
  对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]:** *Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.*
  对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]:** *Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.*
  对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同效应总结 Synergistic Conclusion]:** *Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.*
  对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 超参数敏感性与极端噪声鲁棒性分析段（Sensitivity & Robustness Analysis）(BP20)

Expressive Intent: 我这一段想回应审稿人最常问的：你的模型是不是对超参数很敏感？在缺失值或传感器强噪声下会不会突然崩盘?

Cohesion Formula: Goal → Wide Hyperparameter Plateau → Stress-Test Comparison → Mechanistic Explanation. 绑定逻辑规则：L010, L019.

Publication-Grade Authentic Text

Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption. Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$. Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), pointwise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%. This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分析目的 Objective]:** *Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption.*
  对应步骤：S1 [分析目的 Objective]：We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [超参数平坦区间 Plateau Stability]:** *Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$.*
  对应步骤：S2 [超参数平坦区间 Plateau Stability]：Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [极端边缘工况压力测试 Stress Test under Corruption]:** *Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%.*
  对应步骤：S3 [极端边缘工况压力测试 Stress Test under Corruption]：Under synthetic sensor dropouts up to [40%] and Gaussian noise ($\sigma = 0.3$), baseline accuracy plummets by over [35%], whereas our model degrades by less than [6.5%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [鲁棒性物理归因 Robustness Mechanism]:** *This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.*
  对应步骤：S4 [鲁棒性物理归因 Robustness Mechanism]：This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 局限性（Limitations）客观陈述与未来扩展段（Constructive Limitations）(BP23)

Expressive Intent: 我这一段想写论文末尾或 NeurIPS/ICLR Checklist 要求的 Limitations 段落：诚实指出当前方法的两个客观边界，但绝不贬损论文的核心贡献。

Cohesion Formula: Two Objective Boundaries → [Boundary 1 + Technical Remedy] → [Boundary 2 + Stochastic Extension]. 绑定逻辑规则：L017.

Publication-Grade Authentic Text

Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting. First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy. Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration. Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [边界 1:** *Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting.*
  对应步骤：S1 [边界 1：极端高维或极度稀疏场景 Boundary 1]：Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [边界 1 机理分析与解法 Mechanism & Solution 1]:** *First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy.*
  对应步骤：S2 [边界 1 机理分析与解法 Mechanism & Solution 1]：First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [边界 2:** *Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration.*
  对应步骤：S3 [边界 2：特定假设限制 Boundary 2]：Second, our continuous scale tracker assumes piecewise differentiable drift and may under-estimate instantaneous discontinuous jumps caused by hardware resets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [未来演进方向 Future Extension]:** *Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.*
  对应步骤：S4 [未来演进方向 Future Extension]：Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

19.5 FT10: 学术答辩与组会 6 大高频刁钻提问应答全案（Novelty、超参敏感性、极端噪声、计算复杂度、失败案例与未来扩展）

Composition Architecture & Structural Blueprint Chain

Genre: `Academic_QA_Defense` **Paragraph Count:** 6

Blueprint Chain: `BP44` → `BP46` → `BP20` → `BP21` → `BP22` → `BP23`

Paragraph 1: 演讲总结页（Take-home Message）与问答（Q&A）从容应答段 (BP44)

Expressive Intent: 我这一段想用于演讲最后一页总结（Take-home Messages）以及回答台下教授/审稿人犀利提问时的标准应答模板。

Cohesion Formula: Single Takeaway → Quantitative Triad → Q&A Direct Answer → Empirical Grounding. 绑定逻辑规则：L018, L019.

Publication-Grade Authentic Text

To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics. Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen. — [Q&A Response]: Thank you for that sharp question on irregular missing timestamps. The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [凝练核心带走信息 Core Take-Home Message]:** *To wrap up, the key takeaway from our work is that non-stationary forecasting requires separating shift-invariant phase structure from continuous amplitude dynamics.*
  对应步骤：S1 [凝练核心带走信息 Core Take-Home Message]：To wrap up, if there is one takeaway from our work today, it is that [One-Sentence Core Principle]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [成果三点速览 Rapid Recap]:** *Doing so lets DyTS-Net cut long-horizon error by up to 17% while running at half the computational cost, and our full code and weights are live at the QR code on screen.*
  对应步骤：S2 [成果三点速览 Rapid Recap]：By combining [A] and [B], [Model] delivers [Metric 1], [Metric 2], and [Open-Source/Practical Asset]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]:** *Thank you for that sharp question on irregular missing timestamps.*

对应步骤：S3 [Q&A 礼貌接招与结论前置 Q&A Direct Answer]：(For Q&A) That is a great question regarding [Topic]—the short answer is [Direct Conclusion]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [Q&A 展开与诚实边界 Q&A Elaboration]:** *The short answer is that our neural-ODE branch handles arbitrary time intervals natively: in our 30% random-dropout stress test, MSE degraded by less than 5%, and we can share the backup slide on irregular sampling right here.*
  对应步骤：S4 [Q&A 展开与诚实边界 Q&A Elaboration]：In our current experiments, we observed [Empirical Fact], and extending this to [Questioner's Scenario] can be handled by [Concrete Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 顶会 Rebuttal 针对“创新性存疑（Limited Novelty / Incremental）”的有力澄清段 (BP46)

Expressive Intent: 我这一段想回复审稿人最致命的质疑：'The method seems like a simple combination of existing Module A and Module B (limited novelty).' 既不顶撞审稿人，又能用理论 + 消融数据证明“绝对不是简单拼凑”。

Cohesion Formula: Clarify Distinction → Explain Why Naive Stacking Fails → Highlight Core Coupling Mechanism → Prove with 'Naive A+B' Ablation Data. 绑定逻辑规则：L019, L016.

Publication-Grade Authentic Text

We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs. Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory. Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq. 7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention. To verify this empirically, we implemented a direct 'Naive FFT + ODE' concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over 'Naive FFT + ODE', confirming that the gain stems from our mathematical coupling rather than module stacking.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [礼貌接招并定位误解核心 Courteous Reframing]:** *We appreciate the opportunity to clarify how our architecture differs fundamentally from a naive combination of spectral filtering and neural ODEs.*
  对应步骤：S1 [礼貌接招并定位误解核心 Courteous Reframing]：We appreciate Reviewer X's comment on architectural design and would like to clarify the fundamental distinction between our coupled formulation and a naive combination of [A] and [B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [点破简单拼接为何会失败 Why Naive Combination Fails]:** *Directly stacking an FFT block with a standard neural ODE fails under abrupt regime shifts because spectral normalization strips the exact boundary values required to initialize the ODE trajectory.*
  对应步骤：S2 [点破简单拼接为何会失败 Why Naive Combination Fails]：Directly stacking standard [A] and [B] fails on [Problem] because [Theoretical Conflict / Gradient Interference]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]:** *Our core technical innovation lies in the bidirectional phase-scale coupling operator (Eq.*
  对应步骤：S3 [阐明本文核心数学/机制桥梁 Our Core Coupling Innovation]：Our core contribution lies not in using [A] or [B] in isolation, but in deriving the [Specific Coupling Mechanism C], which resolves [Conflict] by [Mathematical Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]:** *7), which uses instantaneous spectral entropy to gate ODE stiffness while feeding predicted scale bounds back to regularize frequency attention.*
  对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline (‘Naive A+B‘) in Table R2: our coupled formulation outperforms ‘Naive A+B‘ by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]:** *To verify this empirically, we implemented a direct ‘Naive FFT + ODE‘ concatenation baseline in Table R2 below: our coupled formulation reduces 720-step MSE by 15.2% over ‘Naive FFT + ODE‘, confirming that the gain stems from our mathematical coupling rather than module stacking.*
  对应步骤：S4 [用直接对比实验一锤定音 Empirical Proof of Non-Triviality]：To verify this empirically, we implemented the exact naive concatenation baseline (‘Naive A+B‘) in Table R2: our coupled formulation outperforms ‘Naive A+B‘ by [14.6%] in MSE, confirming that the performance gain stems from our coupling mechanism rather than mere module stacking. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 超参数敏感性与极端噪声鲁棒性分析段（Sensitivity & Robustness Analysis）(BP20)

Expressive Intent: 我这一段想回应审稿人最常问的：你的模型是不是对超参数很敏感？在缺失值或传感器强噪声下会不会突然崩盘?

Cohesion Formula: Goal → Wide Hyperparameter Plateau → Stress-Test Comparison → Mechanistic Explanation. 绑定逻辑规则：L010, L019.

Publication-Grade Authentic Text

Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption. Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$. Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%. This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分析目的 Objective]:** *Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption.*
  对应步骤：S1 [分析目的 Objective]：We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [超参数平坦区间 Plateau Stability]:** *Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$.*
  对应步骤：S2 [超参数平坦区间 Plateau Stability]：Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [极端边缘工况压力测试 Stress Test under Corruption]:** *Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%.*
  对应步骤：S3 [极端边缘工况压力测试 Stress Test under Corruption]：Under synthetic sensor dropouts up to [40%] and Gaussian noise ($\sigma = 0.3$), baseline accuracy plummets by over [35%], whereas our model degrades by less than [6.5%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [鲁棒性物理归因 Robustness Mechanism]:** *This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.*
  对应步骤：S4 [鲁棒性物理归因 Robustness Mechanism]：This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 极端/反常情况与失败案例坦诚剖析段（Failure Case & Outlier Analysis）(BP21)

Expressive Intent: 我这一段想通过可视化案例（Case Study / Qualitative Analysis），既展示模型在复杂跳变下的精准跟踪，又坦诚剖析一个模型预测失准的极端案例。

Cohesion Formula: Visual Setup → Success under Shift → Honest Failure Case under Unobserved Exogenous Shock → Actionable Insight. 绑定逻辑规则：L017, L018.

Publication-Grade Authentic Text

Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark. Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error. Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response. Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [定性对比切入 Qualitative Setup]:** *Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark.*
  对应步骤：S1 [定性对比切入 Qualitative Setup]：Figure 5 visualizes predicted trajectories during an abrupt regime shift on [Dataset/Task]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [成功案例亮点 Success Highlight]:** *Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error.*
  对应步骤：S2 [成功案例亮点 Success Highlight]：While [Baseline] lags behind the sudden step change by [K steps] and overshoots, our model rapidly locks onto the new amplitude regime within [M steps]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [失败案例坦诚剖析 Honest Failure Mode]:** *Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response.*
  对应步骤：S3 [失败案例坦诚剖析 Honest Failure Mode]：Nevertheless, inspecting the bottom-right failure case reveals that when an external manual override occurs without historical precursors, all purely endogenous forecasters—including ours—exhibit a transient lag. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [改进启示 Constructive Takeaway]:** *Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.*

对应步骤：S4 [改进启示 Constructive Takeaway]：Incorporating exogenous event logs or textual maintenance records is necessary to anticipate such unscheduled interventions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 计算开销、推理延迟与精度帕累托权衡段（Efficiency-Accuracy Pareto Analysis）(BP22)

Expressive Intent: 我这一段想说明我们的模型不仅准，而且算得快、显存省，在“精度-延迟/显存”帕累托前沿（Pareto Frontier）上显著优于现有大模型。

Cohesion Formula: Hardware Benchmark → Complexity Reduction & Memory Gain → Real-Time Latency Margin → Pareto Conclusion. 绑定逻辑规则：L010, L023.

Publication-Grade Authentic Text

Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module. By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer. At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [硬件与测试基准 Hardware Setup]:** *Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module.*
  对应步骤：S1 [硬件与测试基准 Hardware Setup]：Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [复杂度理论与实测对比 Complexity & Runtime]:** *By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer.*
  对应步骤：S2 [复杂度理论与实测对比 Complexity & Runtime]：By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch by [X%] and peak VRAM by [Y%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实时闭环部署可行性 Real-Time Viability]:** *At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.*
  对应步骤：S3 [实时闭环部署可行性 Real-Time Viability]：At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control

deadline. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 局限性（Limitations）客观陈述与未来扩展段（Constructive Limitations）(BP23)

Expressive Intent: 我这一段想写论文末尾或 NeurIPS/ICLR Checklist 要求的 Limitations 段落：诚实指出当前方法的两个客观边界，但绝不贬损论文的核心贡献。

Cohesion Formula: Two Objective Boundaries → [Boundary 1 + Technical Remedy] → [Boundary 2 + Stochastic Extension]. 绑定逻辑规则：L017.

Publication-Grade Authentic Text

Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting. First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy. Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration. Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [边界 1:** *Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting.*
  对应步骤：S1 [边界 1：极端高维或极度稀疏场景 Boundary 1]：Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [边界 1 机理分析与解法 Mechanism & Solution 1]:** *First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy.*
  对应步骤：S2 [边界 1 机理分析与解法 Mechanism & Solution 1]：First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [边界 2:** *Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration.*

对应步骤：S3 [边界 2：特定假设限制 Boundary 2]：Second, our continuous scale tracker assumes piecewise differentiable drift and may under-estimate instantaneous discontinuous jumps caused by hardware resets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [未来演进方向 Future Extension]:** *Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.*
  对应步骤：S4 [未来演进方向 Future Extension]：Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Chapter 20

Economics, Finance & Management / 经济金融与管理科学

本章精选该学科领域及对应文体的完整学术范文。每一篇范文不仅展示母语级发表标准的正文全貌，更清晰拆解全篇的“段落蓝图链条（Structural Blueprint Chain）”，并对每个自然段逐一剖析其核心交际意图、句际衔接逻辑公式（Cohesion Formula）以及每句话的句法成分与推进机制。

20.1 FT11: 人工智能方向博士/访问学者申请个人陈述（SoP 全文：从时间序列与生成模型迈向具身智能世界模型）

Composition Architecture & Structural Blueprint Chain

Genre: `Statement_of_Purpose` **Paragraph Count:** 6

Blueprint Chain: `BP48` → `BP03` → `BP04` → `BP15` → `BP16` → `BP24`

Paragraph 1: 科研个人陈述（SoP / Research Narrative）：从时间序列/生成建模迈向具身智能的闭环叙事段 (BP48)

Expressive Intent: 我这一段想写申请博士/博后/访问学者或个人学术主页的 Research Statement / SoP 核心段：把我已经做过的深度学习/时间序列/生成模型工作，与我未来想探索的具身智能（Embodied AI）无缝串联成一条充满内驱力的科研主线。

Cohesion Formula: Past Core Thread → Concrete Method & Metric → Intellectual Bridge (Passive Forecasting → Active Closed-Loop Control) → Future Embodied AI Agenda. 绑定逻辑规则：L003, L025, L029.

Publication-Grade Authentic Text

My research focuses on learning reliable representations and generative trajectories from non-stationary sequential data. While investigating multivariate industrial telemetry, I found that conventional attention mechanisms collapse when underlying system dynamics shift—motivating me to design a neural-ODE coupled spectral forecaster that reduced long-range drift by 17.4%. Yet passive sequence forecasting only observes how a system evolves; intelligent physical agents must actively intervene within that loop, where every motor command alters future state distributions. Bridging my background in non-stationary time-series modeling and fast flow-matching samplers, my future research aims to build adaptive visuomotor world models that allow embodied robots to generalize across unfamiliar contact regimes in real time.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [过往研究主线凝练 Past Research Thread]:** *My research focuses on learning reliable representations and generative trajectories from non-stationary sequential data.*
  对应步骤：S1 [过往研究主线凝练 Past Research Thread]：My prior research has centered on modeling non-stationary dynamics and uncertainty in high-dimensional sequential data. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表作攻克过程与硬核成果 Concrete Achievement]:** *While investigating multivariate industrial telemetry, I found that conventional attention mechanisms collapse when underlying system dynamics shift—motivating me to design a neural-ODE coupled spectral forecaster that reduced long-range drift by 17.4%.*
  对应步骤：S2 [代表作攻克过程与硬核成果 Concrete Achievement]：In my recent work on [Time Series / Generative Modeling], I observed that [Core Bottleneck], which led me to formulate [Method]—reducing long-horizon drift by [X%] across industrial telemetry benchmarks. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [向具身智能跃迁的科学顿悟 Scientific Pivot to Embodied AI]:** *Yet passive sequence forecasting only observes how a system evolves; intelligent physical agents must actively intervene within that loop, where every motor command alters future state distributions.*
  对应步骤：S3 [向具身智能跃迁的科学顿悟 Scientific Pivot to Embodied AI]：While passive sequence forecasting predicts how an environment evolves, physical decision-making requires closing the loop: actions actively alter future state distributions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [未来研究愿景 Future Research Agenda]:** *Bridging my background in non-stationary time-series modeling and fast flow-matching samplers, my future research aims to build adaptive visuomotor world models that allow embodied robots to generalize across unfamiliar contact regimes in real time.*
  对应步骤：S4 [未来研究愿景 Future Research Agenda]：Building on my foundation in continuous-time generative modeling, my doctoral/future research aims to develop sample-efficient visuomotor world models that enable robots to adapt online to non-stationary contact dynamics. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 尖锐指出核心技术瓶颈与失效机理（Gap Identification）(BP03)

Expressive Intent: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Cohesion Formula: 因果链式推进（A induces B → This B compounds into C → Naive fix D sacrifices E）。绑定逻辑规则：L003, L004, L024/L025.

Publication-Grade Authentic Text

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*
  对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*
  对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 亮出本文核心洞察与破局方法总览（Key Insight & Method Overview）(BP04)

Expressive Intent: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。

Cohesion Formula: Problem Anchor → Key Insight → Dual-Module Mechanism → Pareto Resolution. 绑定逻辑规则：L002, L023.

Publication-Grade Authentic Text

To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch. Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [方法登场与定位 Method Introduction]:** *To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.*
  对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心直觉洞察 Key Insight]:** *Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.*
  对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [模块协同主干 Module Synergy]:** *Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.*
  对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [闭环收益总结 Closed-Loop Benefit]:** *Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.*
  对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 生成模型专属：条件控制与采样去噪轨迹阐述段（Generative Modeling Mechanism）(BP15)

Expressive Intent: 我这一段想阐述生成模型（Diffusion / Flow Matching / Autoregressive）如何在满足外部物理约束或条件观测的前提下，生成高质量且多样化的序列/轨迹。

Cohesion Formula: Guidance Artifact Problem → Intrinsic Velocity Conditioning → Tangent Projection Step → Fast Exact Convergence. 绑定逻辑规则：L026, L014.

Publication-Grade Authentic Text

Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations. To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field. At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator. This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]:** *Applying post-hoc gradient guidance during diffusion sampling often pushes intermediate noisy states off the data manifold, introducing unphysical high-frequency oscillations.*
  对应步骤：S1 [无条件/朴素引导缺陷 Naive Guidance Bottleneck]：Applying post-hoc classifier guidance to pre-trained diffusion samplers often pushes intermediate noisy states off the data manifold, inducing high-frequency artifacts. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]:** *To enforce physical boundary constraints without manifold deviation, we incorporate observation consistency directly into the conditional flow-matching velocity field.*
  对应步骤：S2 [本文流形内条件注入机制 Manifold-Consistent Conditioning]：To enforce hard boundary constraints without manifold deviation, we embed [Condition/Constraint] directly into the velocity field / denoising kernel. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具体轨迹更新方程 Trajectory Update Formulation]:** *At flow time $\tau \in [0, 1]$, our network predicts a residual velocity $v_\theta(x_\tau, \tau \mid c)$ that projects the transport trajectory onto the null space of the linear constraint operator.*
  对应步骤：S3 [具体轨迹更新方程 Trajectory Update Formulation]：At solver step $\tau \in [0, 1]$, the conditional velocity field $v_\theta(x_\tau, \tau \mid c)$ projects the unconstrained flow onto the tangent space of feasible trajectories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]:** *This projection guarantees exact boundary satisfaction at terminal time $\tau = 1$ while maintaining straight transport paths amenable to 4-step Euler integration.*
  对应步骤：S4 [采样效率与分布保真收益 Sampling & Fidelity Guarantee]：This tangent projection guarantees exact boundary satisfaction at $\tau = 1$ while permitting straight 4-step Euler integration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 具身智能专属：多模态感知-动作闭环与 Sim-to-Real 阐述段（Embodied AI Policy）(BP16)

Expressive Intent: 我这一段想阐述具身智能机器人策略（Embodied Policy）如何融合视觉、本体感觉（Proprioception）或触觉，并克服仿真到真实硬件的动力学差距（Sim-to-Real Gap）。

Cohesion Formula: Visual Occlusion Bottleneck → Asynchronous Multimodal Alignment → Action Chunking → Online Sim-to-Real System Identification. 绑定逻辑规则：L025, L003.

Publication-Grade Authentic Text

Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object. To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings. Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling. At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [单模态开环局限 Open-Loop Visual Limitation]:** *Relying solely on third-person RGB-D observations leaves visuomotor policies vulnerable to severe self-occlusion once the gripper approaches the target object.*
  对应步骤：S1 [单模态开环局限 Open-Loop Visual Limitation]：Relying solely on RGB-D point clouds leaves manipulation policies vulnerable to self-occlusion by the robot gripper during fine-grained contact. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [多模态时空对齐 Multi-Modal Alignment]:** *To preserve observability during contact-rich manipulation, our encoder aligns 30 Hz visual spatial tokens with 500 Hz joint proprioception and wrist force-torque histories via cross-modal rotary embeddings.*
  对应步骤：S2 [多模态时空对齐 Multi-Modal Alignment]：To maintain state observability during contact, our encoder aligns asynchronous 30 Hz visual tokens with 500 Hz joint proprioception and wrist force-

torque histories. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]:** *Rather than issuing single-step commands prone to high-frequency jitter, the policy predicts 16-step SE(3) end-effector action chunks smoothed through exponential temporal ensembling.*
  对应步骤：S3 [动作块预测与平滑执行 Action Chunking & Temporal Ensembling]: Rather than predicting single-step torques prone to jitter, the policy outputs overlapping K-step SE(3) end-effector action chunks smoothed via exponential temporal ensembling. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]:** *At deployment, a recurrent history adapter estimates latent surface friction and payload mass online, bridging the sim-to-real dynamics gap in zero shot.*
  对应步骤：S4 [Sim-to-Real 自适应机制 Sim-to-Real Adaptation]: During physical deployment, a lightweight history adapter infers latent friction and payload parameters online, bridging the sim-to-real dynamics gap without policy fine-tuning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 全文总结与学术影响收尾段（Conclusion & Broader Outlook）(BP24)

Expressive Intent: 我这一段想写整篇小论文的 Conclusion：不简单复制粘贴 Abstract，而是从更高维度总结本文解决的科学矛盾、方法精髓、实证高度与领域启示。

Cohesion Formula: Tension Resolved → Methodological Insight → Empirical Confirmation → Generalizable Blueprint. 绑定逻辑规则：L013, L023, L029.

Publication-Grade Authentic Text

In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting. By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes. Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead. Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心科学问题闭合 Scientific Closure]:** *In this work, we resolved the long-standing tension between shift-invariant normalization and physical amplitude preservation in non-stationary time series forecasting.*

对应步骤：S1 [核心科学问题闭合 Scientific Closure]：In this paper, we addressed the fundamental tension between [Goal A] and [Goal B] in [Domain]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [方法论精髓回顾 Methodological Synthesis]:** *By decoupling frequency-domain phase attention from continuous neural-ODE scale tracking, DyTS-Net captures long-range seasonal dependencies while accurately reconstructing extreme peak magnitudes.*
  对应步骤：S2 [方法论精髓回顾 Methodological Synthesis]：By introducing [Model/Mechanism], we demonstrated that [Core Insight] enables [Capability] without incurring [Traditional Penalty]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实证结果高度概括 Empirical Recap]:** *Empirical evaluations across eight benchmarks and industrial wind-power telemetry demonstrated MSE reductions of up to 17.4% alongside a 45% reduction in computational overhead.*
  对应步骤：S3 [实证结果高度概括 Empirical Recap]：Comprehensive evaluations across [Benchmarks/Hardware] confirmed consistent error reductions of up to [X%] alongside substantial inference speedups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [跨领域迁移展望 Broader Outlook]:** *Beyond forecasting, this structural-scale disentanglement principle provides a generalizable foundation for anomaly detection and closed-loop predictive control in non-stationary cyber-physical systems.*
  对应步骤：S4 [跨领域迁移展望 Broader Outlook]：Beyond [Current Task], our [Core Principle] offers a generalizable blueprint for broader sequential decision-making tasks under non-stationary environments. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

20.2 FT12: 具身智能与非平稳时序建模交叉课题立项申请书（Research Proposal 全文：研究背景、核心科学问题、技术路线与预期成果）

Composition Architecture & Structural Blueprint Chain

Genre: `Research_Proposal` **Paragraph Count:** 7

Blueprint Chain: `BP01` → `BP02` → `BP03` → `BP04` → `BP05` → `BP06` → `BP07`

Paragraph 1: 确立研究背景与真实工程价值（拒绝空喊口号）(BP01)

Expressive Intent: 我这一段想作为 Introduction 第一段：说明多变量时间序列预测或具身智能控制在实际工业系统中的关键作用，以及高可靠性的刚需。

Cohesion Formula: Given-to-New 链：Core Task (S1) → Practical Deployment (S2) → Real-world Disturbances (S3) → Consequence of Failure (S4). 绑定逻辑规则：L001, L011, L029.

Publication-Grade Authentic Text

Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing. In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload. However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise. Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [任务锚定 Topic Anchor]:** *Multivariate time series forecasting underpins safety-critical scheduling across smart power grids, cloud clusters, and automated manufacturing.*
  对应步骤：S1 [任务锚定 Topic Anchor]：直接用紧凑主谓宾点明研究对象及其在真实系统中的核心功能（≤16 词，禁用 In the era of AI）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [应用场景展开 Concrete Scenarios]:** *In these operational settings, control systems rely on multi-horizon predictions to balance supply fluctuations and prevent cascading overload.*
  对应步骤：S2 [应用场景展开 Concrete Scenarios]：列举 2–3 个具体工程场景（如电网负荷调度、灵巧机械手装配）及其决策依赖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [苛刻工况约束 Operational Bottleneck]:** *However, field sensors routinely operate under abrupt regime shifts, irregular sampling intervals, and high-frequency measurement noise.*
  对应步骤：S3 [苛刻工况约束 Operational Bottleneck]：指出真实部署环境下不可避免的物理/统计挑战（如非平稳漂移、传感器缺失、毫秒级延迟约束）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [研究必要性收束 Imperative Hook]:** *Failing to accommodate such non-stationary disturbances degrades long-range forecast reliability and triggers costly control oscillations.*
  对应步骤：S4 [研究必要性收束 Imperative Hook]：点明若无法克服上述约束会造成什么具体系统后果，自然引出下一段对现有方法的审视。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 总结现有主流范式及其隐含假设（欲抑先扬铺垫段）(BP02)

Expressive Intent: 我这一段想客观概括前人主流模型（如 Transformer / Diffusion / MLP）取得了什么进展，但点出它们共同依赖的一个脆弱前提假设。

Cohesion Formula: 先让步肯定（While/Recent paradigms...）→ 后釜底抽薪揭示隐含前提（Yet these architectures implicitly assume...）。绑定逻辑规则：L008, L009.

Publication-Grade Authentic Text

Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition. These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments. Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window. When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [流派归类 Taxonomic Opening]:** *Recent deep forecasters have substantially reduced long-horizon error by replacing point-wise attention with sub-series patch tokenization and frequency-domain decomposition.*
  对应步骤：S1 [流派归类 Taxonomic Opening]：按核心机制归纳现有主流范式及其已解决的子问题。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [代表性机制展开 Mechanism Recap]:** *These architectures effectively capture recurring seasonal patterns when training and test sequences share stationary statistical moments.*
  对应步骤：S2 [代表性机制展开 Mechanism Recap]：简述这些方法如何通过特定架构（如 Patch 分块、频域分解或扩散去噪）提升表征能力。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [转折揭示隐含假设 Pivot to Hidden Assumption]:** *Yet they implicitly assume that inter-channel correlations and spectral energy distributions remain time-invariant across the prediction window.*
  对应步骤：S3 [转折揭示隐含假设 Pivot to Hidden Assumption]：用 Yet / However 锋利转折，指出它们暗中依赖的理想化假设（如平稳分布、静态通道相关性、仿真动力学无偏差）。| 句首锚定旧

信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [假设破裂后果 Breakdown Under Stress]:** *When external shocks alter underlying system dynamics, this stationarity assumption collapses, causing frozen spectral filters to project outdated trends.*
  对应步骤：S4 [假设破裂后果 Breakdown Under Stress]：说明当真实数据违背该假设时，现有架构为何会遭遇性能断崖。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 尖锐指出核心技术瓶颈与失效机理（Gap Identification）(BP03)

Expressive Intent: 我这一段想深入剖析现有方法在特定难题（如非平稳突变、长程自回归误差累积、Sim-to-Real 接触失配）上为什么会失效，把机理讲透。

Cohesion Formula: 因果链式推进（A induces B → This B compounds into C → Naive fix D sacrifices E）。绑定逻辑规则：L003, L004, L024/L025.

Publication-Grade Authentic Text

Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting. While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks. Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons. Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [核心痛点立论 Bottleneck Claim]:** *Bridging non-stationary robustness and amplitude fidelity remains a core bottleneck in long-horizon forecasting.*
  对应步骤：S1 [核心痛点立论 Bottleneck Claim]：一句话直接命名该核心瓶颈（如 Compound error accumulation / Sim-to-real contact mismatch）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层失效机理 Primary Failure Mechanism]:** *While instance normalization strips time-varying mean and variance shifts to stabilize attention weights, it simultaneously erases physical scale cues required to predict extreme load peaks.*
  对应步骤：S2 [第一层失效机理 Primary Failure Mechanism]：从数学或物理传导链条解释误差是如何产生并放大的。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [第二层耦合困境 Secondary Dilemma]:** *Conversely, retaining raw unnormalized inputs exposes self-attention layers to severe covariate shift, causing validation error to diverge over extended horizons.*
  对应步骤：S3 [第二层耦合困境 Secondary Dilemma]：指出为什么简单的修补手段（如单纯加大模型、简单归一化或域随机化）不仅无效还会带来副作用。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [凝练核心研究问题 Research Question Formulation]:** *Resolving this tension requires a mechanism that disentangles intrinsic temporal dynamics from transient scale drift without discarding magnitude information.*
  对应步骤：S4 [凝练核心研究问题 Research Question Formulation]：用 Therefore, a fundamental question arises: how can we... 引出本文目标。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 亮出本文核心洞察与破局方法总览（Key Insight & Method Overview）(BP04)

Expressive Intent: 我这一段想在指出前人痛点后，正式亮出我们的方法（To address this dilemma, we propose...），并用最通俗清晰的直觉告诉审稿人为什么我们的设计能破局。

Cohesion Formula: Problem Anchor → Key Insight → Dual-Module Mechanism → Pareto Resolution. 绑定逻辑规则：L002, L023.

Publication-Grade Authentic Text

To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation. Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system. Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch. Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [方法登场与定位 Method Introduction]:** *To resolve this dilemma, we propose DyTS-Net, a dual-stream forecasting architecture that decouples stationary structural modeling from dynamic scale trajectory estimation.*
  对应步骤：S1 [方法登场与定位 Method Introduction]：To address this dilemma, we propose [Model Name], a [Core Paradigm] framework that [Primary Goal]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [核心直觉洞察 Key Insight]:** *Our key insight is that while high-frequency phase relationships benefit from normalized shift-invariant attention, macro-scale amplitude evolution follows a low-dimensional continuous dynamical system.*
  对应步骤：S2 [核心直觉洞察 Key Insight]：Our key insight is that [Physical/Statistical Principle]，点破破局的底层支点。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [模块协同主干 Module Synergy]:** *Specifically, DyTS-Net first extracts scale-invariant temporal tokens via spectral patch attention, and concurrently models non-stationary mean-variance evolution through a lightweight neural ODE branch.*
  对应步骤：S3 [模块协同主干 Module Synergy]：用 First / Specifically 与 Concurrently / Furthermore 简述两大核心模块如何分工咬合。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [闭环收益总结 Closed-Loop Benefit]:** *Recombining both streams at the decoding stage preserves sharp peak magnitudes without sacrificing training stability.*
  对应步骤：S4 [闭环收益总结 Closed-Loop Benefit]：说明这种协同设计如何同时兼顾原先冲突的两个指标（如兼顾平稳表征与幅值保真，或兼顾多模态多样性与单步实时控制）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 挑战与机制一对一映射段（Challenge-to-Mechanism Mapping）(BP05)

Expressive Intent: 我这一段想说明：把我们的核心思想落地时遇到了两个具体工程/数学难点，我们分别设计了什么精巧机制去攻克它们。

Cohesion Formula: Challenge 1 → Mechanism 1; Challenge 2 → Mechanism 2 → Joint System Property. 绑定逻辑规则：L003, L011.

Publication-Grade Authentic Text

Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges. First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset. Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps. Together, these designs sustain 50 Hz reactive control under partial visual occlusion.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [落地难点总起 Implementation Hurdle]:** *Instantiating closed-loop visuomotor diffusion on physical robots entails two technical challenges.*

对应步骤：S1 [落地难点总起 Implementation Hurdle]：Translating this principle into a scalable architecture entails two technical challenges. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [难点 1 + 靶向设计 1]:** *First, because raw depth camera streams contain occlusion artifacts around reflective object edges, we introduce a tactile-guided cross-attention gate that reweights visual tokens upon contact onset.*
  对应步骤：S2 [难点 1 + 靶向设计 1]: First, because [Challenge 1], we design [Mechanism 1] to [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [难点 2 + 靶向设计 2]:** *Second, to eliminate the 200 ms sampling bottleneck of iterative denoising, we formulate a trajectory consistency distillation objective that maps Gaussian noise directly to collision-free action chunks in two forward steps.*
  对应步骤：S3 [难点 2 + 靶向设计 2]：Second, to prevent [Challenge 2] under [Constraint], we introduce [Mechanism 2] that [Specific Function]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [整体计算/泛化特性 System Property]:** *Together, these designs sustain 50 Hz reactive control under partial visual occlusion.*
  对应步骤：S4 [整体计算/泛化特性 System Property]：Together, these components enable [Model] to operate at [Efficiency/Robustness Metric]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 6: 实验证言与性能突破预览段（Quantitative Empirical Preview）(BP06)

Expressive Intent: 我这一段想在 Introduction 结尾前，用三句话硬核总结我们在哪些基准数据集/真实硬件上做了实验，取得了多大幅度的领先。

Cohesion Formula: Evaluation Scope → Accuracy Gain over Strongest Baselines → Efficiency/Generalization Gain. 绑定逻辑规则：L010, L015.

Publication-Grade Authentic Text

We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors. Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions. Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [评测广度 Scope of Evaluation]:** *We evaluate DyTS-Net across eight long-horizon forecasting benchmarks and a real-world wind-turbine telemetry system spanning 1,400 sensors.*

对应步骤：S1 [评测广度 Scope of Evaluation]：We evaluate [Model] across [N] public benchmarks and [Real-world Hardware/Scenario], spanning [Tasks]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [核心精度突破 Primary Accuracy Gain]:** *Across all prediction horizons up to 720 steps, DyTS-Net reduces MSE by 12.8% over PatchTST and 17.4% over iTransformer, with the largest gains emerging during sudden weather-front transitions.*
  对应步骤：S2 [核心精度突破 Primary Accuracy Gain]：实验主指标对比（必须包含具体百分比、最强基线名称及困难场景表现）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]:** *Furthermore, our architecture requires 45% fewer FLOPs than full channel-mixing Transformers and transfers zero-shot to newly deployed turbine units without fine-tuning.*
  对应步骤：S3 [效率或泛化突破 Efficiency / Zero-Shot Gain]：报告计算速度、显存占用或零样本跨场景泛化收益。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 7: 凝练论文三点核心贡献（Contributions Summary）(BP07)

Expressive Intent: 我这一段想在 Introduction 最末尾清晰列出本文的三点核心贡献（理论/视角贡献 + 核心算法/架构贡献 + 实验与开源实证贡献）。

Cohesion Formula: Parallel Grammatical Structure（严格平行结构：全部用 We + 强及物动词 identify / propose / demonstrate 开头）。绑定逻辑规则：L002, L010.

Publication-Grade Authentic Text

In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability. (2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker. (3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总起引导句 Lead-in]:** *In summary, our main contributions are threefold: (1) We formalize the amplitude-stationarity trade-off in long-horizon forecasting and prove that standard instance normalization bounds peak-load recoverability.*
  对应步骤：S1 [总起引导句 Lead-in]：In summary, our main contributions are threefold: | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [贡献 1:** *(2) We propose DyTS-Net, a hybrid architecture that couples shift-invariant spectral attention with a continuous neural ODE scale tracker.*
  对应步骤：S2 [贡献 1：新视角/问题形式化 Conceptual/Formulation Contribution]：用强动词名物化或动词原形开头（We identify and formalize... / We establish...），点明打破了什么传统局限。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [贡献 2:** *(3) We demonstrate across eight public benchmarks and industrial wind-power telemetry that DyTS-Net lowers long-range MSE by up to 17.4% while cutting inference latency nearly in half.*
  对应步骤：S3 [贡献 2：核心方法与模块 Algorithmic/Architectural Contribution]：We propose [Model/Mechanism], which integrates [A] and [B] to achieve [Theoretical/Functional Guarantee]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

20.3 FT13: 雅思小作文（Task 1）动态多折线图满分范文：四国清洁能源与煤炭发电占比 25 年演进及主次反转

Composition Architecture & Structural Blueprint Chain

Genre: `IELTS_Task1_Dynamic` **Paragraph Count:** 3

Blueprint Chain: `BP25` → `BP26` → `BP32`

Paragraph 1: 图表小作文首段：题目改写与总体趋势概览（Paraphrase + Macro Overview）(BP25)

Expressive Intent: 我这一段想写雅思 Task 1 或数据报告的第一段：用同义句法改写图表标题，并用两句话高度凝练全图最核心的 2 个宏观趋势（极值、反超拐点或主次格局），绝不堆砌零碎数字。

Cohesion Formula: Paraphrase → Overall Trajectory Contrast → Dominant Crossover/Extremum (Zero raw numbers in Overview!). 绑定逻辑规则：L013, L015.

Publication-Grade Authentic Text

The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025. Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily. Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [图表命题同义改写 Prompt Paraphrase]:** *The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025.*
  对应步骤：S1 [图表命题同义改写 Prompt Paraphrase]：The line graph / bar chart compares [Variable/Metric] across [Categories/Regions] over a [N]-year period from [Year A] to [Year B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [宏观趋势 1:** *Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily.*
  对应步骤：S2 [宏观趋势 1：总体走向与反超 Overview Trend 1]：Overall, while [Category A and B] experienced a steady upward trajectory, [Category C] declined markedly, leading to a reversal in dominance around [Mid-Year]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [宏观趋势 2:** *Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.*

对应步骤：S3 [宏观趋势 2：极值或差距特征 Overview Trend 2]：Notably, [Category A] registered the most dramatic growth throughout the period, ultimately overtaking all other sectors to become the primary contributor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 动态时间序列趋势分段描述（暴涨、缓降、平台期与拐点）(BP26)

Expressive Intent: 我这一段想写动态折线图（或模型训练收敛曲线）的主体细节段：把一条经历“初期缓慢爬升 → 中期爆发增长 → 后期高位平台期”的曲线，以及另一条“冲高回落”的曲线对比写活。

Cohesion Formula: Initial Disparity → Multi-Phase Surge & Plateau (Entity A) → Peak-then-Decline (Entity B) → Gap Reversal. 绑定逻辑规则：L010, L013.

Publication-Grade Authentic Text

In 2000, battery-electric vehicles accounted for a mere 4% of municipal fleet purchases, trailing diesel buses at 62%. Following five years of marginal fluctuation, electric adoption accelerated sharply after 2006, quadrupling to 38% by 2018 before leveling off at approximately 41% through 2025. By contrast, diesel procurement peaked briefly at 67% in 2004 but subsequently underwent a sustained contraction, plunging to 19% by the end of the survey. This divergent trajectory reversed the initial 58-percentage-point deficit into a more than twofold lead for electric fleets.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [起始点位对比 Initial Standing]:** *In 2000, battery-electric vehicles accounted for a mere 4% of municipal fleet purchases, trailing diesel buses at 62%.*
  对应步骤：S1 [起始点位对比 Initial Standing]：At the beginning of the period in [2000], [Entity A] stood at a modest [12%],less than a quarter of [Entity B]’s initial figure ([54%]). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [多阶段演进:** *Following five years of marginal fluctuation, electric adoption accelerated sharply after 2006, quadrupling to 38% by 2018 before leveling off at approximately 41% through 2025.*
  对应步骤：S2 [多阶段演进：缓升转暴涨 Phased Acceleration]：After rising marginally over the firstfive years, [Entity A] surged nearly threefold between [2005] and [2018], peaking at [48%] before stabilizing on a high plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [反向曲线对比:** *By contrast, diesel procurement peaked briefly at 67% in 2004 but subsequently underwent a sustained contraction, plunging to 19% by the end of the survey.*
  对应步骤：S3 [反向曲线对比：冲高回落 Counter-Trajectory]：Conversely, [Entity B] climbed briefly to [61%] in [2004] but subsequently entered a prolonged decline, dropping by more than half to [24%] by

[2025]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [终局交叉总结 Terminal Comparison]:** *This divergent trajectory reversed the initial 58-percentage-point deficit into a more than twofold lead for electric fleets.*
  对应步骤：S4 [终局交叉总结 Terminal Comparison]：Consequently, the initial [42]-percentage-point gap between the two categories had completely reversed by the final year. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 异常数据点（Outliers）与局部背离现象精准批注段（Outlier & Anomaly Annotation）(BP32)

Expressive Intent: 我这一段想在图表或实验结果段中，专门挑出违背总体规律的 1–2 个异常点（Outliers）或局部反常尖峰进行精准描述与成因剖析。

Cohesion Formula: General Rule vs. Notable Departure → Quantitative Deviation → Root Cause Attribution → Calibration Takeaway. 绑定逻辑规则：L007, L013.

Publication-Grade Authentic Text

Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset. Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate. Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks. This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总体规律背景下的特例锚定 Outlier Identification]:** *Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset.*
  对应步骤：S1 [总体规律背景下的特例锚定 Outlier Identification]：Although [General Trend holds across most groups/steps], a notable departure from this pattern occurs in [Specific Point/Group]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反常幅度量化 Quantitative Deviation]:** *Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate.*
  对应步骤：S2 [反常幅度量化 Quantitative Deviation]：Whereas peer categories cluster tightly within [A, B], [Outlier Entity] spikes/drops to [Extreme Value], deviating by more than [X%] from the group

median. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [底层物理/统计成因剖析 Root Cause Analysis]:** *Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks.*
  对应步骤：S3 [底层物理/统计成因剖析 Root Cause Analysis]：Closer inspection indicates that this anomaly stems from [Specific Structural/External Factor] rather than random variance. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [方法论启示 Methodological Implication]:** *This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.*
  对应步骤：S4 [方法论启示 Methodological Implication]：Accounting for such localized departures is critical when calibrating [Downstream Model/Policy]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

20.4 FT14: 雅思小作文（Task 1）静态柱状图与饼图组合满分范文：云端推理与机载机器人控制系统的能耗结构对比

Composition Architecture & Structural Blueprint Chain

Genre: `IELTS_Task1_Static` **Paragraph Count:** 3

Blueprint Chain: `BP25` → `BP27` → `BP28`

Paragraph 1: 图表小作文首段：题目改写与总体趋势概览（Paraphrase + Macro Overview）(BP25)

Expressive Intent: 我这一段想写雅思 Task 1 或数据报告的第一段：用同义句法改写图表标题，并用两句话高度凝练全图最核心的 2 个宏观趋势（极值、反超拐点或主次格局），绝不堆砌零碎数字。

Cohesion Formula: Paraphrase → Overall Trajectory Contrast → Dominant Crossover/Extremum (Zero raw numbers in Overview!). 绑定逻辑规则：L013, L015.

Publication-Grade Authentic Text

The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025. Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily. Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [图表命题同义改写 Prompt Paraphrase]:** *The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025.*
  对应步骤：S1 [图表命题同义改写 Prompt Paraphrase]：The line graph / bar chart compares [Variable/Metric] across [Categories/Regions] over a [N]-year period from [Year A] to [Year B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [宏观趋势 1:** *Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily.*
  对应步骤：S2 [宏观趋势 1：总体走向与反超 Overview Trend 1]：Overall, while [Category A and B] experienced a steady upward trajectory, [Category C] declined markedly, leading to a reversal in dominance around [Mid-Year]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [宏观趋势 2:** *Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.*

对应步骤：S3 [宏观趋势 2：极值或差距特征 Overview Trend 2]：Notably, [Category A] registered the most dramatic growth throughout the period, ultimately overtaking all other sectors to become the primary contributor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 静态多组数据横向对比与倍差/占比凸显段（Cross-Group Proportions）(BP27)

Expressive Intent: 我这一段想写静态柱状图或饼图的主体段：横向对比不同国家/不同模型在不同开销维度上的占比差异，灵活运用倍数（double/triple）、分数（nearly a third）与排序结构。

Cohesion Formula: Dominant Item with Multiplier → Sharp Cross-Group Contrast → Pooled Minor Items → Structural Synthesis. 绑定逻辑规则：L009, L010.

Publication-Grade Authentic Text

Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%). A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocessing dominates energy usage at 51%, while attention overhead contracts to just 16%. Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption. This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [主导项锁定 Dominant Category]:** *Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%).*
  对应步骤：S1 [主导项锁定 Dominant Category]：Regarding [Group 1], [Category A] claims the largest share at [46%], nearly double the proportion allocated to [Category B] ([24%]). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [组间横向反差 Cross-Group Contrast]:** *A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocessing dominates energy usage at 51%, while attention overhead contracts to just 16%.*
  对应步骤：S2 [组间横向反差 Cross-Group Contrast]：A markedly different allocation pattern emerges in [Group 2], where [Category C] dominates at [52%], while [Category A] shrinks to just [15%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [次要项合并归纳 Minor Categories Grouping]:** *Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption.*

对应步骤：S3 [次要项合并归纳 Minor Categories Grouping]：Meanwhile, the remaining categories—[Category D and E]—collectively account for less than [12%] across both groups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [结构性差异提炼 Structural Takeaway]:** *This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.*
  对应步骤：S4 [结构性差异提炼 Structural Takeaway]：These figures indicate that whereas [Group 1] prioritizes [Function A], [Group 2] concentrates resources heavily on [Function C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 双图关联与互补/因果印证段（Synthesizing Dual Charts）(BP28)

Expressive Intent: 我这一段想写双图题（如一张图是算力投入/研发经费变化，另一张图是模型误差/碳排放强度变化）的综合关联段，阐明两张图之间的内在协同或背离关系。

Cohesion Formula: Global Correlation → Co-movement Phase → Diminishing Returns / Divergence → Notable Outlier. 绑定逻辑规则：L007, L013.

Publication-Grade Authentic Text

Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold. Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [双图联动锚定 Cross-Chart Link]:** *Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold.*
  对应步骤：S1 [双图联动锚定 Cross-Chart Link]：Synthesizing both charts reveals a clear inverse / positive relationship between [Chart 1 Variable] and [Chart 2 Variable] up to [Threshold Point]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [区间 1 协同变化 Phase 1 Co-movement]:** *Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8% to 0.9% (Figure 2).*
  对应步骤：S2 [区间 1 协同变化 Phase 1 Co-movement]：Specifically, as [Variable X in Chart 1] expanded from [A] to [B] during [Period 1], [Variable Y in Chart 2] dropped correspondingly by [C%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]:** *Beyond $40 million, however, further capital expenditure yielded diminishing returns: doubling investment to $80 million reduced defects by only an additional 0.1 percentage points.*
  对应步骤：S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]：However, beyond [Threshold Point], although [Variable X] continued to double, [Variable Y] plateaued around [Value], indicating diminishing marginal returns. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [例外个体点拨 Exception Highlight]:** *Notably, Plant D defied this expenditure curve, achieving a 0.7% defect rate with just $22 million through closed-loop sensor calibration.*
  对应步骤：S4 [例外个体点拨 Exception Highlight]：An exception to this pattern is [Entity Z], which achieved the lowest [Variable Y] despite maintaining moderate [Variable X]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

20.5 FT15: 雅思小作文（Task 1）闭环工艺与算法流程图满分范文：具身智能遥操作数据采集、双支路增强与主动学习回环

Composition Architecture & Structural Blueprint Chain

Genre: IELTS_Task1_Process **Paragraph Count:** 3

Blueprint Chain: BP25 → BP29 → BP30

Paragraph 1: 图表小作文首段：题目改写与总体趋势概览（Paraphrase + Macro Overview）(BP25)

Expressive Intent: 我这一段想写雅思 Task 1 或数据报告的第一段：用同义句法改写图表标题，并用两句话高度凝练全图最核心的 2 个宏观趋势（极值、反超拐点或主次格局），绝不堆砌零碎数字。

Cohesion Formula: Paraphrase → Overall Trajectory Contrast → Dominant Crossover/Extremum (Zero raw numbers in Overview!). 绑定逻辑规则：L013, L015.

Publication-Grade Authentic Text

The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025. Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily. Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [图表命题同义改写 Prompt Paraphrase]:** *The line graph compares the proportion of electricity generated from four fuel sources across three industrial economies between 2000 and 2025.*
  对应步骤：S1 [图表命题同义改写 Prompt Paraphrase]：The line graph / bar chart compares [Variable/Metric] across [Categories/Regions] over a [N]-year period from [Year A] to [Year B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [宏观趋势 1:** *Overall, renewable sources—particularly solar and wind power—followed a pronounced upward trajectory over the 25-year span, whereas coal reliance contracted steadily.*
  对应步骤：S2 [宏观趋势 1：总体走向与反超 Overview Trend 1]：Overall, while [Category A and B] experienced a steady upward trajectory, [Category C] declined markedly, leading to a reversal in dominance around [Mid-Year]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [宏观趋势 2:** *Most notably, clean energy generation overtook fossil-fuel consumption around 2018, shifting from a marginal supplement to the dominant power supply by the end of the period.*

对应步骤：S3 [宏观趋势 2：极值或差距特征 Overview Trend 2]：Notably, [Category A] registered the most dramatic growth throughout the period, ultimately overtaking all other sectors to become the primary contributor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 流程图/算法流水线与闭环工序衔接段（Process & Closed-Loop Lifecycle）(BP29)

Expressive Intent: 我这一段想写雅思 Task 1 工艺/自然循环流程图（或论文算法数据预处理与闭环训练流水线）：把 5–6 个串联、分叉与回环步骤写得层次分明，绝不机械连用 Then, Then, Then。

Cohesion Formula: Pipeline Overview → Ingestion & Filtering → Parallel Bifurcation → Terminal Merger & Feedback Loop. 绑定逻辑规则：L003, L005.

Publication-Grade Authentic Text

The diagram illustrates the closed-loop data curation pipeline for embodied robot learning, spanning four primary stages from teleoperated collection to failure-driven retraining. The cycle begins when raw multimodal demonstrations are recorded via VR teleoperation and filtered to discard kinematic singularities and dropped frames. Once validated, the trajectory stream bifurcates into two parallel pathways: visual frames undergo background augmentation, while proprioceptive states are normalized into local gripper coordinates. These processed streams are subsequently synchronized to train the visuomotor policy, whose real-world rollout failures are logged and fed back into the teleoperation buffer to close the active-learning loop.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [流程总括与起止界定 Lifecycle Overview]:** *The diagram illustrates the closed-loop data curation pipeline for embodied robot learning, spanning four primary stages from teleoperated collection to failure-driven retraining.*
  对应步骤：S1 [流程总括与起止界定 Lifecycle Overview]：The diagram illustrates the closed-loop pipeline for [Process Name], comprising [N] main stages from initial [Input Stage] to continuous [Feedback/Output Stage]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [前段采集与预处理 Initial Ingestion & Filtering]:** *The cycle begins when raw multimodal demonstrations are recorded via VR teleoperation and filtered to discard kinematic singularities and dropped frames.*
  对应步骤：S2 [前段采集与预处理 Initial Ingestion & Filtering]：The process begins when raw [Material/Data] is collected via [Source] and passed through [Filter/Chamber] to remove [Impurities/Outliers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [中段并行分流 Parallel Branching]:** *Once validated, the trajectory stream bifurcates into two parallel pathways: visual frames undergo background augmentation, while proprioceptive states are normalized into local gripper coordinates.*
  对应步骤：S3 [中段并行分流 Parallel Branching]：Once purified, the stream bifurcates into two parallel pathways: [Fraction A] undergoes [Process A], whereas [Fraction B] is routed to [Process B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [末端合成与闭环回流 Synthesis & Closed-Loop Recycling]:** *These processed streams are subsequently synchronized to train the visuomotor policy, whose real-world rollout failures are logged and fed back into the teleoperation buffer to close the active-learning loop.*
  对应步骤：S4 [末端合成与闭环回流 Synthesis & Closed-Loop Recycling]：The resulting outputs are subsequently merged in [Unit C], with residual by-products / high-error trajectories recycled back into the initial buffer for iterative refinement. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 历史实测与未来预测跨时态衔接段（Historical vs. Projected Trends）(BP30)

Expressive Intent: 我这一段想处理同时包含过去历史数据（如 2010–2025）和未来预测数据（如 2026–2040）的图表段落，严格区分过去实测时态与未来审慎预测表达。

Cohesion Formula: Historical Past (rose/accounted for) → Pivot Year → Future Projection Verbs (is projected/anticipated/expected to) → Moderating Factor. 绑定逻辑规则：L010, L017.

Publication-Grade Authentic Text

Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh. From 2026 onward, total demand is projected to climb at a substantially steeper rate, with baseline forecasts pointing to 1,050 TWh by 2035. Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040. However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [历史实测轨迹 Historical Trajectory (Past Tense)]:** *Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh.*
  对应步骤：S1 [历史实测轨迹 Historical Trajectory (Past Tense)]：Between [2010] and [2025], global data-center electricity demand rose steadily from [200 TWh] to [460 TWh], driven by cloud migration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]:** *From 2026 onward, total demand is projected to climb at a substantially steeper rate, with baseline forecasts pointing to 1,050 TWh by 2035.*
  对应步骤：S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]：From [2026] onward, consumption is projected to accelerate sharply, with forecasts indicating a surge to [1,050 TWh] by [2035]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [结构占比重塑 Structural Shift in Projections]:** *Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040.*
  对应步骤：S3 [结构占比重塑 Structural Shift in Projections]：Within this projected expansion, generative AI training and inference—which accounted for less than [10%] in [2023]—are anticipated to represent over [45%] of total load by [2040]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [条件式审慎收尾 Conditional Hedging]:** *However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.*
  对应步骤：S4 [条件式审慎收尾 Conditional Hedging]：Nevertheless, energy-efficient hardware adoption is expected to moderate the growth slope during the final five years. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Chapter 21

Engineering, Materials & Physical Sciences / 工程材料与物理科学

本章精选该学科领域及对应文体的完整学术范文。每一篇范文不仅展示母语级发表标准的正文全貌，更清晰拆解全篇的“段落蓝图链条（Structural Blueprint Chain）”，并对每个自然段逐一剖析其核心交际意图、句际衔接逻辑公式（Cohesion Formula）以及每句话的句法成分与推进机制。

21.1 FT16: 托福综合写作（Integrated Writing）满分范文：自主无人机配送能否取代城市货车（阅读三论点 vs. 听力逐点反驳）

Composition Architecture & Structural Blueprint Chain

Genre: `TOEFL_Integrated` **Paragraph Count:** 4

Blueprint Chain: `BP31` → `BP27` → `BP28` → `BP32`

Paragraph 1: 托福综合写作 / 学术争议正反交锋段（Point-by-Point Reading vs. Lecture Synthesis）(BP31)

Expressive Intent: 我这一段想写托福综合写作（Integrated Writing）或文献争议交锋段：先用 1 句话精准概括阅读材料（或正方文献）的分论点，再用 2–3 句话详细呈现教授讲座（或反方实证）是如何釜底抽薪推翻该论点的。

Cohesion Formula: Reading Claim → Lecturer’s Direct Challenge → Specific Counter-Mechanism → Reversed Outcome. 绑定逻辑规则：L008, L019.

Publication-Grade Authentic Text

The reading passage contends that replacing urban delivery vans with autonomous quadcopter drones will substantially reduce last-mile transportation energy consumption, citing the lighter weight of aerial vehicles. The lecturer directly challenges this premise by examining energy efficiency per kilogram of delivered freight. Specifically, she points out that unlike ground vehicles supported passively by road surfaces, multi-rotor drones must expend continuous battery power simply to counteract gravity and headwinds while carrying only a single parcel per trip. Consequently, servicing a dense apartment block via individual drone flights consumes nearly three times more energy per package than a single consolidated van route.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [正方/阅读论点概括 Reading Claim & Rationale]:** *The reading passage contends that replacing urban delivery vans with autonomous quadcopter drones will substantially reduce last-mile transportation energy consumption, citing the lighter weight of aerial vehicles.*
  对应步骤：S1 [正方/阅读论点概括 Reading Claim & Rationale]：The reading passage contends that [Claim A], arguing that [Reading's Supporting Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反方/听力直接反驳 Lecture Direct Refutation]:** *The lecturer directly challenges this premise by examining energy efficiency per kilogram of delivered freight.*
  对应步骤：S2 [反方/听力直接反驳 Lecture Direct Refutation]：The lecturer directly challenges this assertion by pointing out that [Core Flaw / Neglected Factor in Reading's Premise]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [反方具体反证展开 Empirical Counter-Evidence]:** *Specifically, she points out that unlike ground vehicles supported passively by road surfaces, multi-rotor drones must expend continuous battery power simply to counteract gravity and headwinds while carrying only a single parcel per trip.*
  对应步骤：S3 [反方具体反证展开 Empirical Counter-Evidence]：Specifically, she explains that when [Real-World Condition], [Mechanism B occurs instead], which [Negates the Claimed Benefit]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [结论锁定 Refutation Closure]:** *Consequently, servicing a dense apartment block via individual drone flights consumes nearly three times more energy per package than a single consolidated van route.*
  对应步骤：S4 [结论锁定 Refutation Closure]：Consequently, rather than achieving [Reading's Goal], the proposed measure actually exacerbates [Negative Outcome]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 静态多组数据横向对比与倍差/占比凸显段（Cross-Group Proportions）(BP27)

Expressive Intent: 我这一段想写静态柱状图或饼图的主体段：横向对比不同国家/不同模型在不同开销维度上的占比差异，灵活运用倍数（double/triple）、分数（nearly a third）与排序结构。

Cohesion Formula: Dominant Item with Multiplier → Sharp Cross-Group Contrast → Pooled Minor Items → Structural Synthesis. 绑定逻辑规则：L009, L010.

Publication-Grade Authentic Text

Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%). A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocess-

ing dominates energy usage at 51%, while attention overhead contracts to just 16%. Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption. This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [主导项锁定 Dominant Category]:** *Within edge-device inference workloads, attention matrix computation consumes the largest fraction of total energy at 48%, exactly double the power drawn by feed-forward layers (24%).*
  对应步骤：S1 [主导项锁定 Dominant Category]：Regarding [Group 1], [Category A] claims the largest share at [46%], nearly double the proportion allocated to [Category B] ([24%]). | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [组间横向反差 Cross-Group Contrast]:** *A strikingly different profile emerges in onboard robotic control loops, where high-frequency sensor preprocessing dominates energy usage at 51%, while attention overhead contracts to just 16%.*
  对应步骤：S2 [组间横向反差 Cross-Group Contrast]：A markedly different allocation pattern emerges in [Group 2], where [Category C] dominates at [52%], while [Category A] shrinks to just [15%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [次要项合并归纳 Minor Categories Grouping]:** *Across both deployment settings, memory offloading and actuation decoding collectively account for under 11% of total consumption.*
  对应步骤：S3 [次要项合并归纳 Minor Categories Grouping]：Meanwhile, the remaining categories—[Category D and E]—collectively account for less than [12%] across both groups. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [结构性差异提炼 Structural Takeaway]:** *This contrast highlights that optimization bottlenecks shift from quadratic token mixing on cloud servers to raw sensory I/O on physical robots.*
  对应步骤：S4 [结构性差异提炼 Structural Takeaway]：These figures indicate that whereas [Group 1] prioritizes [Function A], [Group 2] concentrates resources heavily on [Function C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 双图关联与互补/因果印证段（Synthesizing Dual Charts）(BP28)

Expressive Intent: 我这一段想写双图题（如一张图是算力投入/研发经费变化，另一张图是模型误差/碳排放强度变化）的综合关联段，阐明两张图之间的内在协同或背离关系。

Cohesion Formula: Global Correlation → Co-movement Phase → Diminishing Returns / Divergence → Notable Outlier. 绑定逻辑规则：L007, L013.

Publication-Grade Authentic Text

Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold. Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [双图联动锚定 Cross-Chart Link]:** *Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold.*
  对应步骤：S1 [双图联动锚定 Cross-Chart Link]：Synthesizing both charts reveals a clear inverse / positive relationship between [Chart 1 Variable] and [Chart 2 Variable] up to [Threshold Point]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [区间 1 协同变化 Phase 1 Co-movement]:** *Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8% to 0.9% (Figure 2).*
  对应步骤：S2 [区间 1 协同变化 Phase 1 Co-movement]：Specifically, as [Variable X in Chart 1] expanded from [A] to [B] during [Period 1], [Variable Y in Chart 2] dropped correspondingly by [C%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]:** *Beyond $40 million, however, further capital expenditure yielded diminishing returns: doubling investment to $80 million reduced defects by only an additional 0.1 percentage points.*
  对应步骤：S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]：However, beyond [Threshold Point], although [Variable X] continued to double, [Variable Y] plateaued around [Value], indicating diminishing marginal returns. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [例外个体点拨 Exception Highlight]:** *Notably, Plant D defied this expenditure curve, achieving a 0.7% defect rate with just $22 million through closed-loop sensor calibration.*
  对应步骤：S4 [例外个体点拨 Exception Highlight]：An exception to this pattern is [Entity Z], which achieved the lowest [Variable Y] despite maintaining moderate [Variable X]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 异常数据点（Outliers）与局部背离现象精准批注段（Outlier & Anomaly Annotation）(BP32)

Expressive Intent: 我这一段想在图表或实验结果段中，专门挑出违背总体规律的 1–2 个异常点（Outliers）或局部反常尖峰进行精准描述与成因剖析。

Cohesion Formula: General Rule vs. Notable Departure → Quantitative Deviation → Root Cause Attribution → Calibration Takeaway. 绑定逻辑规则：L007, L013.

Publication-Grade Authentic Text

Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset. Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate. Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks. This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总体规律背景下的特例锚定 Outlier Identification]:** *Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset.*
  对应步骤：S1 [总体规律背景下的特例锚定 Outlier Identification]：Although [General Trend holds across most groups/steps], a notable departure from this pattern occurs in [Specific Point/Group]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反常幅度量化 Quantitative Deviation]:** *Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate.*
  对应步骤：S2 [反常幅度量化 Quantitative Deviation]：Whereas peer categories cluster tightly within $[A, B]$, [Outlier Entity] spikes/drops to [Extreme Value], deviating by more than [X%] from the group median. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [底层物理/统计成因剖析 Root Cause Analysis]:** *Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks.*
  对应步骤：S3 [底层物理/统计成因剖析 Root Cause Analysis]：Closer inspection indicates that this anomaly stems from [Specific Structural/External Factor] rather than random variance. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [方法论启示 Methodological Implication]:** *This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.*
  对应步骤：S4 [方法论启示 Methodological Implication]：Accounting for such localized departures is critical when calibrating [Downstream Model/Policy]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

21.2 FT17: AI 系统评测报告全文：长序列预测与生成模型在历史实测与未来算力扩展下的精度-延迟帕累托报告

Composition Architecture & Structural Blueprint Chain

Genre: Technical_Benchmark_Report **Paragraph Count:** 5

Blueprint Chain: BP17 → BP18 → BP22 → BP28 → BP30

Paragraph 1: 实验设置、数据集特性与强基线选型说明段（Experimental Setup & Baselines）(BP17)

Expressive Intent: 我这一段想在 Experiments 开篇说明：我们为什么选这几类数据集和这几类基线模型，证明我们的实验设计是全面、公平且直击痛点的。

Cohesion Formula: Dataset Diversity → Challenging Regime Highlights → Categorized Strong Baselines → Strict Fairness Protocol. 绑定逻辑规则：L009, L010.

Publication-Grade Authentic Text

To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic). Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability. We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer). All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [数据集选型逻辑 Dataset Selection Rationale]:** *To evaluate robustness across diverse non-stationary regimes, we conduct experiments on seven public benchmarks spanning energy grid load (ETT, ECL), meteorological telemetry (Weather), and urban traffic flow (Traffic).*
  对应步骤：S1 [数据集选型逻辑 Dataset Selection Rationale]：To evaluate performance across diverse non-stationary regimes, we conduct experiments on [N] benchmarks spanning [Domain 1], [Domain 2], and [Domain 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [极端/困难特性突出 Challenging Dataset Properties]:** *Notably, Weather contains abrupt atmospheric pressure transitions that test non-stationary adaptability, whereas Traffic spans 862 synchronized sensors that challenge high-dimensional scalability.*
  对应步骤：S2 [极端/困难特性突出 Challenging Dataset Properties]：Notably, [Dataset A] exhibits [Hard Property 1, e.g., strong covariate shift], whereas [Dataset B] tests scalability across [Hard Property

2, e.g., 862 correlated channels]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [基线流派全覆盖 Baseline Family Coverage]:** *We benchmark our approach against 11 state-of-the-art methods across three paradigms: channel-independent patch Transformers (PatchTST), variate-mixing architectures (iTransformer, Crossformer), and frequency-domain forecasters (TimesNet, FEDformer).*
  对应步骤：S3 [基线流派全覆盖 Baseline Family Coverage]: We compare our method against [K] competitive baselines grouped into three families: [Family 1], [Family 2], and [Family 3]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [公平评测协议 Fair Evaluation Protocol]:** *All baselines are reproduced under identical lookback windows ($L = 96$) and prediction horizons ($H \in \{96, 192, 336, 720\}$) using standard MSE and MAE metrics.*
  对应步骤：S4 [公平评测协议 Fair Evaluation Protocol]: All models are evaluated under identical lookback windows ($L = 96$), prediction horizons ($H \in \{96, 192, 336, 720\}$), and hyperparameter search budgets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 主实验结果整体对比与核心胜出原因剖析段（Main Results Analysis）(BP18)

Expressive Intent: 我这一段想解读主实验大表（Table 1）：既报告全局胜率，又挑出最有代表性的对比案例，解释为什么竞争对手在那里跌倒而我们稳赢。

Cohesion Formula: Global Win Rate → Advantage over Family A → Resilience at Long Horizon vs. Family B → Mechanistic Takeaway. 绑定逻辑规则：L015, L013, L010.

Publication-Grade Authentic Text

Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations. Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical. Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts. By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [全局统治力概述 Overall Dominance]:** *Table 1 reports the multivariate forecasting results, where DyTS-Net secures the lowest error in 25 out of 28 dataset-horizon configurations.*

对应步骤：S1 [全局统治力概述 Overall Dominance]：Table 1 summarizes the multivariate forecasting results, where [Our Model] achieves the lowest error in [X] out of [Y] dataset-horizon configurations. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]:** *Compared with channel-independent PatchTST, DyTS-Net reduces average MSE by 14.2% on the 862-channel Traffic benchmark, where modeling spatial-temporal congestion propagation across adjacent road sensors is critical.*
  对应步骤：S2 [对比流派 1 的差距与原因 Contrast vs. Family 1]：Compared with channel-independent baselines such as [Baseline A], our method reduces average MSE by [X%] on [Dataset], where strong cross-sensor coupling is critical. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]:** *Furthermore, while variate-mixing iTransformer performs competitively at $H = 96$, its MSE escalates by 68% at $H = 720$ on volatile Weather series as unregularized cross-channel attention overfits transient wind gusts.*
  对应步骤：S3 [对比流派 2 在困难场景下的崩溃 Contrast vs. Family 2 under Hard Regime]：Furthermore, while [Baseline B] performs competitively at short horizons ($H = 96$), its error escalates by [Y%] at $H = 720$ due to [Failure Mechanism]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [回扣本文核心机制 Mechanistic Confirmation]:** *By decoupling stationary phase coupling from neural-ODE scale projection, DyTS-Net suppresses long-horizon drift and maintains an 11.6% margin over the strongest baseline at $H = 720$.*
  对应步骤：S4 [回扣本文核心机制 Mechanistic Confirmation]：By contrast, our [Core Module] bounds long-range drift, maintaining a [Z%] margin over the second-best baseline at $H = 720$. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 计算开销、推理延迟与精度帕累托权衡段（Efficiency-Accuracy Pareto Analysis）(BP22)

Expressive Intent: 我这一段想说明我们的模型不仅准，而且算得快、显存省，在“精度-延迟/显存”帕累托前沿（Pareto Frontier）上显著优于现有大模型。

Cohesion Formula: Hardware Benchmark → Complexity Reduction & Memory Gain → Real-Time Latency Margin → Pareto Conclusion. 绑定逻辑规则：L010, L023.

Publication-Grade Authentic Text

Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module. By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch

tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer. At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [硬件与测试基准 Hardware Setup]:** *Figure 6 plots the MSE-latency Pareto frontier measured on an onboard NVIDIA Orin AGX module.*
  对应步骤：S1 [硬件与测试基准 Hardware Setup]：Figure 6 compares the accuracy-efficiency Pareto frontier across all baselines measured on a single NVIDIA RTX 4090 GPU. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [复杂度理论与实测对比 Complexity & Runtime]:** *By replacing $O(L^2)$ point-wise self-attention with $O(L \log L)$ spectral filtering and compact patch tokens, DyTS-Net reduces peak GPU memory by 62% and training wall-clock time by 3.4x compared with iTransformer.*
  对应步骤：S2 [复杂度理论与实测对比 Complexity & Runtime]：By replacing $O(L^2)$ point-wise attention with $O(L \log L)$ spectral filtering and patch aggregation, our model cuts training time per epoch by [X%] and peak VRAM by [Y%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [实时闭环部署可行性 Real-Time Viability]:** *At inference time, generating a 16-step visuomotor trajectory requires only 6.8 ms—well below the 20 ms cycle budget of a 50 Hz industrial robot controller—while outperforming a 200 ms DDPM baseline by 11.4% in task success rate.*
  对应步骤：S3 [实时闭环部署可行性 Real-Time Viability]：At inference time, generating a 720-step forecast (or 16-step robot action chunk) requires only [4.2 ms], well below the [20 ms] real-time control deadline. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 双图关联与互补/因果印证段（Synthesizing Dual Charts）(BP28)

Expressive Intent: 我这一段想写双图题（如一张图是算力投入/研发经费变化，另一张图是模型误差/碳排放强度变化）的综合关联段，阐明两张图之间的内在协同或背离关系。

Cohesion Formula: Global Correlation → Co-movement Phase → Diminishing Returns / Divergence → Notable Outlier. 绑定逻辑规则：L007, L013.

Publication-Grade Authentic Text

Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold. Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [双图联动锚定 Cross-Chart Link]:** *Cross-referencing both figures reveals a strong inverse relationship between automated inspection investment and manufacturing defect rates up to a $40 million threshold.*
  对应步骤：S1 [双图联动锚定 Cross-Chart Link]：Synthesizing both charts reveals a clear inverse / positive relationship between [Chart 1 Variable] and [Chart 2 Variable] up to [Threshold Point]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [区间 1 协同变化 Phase 1 Co-movement]:** *Specifically, across plants investing between $10 million and $40 million (Figure 1), surface defect incidence dropped steeply from 4.8% to 0.9% (Figure 2).*
  对应步骤：S2 [区间 1 协同变化 Phase 1 Co-movement]：Specifically, as [Variable X in Chart 1] expanded from [A] to [B] during [Period 1], [Variable Y in Chart 2] dropped correspondingly by [C%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]:** *Beyond $40 million, however, further capital expenditure yielded diminishing returns: doubling investment to $80 million reduced defects by only an additional 0.1 percentage points.*
  对应步骤：S3 [区间 2 边际递减/背离 Diminishing Returns / Divergence]：However, beyond [Threshold Point], although [Variable X] continued to double, [Variable Y] plateaued around [Value], indicating diminishing marginal returns. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [例外个体点拨 Exception Highlight]:** *Notably, Plant D defied this expenditure curve, achieving a 0.7% defect rate with just $22 million through closed-loop sensor calibration.*
  对应步骤：S4 [例外个体点拨 Exception Highlight]：An exception to this pattern is [Entity Z], which achieved the lowest [Variable Y] despite maintaining moderate [Variable X]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 历史实测与未来预测跨时态衔接段（Historical vs. Projected Trends）(BP30)

Expressive Intent: 我这一段想处理同时包含过去历史数据（如 2010–2025）和未来预测数据（如 2026–2040）的图表段落，严格区分过去实测时态与未来审慎预测表达。

Cohesion Formula: Historical Past (rose/accounted for) → Pivot Year → Future Projection Verbs (is projected/anticipated/expected to) → Moderating Factor. 绑定逻辑规则：L010, L017.

Publication-Grade Authentic Text

Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh. From 2026 onward, total demand is projected to climb at a substantially steeper

rate, with baseline forecasts pointing to 1,050 TWh by 2035. Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040. However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [历史实测轨迹 Historical Trajectory (Past Tense)]:** *Between 2010 and 2025, global data-center electricity consumption more than doubled from 200 TWh to 460 TWh.*
  对应步骤：S1 [历史实测轨迹 Historical Trajectory (Past Tense)]：Between [2010] and [2025], global data-center electricity demand rose steadily from [200 TWh] to [460 TWh], driven by cloud migration. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]:** *From 2026 onward, total demand is projected to climb at a substantially steeper rate, with baseline forecasts pointing to 1,050 TWh by 2035.*
  对应步骤：S2 [当下转折与未来加速 Projected Acceleration (Projection Verbs)]：From [2026] onward, consumption is projected to accelerate sharply, with forecasts indicating a surge to [1,050 TWh] by [2035]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [结构占比重塑 Structural Shift in Projections]:** *Within this expansion, AI model training and inference—which represented under 10% of server load in 2023—are anticipated to account for nearly half (46%) of total consumption by 2040.*
  对应步骤：S3 [结构占比重塑 Structural Shift in Projections]：Within this projected expansion, generative AI training and inference—which accounted for less than [10%] in [2023]—are anticipated to represent over [45%] of total load by [2040]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [条件式审慎收尾 Conditional Hedging]:** *However, widespread deployment of sparsity-aware accelerators is expected to temper the annual growth rate after 2035.*
  对应步骤：S4 [条件式审慎收尾 Conditional Hedging]：Nevertheless, energy-efficient hardware adoption is expected to moderate the growth slope during the final five years. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

21.3 FT18: 科研实验反常现象与极端工况诊断报告全文：为何高容量注意力模型在强随机游走序列上退化及改进方案

Composition Architecture & Structural Blueprint Chain

Genre: `Failure_Mode_Report` **Paragraph Count:** 5

Blueprint Chain: `BP32` → `BP19` → `BP20` → `BP21` → `BP23`

Paragraph 1: 异常数据点（Outliers）与局部背离现象精准批注段（Outlier & Anomaly Annotation）(BP32)

Expressive Intent: 我这一段想在图表或实验结果段中，专门挑出违背总体规律的 1–2 个异常点（Outliers）或局部反常尖峰进行精准描述与成因剖析。

Cohesion Formula: General Rule vs. Notable Departure → Quantitative Deviation → Root Cause Attribution → Calibration Takeaway. 绑定逻辑规则：L007, L013.

Publication-Grade Authentic Text

Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset. Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate. Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks. This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [总体规律背景下的特例锚定 Outlier Identification]:** *Although deep spectral transformers outperform linear baselines across six of the seven benchmarks, a notable departure emerges on the financial Exchange-Rate dataset.*
  对应步骤：S1 [总体规律背景下的特例锚定 Outlier Identification]：Although [General Trend holds across most groups/steps], a notable departure from this pattern occurs in [Specific Point/Group]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反常幅度量化 Quantitative Deviation]:** *Whereas complex attention models reduce MSE by 12%–18% on periodic energy and weather streams, they trail a single-layer DLinear baseline by 4.3% on Exchange-Rate.*
  对应步骤：S2 [反常幅度量化 Quantitative Deviation]：Whereas peer categories cluster tightly within $[A, B]$, [Outlier Entity] spikes/drops to [Extreme Value], deviating by more than [X%] from the group

median. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [底层物理/统计成因剖析 Root Cause Analysis]:** *Autocorrelation analysis reveals that this reversal stems from the near-martingale nature of daily currency returns, where weak seasonal periodicity causes high-capacity attention layers to overfit transient market shocks.*
  对应步骤：S3 [底层物理/统计成因剖析 Root Cause Analysis]：Closer inspection indicates that this anomaly stems from [Specific Structural/External Factor] rather than random variance. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [方法论启示 Methodological Implication]:** *This anomaly confirms that model capacity must be matched to the intrinsic signal-to-noise ratio of the underlying stochastic process.*
  对应步骤：S4 [方法论启示 Methodological Implication]：Accounting for such localized departures is critical when calibrating [Downstream Model/Policy]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 消融实验逐组件归因论证段（Counterfactual Ablation Study）(BP19)

Expressive Intent: 我这一段想汇报消融实验（Table 2）：证明我们的模块 A、模块 B 和正则项 C 每一个都不可或缺，且各自负责解决不同的特定子问题。

Cohesion Formula: Ablation Overview → w/o Module A (Specific Failure) → w/o Module B (Specific Failure) → Cross-Module Synergy. 绑定逻辑规则：L016, L007.

Publication-Grade Authentic Text

To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL. Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures. Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation. Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [消融实验总括 Ablation Setup]:** *To isolate the contribution of each design choice, Table 2 compares full DyTS-Net against three ablated variants across ETTm2 and ECL.*
  对应步骤：S1 [消融实验总括 Ablation Setup]：To isolate the contribution of each architectural component, we evaluate four ablated variants in Table 2 by progressively removing [Mod A], [Mod B], and [Loss C]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]:** *Removing the neural-ODE scale tracker (w/o Scale-ODE) triggers the sharpest degradation during seasonal transition months on ETTm2, raising 720-step MSE by 19.4% as the decoder fails to track shifting baseline temperatures.*
  对应步骤：S2 [移除模块 A 的冲击与归因 Impact of Removing Mod A]：Removing [Mod A] (w/o A) causes the largest degradation on [Non-stationary/Volatile Subset], increasing MSE by [X%] because the model loses [Specific Capability A]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]:** *Conversely, replacing phase-aligned graph attention with standard dot-product attention (w/o Phase-Attn) degrades performance most severely on the 321-channel ECL grid (+13.8% MSE), where unaligned high-frequency load spikes corrupt cross-substation aggregation.*
  对应步骤：S3 [移除模块 B 的冲击与归因 Impact of Removing Mod B]：Similarly, replacing [Mod B] with standard [Naive Operator] degrades performance by [Y%] on high-dimensional benchmarks, confirming its role in [Specific Capability B]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同效应总结 Synergistic Conclusion]:** *Jointly deploying both modules yields complementary gains, as accurate scale tracking prevents amplitude distortion from biasing phase estimation.*
  对应步骤：S4 [协同效应总结 Synergistic Conclusion]：Combining all components yields super-additive gains, as [Mod A] stabilizes input statistics for [Mod B]'s relational reasoning. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 超参数敏感性与极端噪声鲁棒性分析段（Sensitivity & Robustness Analysis）(BP20)

Expressive Intent: 我这一段想回应审稿人最常问的：你的模型是不是对超参数很敏感？在缺失值或传感器强噪声下会不会突然崩盘？

Cohesion Formula: Goal → Wide Hyperparameter Plateau → Stress-Test Comparison → Mechanistic Explanation. 绑定逻辑规则：L010, L019.

Publication-Grade Authentic Text

Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption. Varying the phase regularization weight λ across $[0.05, 0.50]$ alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around $\lambda = 0.15$. Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), pointwise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%. This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分析目的 Objective]:** *Figure 4 evaluates hyperparameter sensitivity and resilience against severe observation corruption.*
  对应步骤：S1 [分析目的 Objective]：We further examine sensitivity to key hyperparameters and resilience against severe sensor corruption in Figure 4. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [超参数平坦区间 Plateau Stability]:** *Varying the phase regularization weight λ across* $[0.05, 0.50]$ *alters validation MSE by less than 1.4%, indicating a wide, tuning-friendly operating plateau around* $\lambda = 0.15$.
  对应步骤：S2 [超参数平坦区间 Plateau Stability]：Varying [Hyperparameter λ] across two orders of magnitude ([0.01, 1.0]) alters validation MSE by less than [1.8%], demonstrating a wide stable operating plateau. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [极端边缘工况压力测试 Stress Test under Corruption]:** *Under stress tests where 30% of sensor readings are randomly masked or corrupted by impulse noise (SNR = 5 dB), point-wise Transformer baselines suffer a 34.2% error surge, whereas DyTS-Net degrades by only 4.8%.*
  对应步骤：S3 [极端边缘工况压力测试 Stress Test under Corruption]：Under synthetic sensor dropouts up to [40%] and Gaussian noise ($\sigma = 0.3$), baseline accuracy plummets by over [35%], whereas our model degrades by less than [6.5%]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [鲁棒性物理归因 Robustness Mechanism]:** *This robustness stems from our spectral energy gate, which automatically attenuates high-entropy corrupted channels prior to cross-variate message passing.*
  对应步骤：S4 [鲁棒性物理归因 Robustness Mechanism]：This resilience stems from [Specific Mechanism], which filters out corrupted channels before global aggregation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 极端/反常情况与失败案例坦诚剖析段（Failure Case & Outlier Analysis）(BP21)

Expressive Intent: 我这一段想通过可视化案例（Case Study / Qualitative Analysis），既展示模型在复杂跳变下的精准跟踪，又坦诚剖析一个模型预测失准的极端案例。

Cohesion Formula: Visual Setup → Success under Shift → Honest Failure Case under Unobserved Exogenous Shock → Actionable Insight. 绑定逻辑规则：L017, L018.

Publication-Grade Authentic Text

Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark. Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization

within 5% amplitude error. Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response. Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [定性对比切入 Qualitative Setup]:** *Figure 5 visualizes 336-step forecasts during an abrupt voltage drop on the ETTm1 transformer load benchmark.*
  对应步骤：S1 [定性对比切入 Qualitative Setup]：Figure 5 visualizes predicted trajectories during an abrupt regime shift on [Dataset/Task]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [成功案例亮点 Success Highlight]:** *Whereas PatchTST and iTransformer exhibit a 14-step phase lag and severe post-recovery overshoot, DyTS-Net tracks both the sharp downward ramp and subsequent stabilization within 5% amplitude error.*
  对应步骤：S2 [成功案例亮点 Success Highlight]：While [Baseline] lags behind the sudden step change by [K steps] and overshoots, our model rapidly locks onto the new amplitude regime within [M steps]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [失败案例坦诚剖析 Honest Failure Mode]:** *Nevertheless, the failure case in Panel (d) reveals that when an unscheduled manual grid curtailment occurs without preceding spectral precursors, our endogenous model experiences a 6-step delayed response.*
  对应步骤：S3 [失败案例坦诚剖析 Honest Failure Mode]：Nevertheless, inspecting the bottom-right failure case reveals that when an external manual override occurs without historical precursors, all purely endogenous forecasters—including ours—exhibit a transient lag. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [改进启示 Constructive Takeaway]:** *Anticipating such zero-precursor manual interventions requires integrating exogenous dispatch logs alongside historical telemetry.*
  对应步骤：S4 [改进启示 Constructive Takeaway]：Incorporating exogenous event logs or textual maintenance records is necessary to anticipate such unscheduled interventions. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 局限性（Limitations）客观陈述与未来扩展段（Constructive Limitations）(BP23)

Expressive Intent: 我这一段想写论文末尾或 NeurIPS/ICLR Checklist 要求的 Limitations 段落：诚实指出当前方法的两个客观边界，但绝不贬损论文的核心贡献。

Cohesion Formula: Two Objective Boundaries → [Boundary 1 + Technical Remedy] → [Boundary 2 + Stochastic Extension]. 绑定逻辑规则：L017.

Publication-Grade Authentic Text

Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting. First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy. Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration. Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [边界 1:** *Despite its strong empirical performance, our framework exhibits two theoretical boundaries worth noting.*
  对应步骤：S1 [边界 1：极端高维或极度稀疏场景 Boundary 1]：Despite its effectiveness across standard benchmarks, our framework currently exhibits two practical boundaries. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [边界 1 机理分析与解法 Mechanism & Solution 1]:** *First, while our spectral branch scales quasi-linearly with sequence length, computing pairwise Hilbert phase differences scales quadratically with channel count ($O(C^2)$), which increases memory footprint when $C > 2,000$; incorporating anchor-based low-rank graph sparsification provides a direct remedy.*
  对应步骤：S2 [边界 1 机理分析与解法 Mechanism & Solution 1]：First, when the variate dimension scales beyond $C > 5,000$ unclustered sensors, constructing the cross-channel phase graph incurs noticeable memory overhead; integrating hierarchical sensor clustering offers a natural remedy. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [边界 2:** *Second, our continuous neural-ODE scale tracker assumes piece-wise smooth drift and may smooth over instantaneous step discontinuities caused by sensor recalibration.*
  对应步骤：S3 [边界 2：特定假设限制 Boundary 2]：Second, our continuous scale tracker assumes piecewise differentiable drift and may under-estimate instantaneous discontinuous jumps caused by hardware resets. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [未来演进方向 Future Extension]:** *Extending the continuous drift prior to jump-diffusion stochastic differential equations offers a principled direction for future investigation.*
  对应步骤：S4 [未来演进方向 Future Extension]：Coupling our neural ODE tracker with jump-diffusion stochastic processes represents a promising avenue for future work. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

21.4 FT19: 六级写作（CET-6）满分范文：高校教学中生成式 AI 辅助与独立批判性思维培养的平衡之道

Composition Architecture & Structural Blueprint Chain

Genre: CET6_Essay **Paragraph Count:** 5

Blueprint Chain: BP33 → BP34 → BP35 → BP37 → BP40

Paragraph 1: 议论文破题首段：具象趋势锚定 + 争议张力 + 明确中心论点（Zero-Template Hook & Thesis）(BP33)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉'With the rapid development of science and technology, more and more people...' 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Cohesion Formula: Concrete Phenomenon (No Cliché) → Balanced Tension → Conditional & Specific Thesis Statement. 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [具象现象锚定 Concrete Trend Anchor]:** *Large language models and automated code assistants have rapidly become routine study aids across university campuses.*
  对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心争议张力 Core Tension / Debate]:** *While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.*
  对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [带条件的中心论点 Nuanced Thesis Statement]:** *In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical*

skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.
对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 核心正方分论点段：主张 → 机制推演 → 具象行业微案例 → 扣题闭环（PEEL 满分段）(BP34)

Expressive Intent: 我这一段想写大作文的第一个主体段（Body Paragraph 1）：论证某个核心优势或必要性，做到有理有据、有具象场景支撑，绝不车轱辘话来回说。

Cohesion Formula: Topic Claim → Causal Mechanism → Concrete Micro-Case (L021) → Warrant Closure. 绑定逻辑规则：L006, L012, L021.

Publication-Grade Authentic Text

Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment. In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis. When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation. Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分论点主题句 Topic Claim]:** *Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment.*
  对应步骤：S1 [分论点主题句 Topic Claim]：一句话清晰亮出本段唯一控制中心（[Policy/Factor X] strengthens [Outcome Y] by [Core Mechanism]）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [因果机制推演 Causal Elaboration]:** *In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis.*
  对应步骤：S2 [因果机制推演 Causal Elaboration]：Explain HOW and WHY：阐明如果没有这层干扰或有了这项机制，认知/经济/技术流程会发生什么具体改变。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具象微观案例 Concrete Micro-Case]:** *When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation.*

对应步骤：S3 [具象微观案例 Concrete Micro-Case]：For instance, when [Specific Group/Industry] [Specific Action], [Measurable/Observable Consequence]（绝不编造假百分比，用具体场景细节说话）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [反事实对比或扣题升华 Warrant / Link Back]:** *Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.*
  对应步骤：S4 [反事实对比或扣题升华 Warrant / Link Back]：Thus, [Summary Noun] shifts human effort from [Low-value Routine] toward [High-value Judgment]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 深层原因剖析段：表层诱因与结构性根源双层挖掘（Dual-Layer Causal Diagnosis）(BP35)

Expressive Intent: 我这一段想在问题分析类作文（Why is this happening?）中，剖析某个社会/科研痛点背后的深层原因（既指出直接表层诱因，又挖出底层的考核激励或结构性根源）。

Cohesion Formula: Dual-Cause Opening → Immediate Operational Driver → Deeper Institutional Root → Systemic Compounding. 绑定逻辑规则：L007, L005.

Publication-Grade Authentic Text

The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design. At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading. More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique. When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [原因总起句 Dual-Cause Thesis]:** *The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design.*
  对应步骤：S1 [原因总起句 Dual-Cause Thesis]：The widespread emergence of [Problem X] stems from both immediate technological pressures and deeper institutional incentives. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层:** *At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading.*
  对应步骤：S2 [第一层：直接操作/技术诱因 Immediate Driver]：At the operational level, [Factor A] lowers the cost of [Behavior], tempting individuals/organizations to [Short-term Action]. | 句首锚定旧

信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [第二层:** *More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique.*
  对应步骤：S3 [第二层：深层制度/激励根源 Structural Root Cause]：More fundamentally, existing evaluation systems reward [Superficial Metric] rather than [Long-term Quality], creating powerful pressure to prioritize speed over rigor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [双因叠加后果 Compounding Effect]:** *When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.*
  对应步骤：S4 [双因叠加后果 Compounding Effect]：When effortless generation meets quantity-driven evaluation, [Problem X] becomes systemic rather than isolated. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 问题诊断与靶向解决方案段：拒绝“政府应该立法”万金油（Targeted Problem-Solution）(BP37)

Expressive Intent: 我这一段想写 Problem-Solution 类作文的解决方案段：针对前文诊断出的两个具体根源，提出两项可操作、有明确执行主体和传导机制的靶向对策，拒绝只喊’The government should make laws and raise awareness’。

Cohesion Formula: Dual-Level Intervention → Actor 1 + Concrete Mechanism → Actor 2 + Concrete Standard → Balanced Closure. 绑定逻辑规则：L002, L010.

Publication-Grade Authentic Text

Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria. First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review. Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies. Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [对策总起句 Targeted Solution Thesis]:** *Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria.*
  对应步骤：S1 [对策总起句 Targeted Solution Thesis]：Mitigating [Problem X] effectively requires coordinated intervention across both [Level A: Technical/Curriculum Design] and [Level B: Institutional

Governance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [对策 1:** *First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review.*
  对应步骤：S2 [对策 1：操作层可验证改革 Actionable Measure 1]：First, [Specific Actor 1, e.g., university departments] should replace [Vulnerable Practice] with [Verifiable Mechanism, e.g., oral defense and in-class iterative drafting], ensuring that [Specific Effect 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对策 2:** *Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies.*
  对应步骤：S3 [对策 2：制度/标准层激励对齐 Actionable Measure 2]：Concurrently, [Specific Actor 2, e.g., regulatory agencies / industry consortia] must establish [Concrete Standard, e.g., mandatory provenance watermarking and audit logs] to penalize [Misconduct]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同闭环效果 Combined Outcome]:** *Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.*
  对应步骤：S4 [协同闭环效果 Combined Outcome]：Together, aligning pedagogical verification with technical traceability curbs misuse without stifling legitimate innovation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 议论文收尾段：跳出简单重复，完成核心权衡整合与边界升华（Synthesis Conclusion）(BP40)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Cohesion Formula: Balanced Restatment → Dual-Pillar Synthesis → Grounded Governance Principle (Zero Slogans!). 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [重申核心权衡与立场 Restate Nuanced Thesis]:** *In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.*
  对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [整合全文两大支柱机制 Synthesize Dual Pillars]:** *By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.*
  对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [理性边界展望（可选）Grounded Forward Look]:** *Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.*
  对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

21.5 FT20: 六级写作（CET-6）满分范文：科研评价体系中抵制数据造假与强化可复现性治理

Composition Architecture & Structural Blueprint Chain

Genre: CET6_Essay **Paragraph Count:** 5

Blueprint Chain: BP33 → BP35 → BP37 → BP39 → BP40

Paragraph 1: 议论文破题首段：具象趋势锚定 + 争议张力 + 明确中心论点（Zero-Template Hook & Thesis）(BP33)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉'With the rapid development of science and technology, more and more people...' 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Cohesion Formula: Concrete Phenomenon (No Cliché) → Balanced Tension → Conditional & Specific Thesis Statement. 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [具象现象锚定 Concrete Trend Anchor]:** *Large language models and automated code assistants have rapidly become routine study aids across university campuses.*
  对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心争议张力 Core Tension / Debate]:** *While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.*
  对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [带条件的中心论点 Nuanced Thesis Statement]:** *In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical*

skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 深层原因剖析段：表层诱因与结构性根源双层挖掘（Dual-Layer Causal Diagnosis）(BP35)

Expressive Intent: 我这一段想在问题分析类作文（Why is this happening?）中，剖析某个社会/科研痛点背后的深层原因（既指出直接表层诱因，又挖出底层的考核激励或结构性根源）。

Cohesion Formula: Dual-Cause Opening → Immediate Operational Driver → Deeper Institutional Root → Systemic Compounding. 绑定逻辑规则：L007, L005.

Publication-Grade Authentic Text

The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design. At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading. More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique. When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [原因总起句 Dual-Cause Thesis]:** *The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design.*

  对应步骤：S1 [原因总起句 Dual-Cause Thesis]：The widespread emergence of [Problem X] stems from both immediate technological pressures and deeper institutional incentives. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层:** *At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading.*

  对应步骤：S2 [第一层：直接操作/技术诱因 Immediate Driver]：At the operational level, [Factor A] lowers the cost of [Behavior], tempting individuals/organizations to [Short-term Action]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [第二层:** *More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique.*
  对应步骤：S3 [第二层：深层制度/激励根源 Structural Root Cause]：More fundamentally, existing evaluation systems reward [Superficial Metric] rather than [Long-term Quality], creating powerful pressure to prioritize speed over rigor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [双因叠加后果 Compounding Effect]:** *When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.*
  对应步骤：S4 [双因叠加后果 Compounding Effect]：When effortless generation meets quantity-driven evaluation, [Problem X] becomes systemic rather than isolated. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 问题诊断与靶向解决方案段：拒绝“政府应该立法”万金油（Targeted Problem-Solution）(BP37)

Expressive Intent: 我这一段想写 Problem-Solution 类作文的解决方案段：针对前文诊断出的两个具体根源，提出两项可操作、有明确执行主体和传导机制的靶向对策，拒绝只喊'The government should make laws and raise awareness'。

Cohesion Formula: Dual-Level Intervention → Actor 1 + Concrete Mechanism → Actor 2 + Concrete Standard → Balanced Closure. 绑定逻辑规则：L002, L010.

Publication-Grade Authentic Text

Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria. First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review. Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies. Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [对策总起句 Targeted Solution Thesis]:** *Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria.*
  对应步骤：S1 [对策总起句 Targeted Solution Thesis]：Mitigating [Problem X] effectively requires coordinated intervention across both [Level A: Technical/Curriculum Design] and [Level B: Institutional Governance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [对策 1:** *First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review.*
  对应步骤：S2 [对策 1：操作层可验证改革 Actionable Measure 1]：First, [Specific Actor 1, e.g., university departments] should replace [Vulnerable Practice] with [Verifiable Mechanism, e.g., oral defense and in-class iterative drafting], ensuring that [Specific Effect 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对策 2:** *Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies.*
  对应步骤：S3 [对策 2：制度/标准层激励对齐 Actionable Measure 2]：Concurrently, [Specific Actor 2, e.g., regulatory agencies / industry consortia] must establish [Concrete Standard, e.g., mandatory provenance watermarking and audit logs] to penalize [Misconduct]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [协同闭环效果 Combined Outcome]:** *Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.*
  对应步骤：S4 [协同闭环效果 Combined Outcome]：Together, aligning pedagogical verification with technical traceability curbs misuse without stifling legitimate innovation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 反事实推理段：如果不采取某项原则，系统将如何一步步滑向失控（Counterfactual Escalation）(BP39)

Expressive Intent: 我这一段想在议论文或科研立项书中，通过严密的“反事实推演（If we fail to... / Absent X...）”强化紧迫感与说服力。

Cohesion Formula: Positive Safeguard → Counterfactual Premise (Absent X...) → Cascading Breakdown → Precedent Confirmation. 绑定逻辑规则：L020, L003.

Publication-Grade Authentic Text

Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models. Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations. In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse. This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [正面基准命题 Positive Baseline]:** *Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models.*
  对应步骤：S1 [正面基准命题 Positive Baseline]：Maintaining rigorous [Principle/Standard X] serves as the primary safeguard against [Systemic Risk Y]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反事实假设切入 Counterfactual Premise]:** *Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations.*
  对应步骤：S2 [反事实假设切入 Counterfactual Premise]：Absent this safeguard—if [Actors] were to rely exclusively on [Unchecked Alternative]—[Immediate Vulnerability] would quickly emerge. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [连锁恶化传导 Cascading Deterioration]:** *In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse.*
  对应步骤：S3 [连锁恶化传导 Cascading Deterioration]：Once [Immediate Vulnerability] takes hold, [Secondary Consequence] compounds across cycles, eventually eroding [Core Foundation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证印证或结论锁定 Empirical Echo]:** *This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.*
  对应步骤：S4 [实证印证或结论锁定 Empirical Echo]：This degradation dynamic mirrors [Real-world/Technical Precedent], underscoring why [Principle X] cannot be compromised for short-term convenience. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 议论文收尾段：跳出简单重复，完成核心权衡整合与边界升华（Synthesis Conclusion）(BP40)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Cohesion Formula: Balanced Restatment → Dual-Pillar Synthesis → Grounded Governance Principle (Zero Slogans!). 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial

long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [重申核心权衡与立场 Restate Nuanced Thesis]:** *In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.*
  对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [整合全文两大支柱机制 Synthesize Dual Pillars]:** *By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.*
  对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [理性边界展望（可选）Grounded Forward Look]:** *Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.*
  对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Chapter 22

Humanities & Social Sciences / 人文社科与学术测评

本章精选该学科领域及对应文体的完整学术范文。每一篇范文不仅展示母语级发表标准的正文全貌，更清晰拆解全篇的“段落蓝图链条（Structural Blueprint Chain）”，并对每个自然段逐一剖析其核心交际意图、句际衔接逻辑公式（Cohesion Formula）以及每句话的句法成分与推进机制。

22.1 FT21: 雅思大作文（Task 2 Discuss Both Views）8.5 分范文：工业自动化机器人普及的就业冲击与技能转型

Composition Architecture & Structural Blueprint Chain

Genre: `IELTS_Task2_DiscussBoth` **Paragraph Count:** 5

Blueprint Chain: `BP33` → `BP34` → `BP36` → `BP37` → `BP40`

Paragraph 1: 议论文破题首段：具象趋势锚定 + 争议张力 + 明确中心论点（Zero-Template Hook & Thesis）(BP33)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉’With the rapid development of science and technology, more and more people...’ 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Cohesion Formula: Concrete Phenomenon (No Cliché) → Balanced Tension → Conditional & Specific Thesis Statement. 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students’ capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [具象现象锚定 Concrete Trend Anchor]:** *Large language models and automated code assistants have rapidly become routine study aids across university campuses.*

对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [核心争议张力 Core Tension / Debate]:** *While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.*
  对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [带条件的中心论点 Nuanced Thesis Statement]:** *In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.*
  对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 核心正方分论点段：主张 → 机制推演 → 具象行业微案例 → 扣题闭环（PEEL 满分段）(BP34)

Expressive Intent: 我这一段想写大作文的第一个主体段（Body Paragraph 1）：论证某个核心优势或必要性，做到有理有据、有具象场景支撑，绝不车轱辘话来回说。

Cohesion Formula: Topic Claim → Causal Mechanism → Concrete Micro-Case (L021) → Warrant Closure. 绑定逻辑规则：L006, L012, L021.

Publication-Grade Authentic Text

Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment. In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis. When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation. Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分论点主题句 Topic Claim]:** *Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment.*

对应步骤：S1 [分论点主题句 Topic Claim]：一句话清晰亮出本段唯一控制中心（[Policy/Factor X] strengthens [Outcome Y] by [Core Mechanism]）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [因果机制推演 Causal Elaboration]:** *In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis.*
  对应步骤：S2 [因果机制推演 Causal Elaboration]：Explain HOW and WHY：阐明如果没有这层干扰或有了这项机制，认知/经济/技术流程会发生什么具体改变。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具象微观案例 Concrete Micro-Case]:** *When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation.*
  对应步骤：S3 [具象微观案例 Concrete Micro-Case]：For instance, when [Specific Group/Industry] [Specific Action], [Measurable/Observable Consequence]（绝不编造假百分比，用具体场景细节说话）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [反事实对比或扣题升华 Warrant / Link Back]:** *Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.*
  对应步骤：S4 [反事实对比或扣题升华 Warrant / Link Back]：Thus, [Summary Noun] shifts human effort from [Low-value Routine] toward [High-value Judgment]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 让步-反驳段：优雅承认反方合理成分并强力扭转权重（Concession & Refutation）(BP36)

Expressive Intent: 我这一段想在议论文中回应反方观点（Some people argue that...）：先大度承认反方担忧在短期或局部确实存在，再用结构性证据或制度设计证明该问题完全可控，从而锁死我方立场。

Cohesion Formula: Admittedly (Valid Concern) → Short-term Reality → However (Zero-Sum Fallacy Pivot) → Historical Analogy & Policy Mitigation. 绑定逻辑规则：L008, L023.

Publication-Grade Authentic Text

Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers. During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules. However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles. Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling in-

dustrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [呈现反方最有力的担忧 Articulate Opposing Concern]:** *Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers.*
  对应步骤：S1 [呈现反方最有力的担忧 Articulate Opposing Concern]：Admittedly, critics raise a valid concern that [Opposing Argument, e.g., automation displaces entry-level clerical and manufacturing workers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [承认局部真实性 Acknowledge Partial Validity]:** *During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules.*
  对应步骤：S2 [承认局部真实性 Acknowledge Partial Validity]：In the short term, rapid deployment of [Technology/Policy] can indeed create transitional friction for workers whose routines are easily codified. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [釜底抽薪式转折 Structural Pivot]:** *However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles.*
  对应步骤：S3 [釜底抽薪式转折 Structural Pivot]：However, viewing [X] purely as a zero-sum threat overlooks how productivity gains generate entirely new categories of skilled employment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [历史/现实反证与制度补救 Evidence & Mitigation]:** *Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.*
  对应步骤：S4 [历史/现实反证与制度补救 Evidence & Mitigation]：Just as spreadsheet software reduced manual bookkeeping while expanding demand for financial analysts, pairing automation with targeted vocational retraining allows displaced workers to transition into system supervision and maintenance roles. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 问题诊断与靶向解决方案段：拒绝“政府应该立法”万金油（Targeted Problem-Solution）(BP37)

Expressive Intent: 我这一段想写 Problem-Solution 类作文的解决方案段：针对前文诊断出的两个具体根源，提出两项可操作、有明确执行主体和传导机制的靶向对策，拒绝只喊’The government should make laws and raise awareness’。

Cohesion Formula: Dual-Level Intervention → Actor 1 + Concrete Mechanism → Actor 2 + Concrete Standard → Balanced Closure. 绑定逻辑规则：L002, L010.

Publication-Grade Authentic Text

Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria. First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review. Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies. Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [对策总起句 Targeted Solution Thesis]:** *Curbing the proliferation of fabricated academic data requires coordinated reforms across both peer-review protocols and institutional promotion criteria.*
  对应步骤：S1 [对策总起句 Targeted Solution Thesis]：Mitigating [Problem X] effectively requires coordinated intervention across both [Level A: Technical/Curriculum Design] and [Level B: Institutional Governance]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [对策 1:** *First, journal editorial boards should mandate that empirical submissions include executable code containers and raw sensor logs, enabling automated reproducibility checks before external review.*
  对应步骤：S2 [对策 1：操作层可验证改革 Actionable Measure 1]：First, [Specific Actor 1, e.g., university departments] should replace [Vulnerable Practice] with [Verifiable Mechanism, e.g., oral defense and in-class iterative drafting], ensuring that [Specific Effect 1]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [对策 2:** *Concurrently, university tenure committees must shift evaluation weight from raw publication counts toward verified open-source contributions and replication studies.*
  对应步骤：S3 [对策 2：制度/标准层激励对齐 Actionable Measure 2]：Concurrently, [Specific Actor 2, e.g., regulatory agencies / industry consortia] must establish [Concrete Standard, e.g., mandatory provenance watermarking and audit logs] to penalize [Misconduct]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [协同闭环效果 Combined Outcome]:** *Aligning technical verification with career incentives removes the underlying pressure to fabricate incremental results.*
  对应步骤：S4 [协同闭环效果 Combined Outcome]：Together, aligning pedagogical verification with technical traceability curbs misuse without stifling legitimate innovation. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 议论文收尾段：跳出简单重复，完成核心权衡整合与边界升华（Synthesis Conclusion）(BP40)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Cohesion Formula: Balanced Restatment → Dual-Pillar Synthesis → Grounded Governance Principle (Zero Slogans!). 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [重申核心权衡与立场 Restate Nuanced Thesis]:** *In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.*
  对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [整合全文两大支柱机制 Synthesize Dual Pillars]:** *By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.*
  对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [理性边界展望（可选）Grounded Forward Look]:** *Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.*
  对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

22.2 FT22: 雅思大作文（Task 2 To What Extent Agree）8.5 分范文：开源大模型权重的科学透明收益是否远超其滥用风险

Composition Architecture & Structural Blueprint Chain

Genre: `IELTS_Task2_AgreeDisagree` **Paragraph Count:** 5

Blueprint Chain: `BP33` → `BP38` → `BP36` → `BP39` → `BP40`

Paragraph 1: 议论文破题首段：具象趋势锚定 + 争议张力 + 明确中心论点（Zero-Template Hook & Thesis）(BP33)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉'With the rapid development of science and technology, more and more people...' 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Cohesion Formula: Concrete Phenomenon (No Cliché) → Balanced Tension → Conditional & Specific Thesis Statement. 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [具象现象锚定 Concrete Trend Anchor]:** *Large language models and automated code assistants have rapidly become routine study aids across university campuses.*
  对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心争议张力 Core Tension / Debate]:** *While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.*
  对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [带条件的中心论点 Nuanced Thesis Statement]:** *In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical*

skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 利弊权衡段：证明结构性长期收益远超短期可管可控成本（Outweigh / Trade-off Analysis）(BP38)

Expressive Intent: 我这一段想回应'Do the advantages outweigh the disadvantages?' 题型：把弊端定性为"暂时的、可通过技术/规则收敛的局部成本"，把优势定性为"不可替代的结构性长期突破"。

Cohesion Formula: Asymmetry Thesis → Bound the Drawback (Temporary & Mitigable) → Highlight Compounding Advantage (Irreplaceable) → Rational Verdict. 绑定逻辑规则：L008, L023.

Publication-Grade Authentic Text

Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks. While 公开 releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters. By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics. Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [权衡定性主题句 Asymmetry Thesis]:** *Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks.*

  对应步骤：S1 [权衡定性主题句 Asymmetry Thesis]：Weighing both dimensions reveals that the long-term structural benefits of [X] far outweigh its manageable transitional costs. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [将弊端限定为可收敛成本 Bounding the Drawback]:** *While* 公开 *releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters.*

  对应步骤：S2 [将弊端限定为可收敛成本 Bounding the Drawback]：While [Drawback A, e.g., initial capital outlay / training overhead] poses a tangible hurdle, this cost is bounded and diminishes rapidly as [Mitigating Factor, e.g., hardware amortization / open-source standardization] matures. | 句首锚定旧信

息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]:** *By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics.*
  对应步骤：S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]：By contrast, the core advantage—[Benefit B, e.g., eliminating human exposure to hazardous environments / discovering non-linear scientific laws]—delivers compounding value that manual methods cannot replicate at any cost. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [权衡结论 Trade-off Verdict]:** *Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.*
  对应步骤：S4 [权衡结论 Trade-off Verdict]：Therefore, addressing the friction through targeted safeguards is far more rational than forgoing the transformative gain. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 让步-反驳段：优雅承认反方合理成分并强力扭转权重（Concession & Refutation）(BP36)

Expressive Intent: 我这一段想在议论文中回应反方观点（Some people argue that...）：先大度承认反方担忧在短期或局部确实存在，再用结构性证据或制度设计证明该问题完全可控，从而锁死我方立场。

Cohesion Formula: Admittedly (Valid Concern) → Short-term Reality → However (Zero-Sum Fallacy Pivot) → Historical Analogy & Policy Mitigation. 绑定逻辑规则：L008, L023.

Publication-Grade Authentic Text

Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers. During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules. However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles. Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [呈现反方最有力的担忧 Articulate Opposing Concern]:** *Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers.*
  对应步骤：S1 [呈现反方最有力的担忧 Articulate Opposing Concern]：Admittedly, critics raise a valid concern that [Opposing Argument, e.g., automation displaces entry-level clerical and manufacturing workers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [承认局部真实性 Acknowledge Partial Validity]:** *During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules.*
  对应步骤：S2 [承认局部真实性 Acknowledge Partial Validity]：In the short term, rapid deployment of [Technology/Policy] can indeed create transitional friction for workers whose routines are easily codified. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [釜底抽薪式转折 Structural Pivot]:** *However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles.*
  对应步骤：S3 [釜底抽薪式转折 Structural Pivot]：However, viewing [X] purely as a zero-sum threat overlooks how productivity gains generate entirely new categories of skilled employment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [历史/现实反证与制度补救 Evidence & Mitigation]:** *Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.*
  对应步骤：S4 [历史/现实反证与制度补救 Evidence & Mitigation]：Just as spreadsheet software reduced manual bookkeeping while expanding demand for financial analysts, pairing automation with targeted vocational retraining allows displaced workers to transition into system supervision and maintenance roles. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 反事实推理段：如果不采取某项原则，系统将如何一步步滑向失控（Counterfactual Escalation）(BP39)

Expressive Intent: 我这一段想在议论文或科研立项书中，通过严密的“反事实推演（If we fail to... / Absent X...）”强化紧迫感与说服力。

Cohesion Formula: Positive Safeguard → Counterfactual Premise (Absent X...) → Cascading Breakdown → Precedent Confirmation. 绑定逻辑规则：L020, L003.

Publication-Grade Authentic Text

Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models. Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations. In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse. This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [正面基准命题 Positive Baseline]:** *Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models.*
  对应步骤：S1 [正面基准命题 Positive Baseline]：Maintaining rigorous [Principle/Standard X] serves as the primary safeguard against [Systemic Risk Y]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反事实假设切入 Counterfactual Premise]:** *Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations.*
  对应步骤：S2 [反事实假设切入 Counterfactual Premise]：Absent this safeguard—if [Actors] were to rely exclusively on [Unchecked Alternative]—[Immediate Vulnerability] would quickly emerge. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [连锁恶化传导 Cascading Deterioration]:** *In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse.*
  对应步骤：S3 [连锁恶化传导 Cascading Deterioration]：Once [Immediate Vulnerability] takes hold, [Secondary Consequence] compounds across cycles, eventually eroding [Core Foundation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证印证或结论锁定 Empirical Echo]:** *This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.*
  对应步骤：S4 [实证印证或结论锁定 Empirical Echo]：This degradation dynamic mirrors [Real-world/Technical Precedent], underscoring why [Principle X] cannot be compromised for short-term convenience. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 议论文收尾段：跳出简单重复，完成核心权衡整合与边界升华（Synthesis Conclusion）(BP40)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Cohesion Formula: Balanced Restatment → Dual-Pillar Synthesis → Grounded Governance Principle (Zero Slogans!). 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [重申核心权衡与立场 Restate Nuanced Thesis]:** *In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.*
  对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [整合全文两大支柱机制 Synthesize Dual Pillars]:** *By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.*
  对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [理性边界展望（可选）Grounded Forward Look]:** *Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.*
  对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

22.3 FT23: 托福独立写作满分范文：计算机高等教育应孤立偏向工程编码还是将数学基础与项目实践深度融合

Composition Architecture & Structural Blueprint Chain

Genre: TOEFL_Independent **Paragraph Count:** 5

Blueprint Chain: BP33 → BP34 → BP36 → BP39 → BP40

Paragraph 1: 议论文破题首段：具象趋势锚定 + 争议张力 + 明确中心论点（Zero-Template Hook & Thesis）(BP33)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉'With the rapid development of science and technology, more and more people...' 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Cohesion Formula: Concrete Phenomenon (No Cliché) → Balanced Tension → Conditional & Specific Thesis Statement. 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [具象现象锚定 Concrete Trend Anchor]:** *Large language models and automated code assistants have rapidly become routine study aids across university campuses.*
  对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心争议张力 Core Tension / Debate]:** *While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.*
  对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [带条件的中心论点 Nuanced Thesis Statement]:** *In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical*

skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 核心正方分论点段：主张 → 机制推演 → 具象行业微案例 → 扣题闭环（PEEL 满分段）(BP34)

Expressive Intent: 我这一段想写大作文的第一个主体段（Body Paragraph 1）：论证某个核心优势或必要性，做到有理有据、有具象场景支撑，绝不车轱辘话来回说。

Cohesion Formula: Topic Claim → Causal Mechanism → Concrete Micro-Case (L021) → Warrant Closure. 绑定逻辑规则：L006, L012, L021.

Publication-Grade Authentic Text

Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment. In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis. When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation. Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [分论点主题句 Topic Claim]:** *Delegating repetitive data compilation to automated systems enables professionals to concentrate on high-level diagnostic judgment.*
  对应步骤：S1 [分论点主题句 Topic Claim]：一句话清晰亮出本段唯一控制中心（[Policy/Factor X] strengthens [Outcome Y] by [Core Mechanism]）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [因果机制推演 Causal Elaboration]:** *In data-intensive fields such as clinical medicine and structural engineering, practitioners traditionally spend hours cross-checking routine telemetry logs before formulating a hypothesis.*
  对应步骤：S2 [因果机制推演 Causal Elaboration]：Explain HOW and WHY：阐明如果没有这层干扰或有了这项机制，认知/经济/技术流程会发生什么具体改变。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [具象微观案例 Concrete Micro-Case]:** *When hospital radiology departments deploy automated screening algorithms to flag normal chest scans, physicians can redirect their cognitive bandwidth toward ambiguous pathology cases and patient consultation.*

对应步骤：S3 [具象微观案例 Concrete Micro-Case]：For instance, when [Specific Group/Industry] [Specific Action], [Measurable/Observable Consequence]（绝不编造假百分比，用具体场景细节说话）。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S4 [反事实对比或扣题升华 Warrant / Link Back]:** *Consequently, intelligent triage elevates both diagnostic throughput and clinical care quality.*
  对应步骤：S4 [反事实对比或扣题升华 Warrant / Link Back]：Thus, [Summary Noun] shifts human effort from [Low-value Routine] toward [High-value Judgment]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 让步-反驳段：优雅承认反方合理成分并强力扭转权重（Concession & Refutation）(BP36)

Expressive Intent: 我这一段想在议论文中回应反方观点（Some people argue that...）：先大度承认反方担忧在短期或局部确实存在，再用结构性证据或制度设计证明该问题完全可控，从而锁死我方立场。

Cohesion Formula: Admittedly (Valid Concern) → Short-term Reality → However (Zero-Sum Fallacy Pivot) → Historical Analogy & Policy Mitigation. 绑定逻辑规则：L008, L023.

Publication-Grade Authentic Text

Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers. During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules. However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles. Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [呈现反方最有力的担忧 Articulate Opposing Concern]:** *Admittedly, critics raise a legitimate concern that autonomous robotics and generative software can displace routine clerical and assembly-line workers.*
  对应步骤：S1 [呈现反方最有力的担忧 Articulate Opposing Concern]：Admittedly, critics raise a valid concern that [Opposing Argument, e.g., automation displaces entry-level clerical and manufacturing workers]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [承认局部真实性 Acknowledge Partial Validity]:** *During initial transition phases, rapid automation does create localized employment friction for personnel whose tasks follow rigid, codifiable rules.*
  对应步骤：S2 [承认局部真实性 Acknowledge Partial Validity]：In the short term, rapid deployment of [Technology/Policy] can indeed create transitional friction for workers whose routines are easily codified. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [釜底抽薪式转折 Structural Pivot]:** *However, treating technological upgrading as a fixed-pie zero-sum game overlooks how productivity gains expand downstream service and engineering roles.*
  对应步骤：S3 [釜底抽薪式转折 Structural Pivot]：However, viewing [X] purely as a zero-sum threat overlooks how productivity gains generate entirely new categories of skilled employment. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [历史/现实反证与制度补救 Evidence & Mitigation]:** *Much as spreadsheet software eliminated manual ledger keeping while multiplying demand for financial data analysts, coupling industrial automation with subsidized reskilling programs enables displaced workers to transition into robot fleet supervision, calibration, and exception handling.*
  对应步骤：S4 [历史/现实反证与制度补救 Evidence & Mitigation]：Just as spreadsheet software reduced manual bookkeeping while expanding demand for financial analysts, pairing automation with targeted vocational retraining allows displaced workers to transition into system supervision and maintenance roles. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 反事实推理段：如果不采取某项原则，系统将如何一步步滑向失控（Counterfactual Escalation）(BP39)

Expressive Intent: 我这一段想在议论文或科研立项书中，通过严密的“反事实推演（If we fail to... / Absent X...）”强化紧迫感与说服力。

Cohesion Formula: Positive Safeguard → Counterfactual Premise (Absent X...) → Cascading Breakdown → Precedent Confirmation. 绑定逻辑规则：L020, L003.

Publication-Grade Authentic Text

Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models. Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations. In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse. This degenerative feedback loop—known empirically as model autophagy—underscores

why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [正面基准命题 Positive Baseline]:** *Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models.*
  对应步骤：S1 [正面基准命题 Positive Baseline]：Maintaining rigorous [Principle/Standard X] serves as the primary safeguard against [Systemic Risk Y]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [反事实假设切入 Counterfactual Premise]:** *Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations.*
  对应步骤：S2 [反事实假设切入 Counterfactual Premise]：Absent this safeguard—if [Actors] were to rely exclusively on [Unchecked Alternative]—[Immediate Vulnerability] would quickly emerge. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [连锁恶化传导 Cascading Deterioration]:** *In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse.*
  对应步骤：S3 [连锁恶化传导 Cascading Deterioration]：Once [Immediate Vulnerability] takes hold, [Secondary Consequence] compounds across cycles, eventually eroding [Core Foundation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证印证或结论锁定 Empirical Echo]:** *This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.*
  对应步骤：S4 [实证印证或结论锁定 Empirical Echo]：This degradation dynamic mirrors [Real-world/Technical Precedent], underscoring why [Principle X] cannot be compromised for short-term convenience. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 议论文收尾段：跳出简单重复，完成核心权衡整合与边界升华（Synthesis Conclusion）(BP40)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Cohesion Formula: Balanced Restatment → Dual-Pillar Synthesis → Grounded Governance Principle (Zero Slogans!). 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [重申核心权衡与立场 Restate Nuanced Thesis]:** *In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.*
  对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [整合全文两大支柱机制 Synthesize Dual Pillars]:** *By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.*
  对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [理性边界展望（可选）Grounded Forward Look]:** *Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.*
  对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

22.4 FT24: 托福独立写作满分范文：生成模型训练中坚守真实人类语料锚点与防止模型自噬（Model Autophagy）

Composition Architecture & Structural Blueprint Chain

Genre: TOEFL_Independent **Paragraph Count:** 5

Blueprint Chain: BP33 → BP35 → BP39 → BP38 → BP40

Paragraph 1: 议论文破题首段：具象趋势锚定 + 争议张力 + 明确中心论点（Zero-Template Hook & Thesis）(BP33)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的第一段（Introduction）：彻底扔掉'With the rapid development of science and technology, more and more people...' 等烂大街模板，用 3 句话写出考官眼前一亮的母语级破题段。

Cohesion Formula: Concrete Phenomenon (No Cliché) → Balanced Tension → Conditional & Specific Thesis Statement. 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

Large language models and automated code assistants have rapidly become routine study aids across university campuses. While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation. In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [具象现象锚定 Concrete Trend Anchor]:** *Large language models and automated code assistants have rapidly become routine study aids across university campuses.*
  对应步骤：S1 [具象现象锚定 Concrete Trend Anchor]：直接从具体技术变革或制度调整切入（如 Generative AI tools now draft code and summarize literature across university campuses），绝不用 With the development of...。| 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [核心争议张力 Core Tension / Debate]:** *While proponents contend that these tools accelerate literature synthesis and lower programming barriers, critics warn that unchecked reliance erodes students' capacity for independent problem formulation.*
  对应步骤：S2 [核心争议张力 Core Tension / Debate]：While proponents argue that [View A: efficiency/accessibility], critics warn that [View B: skill erosion/inequality]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [带条件的中心论点 Nuanced Thesis Statement]:** *In my view, universities should integrate generative tools into advanced coursework only after students master foundational analytical*

skills, as premature delegation weakens debugging competence and obscures underlying mathematical principles.

对应步骤：S3 [带条件的中心论点 Nuanced Thesis Statement]：In my view, rather than banning [X] outright or adopting it uncritically, [Institution/Society] should [Core Stance], because [Primary Reason 1] and [Primary Reason 2]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 2: 深层原因剖析段：表层诱因与结构性根源双层挖掘（Dual-Layer Causal Diagnosis）(BP35)

Expressive Intent: 我这一段想在问题分析类作文（Why is this happening?）中，剖析某个社会/科研痛点背后的深层原因（既指出直接表层诱因，又挖出底层的考核激励或结构性根源）。

Cohesion Formula: Dual-Cause Opening → Immediate Operational Driver → Deeper Institutional Root → Systemic Compounding. 绑定逻辑规则：L007, L005.

Publication-Grade Authentic Text

The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design. At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading. More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique. When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [原因总起句 Dual-Cause Thesis]:** *The growing tendency among undergraduates to submit unverified AI-generated essays stems from both lowered friction costs and flawed assessment design.*

  对应步骤：S1 [原因总起句 Dual-Cause Thesis]：The widespread emergence of [Problem X] stems from both immediate technological pressures and deeper institutional incentives. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [第一层:** *At an immediate level, conversational models produce grammatically polished drafts within seconds, tempting time-pressed students to bypass the arduous process of outlining and primary reading.*

  对应步骤：S2 [第一层：直接操作/技术诱因 Immediate Driver]：At the operational level, [Factor A] lowers the cost of [Behavior], tempting individuals/organizations to [Short-term Action]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S3 [第二层:** *More fundamentally, many university courses still rely on generic, formulaic essay prompts that reward surface fluency rather than original empirical critique.*
  对应步骤：S3 [第二层：深层制度/激励根源 Structural Root Cause]：More fundamentally, existing evaluation systems reward [Superficial Metric] rather than [Long-term Quality], creating powerful pressure to prioritize speed over rigor. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [双因叠加后果 Compounding Effect]:** *When effortless text generation meets standardized grading rubrics that fail to penalize hollow reasoning, superficial compliance inevitably crowds out deep inquiry.*
  对应步骤：S4 [双因叠加后果 Compounding Effect]：When effortless generation meets quantity-driven evaluation, [Problem X] becomes systemic rather than isolated. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 3: 反事实推理段：如果不采取某项原则，系统将如何一步步滑向失控（Counterfactual Escalation）(BP39)

Expressive Intent: 我这一段想在议论文或科研立项书中，通过严密的“反事实推演（If we fail to... / Absent X...）”强化紧迫感与说服力。

Cohesion Formula: Positive Safeguard → Counterfactual Premise (Absent X...) → Cascading Breakdown → Precedent Confirmation. 绑定逻辑规则：L020, L003.

Publication-Grade Authentic Text

Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models. Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations. In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse. This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [正面基准命题 Positive Baseline]:** *Preserving verified human-authored corpora serves as an indispensable anchor for training successive generations of generative models.*
  对应步骤：S1 [正面基准命题 Positive Baseline]：Maintaining rigorous [Principle/Standard X] serves as the primary safeguard against [Systemic Risk Y]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

- **S2 [反事实假设切入 Counterfactual Premise]:** *Absent this human data anchor—if future models were trained recursively on synthetic outputs generated by their predecessors—subtle approximation errors would compound across iterations.*
  对应步骤：S2 [反事实假设切入 Counterfactual Premise]：Absent this safeguard—if [Actors] were to rely exclusively on [Unchecked Alternative]—[Immediate Vulnerability] would quickly emerge. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [连锁恶化传导 Cascading Deterioration]:** *In each recursive cycle, low-probability tail events and nuanced stylistic variations are progressively truncated, driving the learned distribution toward high-density mode collapse.*
  对应步骤：S3 [连锁恶化传导 Cascading Deterioration]：Once [Immediate Vulnerability] takes hold, [Secondary Consequence] compounds across cycles, eventually eroding [Core Foundation]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [实证印证或结论锁定 Empirical Echo]:** *This degenerative feedback loop—known empirically as model autophagy—underscores why synthetic data augmentation can complement, but never wholly replace, curated real-world observations.*
  对应步骤：S4 [实证印证或结论锁定 Empirical Echo]：This degradation dynamic mirrors [Real-world/Technical Precedent], underscoring why [Principle X] cannot be compromised for short-term convenience. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 4: 利弊权衡段：证明结构性长期收益远超短期可管可控成本（Outweigh / Trade-off Analysis）(BP38)

Expressive Intent: 我这一段想回应'Do the advantages outweigh the disadvantages?' 题型：把弊端定性为“暂时的、可通过技术/规则收敛的局部成本”，把优势定性为“不可替代的结构性长期突破”。

Cohesion Formula: Asymmetry Thesis → Bound the Drawback (Temporary & Mitigable) → Highlight Compounding Advantage (Irreplaceable) → Rational Verdict. 绑定逻辑规则：L008, L023.

Publication-Grade Authentic Text

Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks. While 公开 releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters. By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics. Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [权衡定性主题句 Asymmetry Thesis]:** *Weighing both dimensions demonstrates that the structural scientific gains of open-weight foundation models far outweigh their manageable misuse risks.*
  对应步骤：S1 [权衡定性主题句 Asymmetry Thesis]：Weighing both dimensions reveals that the long-term structural benefits of [X] far outweigh its manageable transitional costs. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [将弊端限定为可收敛成本 Bounding the Drawback]:** *While* 公开 *releasing model weights lowers the barrier for generating deceptive phishing content, such downstream abuse can be substantially contained through cryptographic provenance standards and endpoint spam filters.*
  对应步骤：S2 [将弊端限定为可收敛成本 Bounding the Drawback]：While [Drawback A, e.g., initial capital outlay / training overhead] poses a tangible hurdle, this cost is bounded and diminishes rapidly as [Mitigating Factor, e.g., hardware amortization / open-source standardization] matures. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]:** *By contrast, open weights provide academic laboratories and resource-constrained hospitals with transparent, auditable architectures that proprietary black-box APIs cannot substitute—accelerating reproducible breakthroughs in protein folding and rare-disease diagnostics.*
  对应步骤：S3 [凸显优势的不可替代性 Highlighting Irreversible Gain]：By contrast, the core advantage—[Benefit B, e.g., eliminating human exposure to hazardous environments / discovering non-linear scientific laws]—delivers compounding value that manual methods cannot replicate at any cost. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S4 [权衡结论 Trade-off Verdict]:** *Because the security risks are mitigable via defense-in-depth whereas the scientific transparency is irreplaceable, open research ecosystems yield a decisive net benefit.*
  对应步骤：S4 [权衡结论 Trade-off Verdict]：Therefore, addressing the friction through targeted safeguards is far more rational than forgoing the transformative gain. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。

Paragraph 5: 议论文收尾段：跳出简单重复，完成核心权衡整合与边界升华（Synthesis Conclusion）(BP40)

Expressive Intent: 我这一段想写六级、雅思 Task 2 或托福独立写作的结尾段（Conclusion）：用 2–3 句话收束全文，做到既呼应中心论点、整合正反权衡，又绝不喊假大空口号（Let us work together for a better tomorrow）。

Cohesion Formula: Balanced Restatment → Dual-Pillar Synthesis → Grounded Governance Principle (Zero Slogans!). 绑定逻辑规则：L023, L029.

Publication-Grade Authentic Text

In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains. By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight. Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.

Rhetorical Move Breakdown & Constituent Analysis

- **S1 [重申核心权衡与立场 Restate Nuanced Thesis]:** *In conclusion, while automated decision systems inevitably introduce transitional labor friction and verification challenges, integrating them into industrial and academic workflows yields substantial long-term productivity gains.*
  对应步骤：S1 [重申核心权衡与立场 Restate Nuanced Thesis]：In conclusion, while [Acknowledged Factor A] introduces real challenges in [Specific Aspect], [Core Thesis Stance] remains the most viable path forward. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S2 [整合全文两大支柱机制 Synthesize Dual Pillars]:** *By pairing algorithmic deployment with transparent audit trails and targeted workforce retraining, institutions can capture efficiency breakthroughs without eroding human analytical oversight.*
  对应步骤：S2 [整合全文两大支柱机制 Synthesize Dual Pillars]：By combining [Pillar 1 from Body 1] with [Pillar 2 / Safeguard from Body 2], [Actors] can harness [Core Benefit] while containing [Primary Risk]. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。
- **S3 [理性边界展望（可选）Grounded Forward Look]:** *Ultimately, the value of intelligent automation rests not on replacing human judgment, but on elevating the precision of the evidence upon which human judgment acts.*
  对应步骤：S3 [理性边界展望（可选）Grounded Forward Look]：Ultimately, the success of [Technology/Policy] depends not on its standalone capability, but on the institutional rigor with which it is governed. | 句首锚定旧信息/逻辑视角，主干动词紧跟主语，复杂修饰置于宾语之后（符合尾重原则 L011 与旧新信息流 L001）。